

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```

def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:

```



```

        dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)

    return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}

```

```

    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_rabbit, id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_rabbit, id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test

- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

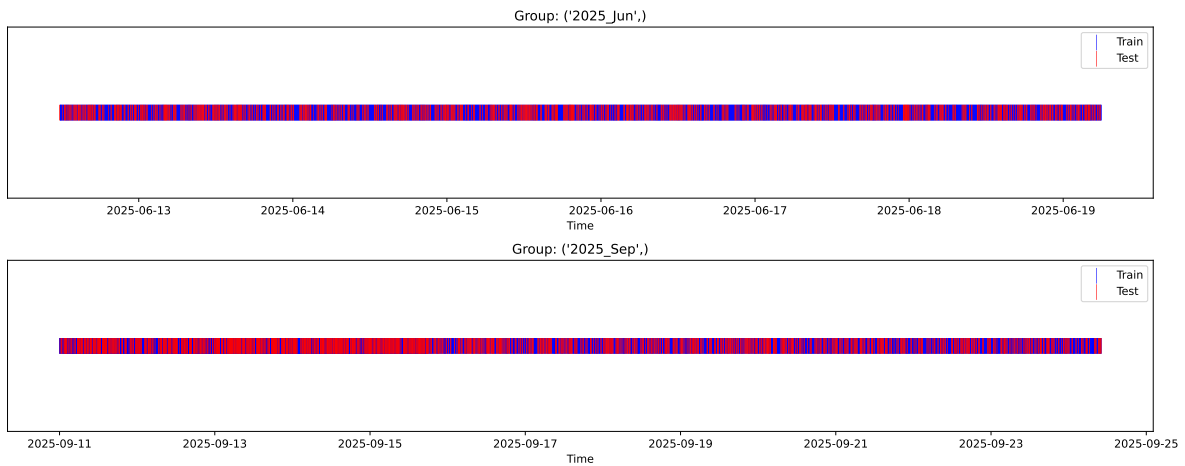
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

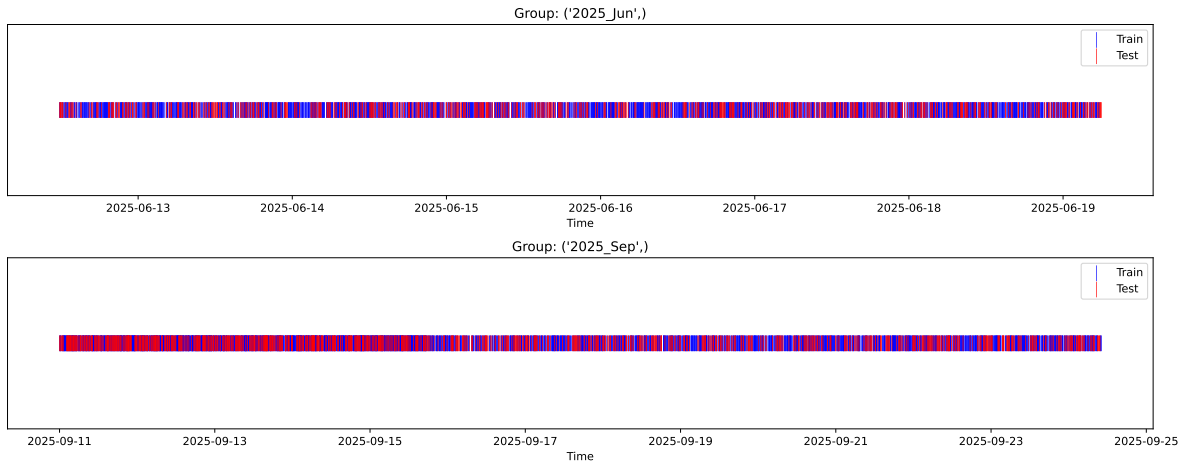
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

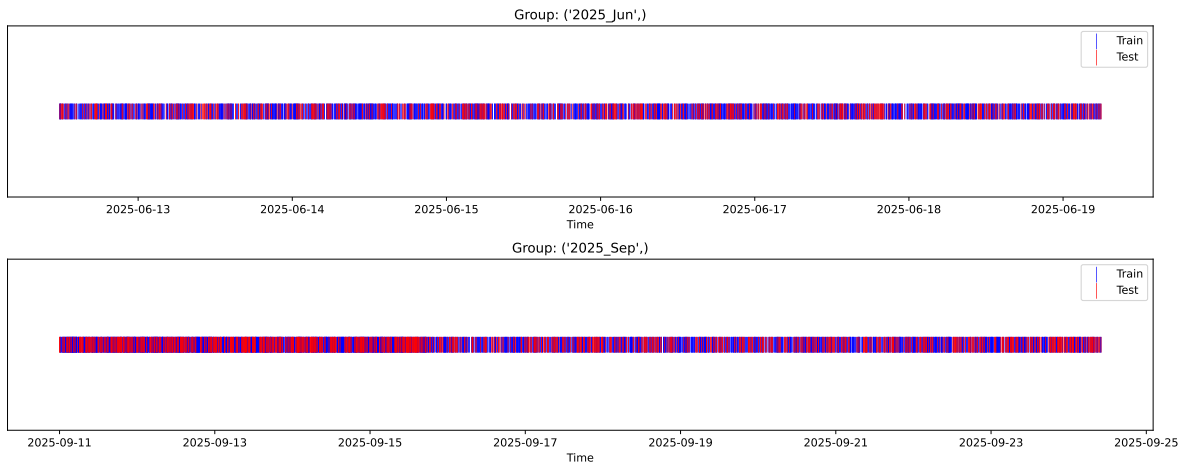
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



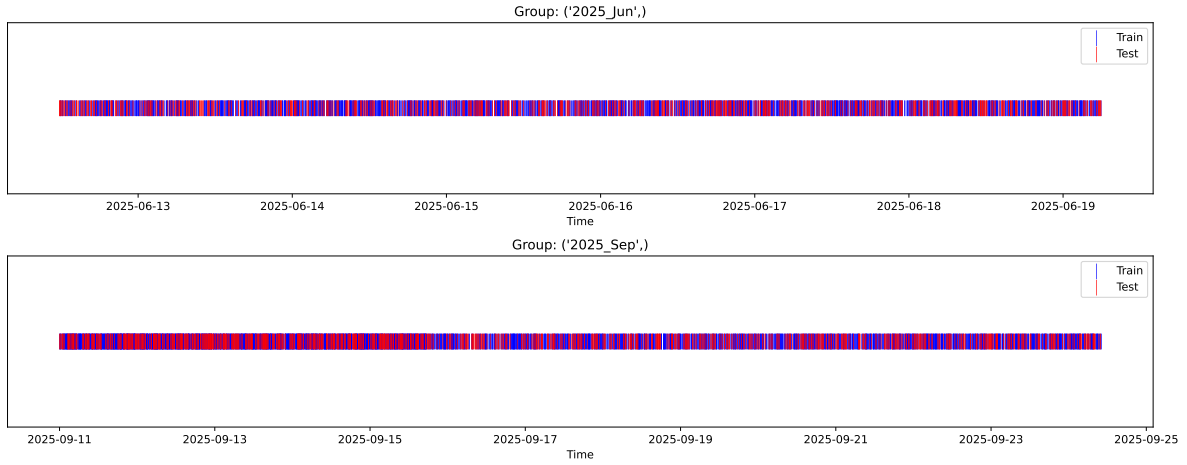
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



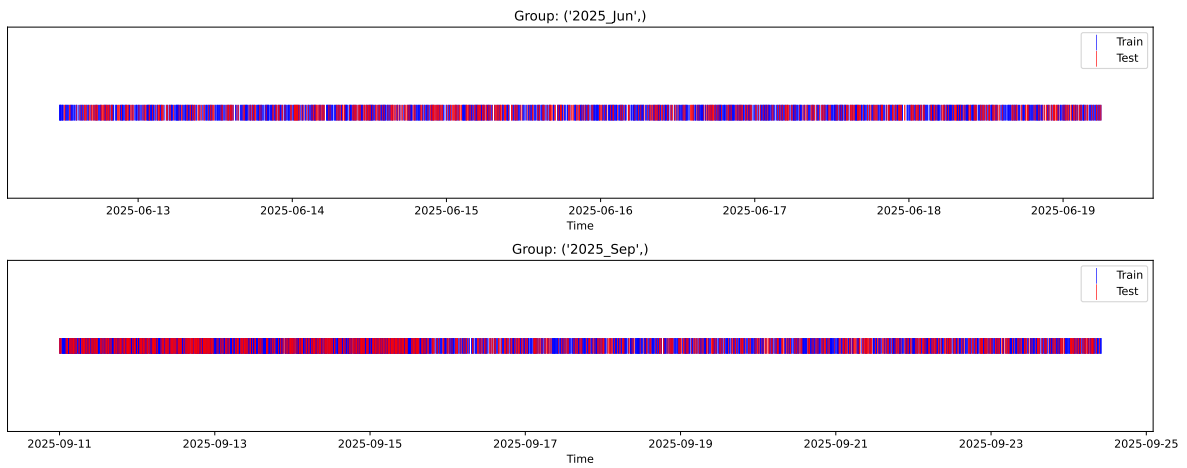
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



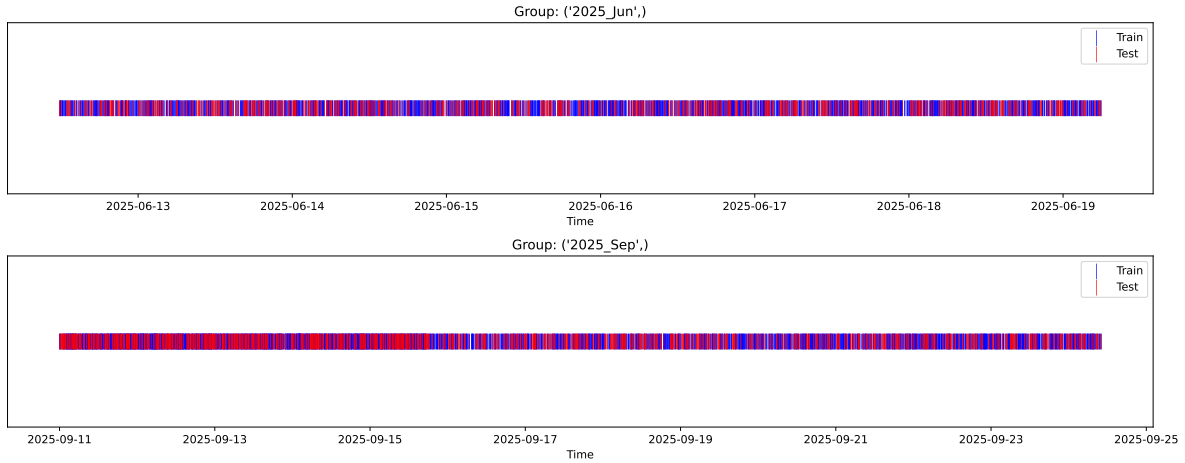
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 1_3

Len(group) : 795, Len(training) : 557, Len(test) : 238

Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

Fold 0 — y_test: mean=2.57, std=0.62, y_train: mean=2.61, std=0.56

Fold 1 — y_test: mean=2.53, std=0.55, y_train: mean=2.63, std=0.59

Fold 2 — y_test: mean=2.65, std=0.55, y_train: mean=2.58, std=0.59

Fold 3 — y_test: mean=2.64, std=0.59, y_train: mean=2.58, std=0.57

Fold 4 — y_test: mean=2.56, std=0.54, y_train: mean=2.61, std=0.59

train scores CV: [0.56730715 0.5334636 0.56784944 0.59353423 0.55417991]

test scores CV: [0.5431311 0.6179077 0.53621024 0.4830173 0.57408525]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5633	0.5509	0.0445	0.4979	0.5003	-0.0505	0.9082	1.0225

Coefficients:

$\text{coef_r_CO2} = -0.0079$

$\text{coef_r_NH3} = 0.0524$

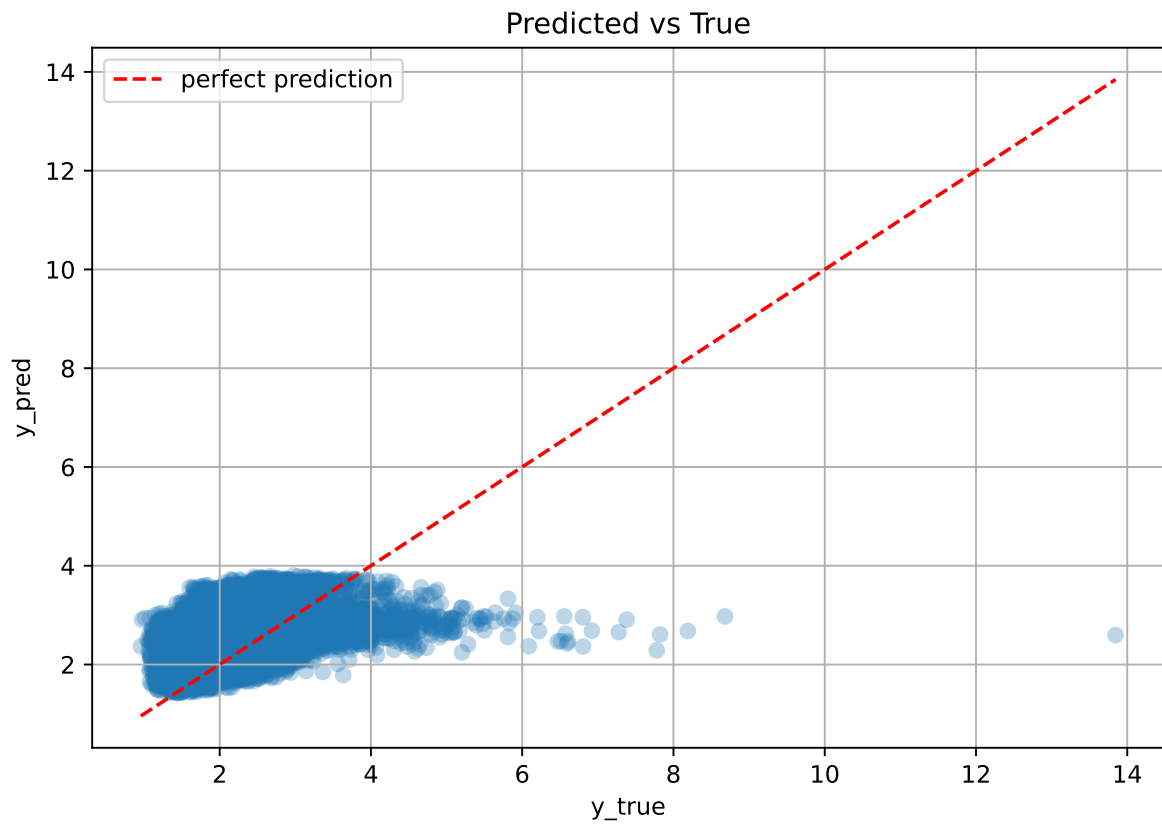
$\text{coef_r_THI} = 0.5946$

$\text{coef_r_temperature} = -0.3581$

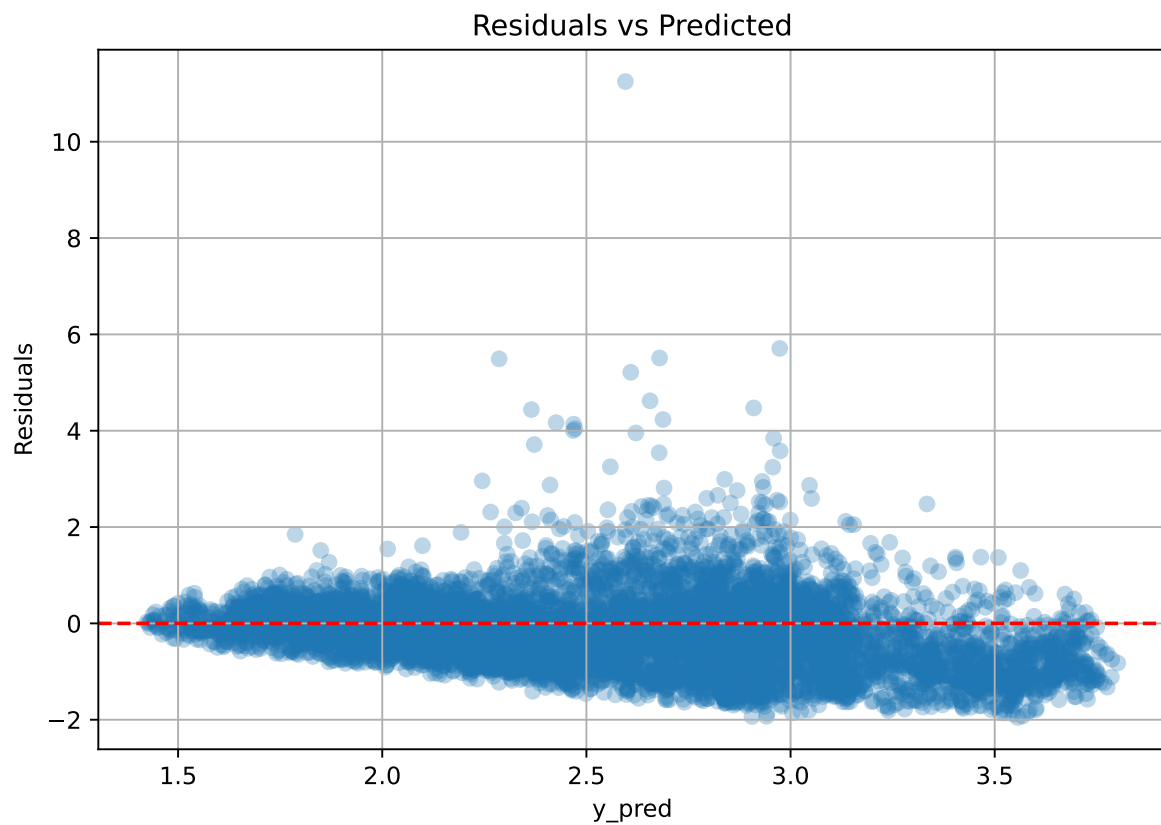
$\text{coef_r_umidity} = -0.2680$

$\text{intercept} = 2.5975$

2.3.2 Scatter



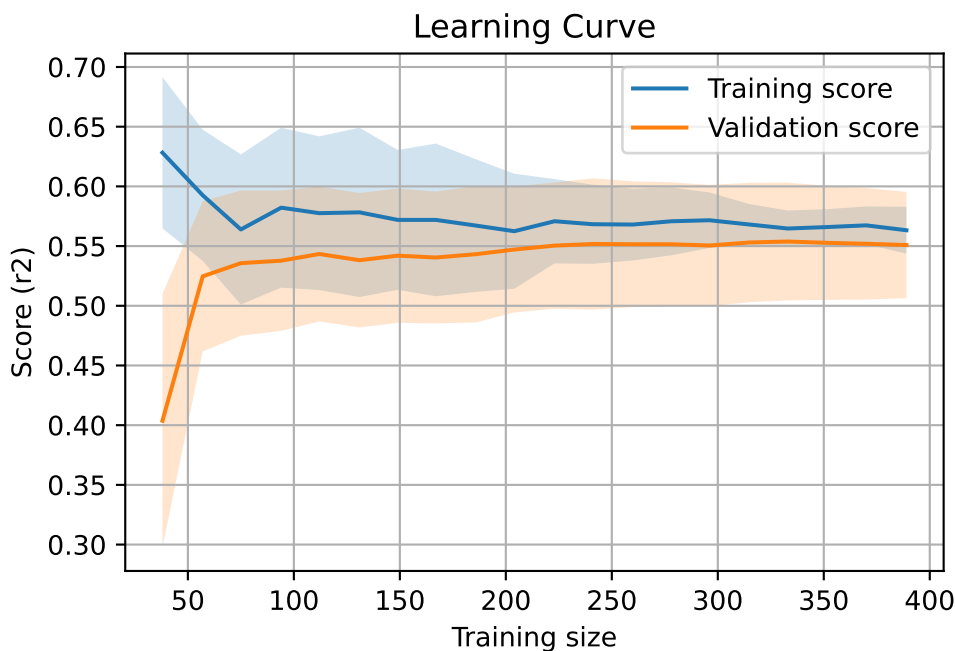
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 1_3

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238



2.3.5 Gridsearch — Group: 2_11

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

Fold 0 — y_test: mean=2.58, std=0.86, y_train: mean=2.52, std=0.86

Fold 1 — y_test: mean=2.56, std=0.84, y_train: mean=2.53, std=0.87

Fold 2 — y_test: mean=2.50, std=0.84, y_train: mean=2.55, std=0.87

Fold 3 — y_test: mean=2.53, std=0.87, y_train: mean=2.55, std=0.85

Fold 4 — y_test: mean=2.53, std=0.82, y_train: mean=2.54, std=0.87

train scores CV: [0.46497694 0.46580479 0.46495948 0.46844345 0.45091902]

test scores CV: [0.45757013 0.45504757 0.45666765 0.44978339 0.49647352]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.463	0.4631	0.0169	0.3942	0.3945	-0.0686	0.8519	0.9998

Coefficients:

coef_r_CO2 = 0.0009

coef_r_NH3 = 0.0485

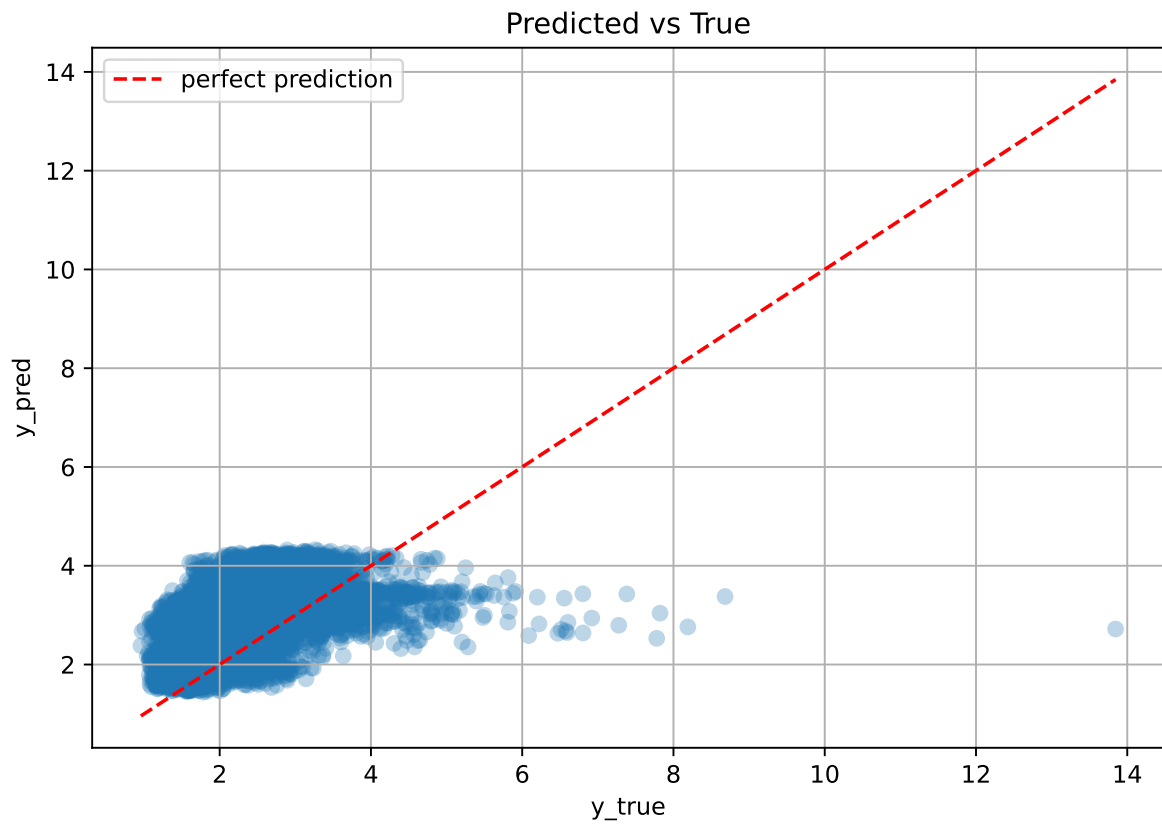
coef_r_THI = 0.8323

coef_r_temperature = -0.9731

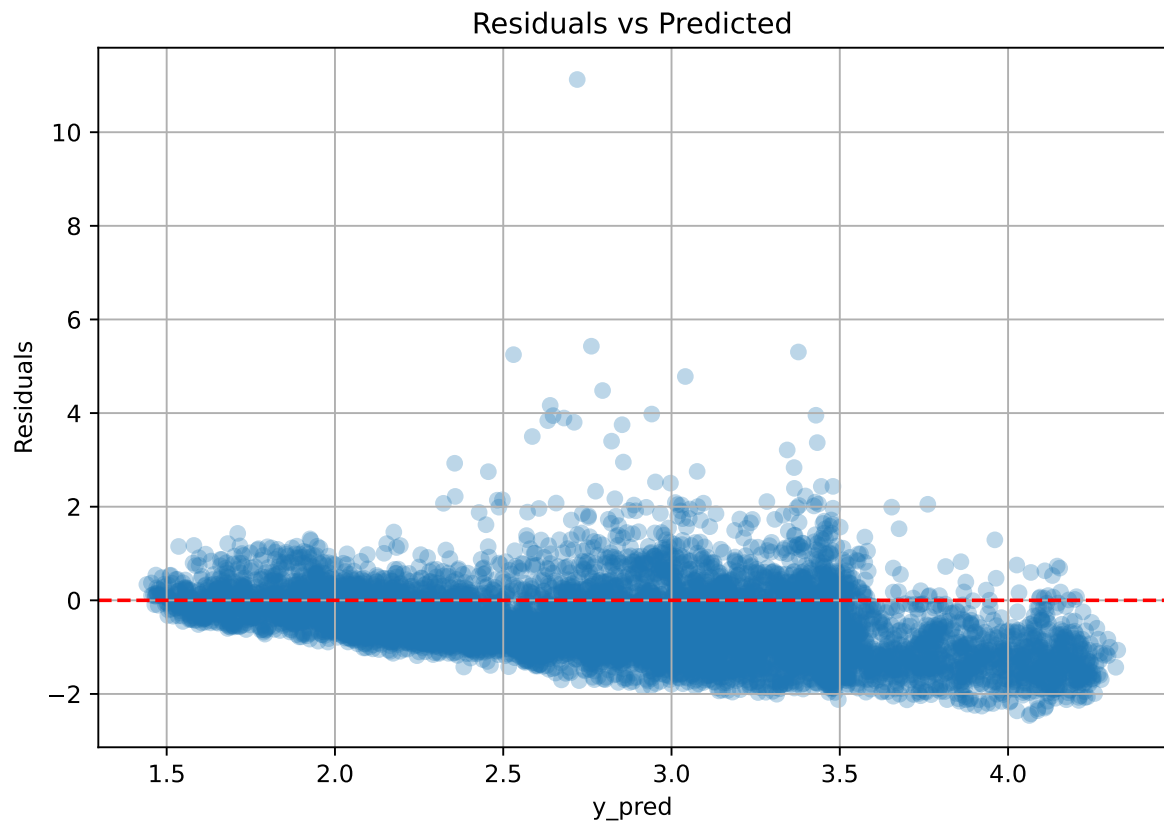
coef_r_umidity = -0.7852

intercept = 2.5398

2.3.6 Scatter



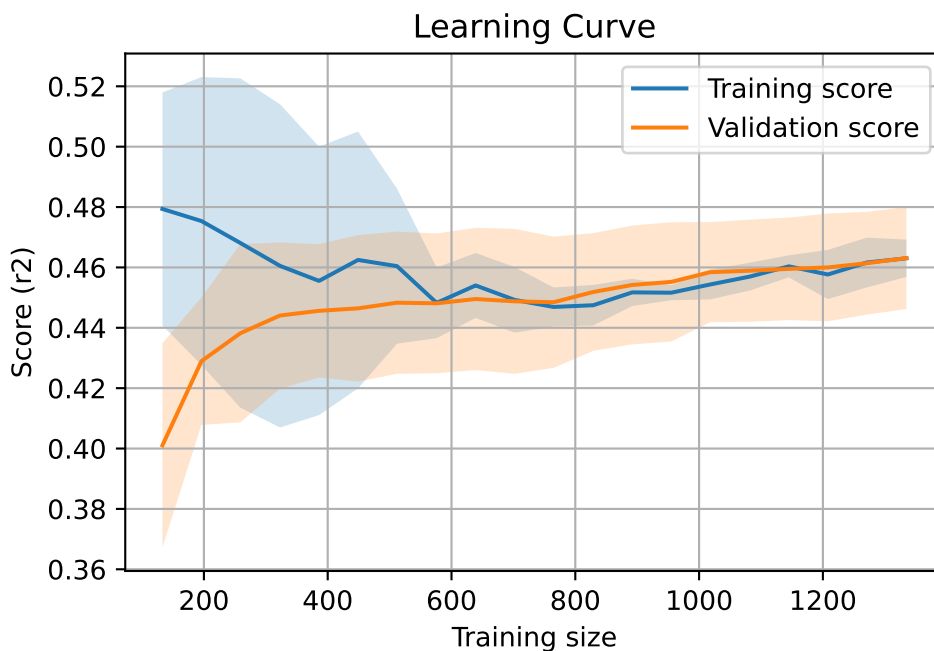
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2_11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.9 Gridsearch — Group: 3_4

Len(group) : 568, Len(training) : 398, Len(test) : 170

Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

Fold 0 — y_test: mean=1.93, std=0.75, y_train: mean=1.93, std=0.84

Fold 1 — y_test: mean=1.85, std=0.67, y_train: mean=1.96, std=0.87

Fold 2 — y_test: mean=2.00, std=0.92, y_train: mean=1.90, std=0.76

Fold 3 — y_test: mean=1.92, std=0.88, y_train: mean=1.93, std=0.78

Fold 4 — y_test: mean=1.92, std=0.86, y_train: mean=1.93, std=0.80

train scores CV: [0.3155034 0.29364485 0.29854522 0.32073505 0.30855581]

test scores CV: [0.17929864 0.27024886 0.27717852 0.24665152 0.26532812]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3074	0.2477	0.0357	0.1869	0.2027	-0.045	0.8182	1.2408

Coefficients:

$\text{coef_r_CO2} = 0.1066$

$\text{coef_r_NH3} = -0.2625$

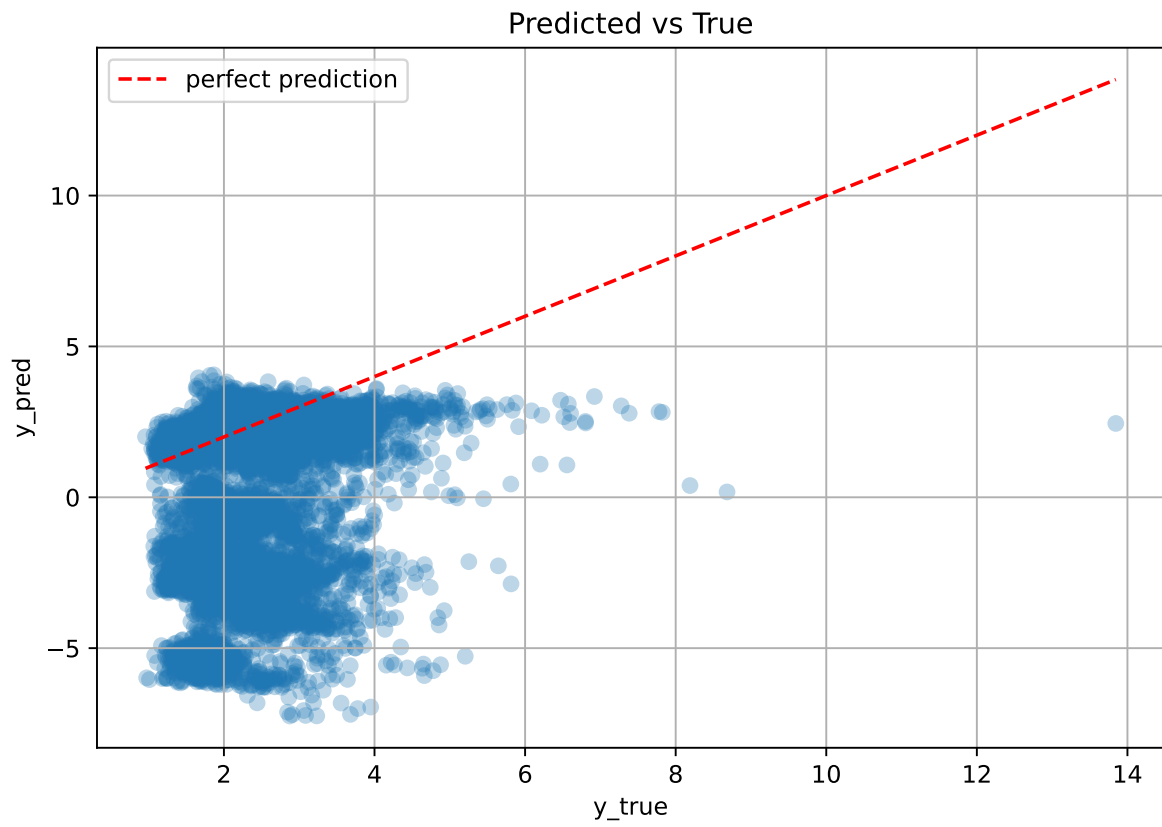
$\text{coef_r_THI} = 2.6048$

$\text{coef_r_temperature} = -3.7070$

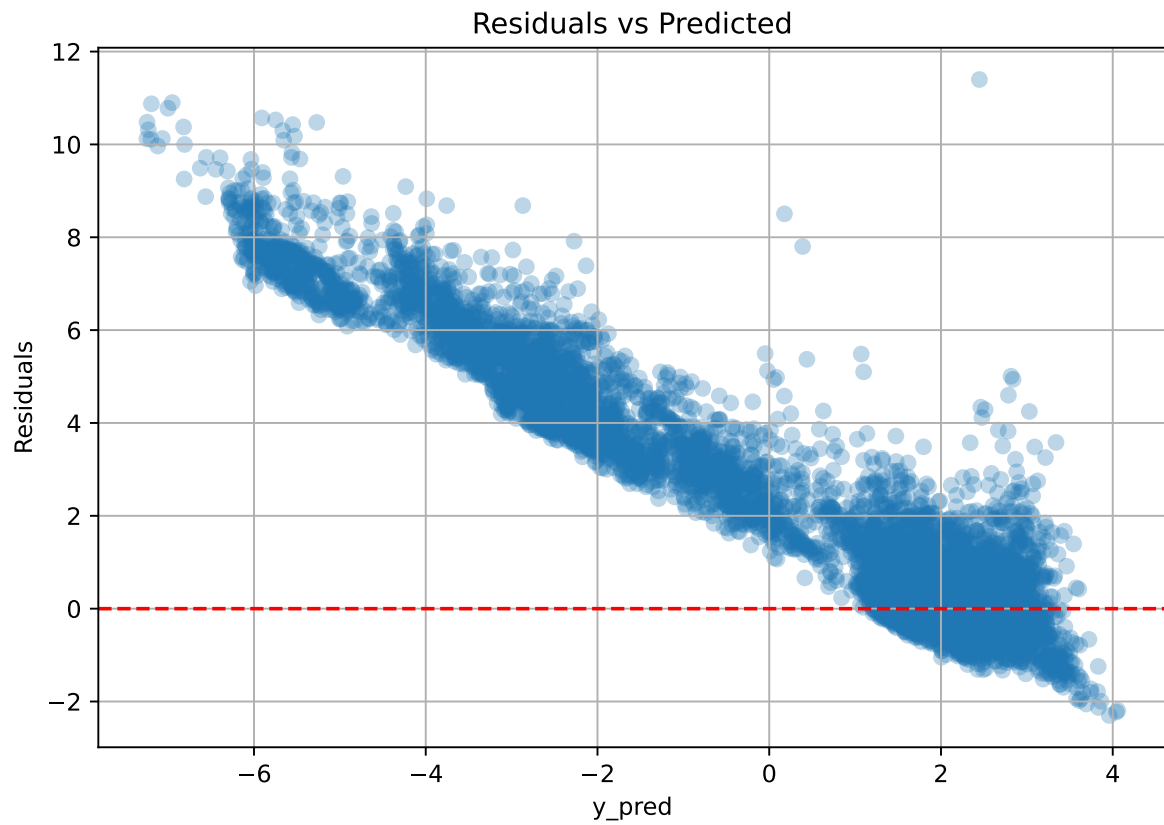
$\text{coef_r_umidity} = -1.4591$

$\text{intercept} = 1.9278$

2.3.10 Scatter



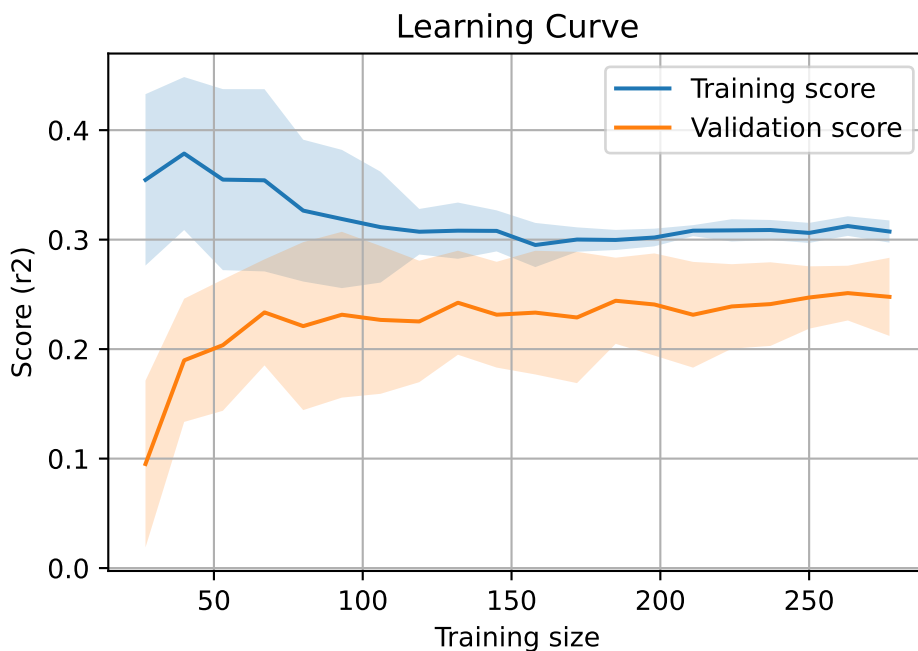
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 3_4

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170



2.3.13 Gridsearch — Group: 4_5

Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

Fold 0 — y_test: mean=2.27, std=0.68, y_train: mean=2.22, std=0.68

Fold 1 — y_test: mean=2.25, std=0.71, y_train: mean=2.23, std=0.67

Fold 2 — y_test: mean=2.20, std=0.65, y_train: mean=2.25, std=0.69

Fold 3 — y_test: mean=2.21, std=0.65, y_train: mean=2.25, std=0.70

Fold 4 — y_test: mean=2.22, std=0.63, y_train: mean=2.24, std=0.70

train scores CV: [0.4277706 0.44021497 0.41172833 0.43714743 0.44005866]

test scores CV: [0.41299685 0.39069188 0.45040747 0.3878951 0.37549212]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4314	0.4035	0.0264	0.4566	0.4572	0.0537	1.1332	1.0691

Coefficients:

$\text{coef_r_CO2} = -0.0952$

$\text{coef_r_NH3} = -0.1254$

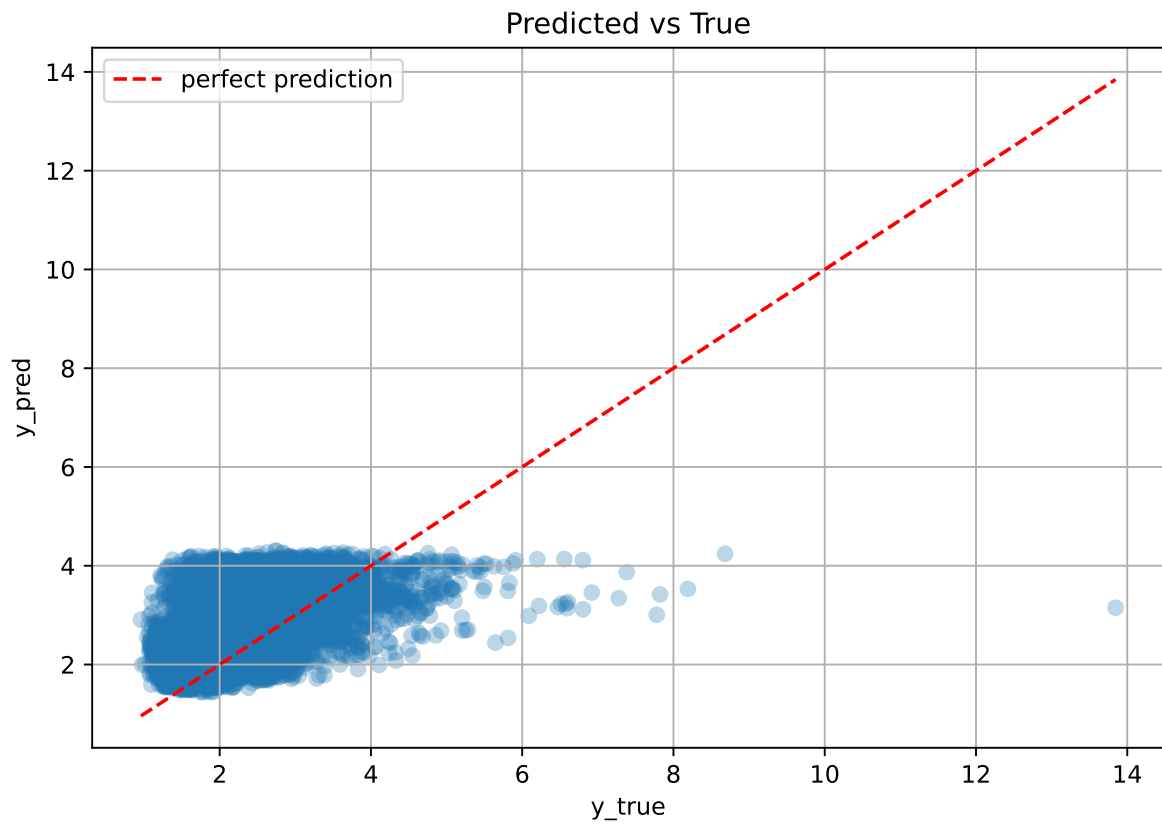
$\text{coef_r_THI} = 0.0994$

$\text{coef_r_temperature} = 0.0362$

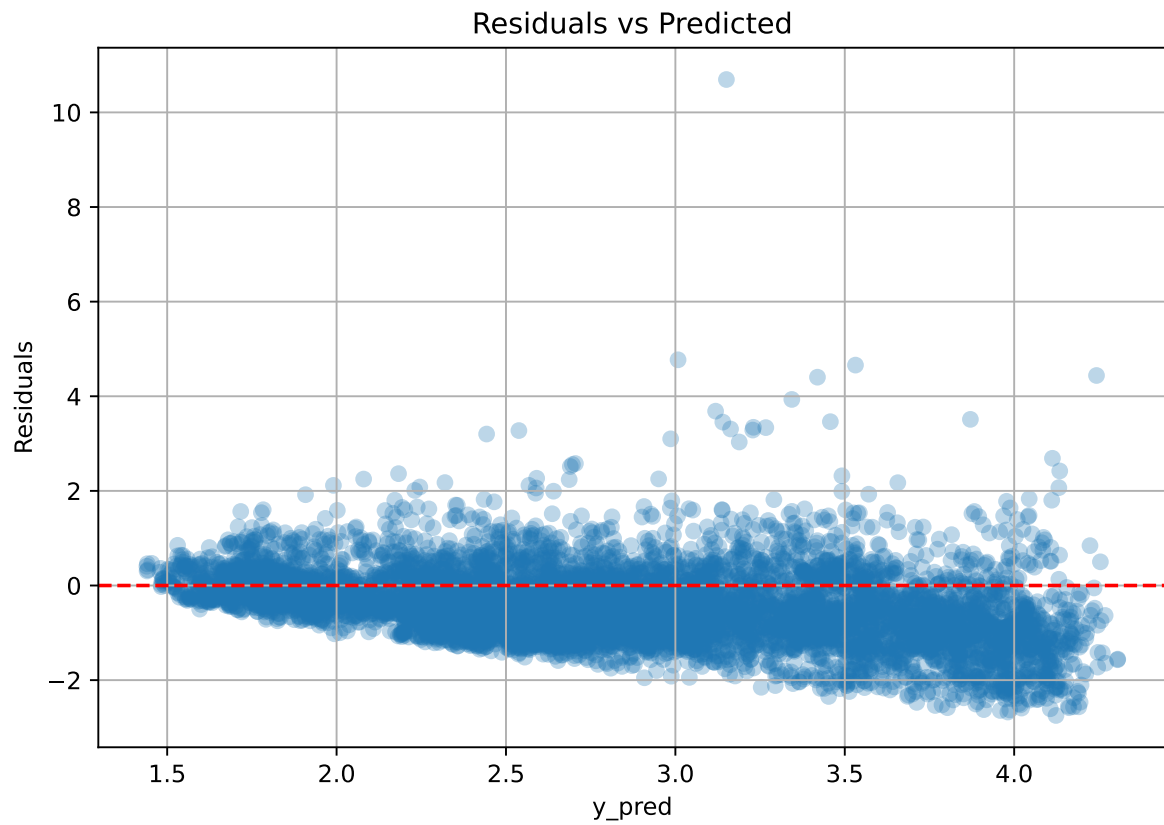
$\text{coef_r_umidity} = -0.4502$

$\text{intercept} = 2.2379$

2.3.14 Scatter



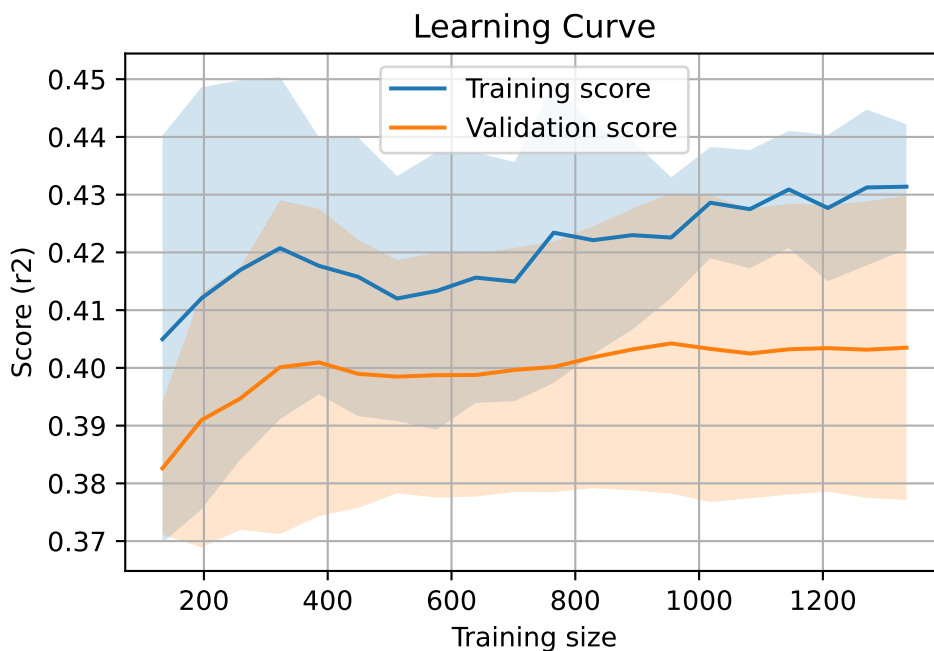
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 4_5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.3.17 Gridsearch — Group: 6_3

Len(group) : 561, Len(training) : 393, Len(test) : 168

Distribution y_train vs y_test: y_train: mean=1.935, std=0.720, min=1.041, max=6.086
y_test: mean=2.031, std=0.744, min=1.078, max=4.912

Fold 0 — y_test: mean=1.85, std=0.64, y_train: mean=1.97, std=0.75

Fold 1 — y_test: mean=1.98, std=0.75, y_train: mean=1.91, std=0.70

Fold 2 — y_test: mean=1.91, std=0.77, y_train: mean=1.95, std=0.70

Fold 3 — y_test: mean=1.92, std=0.68, y_train: mean=1.94, std=0.73

Fold 4 — y_test: mean=1.99, std=0.74, y_train: mean=1.91, std=0.71

train scores CV: [0.60490696 0.59617869 0.68479321 0.61316376 0.62706164]

test scores CV: [0.74052864 0.71853853 0.55240443 0.71109031 0.66169535]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.6252	0.6769	0.0674	0.6159	0.6233	-0.0536	0.9209	0.9237

Coefficients:

$\text{coef_r_CO2} = -0.0551$

$\text{coef_r_NH3} = -0.2103$

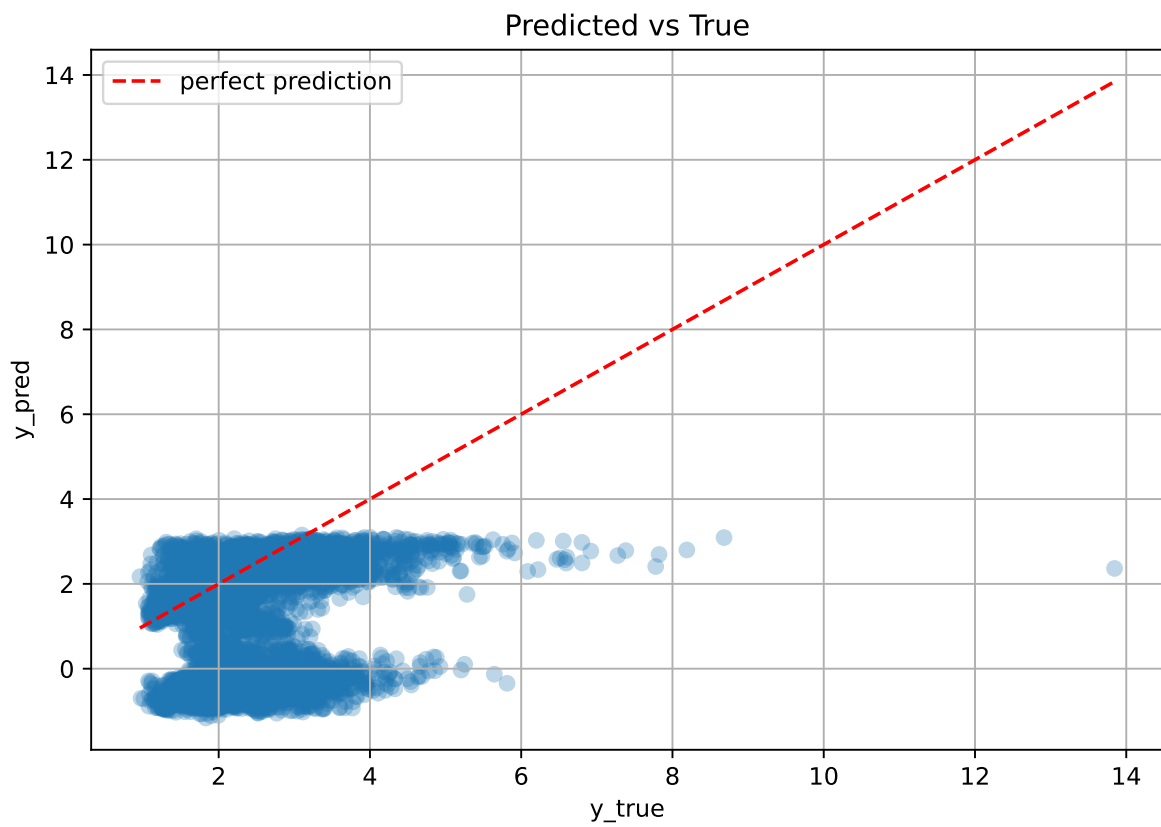
$\text{coef_r_THI} = 1.2572$

$\text{coef_r_temperature} = -1.5809$

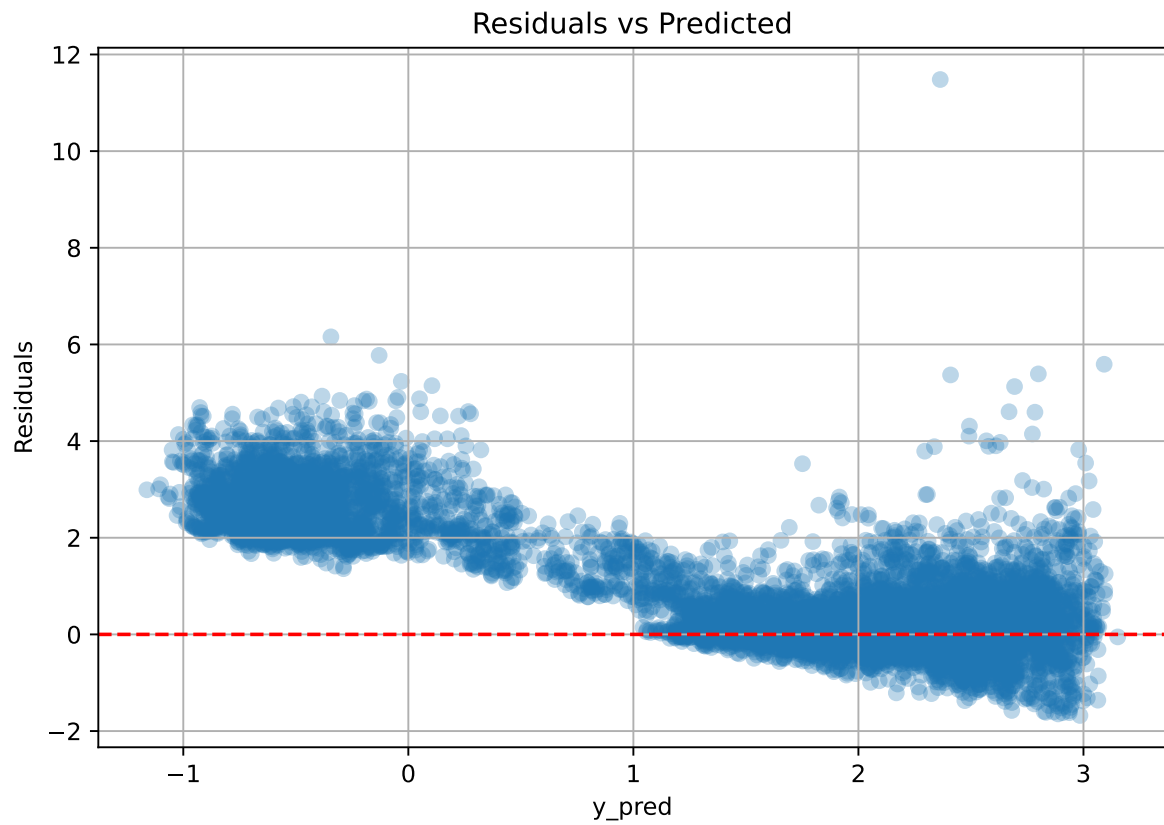
$\text{coef_r_umidity} = -0.8280$

$\text{intercept} = 1.9349$

2.3.18 Scatter



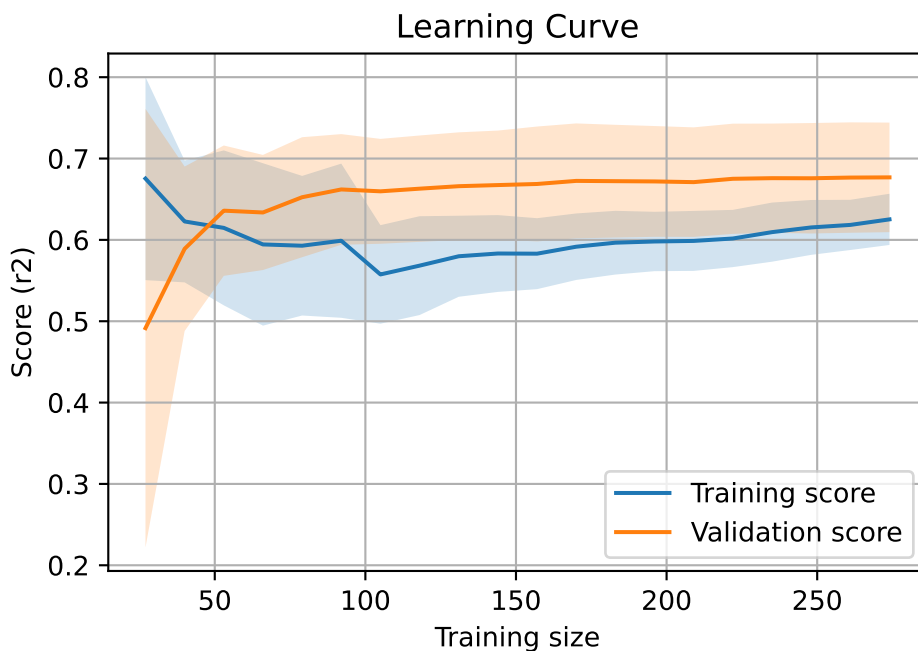
2.3.19 Scatter residuals



2.3.20 Learning curve — Group: 6_3

`Len(group)` : 561, `Len(training)` : 393,

`Len(training_real)` : 276, `Len(test_of_training)` : 117, `Len(test)` : 168



2.3.21 Gridsearch — Group: 6_4

Len(group) : 792, Len(training) : 555, Len(test) : 237

Distribution y_train vs y_test: y_train: mean=2.486, std=0.782, min=1.253, max=7.383
y_test: mean=2.410, std=0.685, min=1.398, max=5.144

Fold 0 — y_test: mean=2.46, std=0.72, y_train: mean=2.50, std=0.81

Fold 1 — y_test: mean=2.48, std=0.72, y_train: mean=2.49, std=0.81

Fold 2 — y_test: mean=2.47, std=0.79, y_train: mean=2.49, std=0.78

Fold 3 — y_test: mean=2.55, std=0.82, y_train: mean=2.46, std=0.76

Fold 4 — y_test: mean=2.50, std=0.78, y_train: mean=2.48, std=0.78

train scores CV: [0.15506482 0.17197221 0.15800044 0.17447249 0.20192166]

test scores CV: [0.23114143 0.18901692 0.21005333 0.16444206 0.10988493]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.1723	0.1809	0.0418	0.2601	0.2639	0.083	1.4589	0.9523

Coefficients:

$\text{coef_r_CO2} = -0.1789$

$\text{coef_r_NH3} = 0.0279$

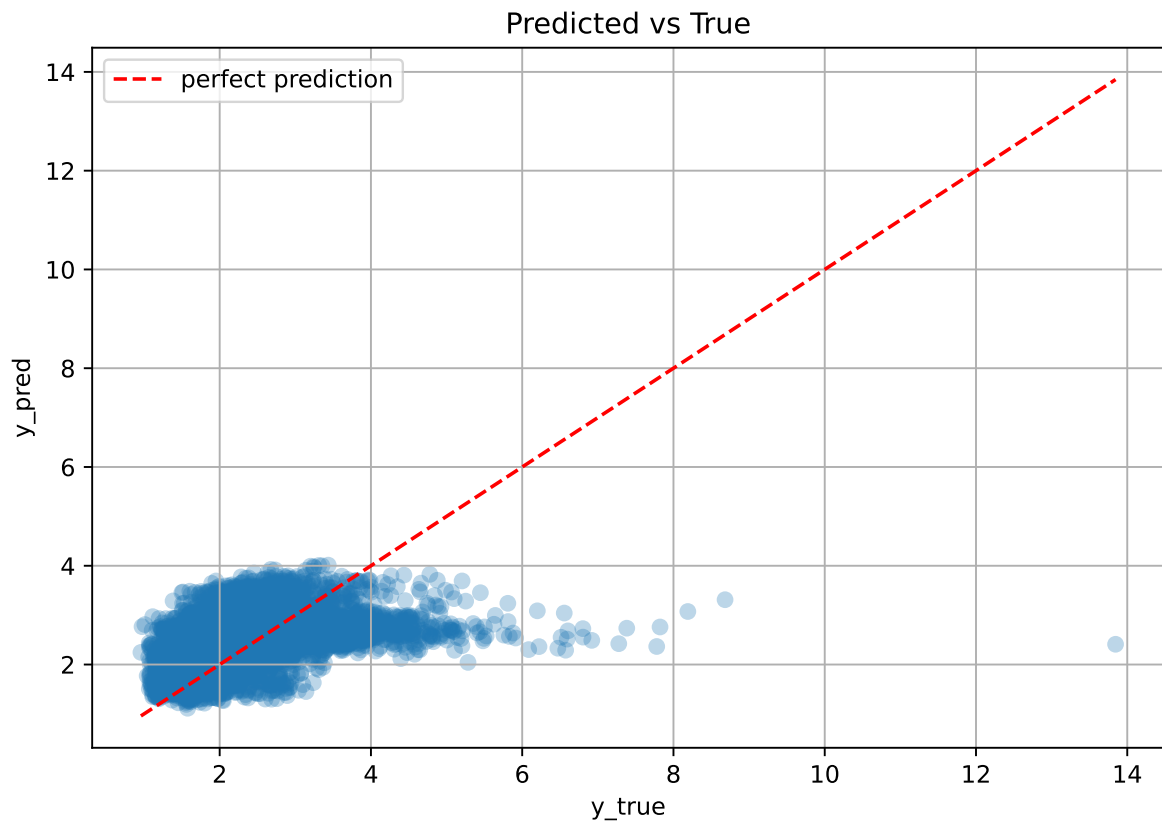
$\text{coef_r_THI} = 1.6225$

$\text{coef_r_temperature} = -2.4188$

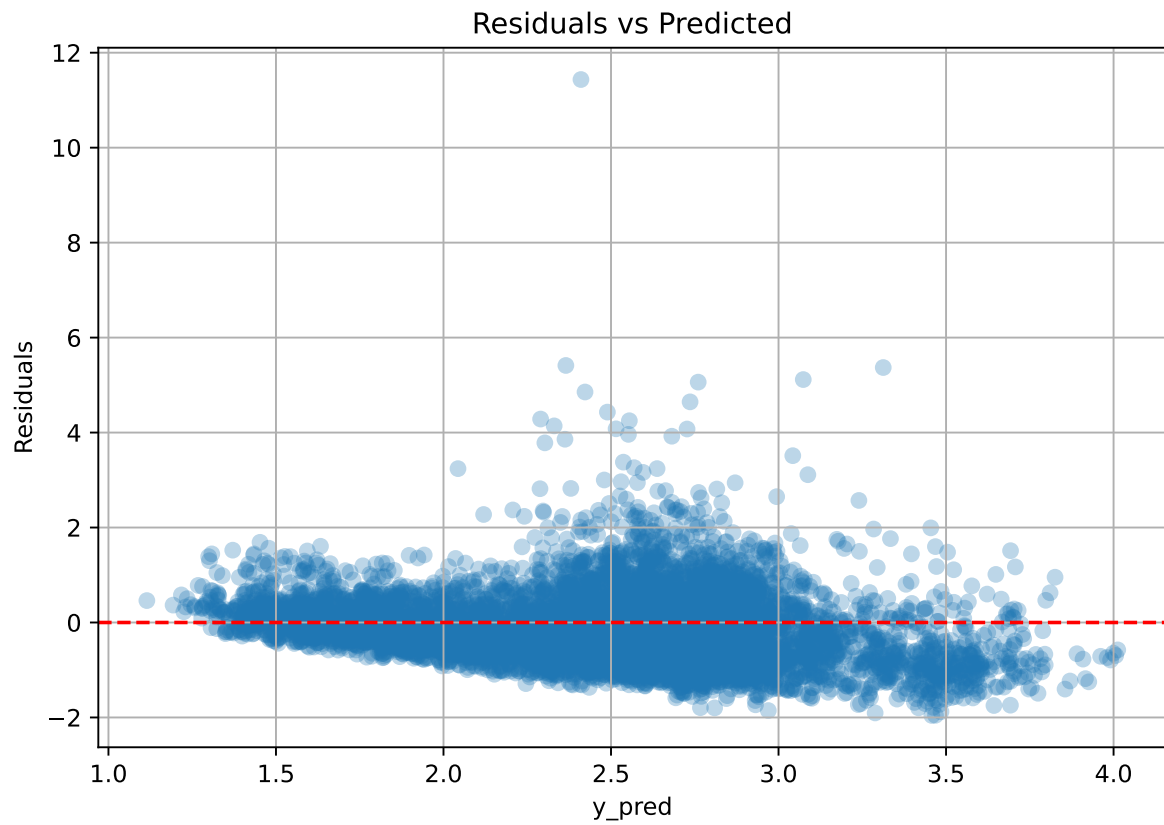
$\text{coef_r_umidity} = -1.1321$

$\text{intercept} = 2.4860$

2.3.22 Scatter



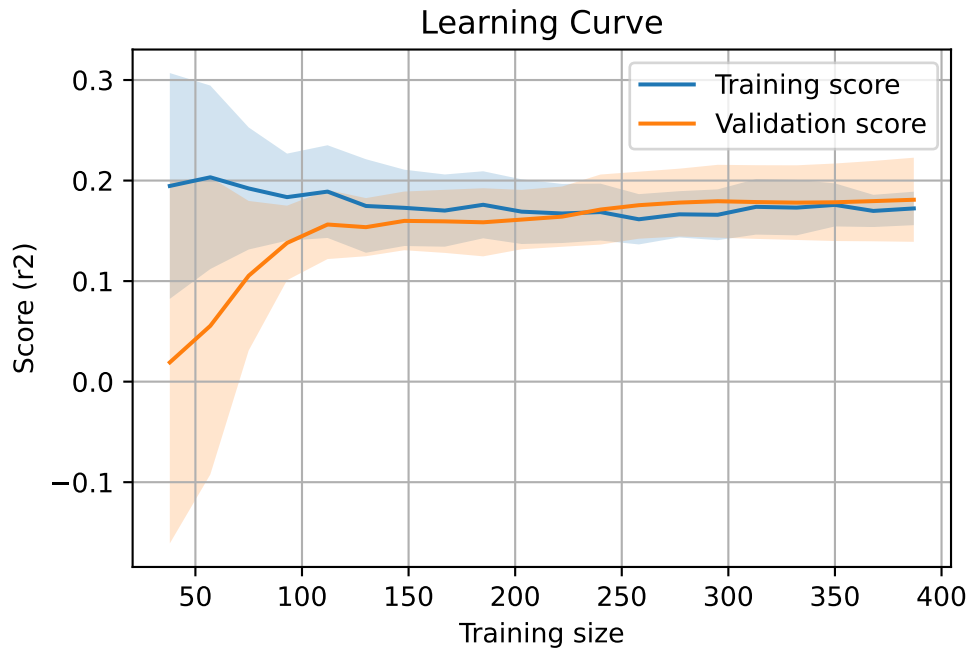
2.3.23 Scatter residuals



2.3.24 Learning curve — Group: 6_4

`Len(group)` : 792, `Len(training)` : 555,

`Len(training_real)` : 389, `Len(test_of_training)` : 166, `Len(test)` : 237



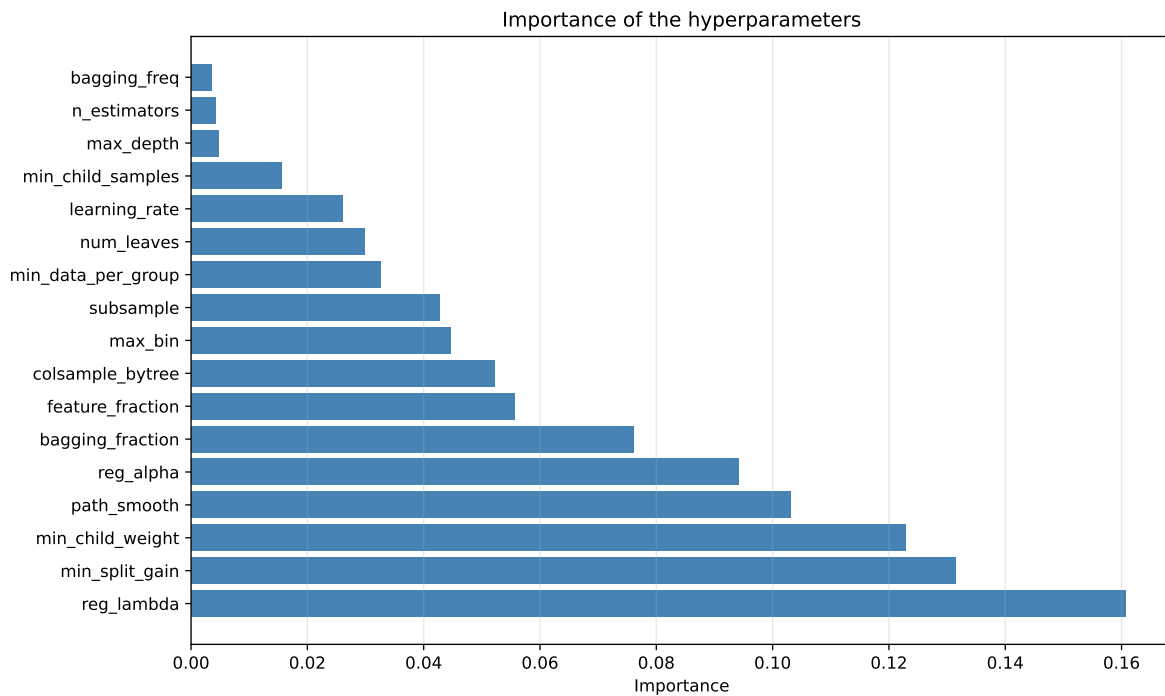
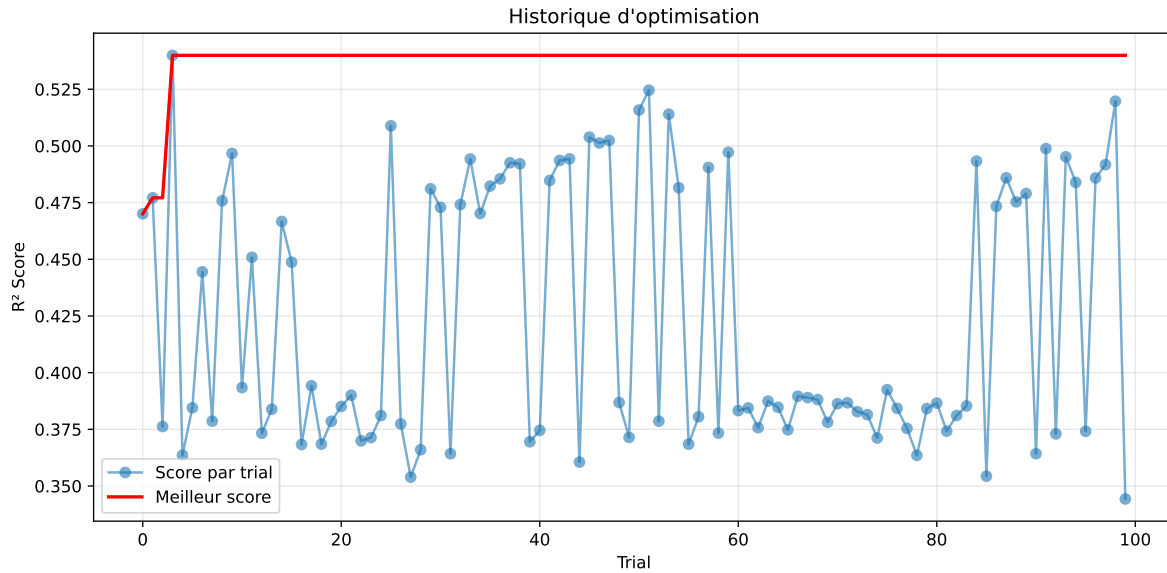
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 1_3

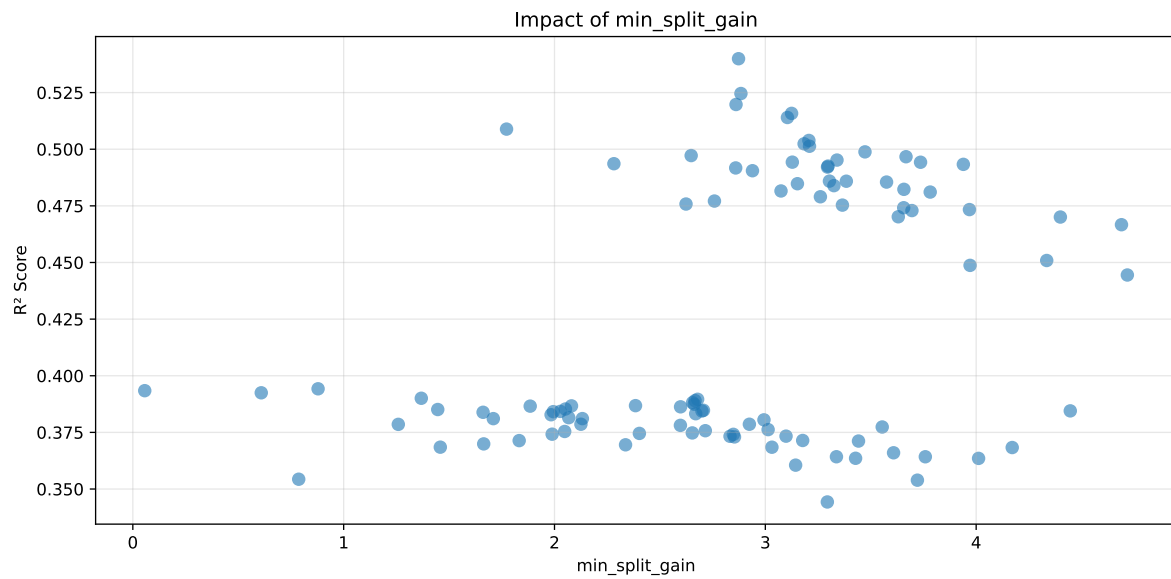
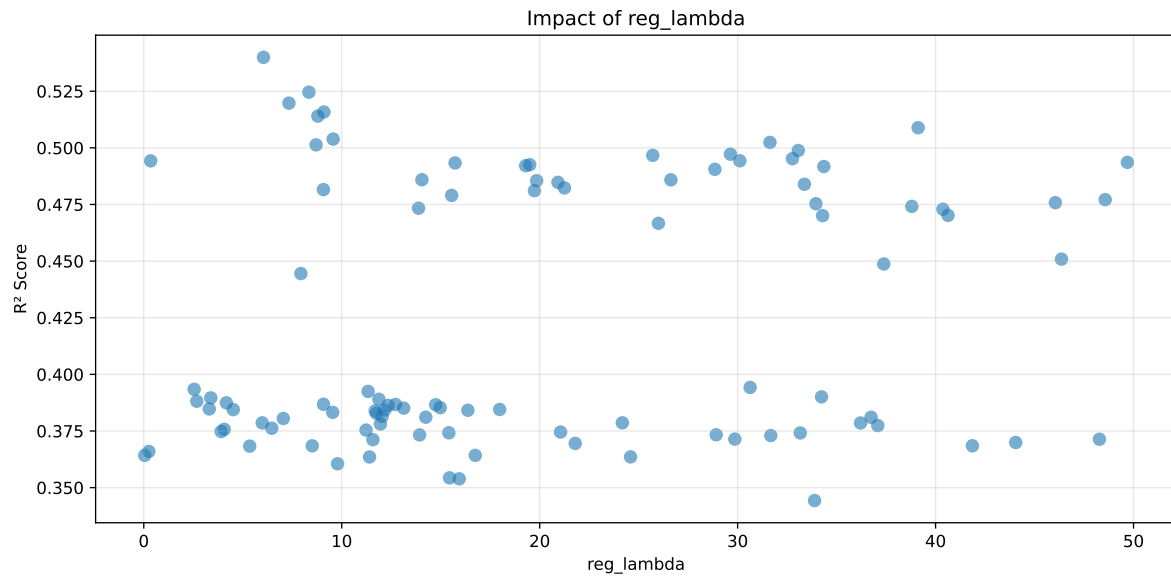
Len(group) : 795, Len(training) : 557, Len(test) : 238

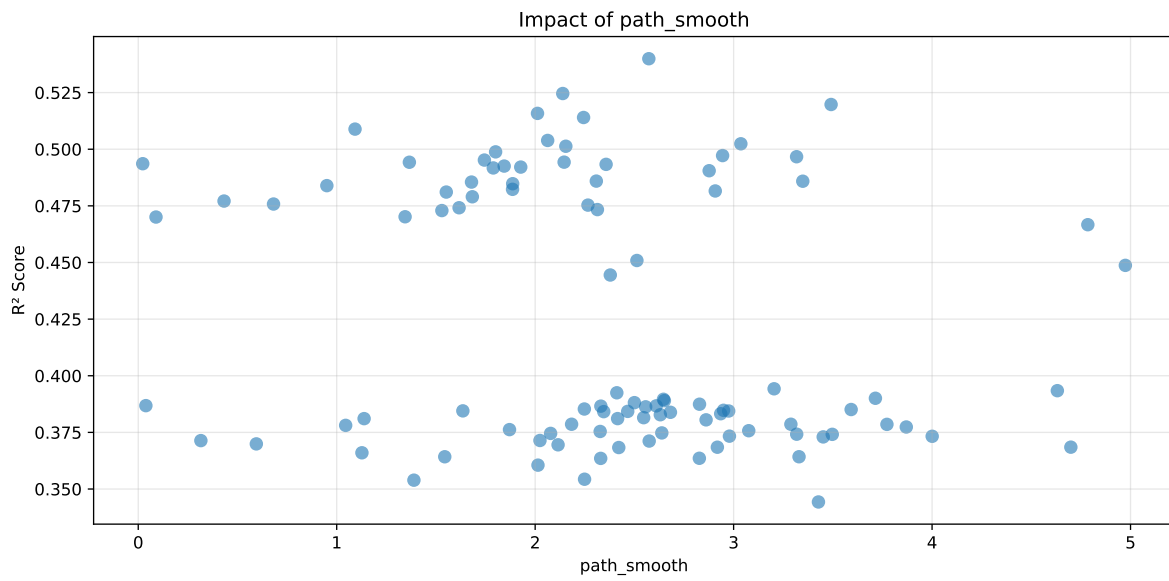
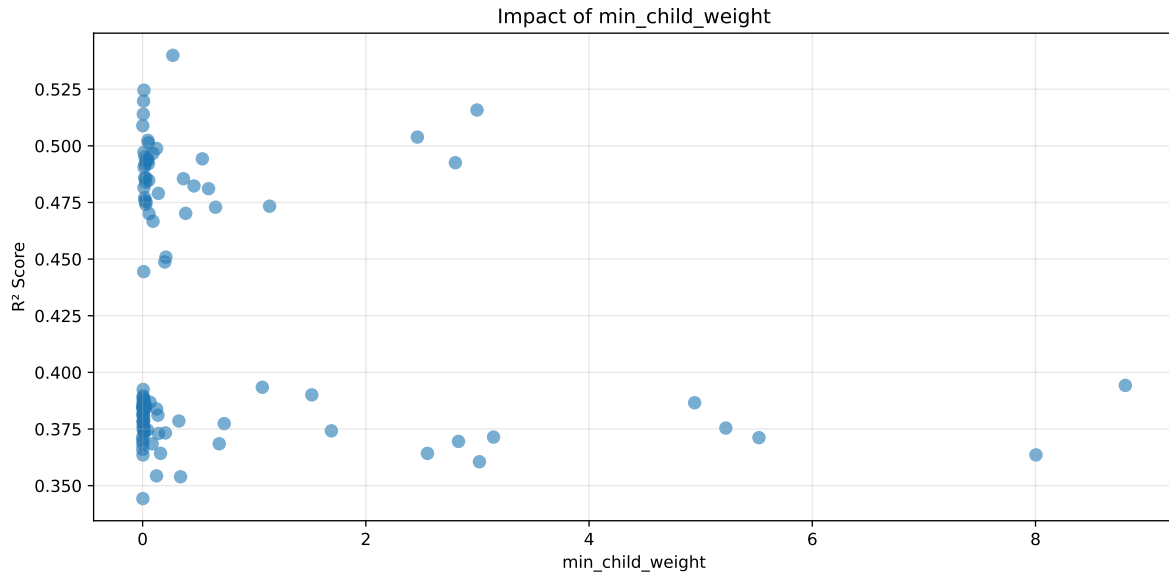
Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

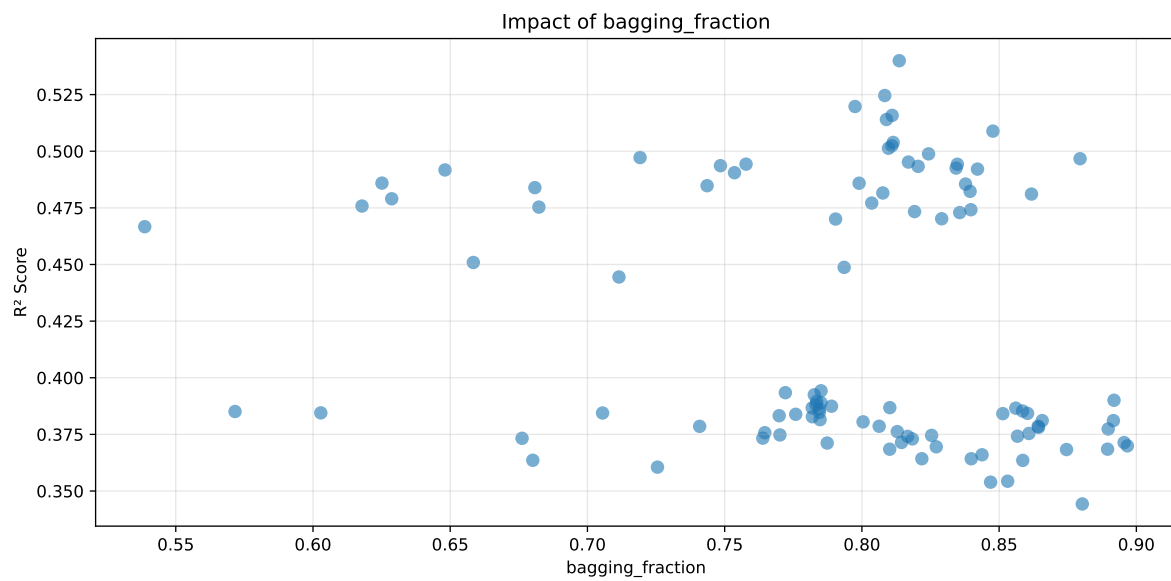
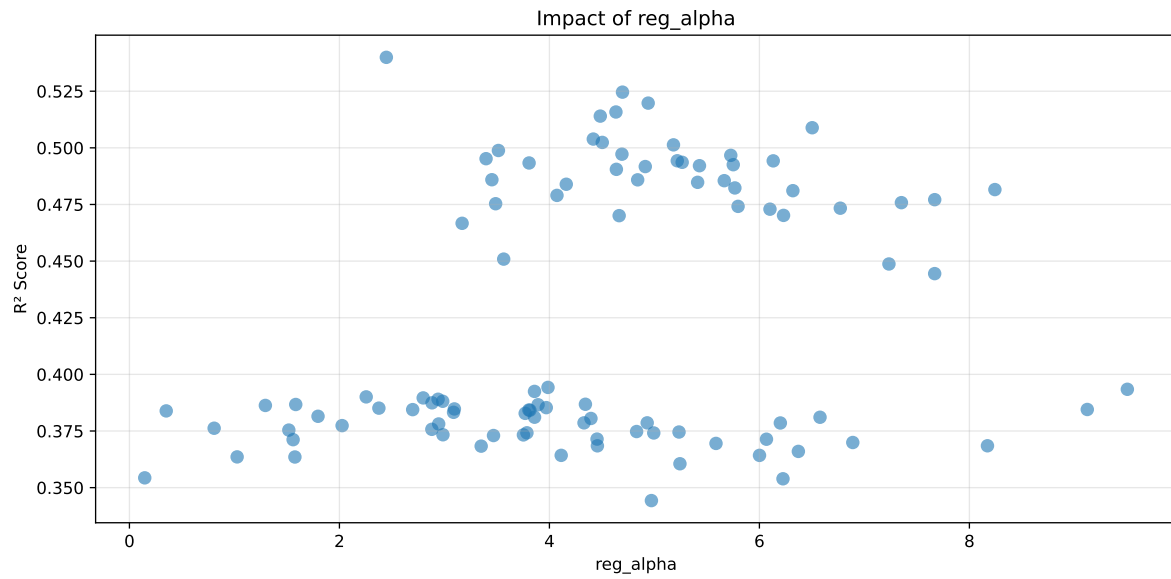
===== TOP Importance FEATURES =====
feature importance r_temperature 16 r_umidity 10 r_CO2 5 r_THI 4 r_NH3 0

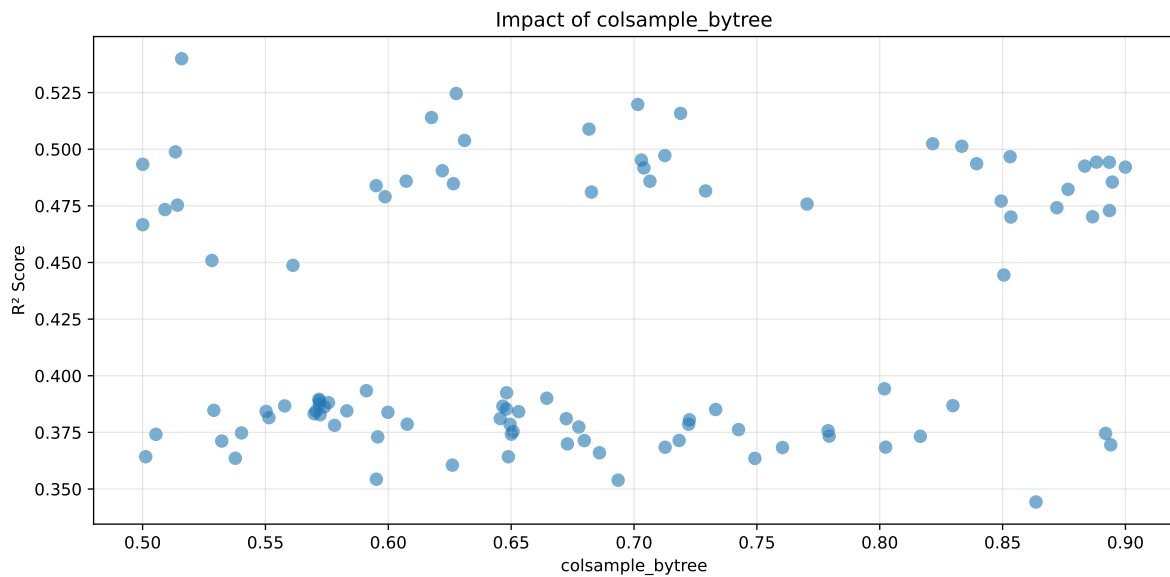
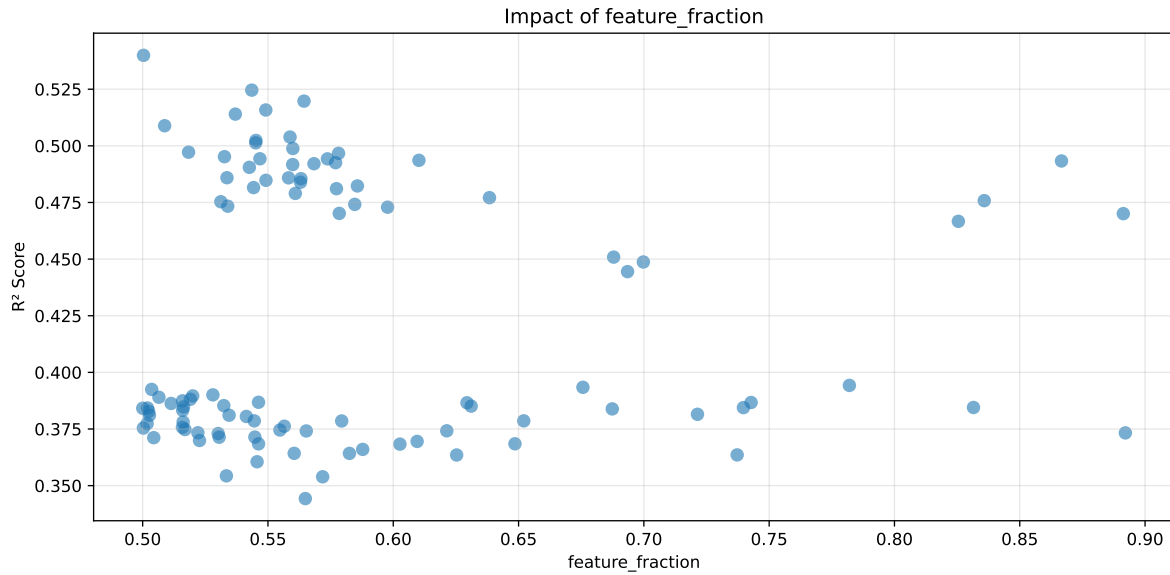


Paramètres avec >5% d'importance : ['reg_lambda', 'min_split_gain', 'min_child_weight', 'path_smooth', 'reg_alpha', 'bagging_fraction', 'feature_fraction', 'colsample_bytree']



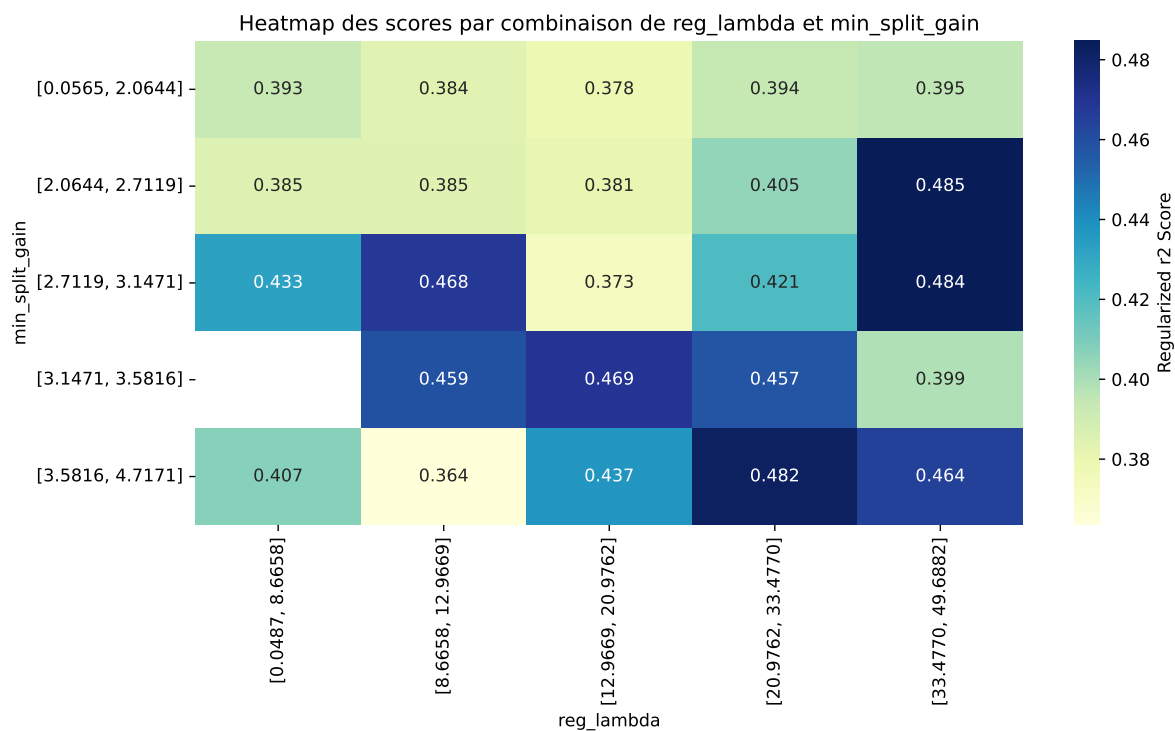






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5668	0.54	0.5029	0.0443	0.54	0.5189	-0.0211	0.9609	1.0498
2	0.549	0.5246	0.4847	0.0406	0.5246	0.5128	-0.0118	0.9775	1.0466
3	0.5448	0.5197	0.4817	0.0373	0.5197	0.511	-0.0088	0.9832	1.0482
4	0.5407	0.5158	0.4754	0.0338	0.5158	0.5032	-0.0126	0.9756	1.0482
5	0.5392	0.514	0.4741	0.0324	0.514	0.5106	-0.0034	0.9934	1.0489
6	0.5338	0.5089	0.4697	0.0361	0.5089	0.5055	-0.0034	0.9933	1.0489
7	0.5287	0.5039	0.4653	0.0328	0.5039	0.4911	-0.0128	0.9746	1.0492
8	0.5249	0.5024	0.4586	0.031	0.5024	0.5076	0.0052	1.0103	1.0447
9	0.5246	0.5013	0.4618	0.0316	0.5013	0.4887	-0.0126	0.9749	1.0465
10	0.5211	0.4988	0.4591	0.0321	0.4988	0.4858	-0.0131	0.9738	1.0446
---	---	---	---	---	---	---	---	---	---
100	0.5262	0.4959	0.4602	0.0408	0.3443	0.4896	-0.0062	0.9874	1.0611

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.07990224657331868, 'reg__num_leaves': 24, 'reg__subsample': 0.8317731204434745, 'reg__colsample_bytree': 0.515891358638138, 'reg__min_child_samples': 45, 'reg__reg_alpha': 2.4469447655676335, 'reg__reg_lambda': 6.048833422417238, 'reg__min_split_gain': 2.8728903846422007, 'reg__path_smooth': 2.572769434503953, 'reg__min_child_weight':

0.27088878048308285, 'reg__feature_fraction': 0.5003379484179248, 'reg__bagging_fraction': 0.8136118103966077, 'reg__bagging_freq': 6, 'reg__max_bin': 90, 'reg__min_data_per_group': 50}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.0748732305731291, 'reg__num_leaves': 42, 'reg__subsample': 0.8501995659785112, 'reg__colsample_bytree': 0.6276563778412565, 'reg__min_child_samples': 19, 'reg__reg_alpha': 4.695989515439477, 'reg__reg_lambda': 8.340006507622917, 'reg__min_split_gain': 2.8842667004151656, 'reg__path_smooth': 2.139040843942623, 'reg__min_child_weight': 0.011450694389272506, 'reg__feature_fraction': 0.5435392689386364, 'reg__bagging_fraction': 0.80832900635403, 'reg__bagging_freq': 2, 'reg__max_bin': 83, 'reg__min_data_per_group': 163}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.09173253895251987, 'reg__num_leaves': 44, 'reg__subsample': 0.5579238446203791, 'reg__colsample_bytree': 0.7015135603744471, 'reg__min_child_samples': 32, 'reg__reg_alpha': 4.9407008901528835, 'reg__reg_lambda': 7.33418938587727, 'reg__min_split_gain': 2.8610206093466175, 'reg__path_smooth': 3.4911787256260207, 'reg__min_child_weight': 0.008277238830296574, 'reg__feature_fraction': 0.5643883911188871, 'reg__bagging_fraction': 0.7975047941797653, 'reg__bagging_freq': 2, 'reg__max_bin': 88, 'reg__min_data_per_group': 52}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.0725376592809096, 'reg__num_leaves': 39, 'reg__subsample': 0.8658856209963978, 'reg__colsample_bytree': 0.7189465479860742, 'reg__min_child_samples': 49, 'reg__reg_alpha': 4.633583352102324, 'reg__reg_lambda': 9.100473300951434, 'reg__min_split_gain': 3.1245572880512933, 'reg__path_smooth': 2.011832808956133, 'reg__min_child_weight': 2.996324877061134, 'reg__feature_fraction': 0.5491314841915325, 'reg__bagging_fraction': 0.8110297683360009, 'reg__bagging_freq': 2, 'reg__max_bin': 81, 'reg__min_data_per_group': 159}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.07655287840823959, 'reg__num_leaves': 42, 'reg__subsample': 0.8658482097385078, 'reg__colsample_bytree': 0.6175812490306534, 'reg__min_child_samples': 18, 'reg__reg_alpha': 4.484556810919733, 'reg__reg_lambda': 8.794298472317303, 'reg__min_split_gain': 3.1048178766769925, 'reg__path_smooth': 2.2439636029217414, 'reg__min_child_weight': 0.006050033523067117, 'reg__feature_fraction': 0.5369785676547454, 'reg__bagging_fraction': 0.8089391032075559, 'reg__bagging_freq': 2, 'reg__max_bin': 80, 'reg__min_data_per_group': 158}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.09712833695023625, 'reg__num_leaves': 31, 'reg__subsample': 0.5079927970680983, 'reg__colsample_bytree': 0.6816690631303608, 'reg__min_child_samples': 25, 'reg__reg_alpha': 6.5028354435217555, 'reg__reg_lambda': 39.119862732249786, 'reg__min_split_gain': 1.7725348885031136, 'reg__path_smooth': 1.0923647407142103, 'reg__min_child_weight':

0.0015312723056720699, 'reg__feature_fraction': 0.5087651722695445, 'reg__bagging_fraction': 0.847747476697043, 'reg__bagging_freq': 5, 'reg__max_bin': 67, 'reg__min_data_per_group': 103}

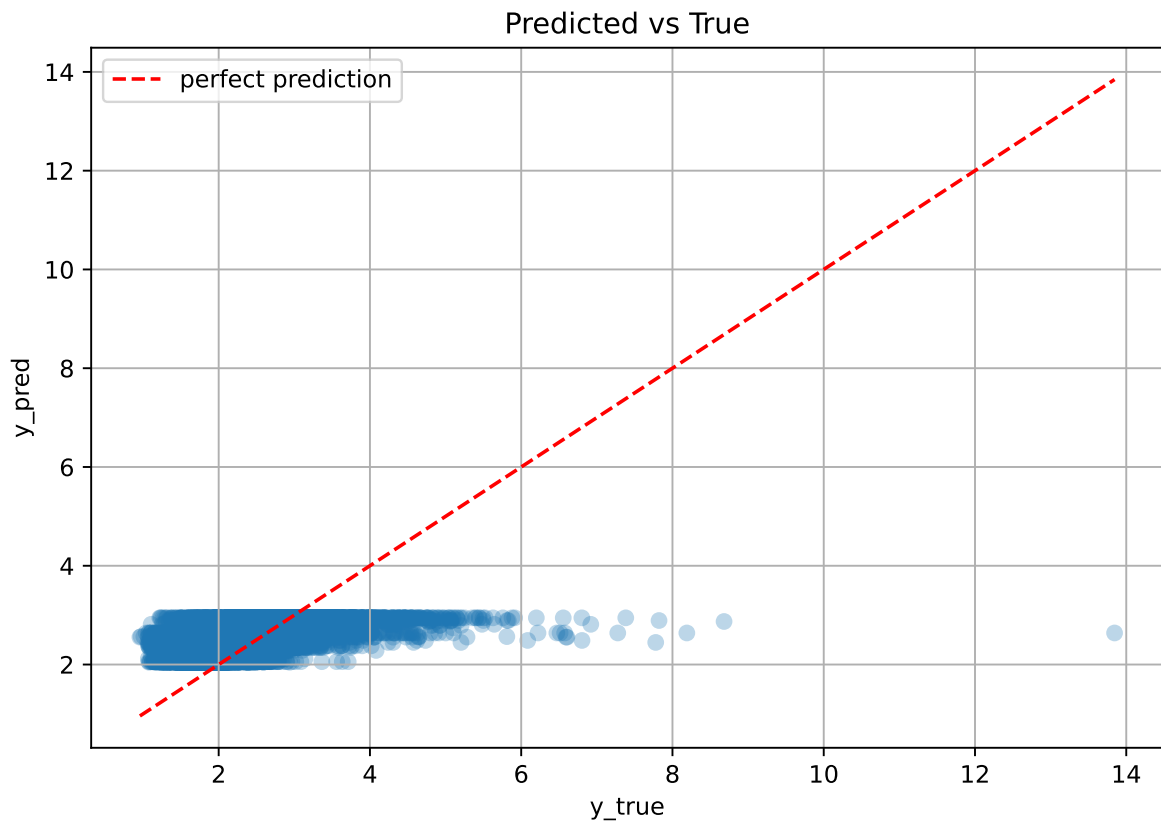
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 7, 'reg__learning_rate': 0.07568971932721212, 'reg__num_leaves': 44, 'reg__subsample': 0.8590737528378741, 'reg__colsample_bytree': 0.6310016495192613, 'reg__min_child_samples': 20, 'reg__reg_alpha': 4.418003014406336, 'reg__reg_lambda': 9.567298422191579, 'reg__min_split_gain': 3.2065674877337806, 'reg__path_smooth': 2.0630301596502503, 'reg__min_child_weight': 2.4617651126708204, 'reg__feature_fraction': 0.5588041761789271, 'reg__bagging_fraction': 0.8114117920538673, 'reg__bagging_freq': 3, 'reg__max_bin': 83, 'reg__min_data_per_group': 136}

Rank 8: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.04802868252136322, 'reg__num_leaves': 45, 'reg__subsample': 0.8595800735782624, 'reg__colsample_bytree': 0.8215438962061296, 'reg__min_child_samples': 19, 'reg__reg_alpha': 4.503485425721733, 'reg__reg_lambda': 31.633552388984388, 'reg__min_split_gain': 3.1838682229738993, 'reg__path_smooth': 3.036457407249575, 'reg__min_child_weight': 0.04713532052432486, 'reg__feature_fraction': 0.5452135784922287, 'reg__bagging_fraction': 0.8109240247480058, 'reg__bagging_freq': 2, 'reg__max_bin': 80, 'reg__min_data_per_group': 136}

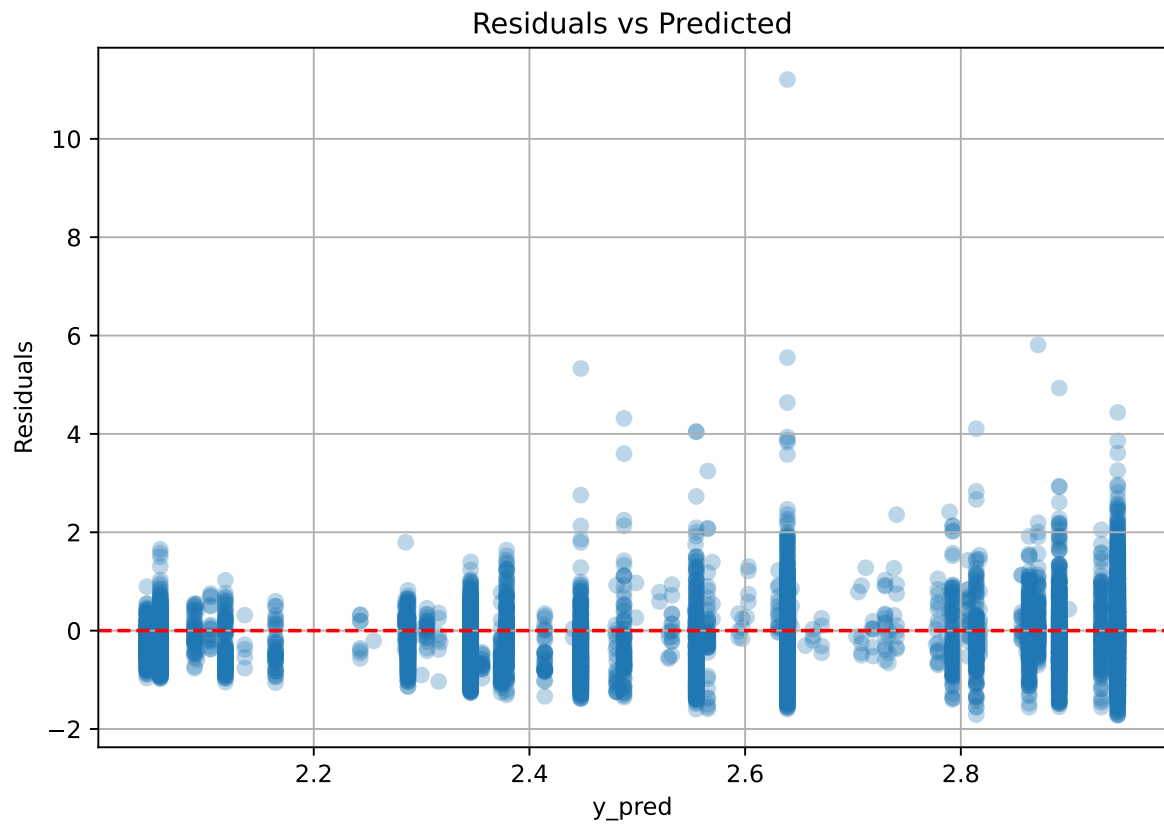
Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.07680355959763387, 'reg__num_leaves': 43, 'reg__subsample': 0.8556665875446855, 'reg__colsample_bytree': 0.8333740673417842, 'reg__min_child_samples': 20, 'reg__reg_alpha': 5.181557646519674, 'reg__reg_lambda': 8.70435960684312, 'reg__min_split_gain': 3.2095209151068147, 'reg__path_smooth': 2.154558515105919, 'reg__min_child_weight': 0.05239684099142727, 'reg__feature_fraction': 0.5451119786879913, 'reg__bagging_fraction': 0.80968318338876, 'reg__bagging_freq': 3, 'reg__max_bin': 82, 'reg__min_data_per_group': 135}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.046938961380451574, 'reg__num_leaves': 47, 'reg__subsample': 0.8396881736136698, 'reg__colsample_bytree': 0.5133632369724888, 'reg__min_child_samples': 38, 'reg__reg_alpha': 3.5143413642266674, 'reg__reg_lambda': 33.064059440577736, 'reg__min_split_gain': 3.4726615338436373, 'reg__path_smooth': 1.8006930002890038, 'reg__min_child_weight': 0.1251606566034408, 'reg__feature_fraction': 0.5599250782152192, 'reg__bagging_fraction': 0.8242857634120354, 'reg__bagging_freq': 3, 'reg__max_bin': 89, 'reg__min_data_per_group': 150}

2.4.2 Scatter



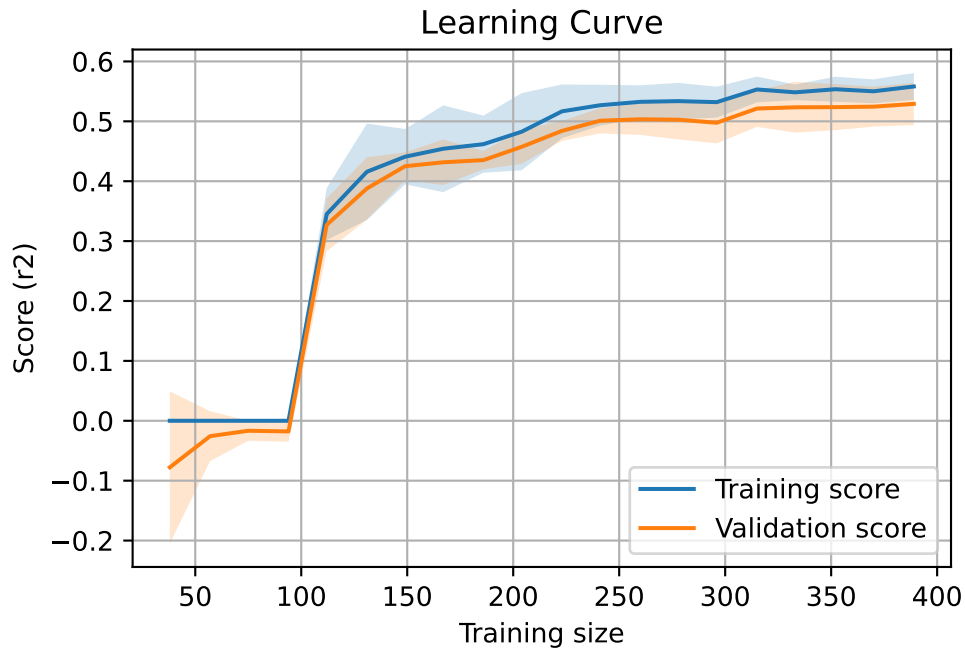
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 1_3

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238

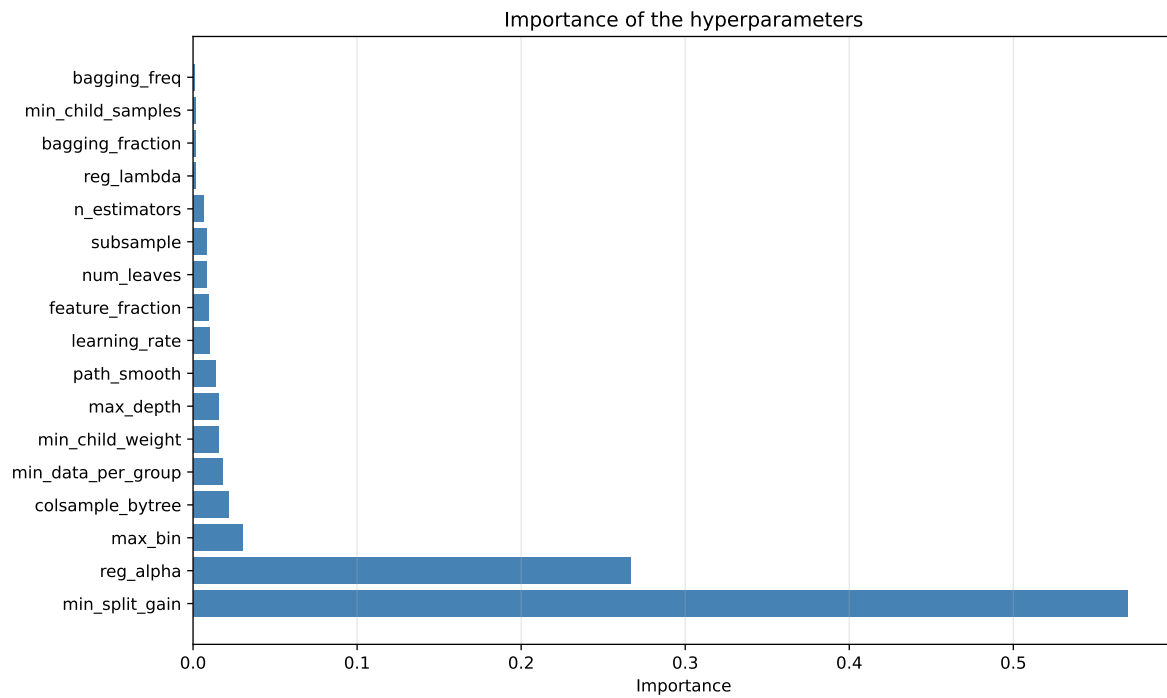
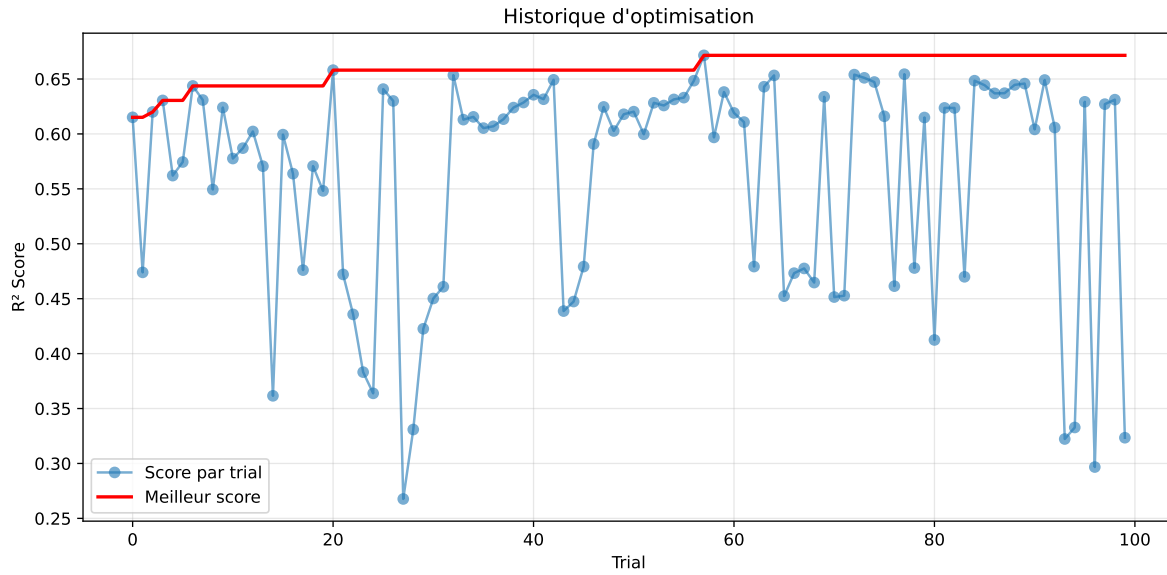


2.4.5 Gridsearch — Group: 2_11

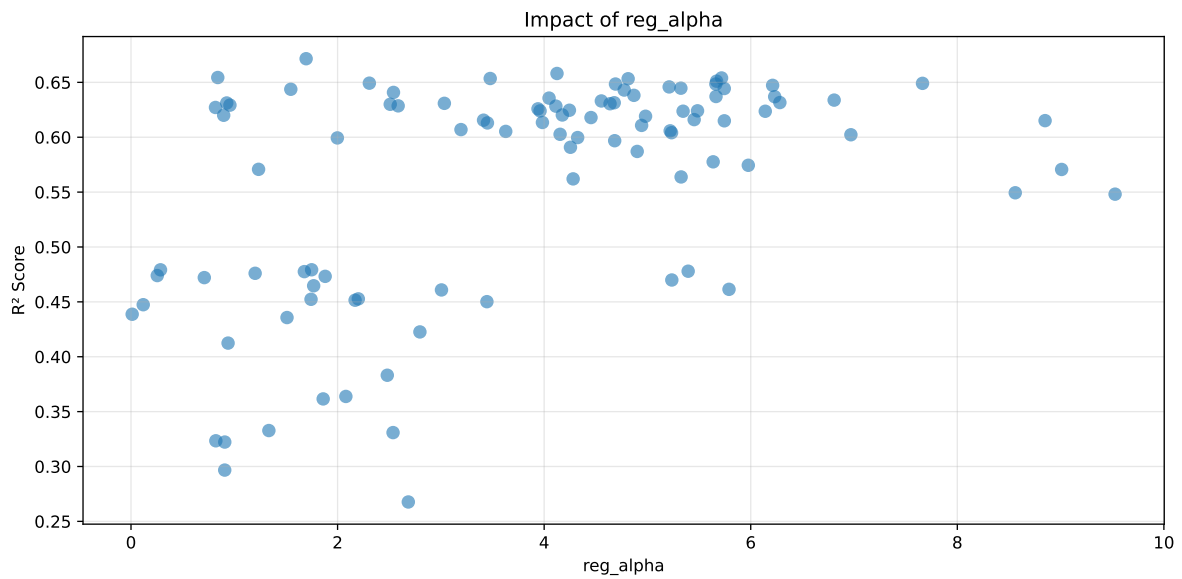
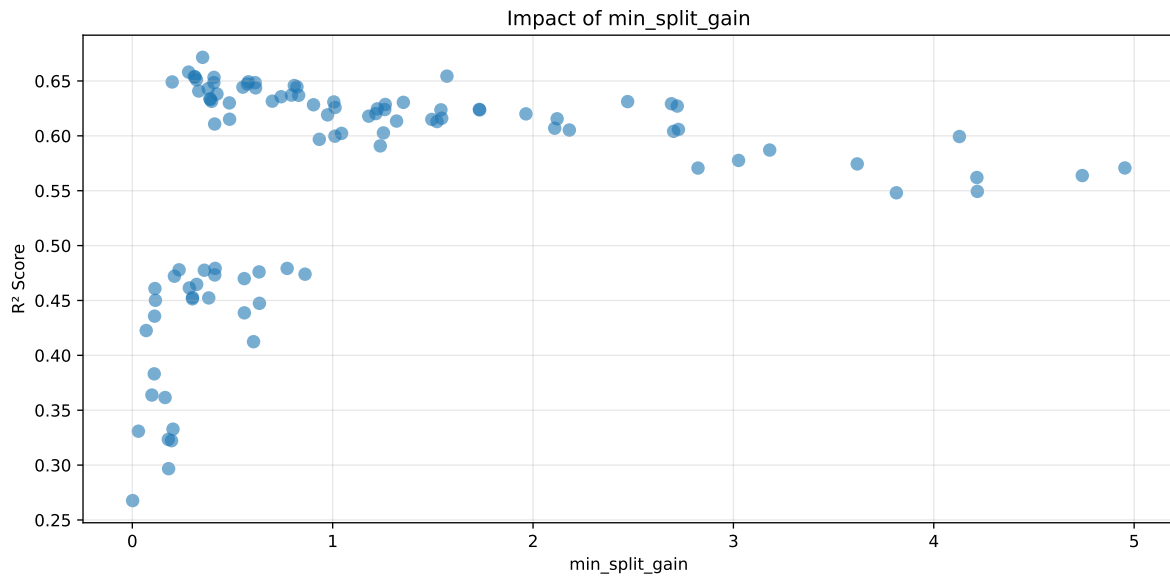
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_NH3 449 r_umidity 446 r_CO2 360 r_THI 215 r_temperature 190

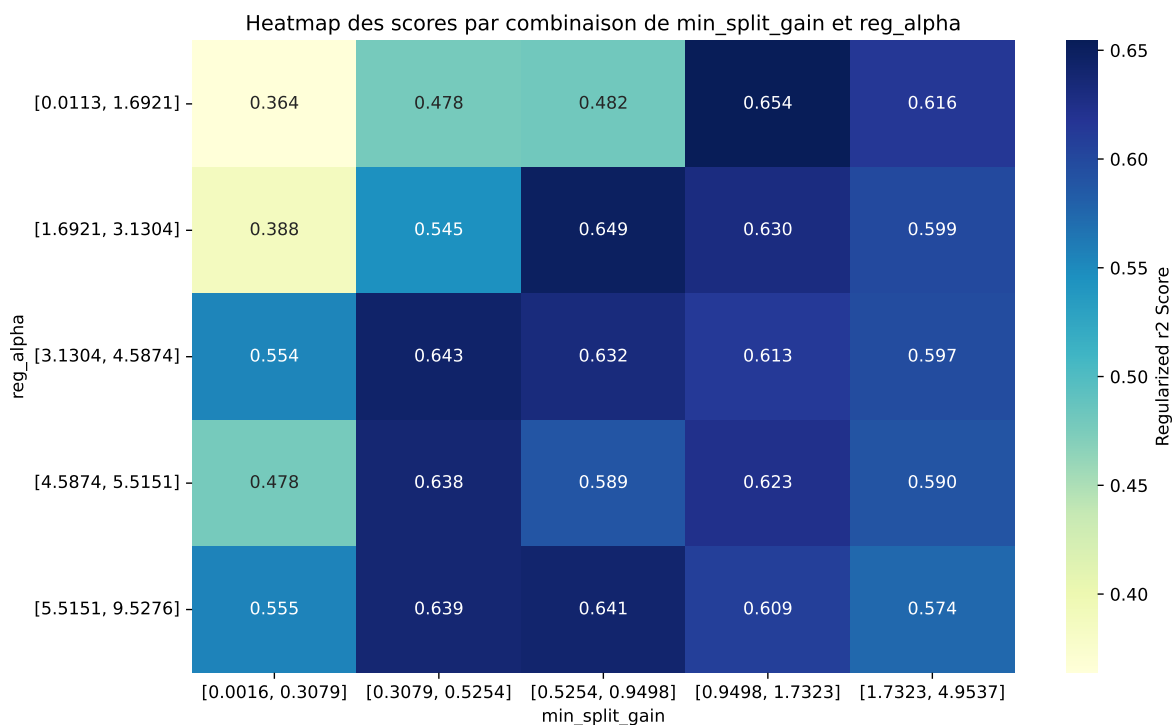


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7048	0.6715	0.5698	0.0151	0.6715	0.5802	-0.0913	0.864	1.0496
2	0.6892	0.6581	0.5563	0.0136	0.6581	0.5755	-0.0826	0.8745	1.0473
3	0.6828	0.6544	0.5539	0.01	0.6544	0.5789	-0.0755	0.8846	1.0434
4	0.6847	0.654	0.5609	0.0107	0.654	0.5697	-0.0843	0.8711	1.0469
5	0.6844	0.6535	0.5581	0.0096	0.6535	0.5751	-0.0785	0.88	1.0473
6	0.6849	0.6533	0.5583	0.0131	0.6533	0.5708	-0.0825	0.8737	1.0484
7	0.6834	0.6511	0.5563	0.0176	0.6511	0.5657	-0.0854	0.8689	1.0497
8	0.6806	0.6493	0.5574	0.0078	0.6493	0.5727	-0.0766	0.882	1.0482
9	0.6774	0.6491	0.557	0.013	0.6491	0.5705	-0.0786	0.879	1.0436
10	0.6723	0.6485	0.5539	0.0119	0.6485	0.5689	-0.0795	0.8774	1.0368
---	---	---	---	---	---	---	---	---	---
100	0.7703	0.6865	0.5785	0.0094	0.2677	0.5905	-0.096	0.8601	1.122

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.02506073906483933, 'reg__num_leaves': 52, 'reg__subsample': 0.7750138891995213, 'reg__colsample_bytree': 0.7525000090307737, 'reg__min_child_samples': 31, 'reg__reg_alpha': 1.6955561063882088, 'reg__reg_lambda': 10.14100451578913, 'reg__min_split_gain': 0.35088123868518706, 'reg__path_smooth': 3.714098766050947, 'reg__min_child_weight':

0.11244818266307191, 'reg__feature_fraction': 0.7968789161322569, 'reg__bagging_fraction': 0.7132237398638704, 'reg__bagging_freq': 7, 'reg__max_bin': 162, 'reg__min_data_per_group': 97}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.09115522266623491, 'reg__num_leaves': 48, 'reg__subsample': 0.7646528086614623, 'reg__colsample_bytree': 0.6629518834281192, 'reg__min_child_samples': 38, 'reg__reg_alpha': 4.1242935847224915, 'reg__reg_lambda': 5.062429388658762, 'reg__min_split_gain': 0.2809813910043346, 'reg__path_smooth': 4.915840724445432, 'reg__min_child_weight': 0.013596850944833067, 'reg__feature_fraction': 0.6511437580637881, 'reg__bagging_fraction': 0.8113075517571684, 'reg__bagging_freq': 7, 'reg__max_bin': 206, 'reg__min_data_per_group': 152}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.019513815643427967, 'reg__num_leaves': 30, 'reg__subsample': 0.7285273544302782, 'reg__colsample_bytree': 0.6011950679660146, 'reg__min_child_samples': 29, 'reg__reg_alpha': 0.8401567729086953, 'reg__reg_lambda': 8.10503249320582, 'reg__min_split_gain': 1.5701844926568802, 'reg__path_smooth': 1.6805711687789089, 'reg__min_child_weight': 0.02675419960363769, 'reg__feature_fraction': 0.8357849878226027, 'reg__bagging_fraction': 0.8353245838619471, 'reg__bagging_freq': 7, 'reg__max_bin': 135, 'reg__min_data_per_group': 112}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.054697867292386786, 'reg__num_leaves': 57, 'reg__subsample': 0.8605926496357786, 'reg__colsample_bytree': 0.794449842883156, 'reg__min_child_samples': 29, 'reg__reg_alpha': 5.717579074504424, 'reg__reg_lambda': 3.6068034289405526, 'reg__min_split_gain': 0.30980872081406785, 'reg__path_smooth': 3.3424260596066278, 'reg__min_child_weight': 0.16582839552371412, 'reg__feature_fraction': 0.5763536902132212, 'reg__bagging_fraction': 0.7457971570821975, 'reg__bagging_freq': 2, 'reg__max_bin': 127, 'reg__min_data_per_group': 112}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.05022143927140124, 'reg__num_leaves': 34, 'reg__subsample': 0.8099094892920691, 'reg__colsample_bytree': 0.682558928081841, 'reg__min_child_samples': 36, 'reg__reg_alpha': 3.47777780155916, 'reg__reg_lambda': 33.13897444843016, 'reg__min_split_gain': 0.3134948316309627, 'reg__path_smooth': 3.9200516576600446, 'reg__min_child_weight': 2.338833769944953, 'reg__feature_fraction': 0.6592020897861481, 'reg__bagging_fraction': 0.7964112015543963, 'reg__bagging_freq': 6, 'reg__max_bin': 179, 'reg__min_data_per_group': 132}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 7, 'reg__learning_rate': 0.04714976190672968, 'reg__num_leaves': 30, 'reg__subsample': 0.7523165662226614, 'reg__colsample_bytree': 0.6098064417658369, 'reg__min_child_samples': 29, 'reg__reg_alpha': 4.813166248498061, 'reg__reg_lambda': 19.561291898951932, 'reg__min_split_gain': 0.4076861649924055, 'reg__path_smooth': 2.137233919681825, 'reg__min_child_weight':

0.018570235455409267, 'reg__feature_fraction': 0.625853415959434, 'reg__bagging_fraction': 0.8407359264063607, 'reg__bagging_freq': 5, 'reg__max_bin': 126, 'reg__min_data_per_group': 106}

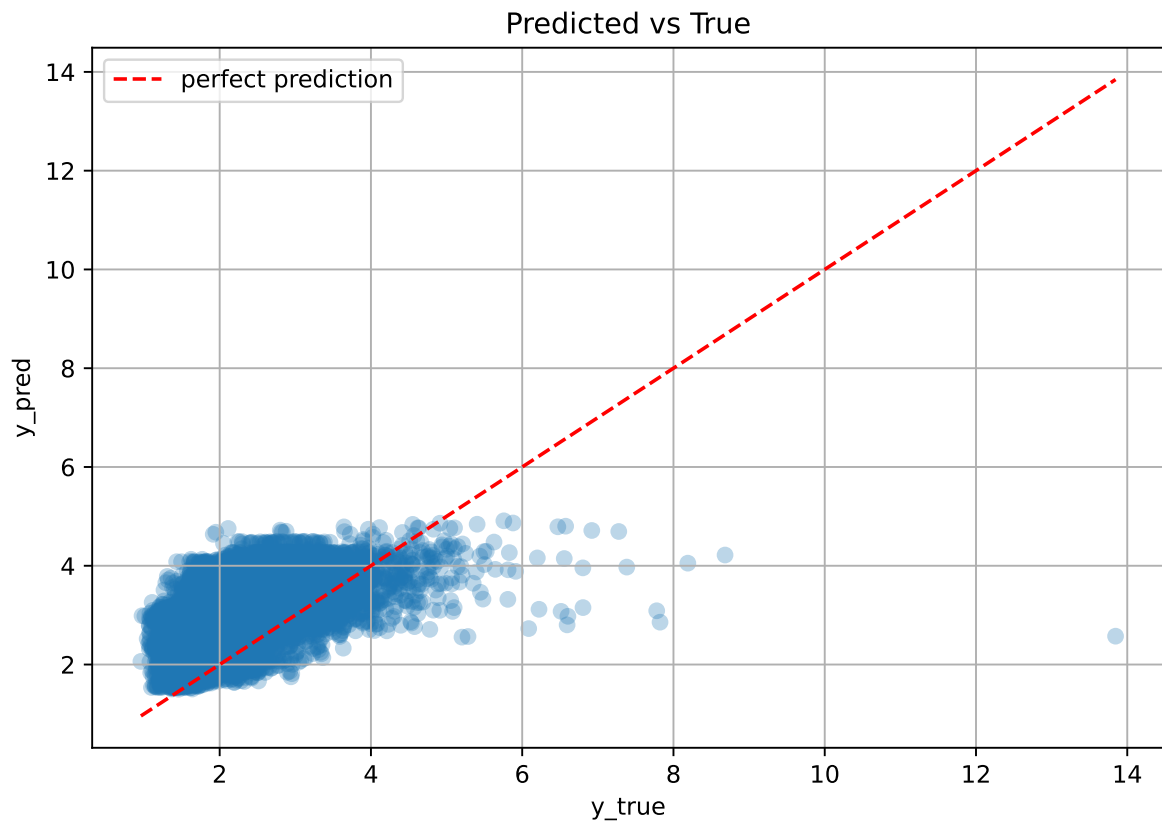
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.08851560645119906, 'reg__num_leaves': 58, 'reg__subsample': 0.7548582405031575, 'reg__colsample_bytree': 0.7947213391651522, 'reg__min_child_samples': 29, 'reg__reg_alpha': 5.669169945618118, 'reg__reg_lambda': 4.686884372830253, 'reg__min_split_gain': 0.32015764340574737, 'reg__path_smooth': 3.2379359239326435, 'reg__min_child_weight': 0.0010328321663218365, 'reg__feature_fraction': 0.5539698836903503, 'reg__bagging_fraction': 0.8383890234594774, 'reg__bagging_freq': 2, 'reg__max_bin': 102, 'reg__min_data_per_group': 109}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.029917084822108947, 'reg__num_leaves': 52, 'reg__subsample': 0.8383948232601273, 'reg__colsample_bytree': 0.7299963709026247, 'reg__min_child_samples': 33, 'reg__reg_alpha': 2.307600317452381, 'reg__reg_lambda': 39.72450547948174, 'reg__min_split_gain': 0.5794420169888491, 'reg__path_smooth': 2.4433868459143593, 'reg__min_child_weight': 0.005636857159054257, 'reg__feature_fraction': 0.7276118358611808, 'reg__bagging_fraction': 0.7688043258815208, 'reg__bagging_freq': 7, 'reg__max_bin': 112, 'reg__min_data_per_group': 120}

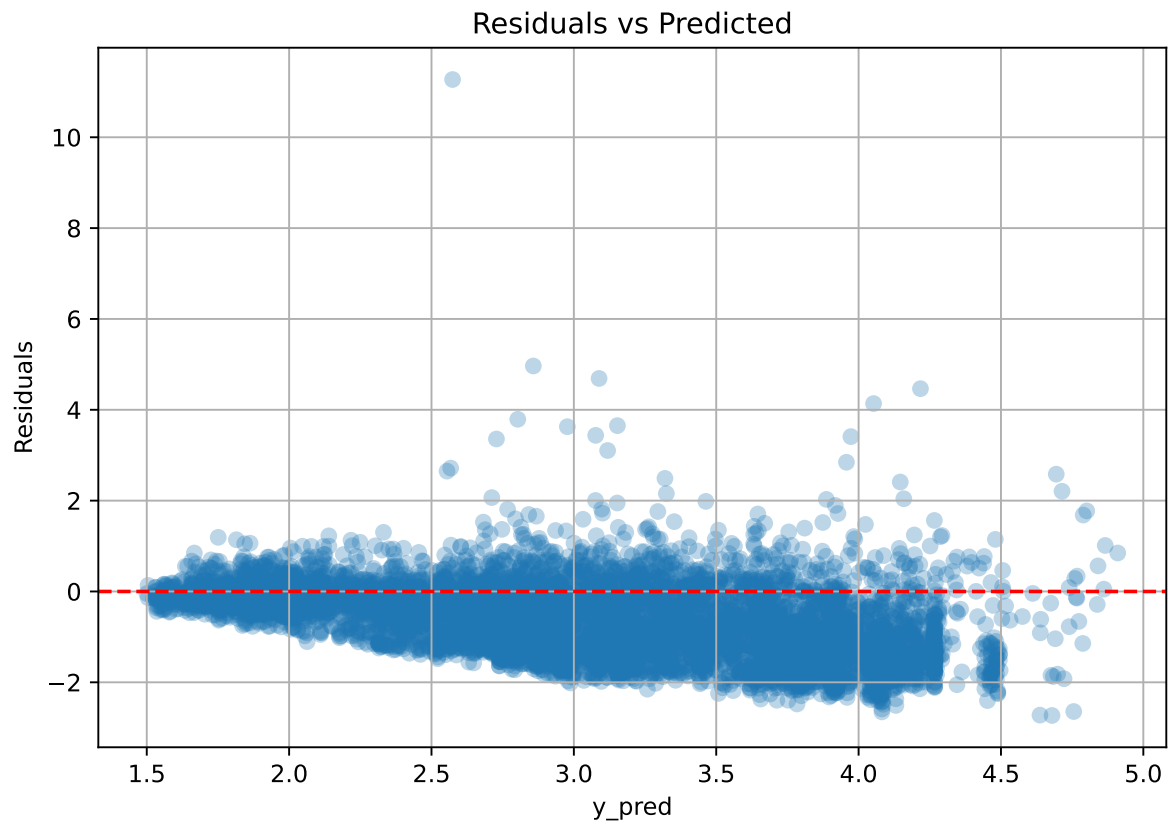
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.021607414334453377, 'reg__num_leaves': 60, 'reg__subsample': 0.7961329056908558, 'reg__colsample_bytree': 0.7386834887054834, 'reg__min_child_samples': 24, 'reg__reg_alpha': 7.663109539734248, 'reg__reg_lambda': 2.445712263170279, 'reg__min_split_gain': 0.19903343076217125, 'reg__path_smooth': 4.424960456442006, 'reg__min_child_weight': 0.3596047540602059, 'reg__feature_fraction': 0.556214952505467, 'reg__bagging_fraction': 0.6735603392567204, 'reg__bagging_freq': 2, 'reg__max_bin': 177, 'reg__min_data_per_group': 125}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.04706952391030352, 'reg__num_leaves': 63, 'reg__subsample': 0.7686381401703304, 'reg__colsample_bytree': 0.757541033017736, 'reg__min_child_samples': 32, 'reg__reg_alpha': 4.690626005579727, 'reg__reg_lambda': 8.795728654346117, 'reg__min_split_gain': 0.4069599385668166, 'reg__path_smooth': 3.63334432397256, 'reg__min_child_weight': 0.11198749307541604, 'reg__feature_fraction': 0.5760049390065212, 'reg__bagging_fraction': 0.7265757824880216, 'reg__bagging_freq': 7, 'reg__max_bin': 157, 'reg__min_data_per_group': 99}

2.4.6 Scatter



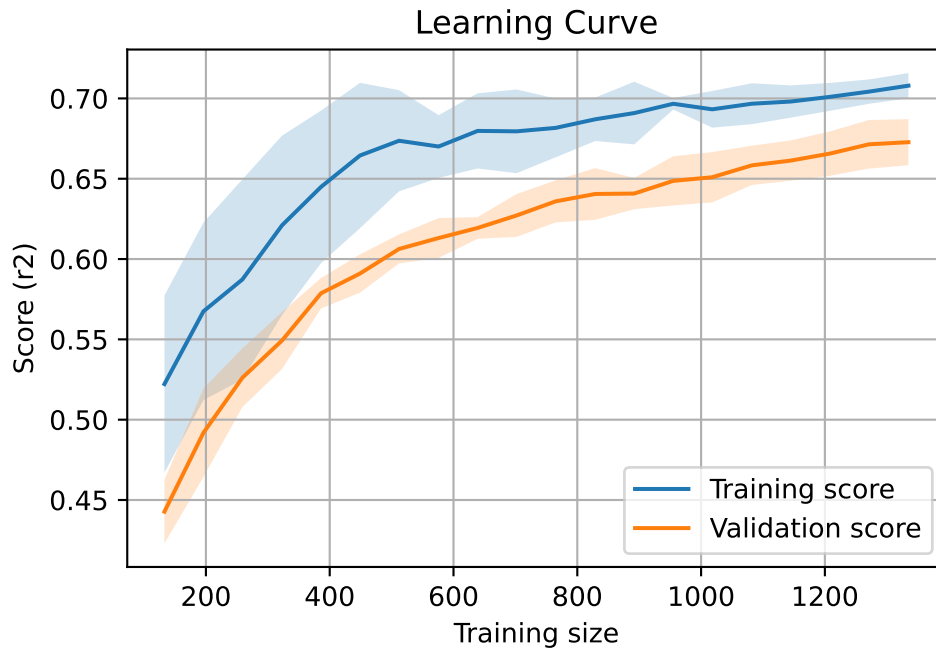
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2_11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

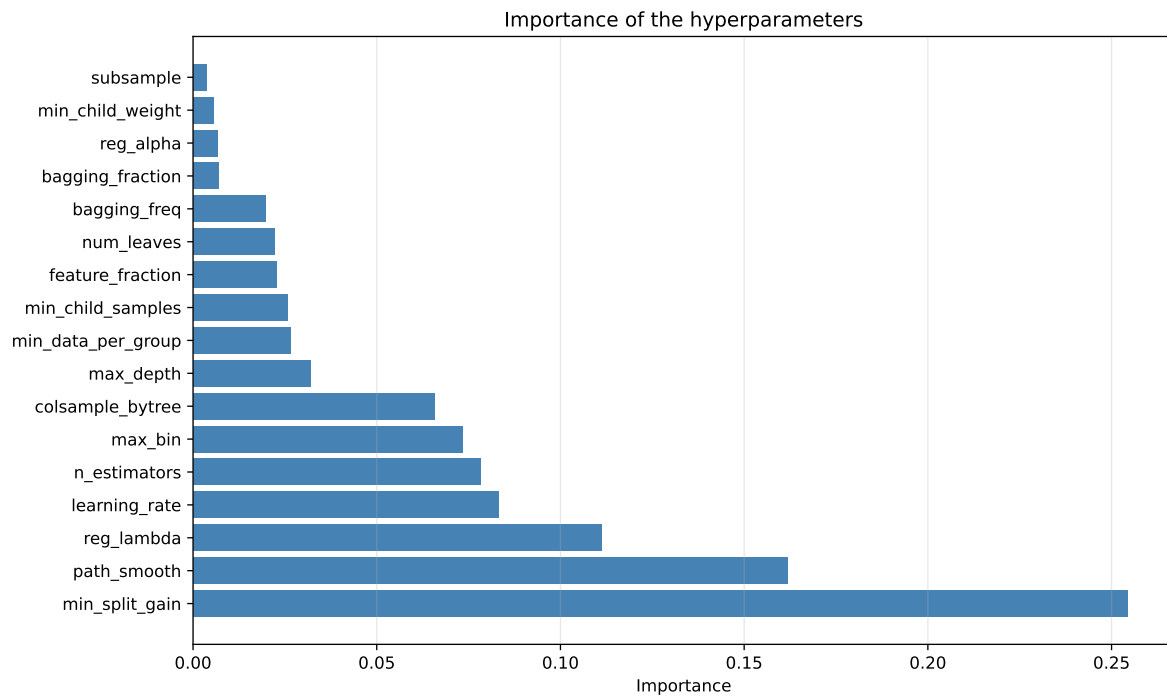
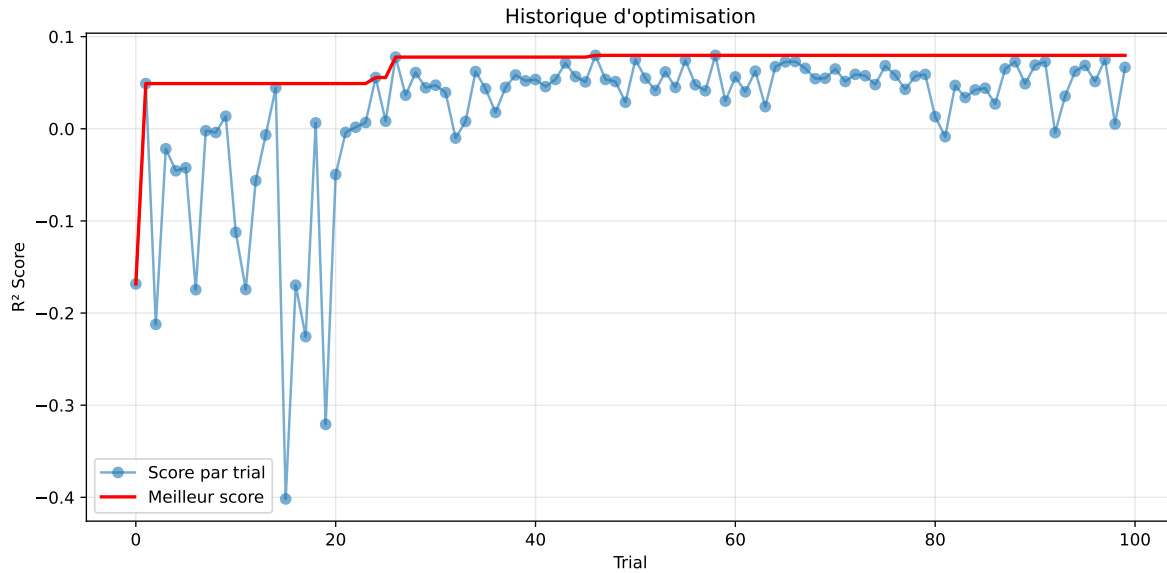


2.4.9 Gridsearch — Group: 3_4

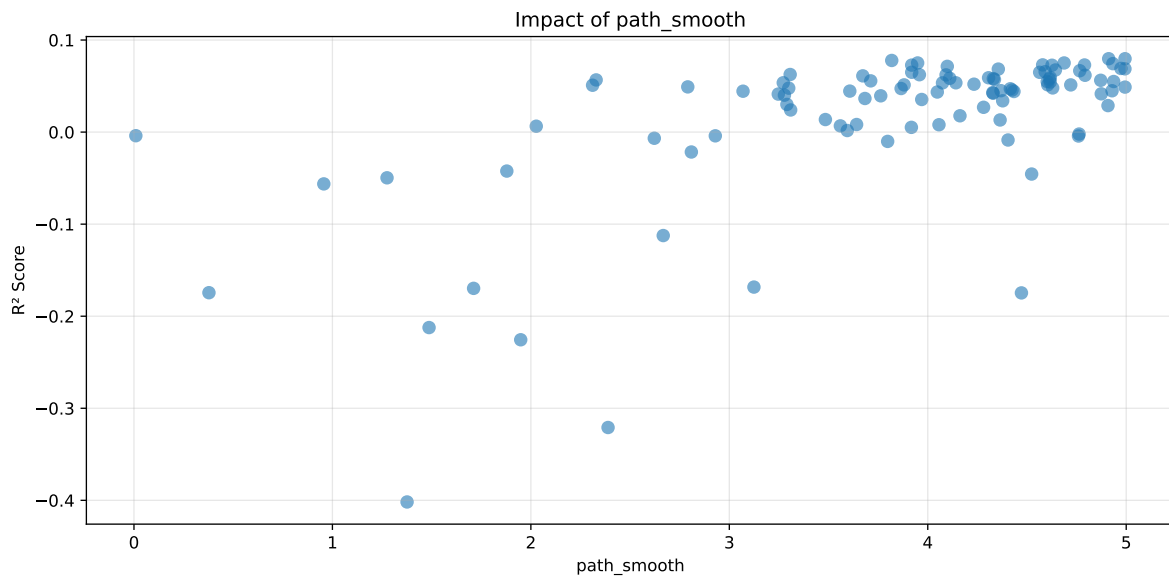
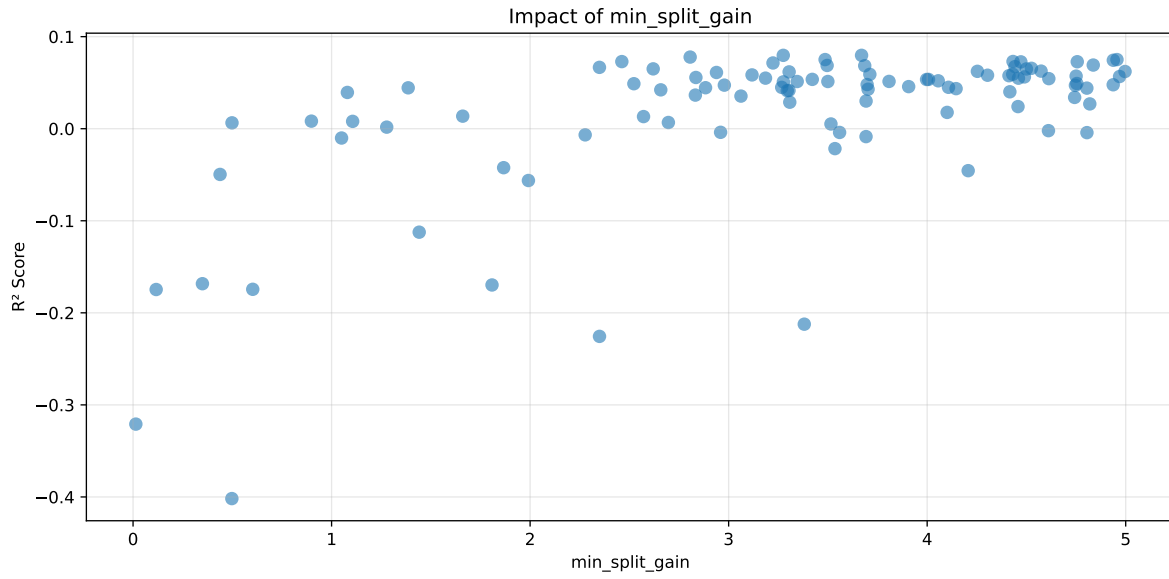
Len(group) : 568, Len(training) : 398, Len(test) : 170

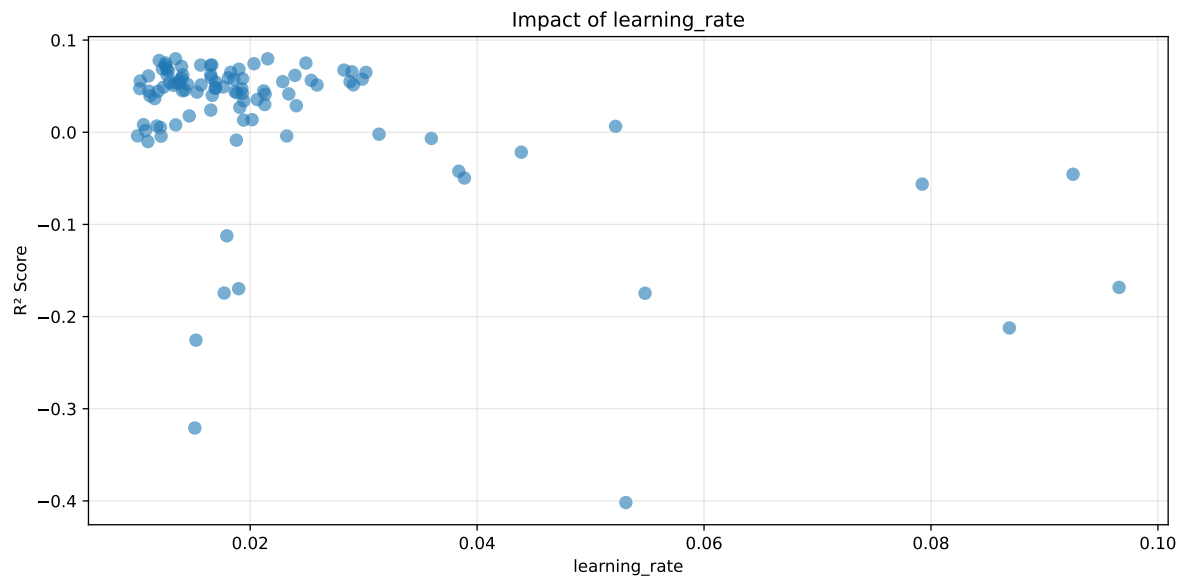
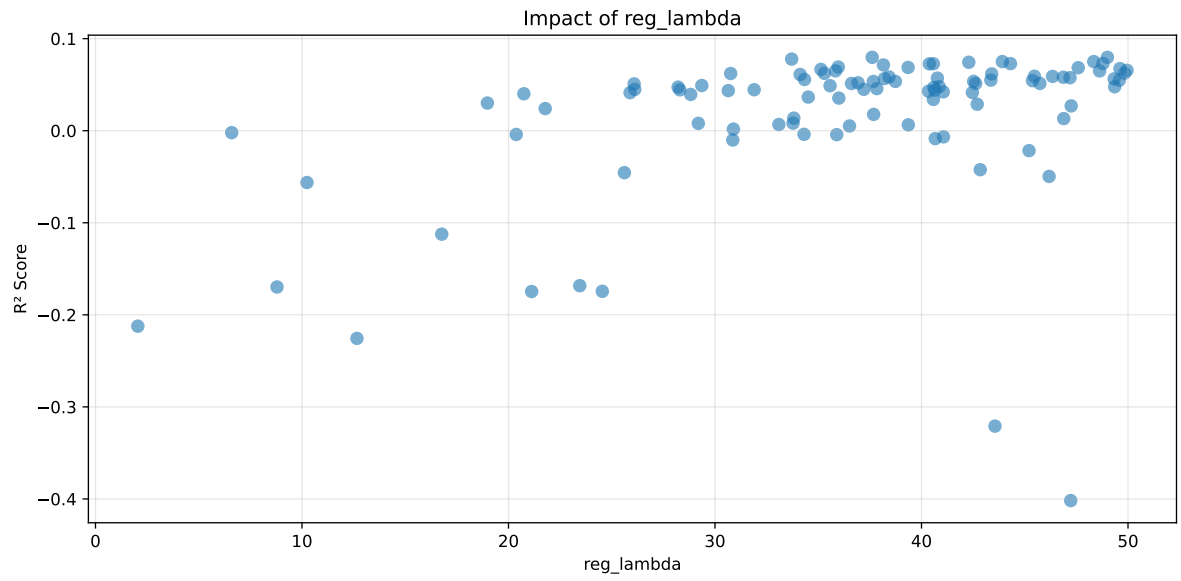
Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

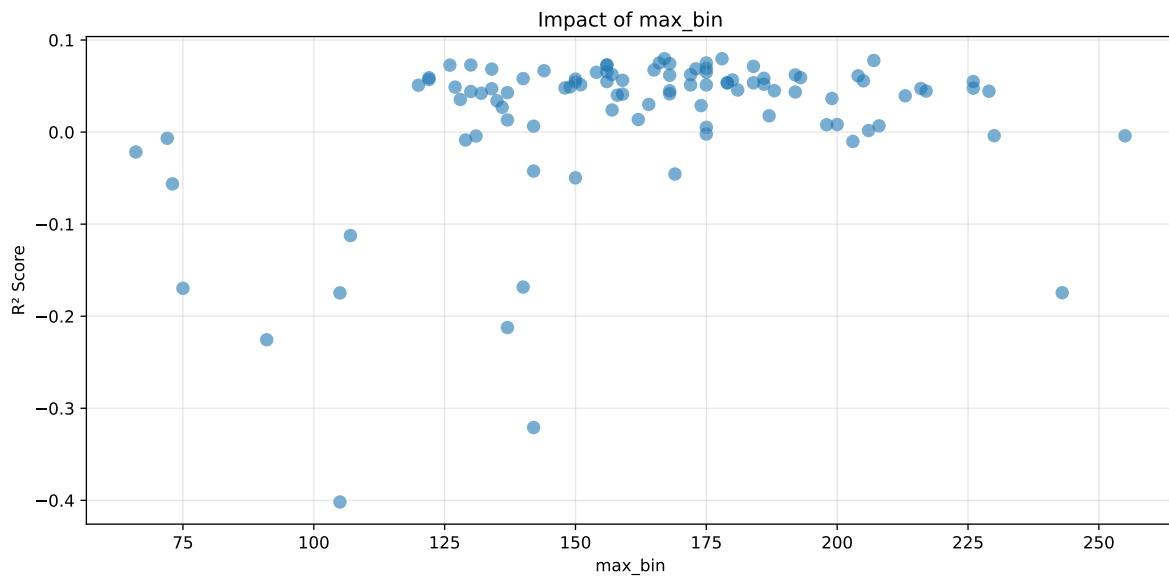
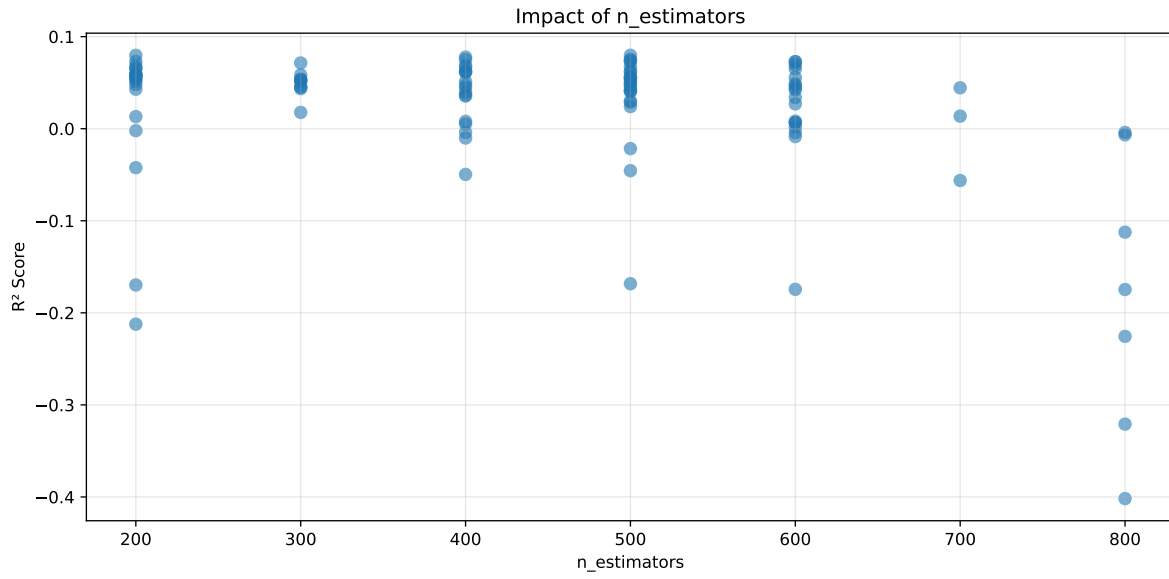
===== TOP Importance FEATURES =====
feature importance r_umidity 54 r_NH3 16 r_temperature 7 r_THI 6 r_CO2 2

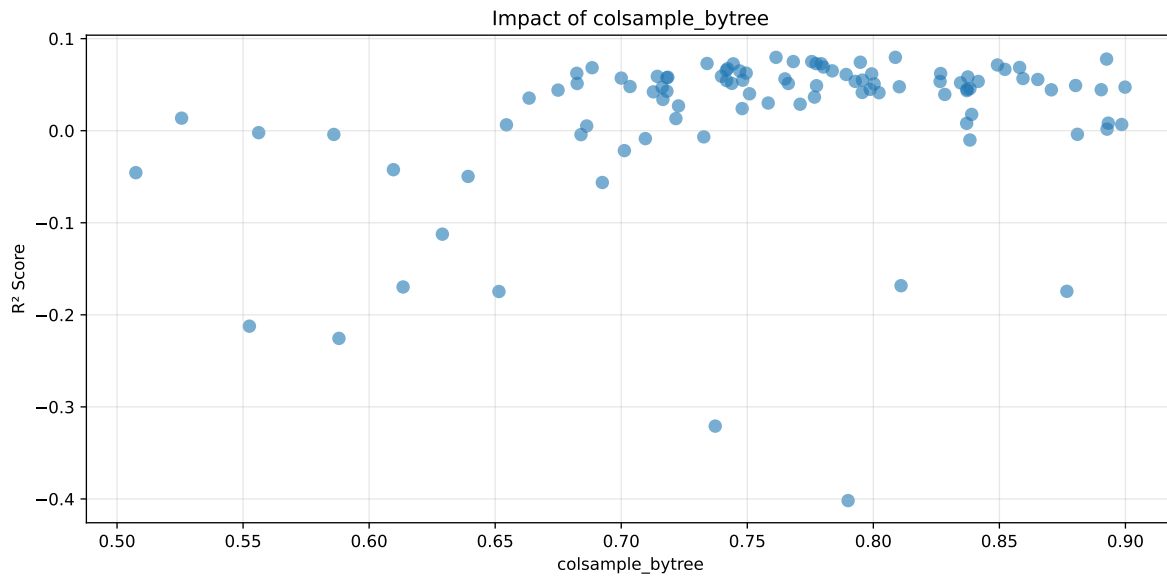


Paramètres avec >5% d'importance : ['min_split_gain', 'path_smooth', 'reg_lambda', 'learning_rate', 'n_estimators', 'max_bin', 'colsample_bytree']



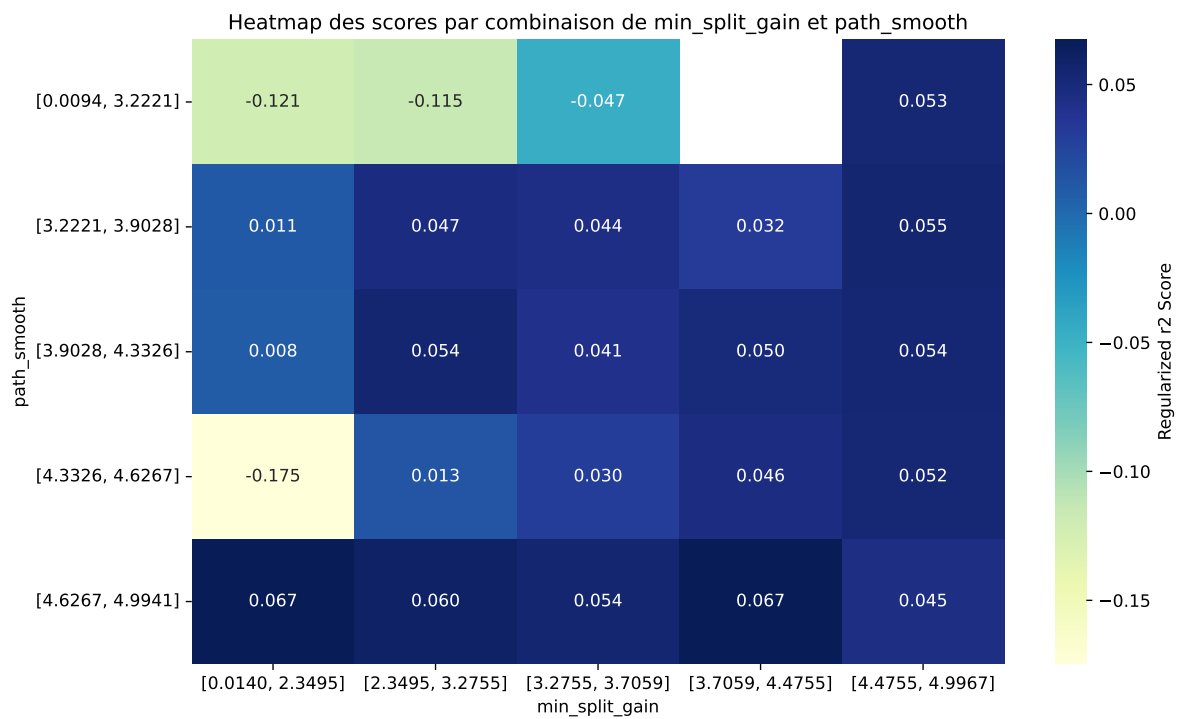






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.1208	0.114	0.0404	0.0473	0.0797	0.0655	-0.0485	0.5749	1.0601
2	0.1962	0.1768	0.0604	0.0466	0.0797	0.1063	-0.0705	0.6012	1.1099
3	0.1684	0.1533	0.0378	0.0461	0.0778	0.0843	-0.069	0.5498	1.0985
4	0.13	0.1209	0.0398	0.0378	0.0751	0.0777	-0.0432	0.6429	1.0757
5	0.144	0.1325	0.0448	0.0408	0.0751	0.0708	-0.0617	0.5344	1.0866
6	0.1372	0.1267	0.0436	0.0313	0.0744	0.0659	-0.0609	0.5197	1.0826
7	0.1598	0.1453	0.0448	0.0317	0.073	0.0714	-0.0739	0.4916	1.0995
8	0.2086	0.186	0.0689	0.0487	0.0729	0.1074	-0.0786	0.5774	1.1216
9	0.1274	0.1183	0.0436	0.0392	0.0728	0.0797	-0.0386	0.6735	1.077
10	0.1742	0.1573	0.0404	0.0329	0.0726	0.0603	-0.097	0.3834	1.1076
---	---	---	---	---	---	---	---	---	---
100	0.4399	0.2996	0.133	0.0322	-0.4018	0.1852	-0.1145	0.6179	1.4682

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.021542980001210695, 'reg__num_leaves': 22, 'reg__subsample': 0.5401537994646601, 'reg__colsample_bytree': 0.8087483663558098, 'reg__min_child_samples': 25, 'reg__reg_alpha': 5.6492815258471385, 'reg__reg_lambda': 49.00261575992393, 'reg__min_split_gain': 3.669531718656025, 'reg__path_smooth': 4.9940526958278975, 'reg__min_child_weight': 0.13382749236195784, 'reg__feature_fraction': 0.6958610880048173, 'reg__bagging_fraction': 0.6186227658678753, 'reg__bagging_freq': 2, 'reg__max_bin': 167, 'reg__min_data_per_group': 185}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.013425845063623187, 'reg__num_leaves': 28, 'reg__subsample': 0.6227815389112461, 'reg__colsample_bytree': 0.7614012816185485, 'reg__min_child_samples': 38, 'reg__reg_alpha': 4.824075527249937, 'reg__reg_lambda': 37.61194228269758, 'reg__min_split_gain': 3.275145691769013, 'reg__path_smooth': 4.911648414086385, 'reg__min_child_weight': 0.5897598815015695, 'reg__feature_fraction': 0.755728543662109, 'reg__bagging_fraction': 0.7332940017657528, 'reg__bagging_freq': 2, 'reg__max_bin': 178, 'reg__min_data_per_group': 173}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.011982471870867956, 'reg__num_leaves': 31, 'reg__subsample': 0.5006508068964897, 'reg__colsample_bytree': 0.8925194407836098, 'reg__min_child_samples': 49, 'reg__reg_alpha': 7.009899967741214, 'reg__reg_lambda': 33.71246287126011, 'reg__min_split_gain': 2.8061635027888023, 'reg__path_smooth': 3.818427886385372, 'reg__min_child_weight': 1.4478108619857397, 'reg__feature_fraction': 0.8157664383511234, 'reg__bagging_fraction': 0.7049202075282904, 'reg__bagging_freq': 3, 'reg__max_bin': 207, 'reg__min_data_per_group': 96}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.012541091274146566, 'reg__num_leaves': 25, 'reg__subsample': 0.5475570879848484, 'reg__colsample_bytree': 0.7755626429940727, 'reg__min_child_samples': 35, 'reg__reg_alpha': 5.019364036648363, 'reg__reg_lambda': 48.34049036598373, 'reg__min_split_gain': 3.4858704969857417, 'reg__path_smooth': 3.949007213727359, 'reg__min_child_weight': 0.9083127346203533, 'reg__feature_fraction': 0.8023006683705604, 'reg__bagging_fraction': 0.5492059899166796, 'reg__bagging_freq': 3, 'reg__max_bin': 175, 'reg__min_data_per_group': 176}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.024909438658515772, 'reg__num_leaves': 22, 'reg__subsample': 0.6582010889468992, 'reg__colsample_bytree': 0.7682475396541562, 'reg__min_child_samples': 27, 'reg__reg_alpha': 4.7558123846701115, 'reg__reg_lambda': 43.916033762353585, 'reg__min_split_gain': 4.956059685940214, 'reg__path_smooth': 4.687115540525264, 'reg__min_child_weight': 0.6843792440257433, 'reg__feature_fraction': 0.6931610649653867, 'reg__bagging_fraction': 0.7299737461983811, 'reg__bagging_freq': 2, 'reg__max_bin': 166, 'reg__min_data_per_group': 173}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.020335369959787648, 'reg__num_leaves': 23, 'reg__subsample': 0.5330299075678953, 'reg__colsample_bytree': 0.794845504376179, 'reg__min_child_samples': 26, 'reg__reg_alpha': 5.7309950194453165, 'reg__reg_lambda': 42.29068002708768, 'reg__min_split_gain': 4.9376701820057205, 'reg__path_smooth': 4.932634405503341, 'reg__min_child_weight': 0.6273094163258155, 'reg__feature_fraction': 0.6848119217277968, 'reg__bagging_fraction': 0.7195697243558317, 'reg__bagging_freq': 2, 'reg__max_bin': 168, 'reg__min_data_per_group': 179}

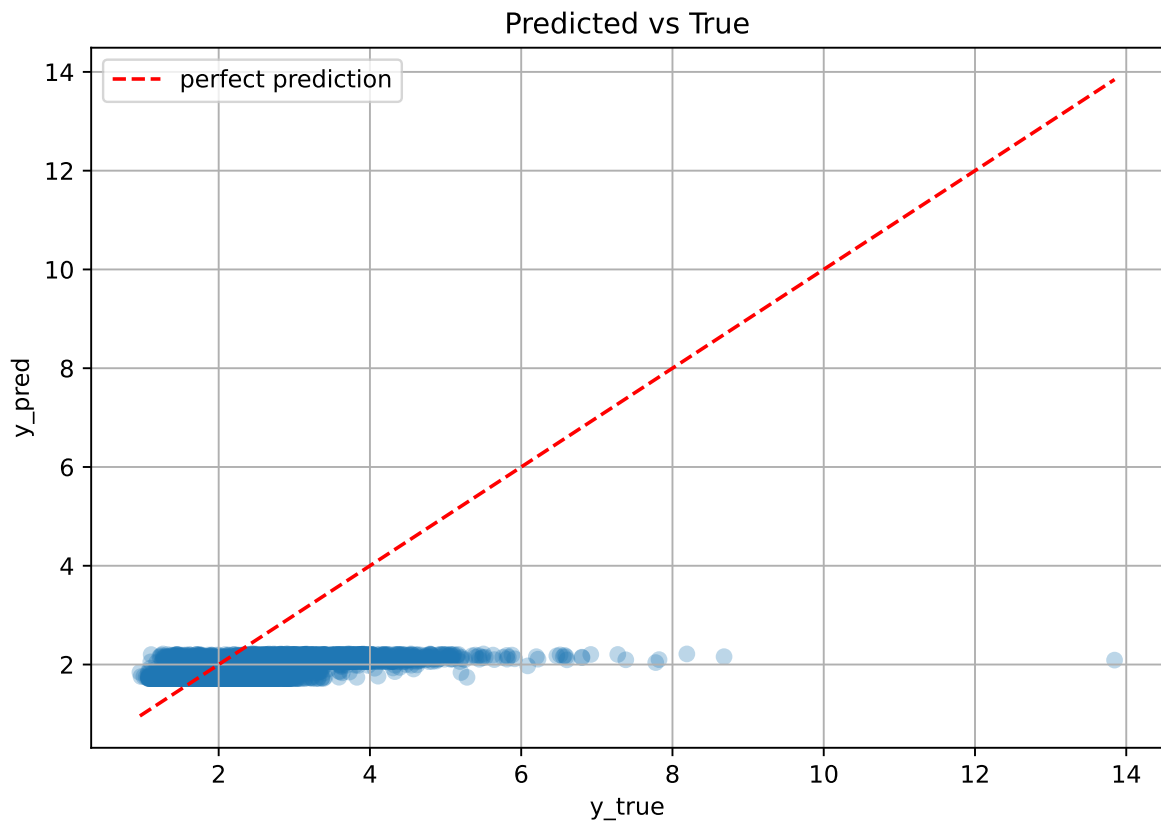
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.01662209286881076, 'reg__num_leaves': 21, 'reg__subsample': 0.5358923886076813, 'reg__colsample_bytree': 0.7340463491686904, 'reg__min_child_samples': 33, 'reg__reg_alpha': 4.217642229838263, 'reg__reg_lambda': 48.78572505628311, 'reg__min_split_gain': 4.432467767102356, 'reg__path_smooth': 4.578877289230253, 'reg__min_child_weight': 0.17552865689187086, 'reg__feature_fraction': 0.7010339893874381, 'reg__bagging_fraction': 0.6179717466640545, 'reg__bagging_freq': 1, 'reg__max_bin': 156, 'reg__min_data_per_group': 159}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.015630861806905427, 'reg__num_leaves': 25, 'reg__subsample': 0.7078443122279559, 'reg__colsample_bytree': 0.77727099246361, 'reg__min_child_samples': 28, 'reg__reg_alpha': 3.5713611325892387, 'reg__reg_lambda': 44.30825308476297, 'reg__min_split_gain': 2.4620079851079497, 'reg__path_smooth': 4.789913123986523, 'reg__min_child_weight': 0.055414993639007815, 'reg__feature_fraction': 0.7820994606036714, 'reg__bagging_fraction': 0.5942397716016478, 'reg__bagging_freq': 2, 'reg__max_bin': 130, 'reg__min_data_per_group': 178}

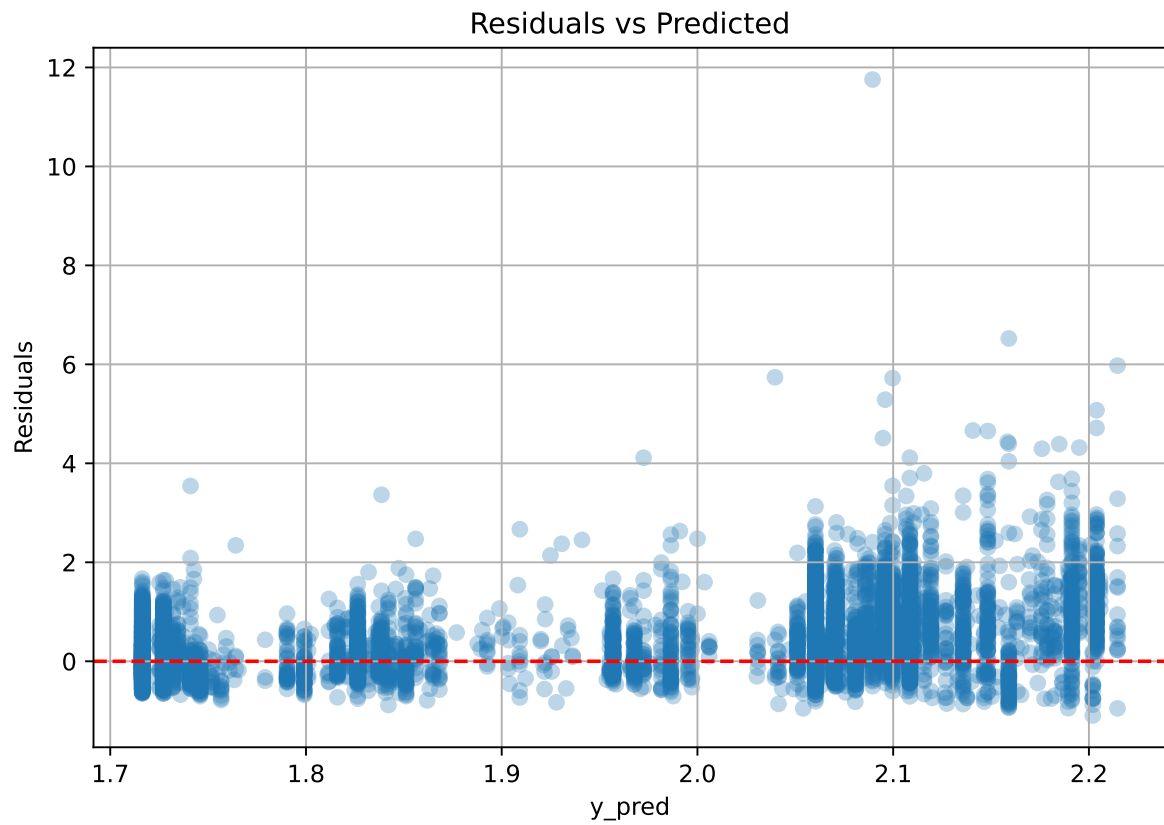
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.01250677230484856, 'reg__num_leaves': 24, 'reg__subsample': 0.6900133019521095, 'reg__colsample_bytree': 0.7793490927180844, 'reg__min_child_samples': 21, 'reg__reg_alpha': 5.086063892469309, 'reg__reg_lambda': 40.57460341304551, 'reg__min_split_gain': 4.756009356797355, 'reg__path_smooth': 3.9189228038079116, 'reg__min_child_weight': 0.06091502440162609, 'reg__feature_fraction': 0.7676546353062481, 'reg__bagging_fraction': 0.7168613379807811, 'reg__bagging_freq': 2, 'reg__max_bin': 126, 'reg__min_data_per_group': 177}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.016512104013730267, 'reg__num_leaves': 21, 'reg__subsample': 0.6676299971921564, 'reg__colsample_bytree': 0.7444338082220165, 'reg__min_child_samples': 22, 'reg__reg_alpha': 4.156104389686481, 'reg__reg_lambda': 40.3814869513771, 'reg__min_split_gain': 4.471966975656571, 'reg__path_smooth': 4.625960586560135, 'reg__min_child_weight': 0.3379542323637709, 'reg__feature_fraction': 0.7101495754927339, 'reg__bagging_fraction': 0.5355458595141719, 'reg__bagging_freq': 1, 'reg__max_bin': 156, 'reg__min_data_per_group': 158}

2.4.10 Scatter



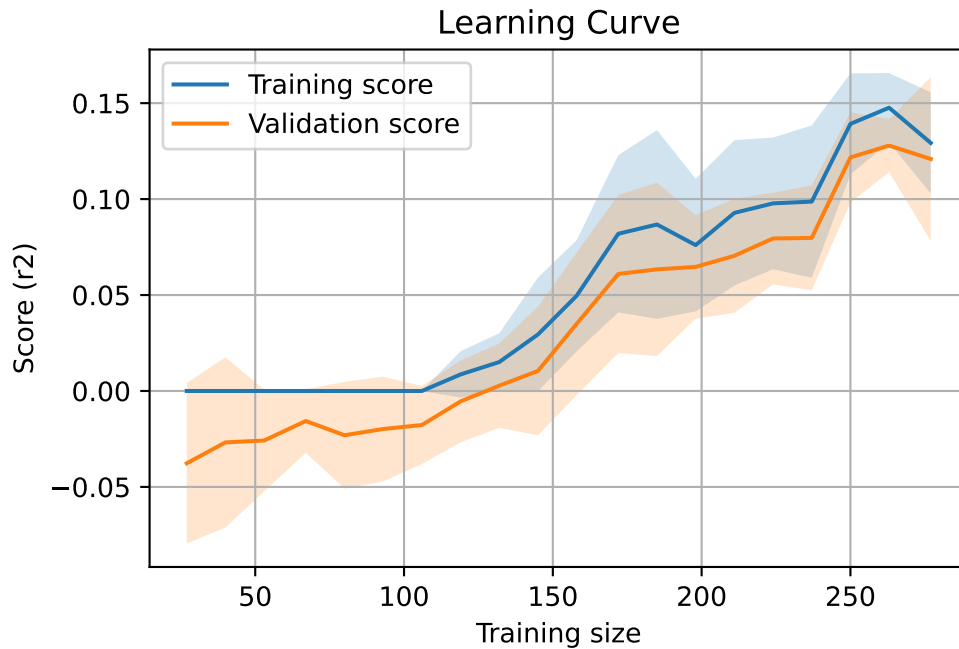
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 3_4

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170

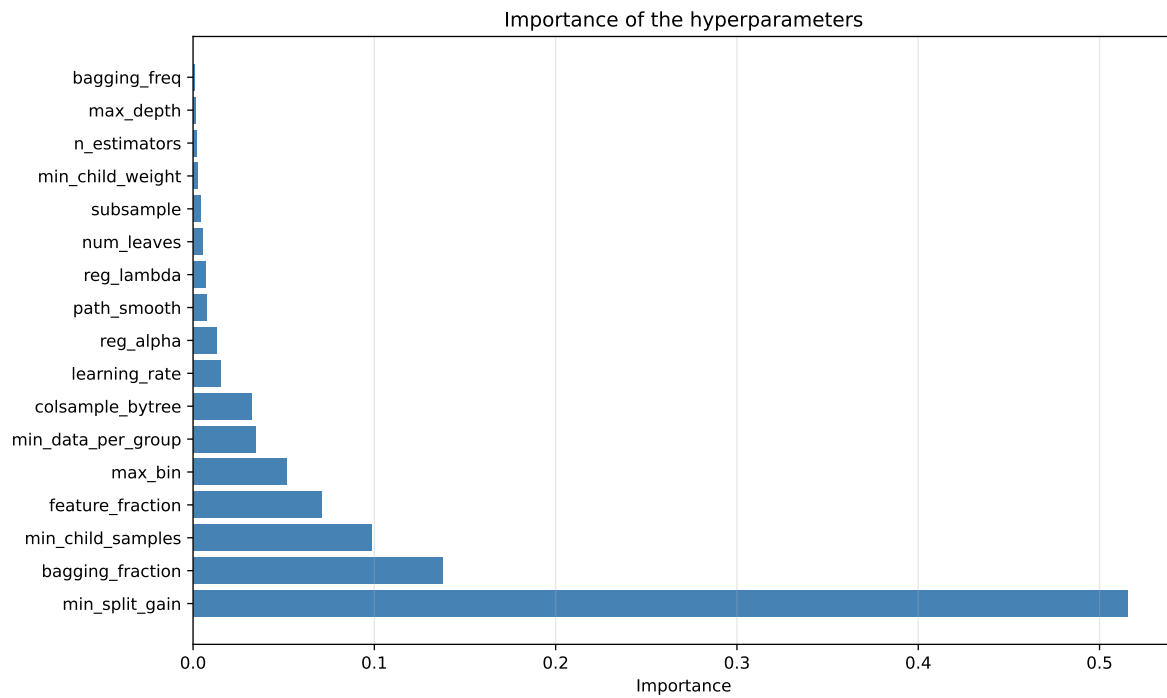
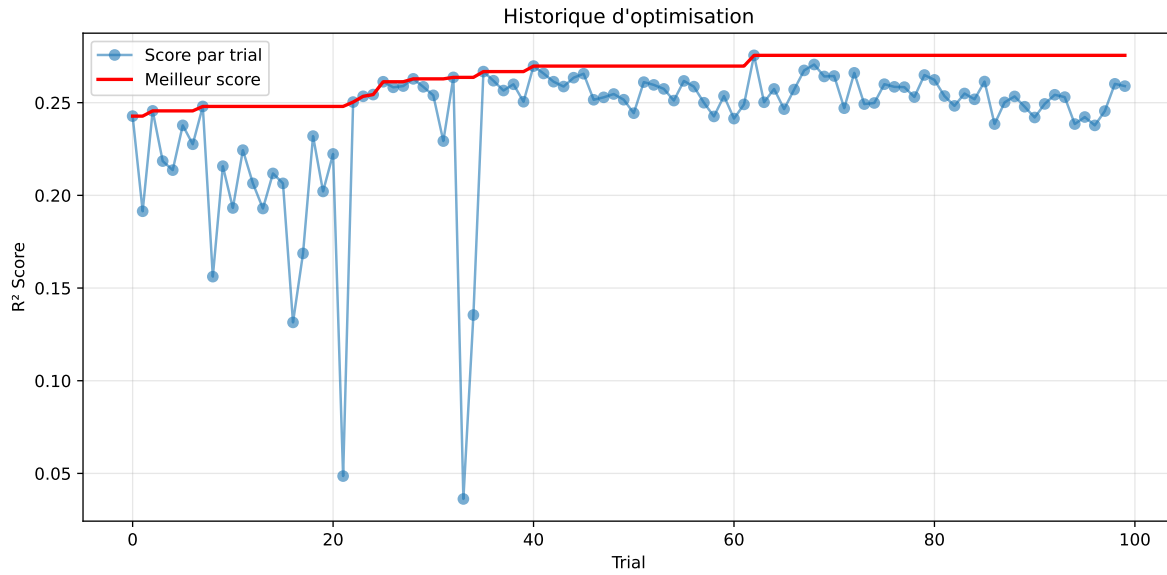


2.4.13 Gridsearch — Group: 4_5

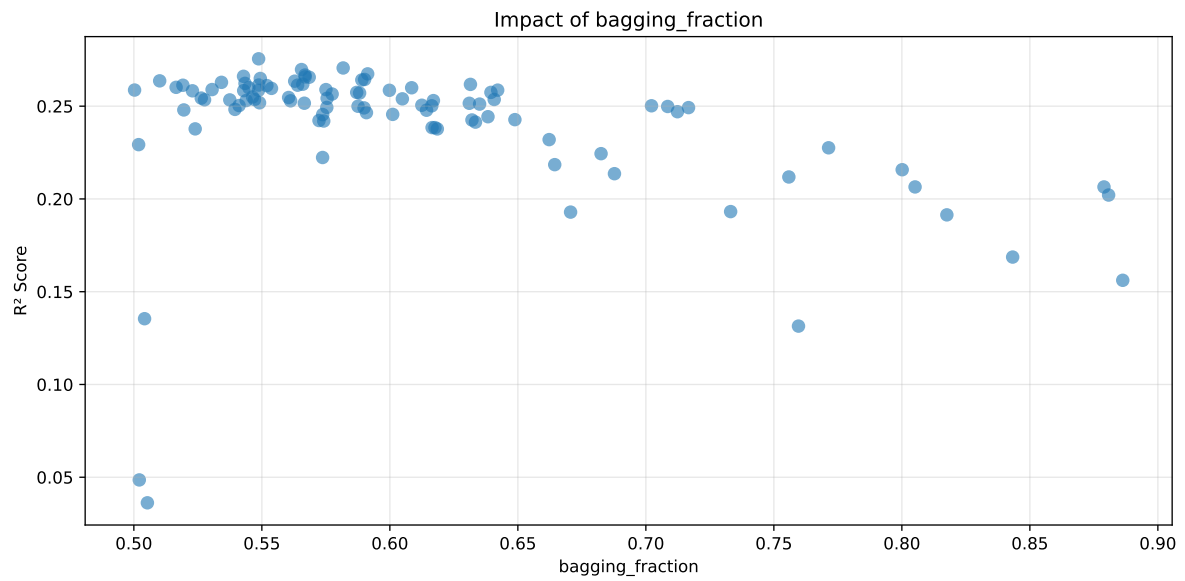
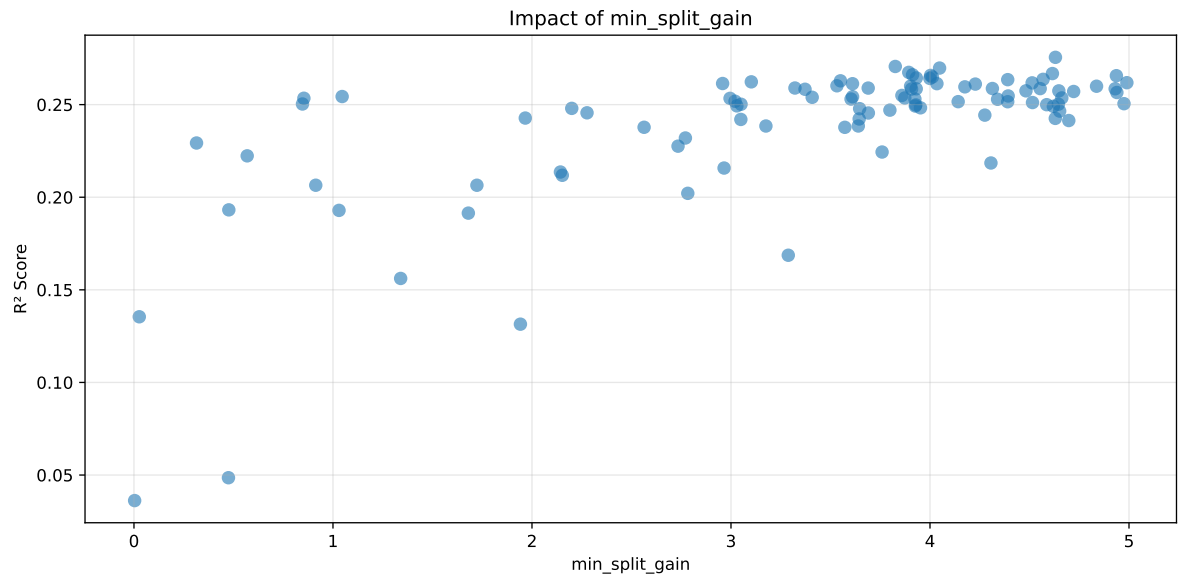
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

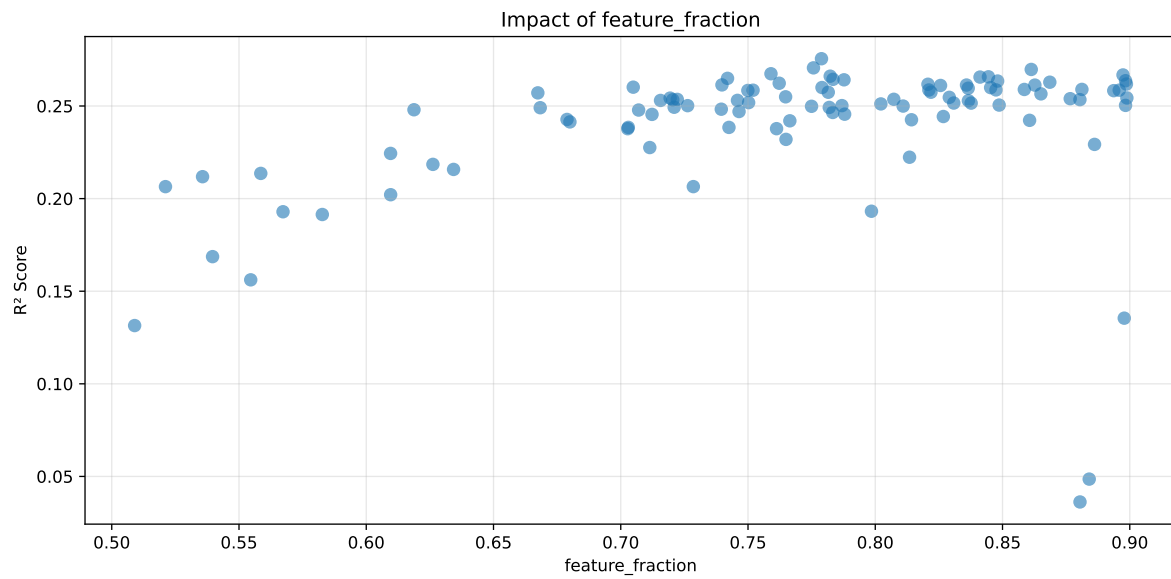
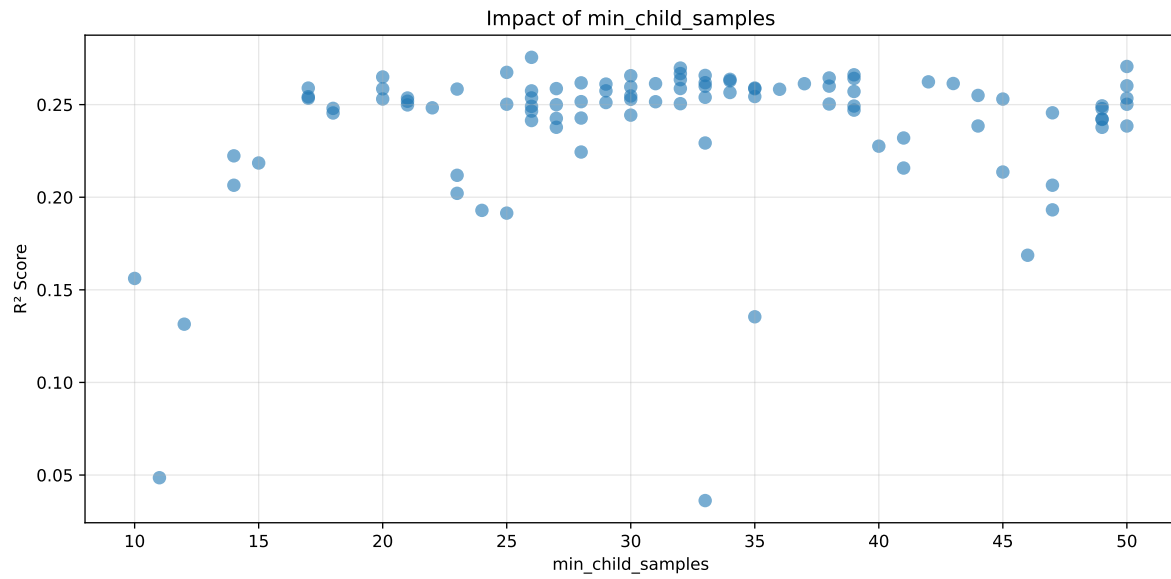
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

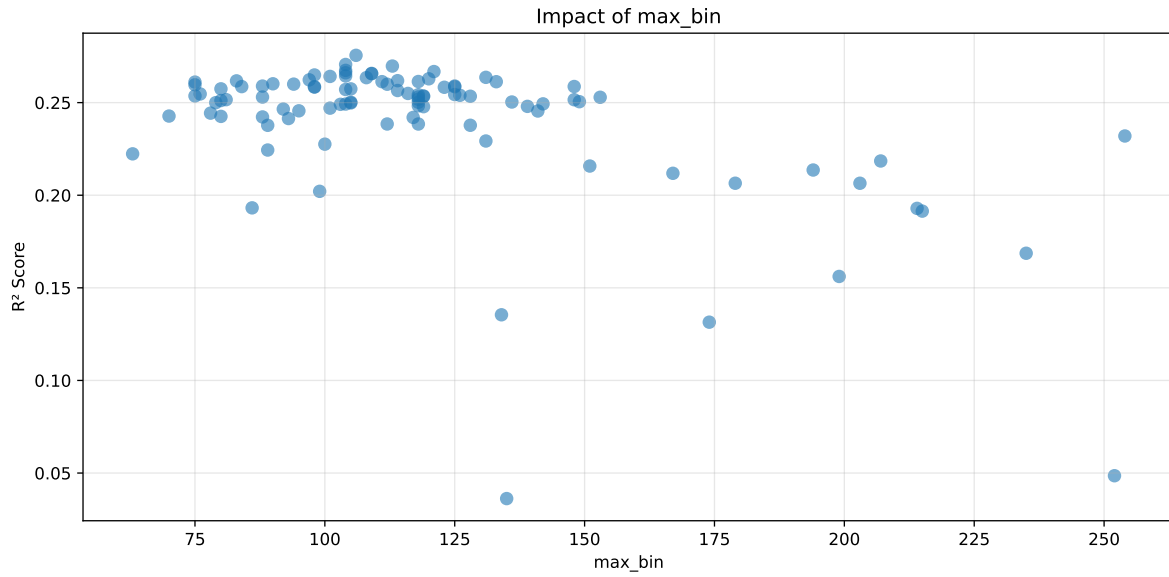
===== TOP Importance FEATURES =====
feature importance r_umidity 28 r_NH3 3 r_temperature 3 r_CO2 0 r_THI 0



Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction', 'min_child_samples', 'feature_fraction', 'max_bin']

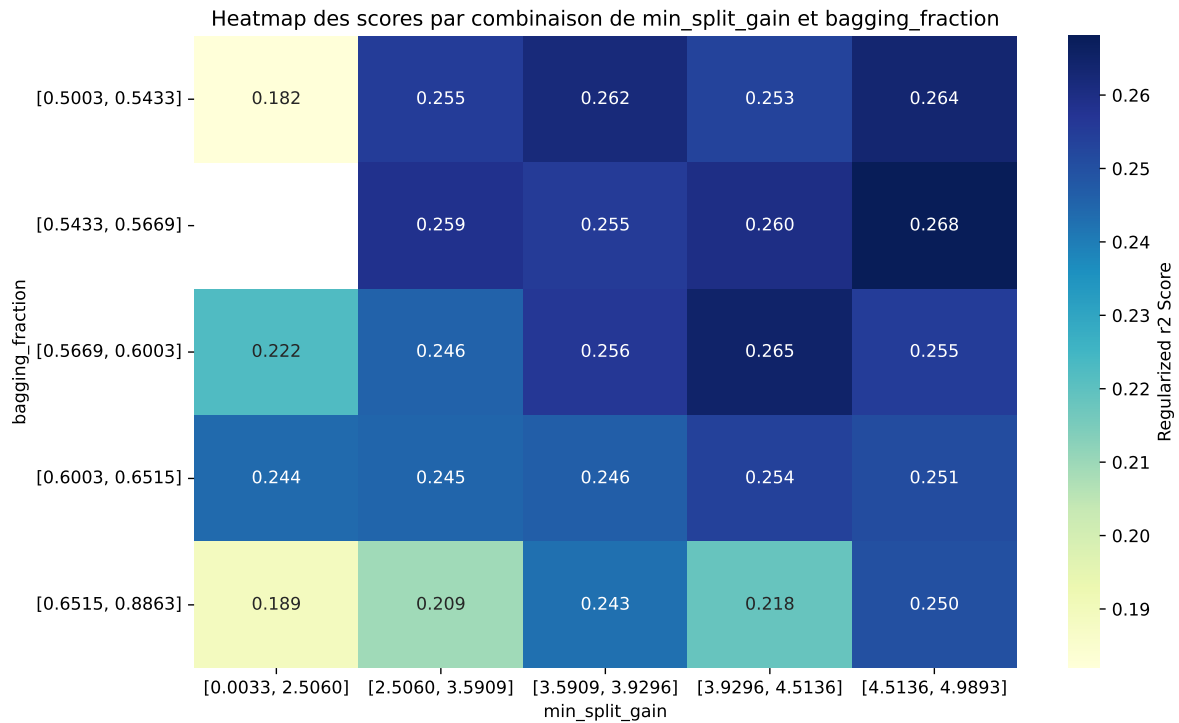






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.392	0.3726	0.4226	0.0224	0.2755	0.4386	0.066	1.177	1.0521
2	0.3974	0.3763	0.428	0.024	0.2706	0.4467	0.0704	1.1871	1.0562
3	0.4021	0.3801	0.4314	0.023	0.2697	0.4464	0.0663	1.1744	1.0581
4	0.4011	0.3788	0.4318	0.0218	0.2674	0.4559	0.0771	1.2035	1.0588
5	0.3909	0.3702	0.4207	0.0238	0.2668	0.436	0.0658	1.1777	1.0559
6	0.3987	0.3766	0.4294	0.0228	0.2661	0.445	0.0684	1.1817	1.0587
7	0.3941	0.3727	0.4241	0.0248	0.2658	0.4366	0.0639	1.1714	1.0574
8	0.3898	0.3691	0.4198	0.0248	0.2656	0.4358	0.0667	1.1807	1.0561
9	0.4019	0.379	0.4339	0.0218	0.2649	0.449	0.07	1.1847	1.0602
10	0.3977	0.3755	0.4267	0.0198	0.2644	0.4332	0.0577	1.1537	1.0592
---	---	---	---	---	---	---	---	---	---
100	0.5608	0.4734	0.5392	0.0208	0.0362	0.5557	0.0823	1.174	1.1847

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.0956014857143643, 'reg__num_leaves': 41, 'reg__subsample': 0.85666127366791, 'reg__colsample_bytree': 0.8172153524823681, 'reg__min_child_samples': 26, 'reg__reg_alpha': 6.992148591719106, 'reg__reg_lambda': 18.88020665607728, 'reg__min_split_gain': 4.630436066502336, 'reg__path_smooth': 2.259328229807912, 'reg__min_child_weight': 0.0010768404969732543, 'reg__feature_fraction': 0.77886959135914, 'reg__bagging_fraction': 0.5487370977767367, 'reg__bagging_freq': 4, 'reg__max_bin': 106, 'reg__min_data_per_group': 159}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.09758325663216173, 'reg__num_leaves': 48, 'reg__subsample': 0.8725234063829723, 'reg__colsample_bytree': 0.8448654211537614, 'reg__min_child_samples': 50, 'reg__reg_alpha': 9.33383264405423, 'reg__reg_lambda': 19.780886094596816, 'reg__min_split_gain': 3.825611709355474, 'reg__path_smooth': 2.3678080722367016, 'reg__min_child_weight': 0.22232310690546664, 'reg__feature_fraction': 0.7757190505975511, 'reg__bagging_fraction': 0.5817619431542468, 'reg__bagging_freq': 3, 'reg__max_bin': 104, 'reg__min_data_per_group': 158}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.0639415134276506, 'reg__num_leaves': 46, 'reg__subsample': 0.6667559074271373, 'reg__colsample_bytree': 0.7858644245533021, 'reg__min_child_samples': 32, 'reg__reg_alpha': 8.244275208339475, 'reg__reg_lambda': 1.719610365246801, 'reg__min_split_gain': 4.048179434970204, 'reg__path_smooth': 2.6421589617137853, 'reg__min_child_weight': 0.012650560597505434, 'reg__feature_fraction': 0.86128877699917, 'reg__bagging_fraction': 0.5654957435385843, 'reg__bagging_freq': 5, 'reg__max_bin': 113, 'reg__min_data_per_group': 194}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.09717244272613904, 'reg__num_leaves': 49, 'reg__subsample': 0.8501810978474623, 'reg__colsample_bytree': 0.8274081979926258, 'reg__min_child_samples': 25, 'reg__reg_alpha': 6.255522213656845, 'reg__reg_lambda': 36.58844424992063, 'reg__min_split_gain': 3.8932619456229833, 'reg__path_smooth': 2.191894236611198, 'reg__min_child_weight': 0.0010136980336757292, 'reg__feature_fraction': 0.7590191263566802, 'reg__bagging_fraction': 0.5913368240396856, 'reg__bagging_freq': 3, 'reg__max_bin': 104, 'reg__min_data_per_group': 159}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.06266999039398259, 'reg__num_leaves': 62, 'reg__subsample': 0.7845736209365926, 'reg__colsample_bytree': 0.8943947129410992, 'reg__min_child_samples': 32, 'reg__reg_alpha': 8.693771560583254, 'reg__reg_lambda': 31.116579876515065, 'reg__min_split_gain': 4.615174349801878, 'reg__path_smooth': 2.912219940806775, 'reg__min_child_weight': 0.007578379672648201, 'reg__feature_fraction': 0.8973424179161805, 'reg__bagging_fraction': 0.5668631567576337, 'reg__bagging_freq': 5, 'reg__max_bin': 121, 'reg__min_data_per_group': 147}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.06945539878327894, 'reg__num_leaves': 48, 'reg__subsample': 0.842601094822102, 'reg__colsample_bytree': 0.8429047703920105, 'reg__min_child_samples': 39, 'reg__reg_alpha': 6.278690973114795, 'reg__reg_lambda': 35.519138373751716, 'reg__min_split_gain': 3.9123154413330106, 'reg__path_smooth': 3.321457250979528, 'reg__min_child_weight': 0.0037488205336439835, 'reg__feature_fraction': 0.7822944793598184, 'reg__bagging_fraction': 0.5428423504001003, 'reg__bagging_freq': 5, 'reg__max_bin': 104, 'reg__min_data_per_group': 159}

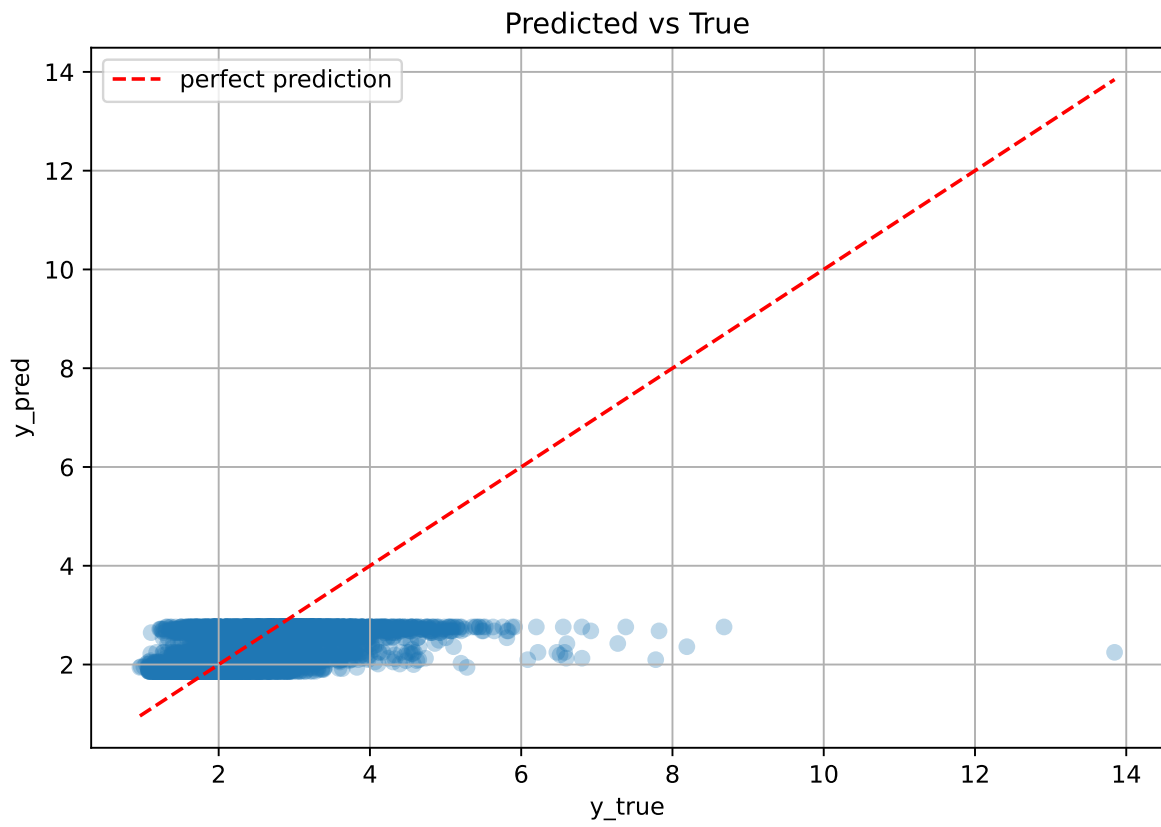
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.06394621476327837, 'reg__num_leaves': 47, 'reg__subsample': 0.6675402038187306, 'reg__colsample_bytree': 0.7939186447083938, 'reg__min_child_samples': 33, 'reg__reg_alpha': 8.227749903384886, 'reg__reg_lambda': 26.659666723082125, 'reg__min_split_gain': 4.002956737710421, 'reg__path_smooth': 2.662313232341151, 'reg__min_child_weight': 0.0741048520485294, 'reg__feature_fraction': 0.8444668262139667, 'reg__bagging_fraction': 0.5669075996716262, 'reg__bagging_freq': 5, 'reg__max_bin': 109, 'reg__min_data_per_group': 173}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.06243549759296741, 'reg__num_leaves': 46, 'reg__subsample': 0.8406942715845026, 'reg__colsample_bytree': 0.8072935470260268, 'reg__min_child_samples': 30, 'reg__reg_alpha': 8.168421101538554, 'reg__reg_lambda': 26.522114943633085, 'reg__min_split_gain': 4.9366167419736, 'reg__path_smooth': 2.5177186323731084, 'reg__min_child_weight': 0.0028450283835499547, 'reg__feature_fraction': 0.8411615983497774, 'reg__bagging_fraction': 0.5685042456612337, 'reg__bagging_freq': 5, 'reg__max_bin': 109, 'reg__min_data_per_group': 169}

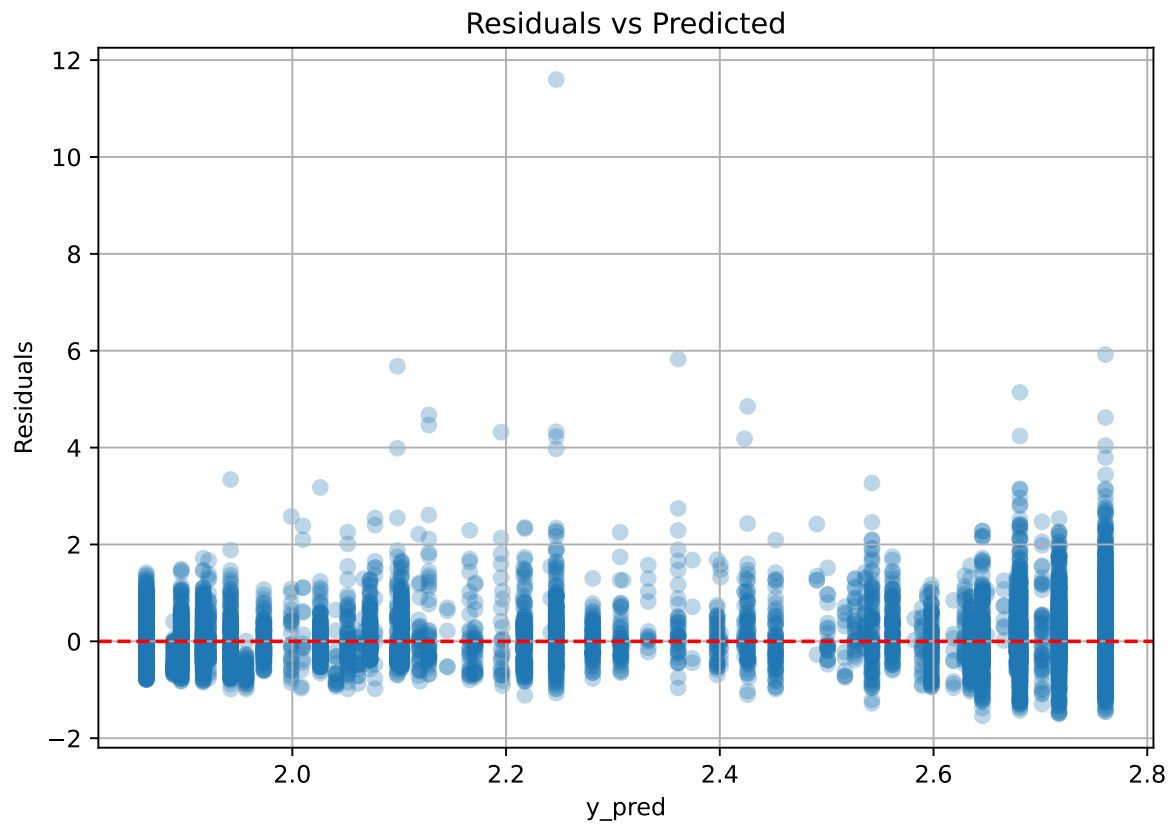
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.032441626602811656, 'reg__num_leaves': 37, 'reg__subsample': 0.694673184969672, 'reg__colsample_bytree': 0.7946614858188744, 'reg__min_child_samples': 20, 'reg__reg_alpha': 6.299814895132428, 'reg__reg_lambda': 39.40972311820233, 'reg__min_split_gain': 4.012063163495463, 'reg__path_smooth': 1.8887142021435954, 'reg__min_child_weight': 0.0018486594672701962, 'reg__feature_fraction': 0.7419533900216724, 'reg__bagging_fraction': 0.5493915659250579, 'reg__bagging_freq': 2, 'reg__max_bin': 98, 'reg__min_data_per_group': 131}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.07095222076665018, 'reg__num_leaves': 48, 'reg__subsample': 0.8467821423180487, 'reg__colsample_bytree': 0.8469569867310689, 'reg__min_child_samples': 38, 'reg__reg_alpha': 9.62732512533369, 'reg__reg_lambda': 19.747033527338306, 'reg__min_split_gain': 3.9312884052764163, 'reg__path_smooth': 3.487305800697689, 'reg__min_child_weight': 0.02314767277870399, 'reg__feature_fraction': 0.7833915899052587, 'reg__bagging_fraction': 0.5901310136587075, 'reg__bagging_freq': 3, 'reg__max_bin': 104, 'reg__min_data_per_group': 159}

2.4.14 Scatter



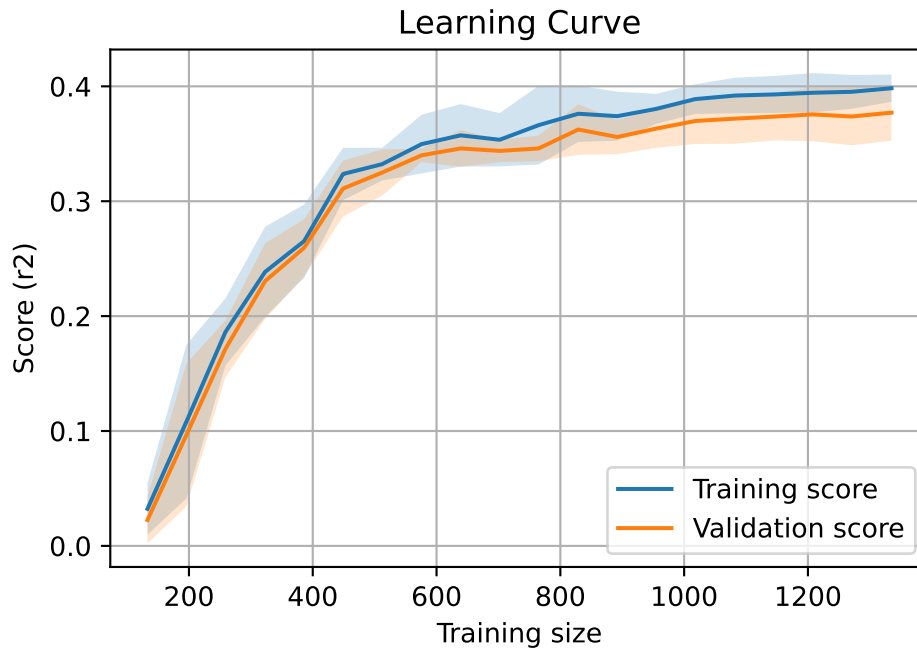
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 4_5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

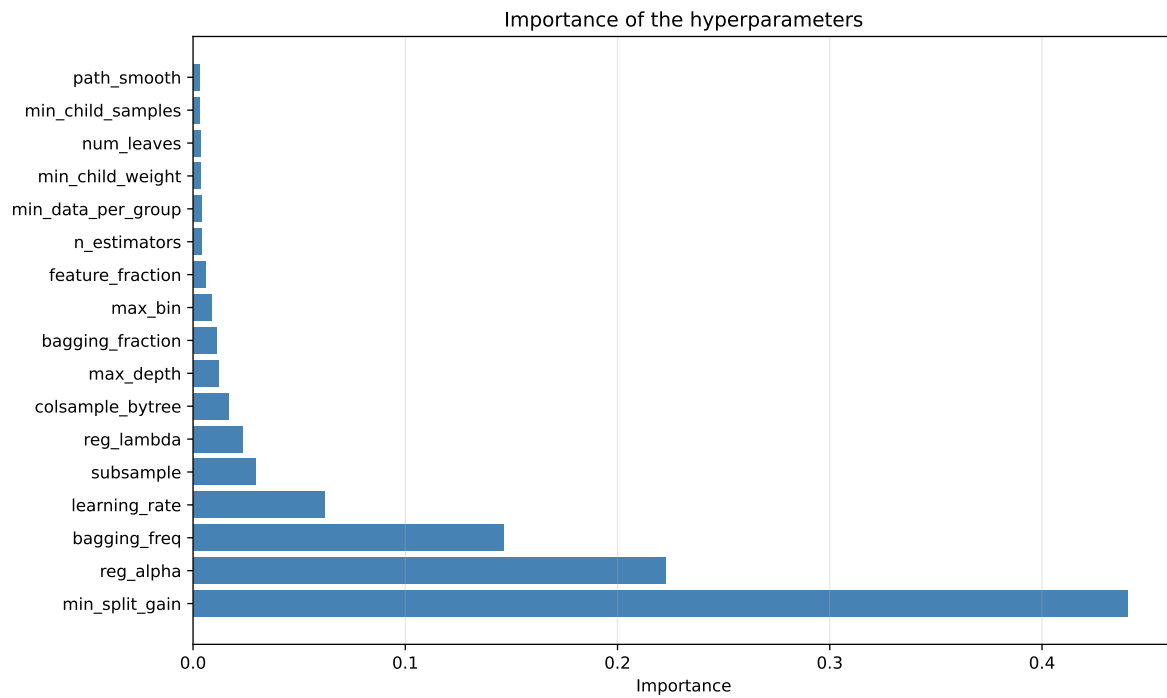
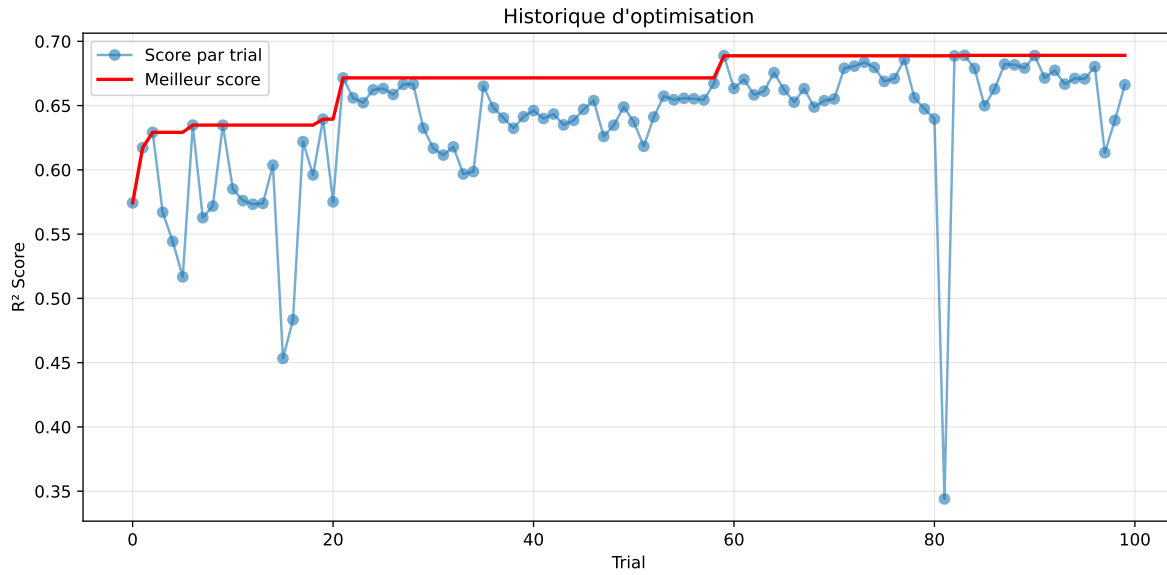


2.4.17 Gridsearch — Group: 6_3

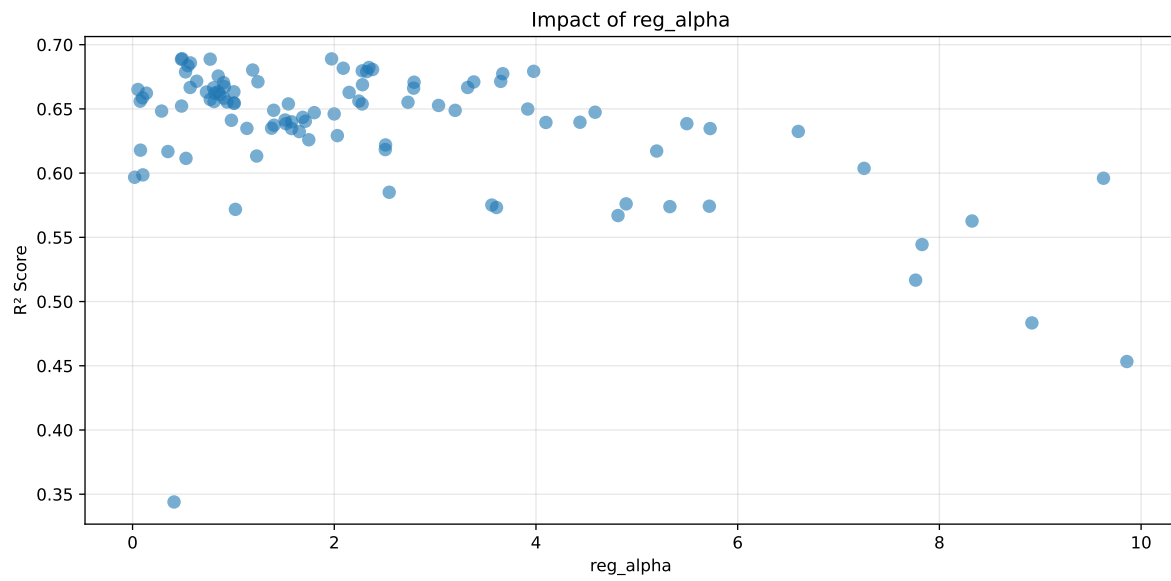
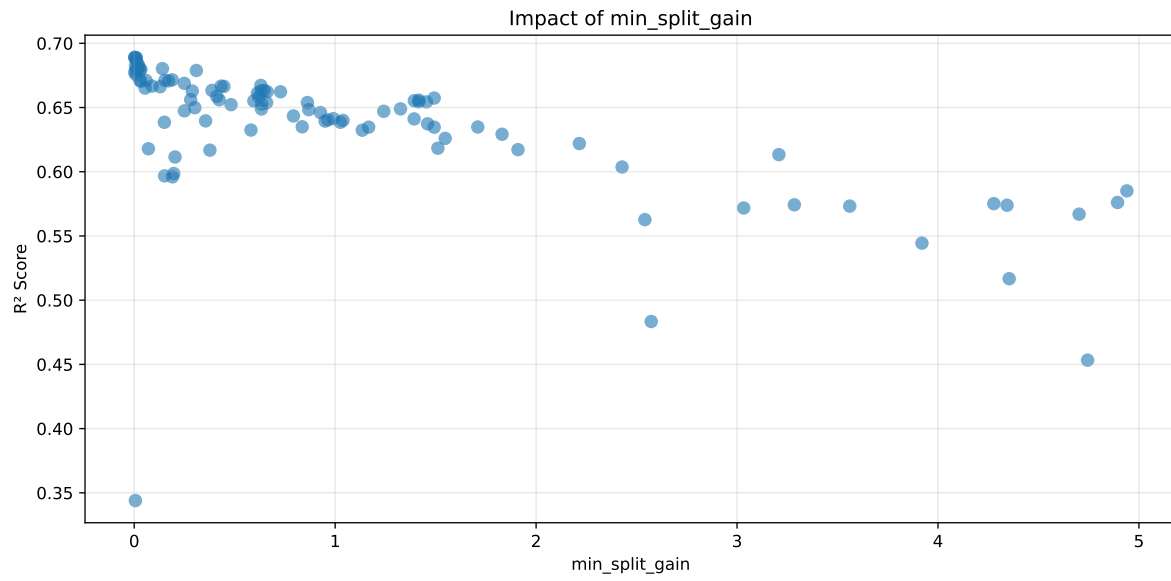
Len(group) : 561, Len(training) : 393, Len(test) : 168

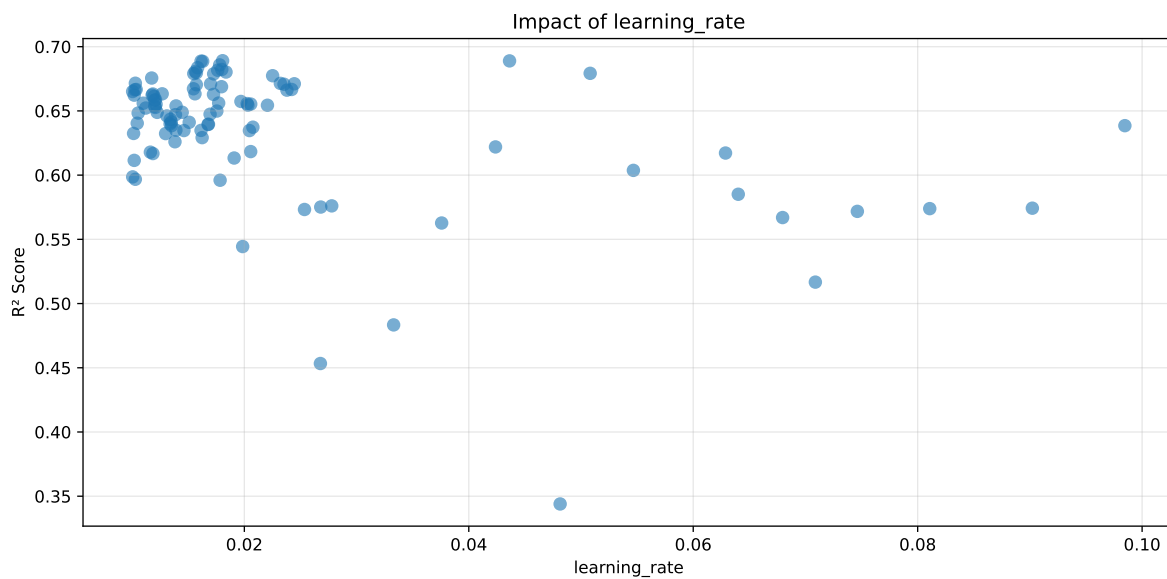
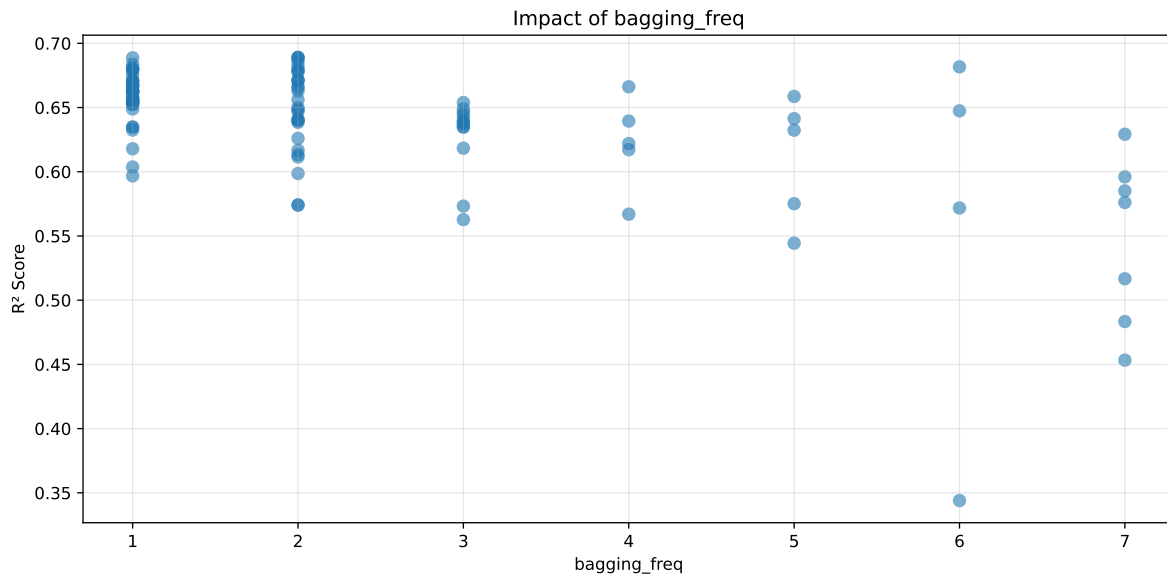
Distribution y_train vs y_test: y_train: mean=1.935, std=0.720, min=1.041, max=6.086
y_test: mean=2.031, std=0.744, min=1.078, max=4.912

===== TOP Importance FEATURES =====
feature importance r_umidity 589 r_NH3 426 r_THI 392 r_temperature 349 r_CO2 301



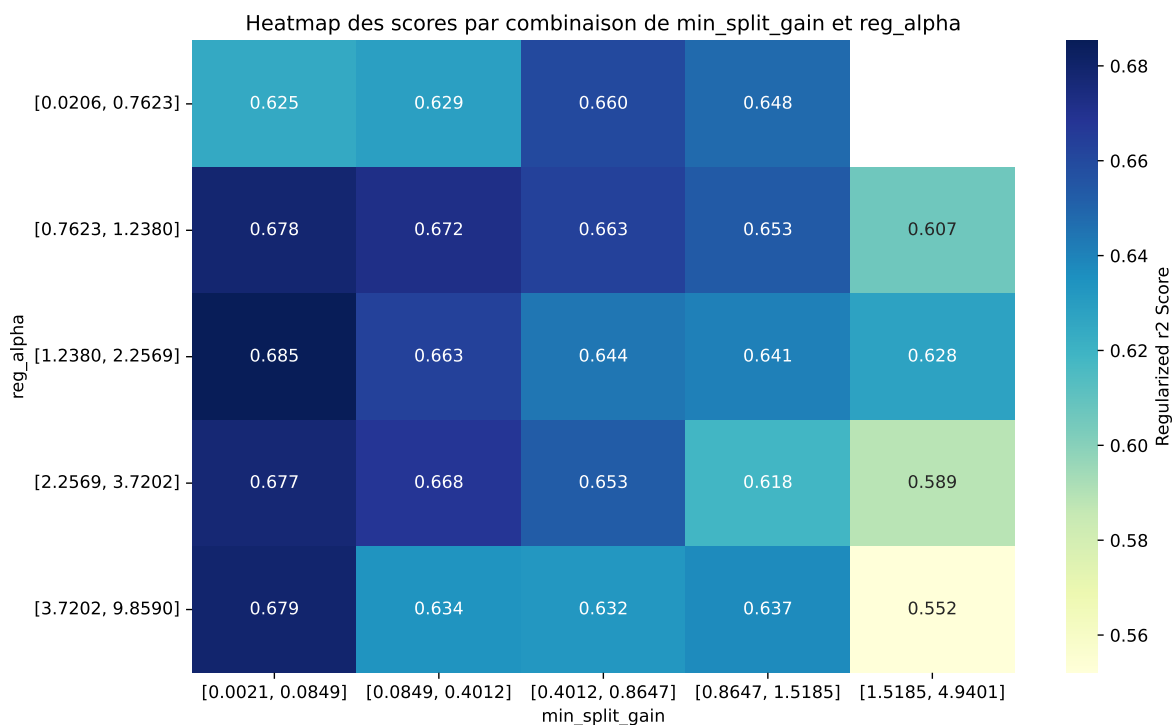
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'bagging_freq', 'learning_rate']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7032	0.6891	0.6465	0.0564	0.6891	0.7023	0.0133	1.0192	1.0205
2	0.7126	0.6889	0.6437	0.0569	0.6889	0.6986	0.0097	1.014	1.0343
3	0.7142	0.6887	0.648	0.0588	0.6887	0.6866	-0.0022	0.9968	1.0369
4	0.7001	0.6886	0.642	0.0599	0.6886	0.6865	-0.0021	0.9969	1.0167
5	0.6926	0.6858	0.6338	0.0594	0.6858	0.6788	-0.007	0.9898	1.0099
6	0.6856	0.6836	0.6273	0.0601	0.6836	0.6649	-0.0188	0.9726	1.0028
7	0.68	0.6822	0.6202	0.059	0.6822	0.6685	-0.0137	0.9799	0.9967
8	0.688	0.6817	0.6248	0.0594	0.6817	0.6707	-0.011	0.9839	1.0093
9	0.6799	0.6807	0.6234	0.0596	0.6807	0.6617	-0.019	0.972	0.9988
10	0.6786	0.6803	0.6152	0.0635	0.6803	0.6369	-0.0433	0.9363	0.9975
---	---	---	---	---	---	---	---	---	---
100	0.7596	0.6903	0.6823	0.0565	0.344	0.7308	0.0404	1.0586	1.1003

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.018061932370893662, 'reg__num_leaves': 35, 'reg__subsample': 0.5413098853323703, 'reg__colsample_bytree': 0.5612110552244485, 'reg__min_child_samples': 33, 'reg__reg_alpha': 0.4916558736329236, 'reg__reg_lambda': 19.774528281137567, 'reg__min_split_gain': 0.002081660168505012, 'reg__path_smooth': 1.8051985449415684, 'reg__min_child_weight':

0.38468998852741415, 'reg__feature_fraction': 0.5953324640507076, 'reg__bagging_fraction': 0.5506868223083587, 'reg__bagging_freq': 2, 'reg__max_bin': 72, 'reg__min_data_per_group': 77}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.04363195220170905, 'reg__num_leaves': 28, 'reg__subsample': 0.5520465755559715, 'reg__colsample_bytree': 0.5749063598436338, 'reg__min_child_samples': 35, 'reg__reg_alpha': 1.9735238067332654, 'reg__reg_lambda': 15.880503267401046, 'reg__min_split_gain': 0.003336870017331791, 'reg__path_smooth': 2.400458427196276, 'reg__min_child_weight': 0.06874157183880071, 'reg__feature_fraction': 0.528312415013606, 'reg__bagging_fraction': 0.5185606075477777, 'reg__bagging_freq': 2, 'reg__max_bin': 68, 'reg__min_data_per_group': 87}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.016142681574482884, 'reg__num_leaves': 40, 'reg__subsample': 0.5589946712568653, 'reg__colsample_bytree': 0.6250172407593653, 'reg__min_child_samples': 29, 'reg__reg_alpha': 0.7697281798719797, 'reg__reg_lambda': 21.597165825645355, 'reg__min_split_gain': 0.010726239667578297, 'reg__path_smooth': 1.907454857921788, 'reg__min_child_weight': 0.5646359878844187, 'reg__feature_fraction': 0.6137041868412316, 'reg__bagging_fraction': 0.5902421367990116, 'reg__bagging_freq': 1, 'reg__max_bin': 147, 'reg__min_data_per_group': 94}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.01628700820294575, 'reg__num_leaves': 35, 'reg__subsample': 0.5358971429396631, 'reg__colsample_bytree': 0.6427072760993051, 'reg__min_child_samples': 33, 'reg__reg_alpha': 0.4844541034780112, 'reg__reg_lambda': 19.2997011825584, 'reg__min_split_gain': 0.008205498197867315, 'reg__path_smooth': 1.8224482909873612, 'reg__min_child_weight': 0.33111466182354754, 'reg__feature_fraction': 0.6287237160116566, 'reg__bagging_fraction': 0.548666276922721, 'reg__bagging_freq': 2, 'reg__max_bin': 166, 'reg__min_data_per_group': 101}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.01780185410453469, 'reg__num_leaves': 35, 'reg__subsample': 0.5793217931744634, 'reg__colsample_bytree': 0.5555630107494782, 'reg__min_child_samples': 32, 'reg__reg_alpha': 0.5737354805278163, 'reg__reg_lambda': 22.91923390532057, 'reg__min_split_gain': 0.011302966314468255, 'reg__path_smooth': 1.6967590701064397, 'reg__min_child_weight': 0.35284043396571596, 'reg__feature_fraction': 0.5943705031932621, 'reg__bagging_fraction': 0.5443655189221438, 'reg__bagging_freq': 2, 'reg__max_bin': 171, 'reg__min_data_per_group': 50}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.015836602068893108, 'reg__num_leaves': 31, 'reg__subsample': 0.5668403747002976, 'reg__colsample_bytree': 0.5900425842843176, 'reg__min_child_samples': 40, 'reg__reg_alpha': 0.5491659327068344, 'reg__reg_lambda': 20.923486165309676, 'reg__min_split_gain': 0.019411053253034327, 'reg__path_smooth': 1.6984076698300001, 'reg__min_child_weight':

0.4268413974283803, 'reg__feature_fraction': 0.5536152146888877, 'reg__bagging_fraction': 0.5394208260324278, 'reg__bagging_freq': 1, 'reg__max_bin': 94, 'reg__min_data_per_group': 97}

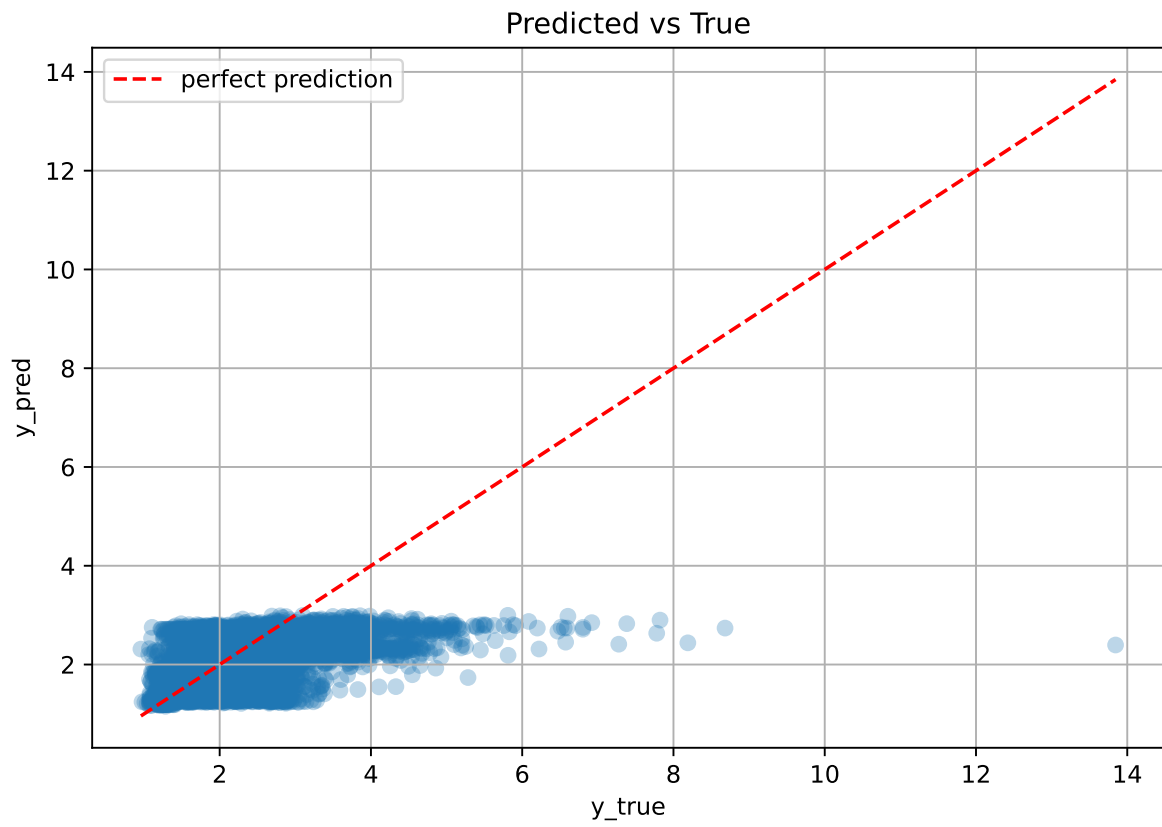
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.01798825891338473, 'reg__num_leaves': 29, 'reg__subsample': 0.5512316068741918, 'reg__colsample_bytree': 0.561770516134588, 'reg__min_child_samples': 28, 'reg__reg_alpha': 2.343663520650517, 'reg__reg_lambda': 18.865870558724115, 'reg__min_split_gain': 0.006591200941473391, 'reg__path_smooth': 1.8693553444670372, 'reg__min_child_weight': 0.3150618951497682, 'reg__feature_fraction': 0.521991825248497, 'reg__bagging_fraction': 0.5164465783487912, 'reg__bagging_freq': 2, 'reg__max_bin': 65, 'reg__min_data_per_group': 99}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.017624528741360623, 'reg__num_leaves': 25, 'reg__subsample': 0.5501453970946557, 'reg__colsample_bytree': 0.5725539178450002, 'reg__min_child_samples': 28, 'reg__reg_alpha': 2.0890557839619874, 'reg__reg_lambda': 16.908009774591008, 'reg__min_split_gain': 0.024376507947170178, 'reg__path_smooth': 1.8608957643040869, 'reg__min_child_weight': 0.28963675940972666, 'reg__feature_fraction': 0.5193260285949601, 'reg__bagging_fraction': 0.5526539293629528, 'reg__bagging_freq': 6, 'reg__max_bin': 67, 'reg__min_data_per_group': 98}

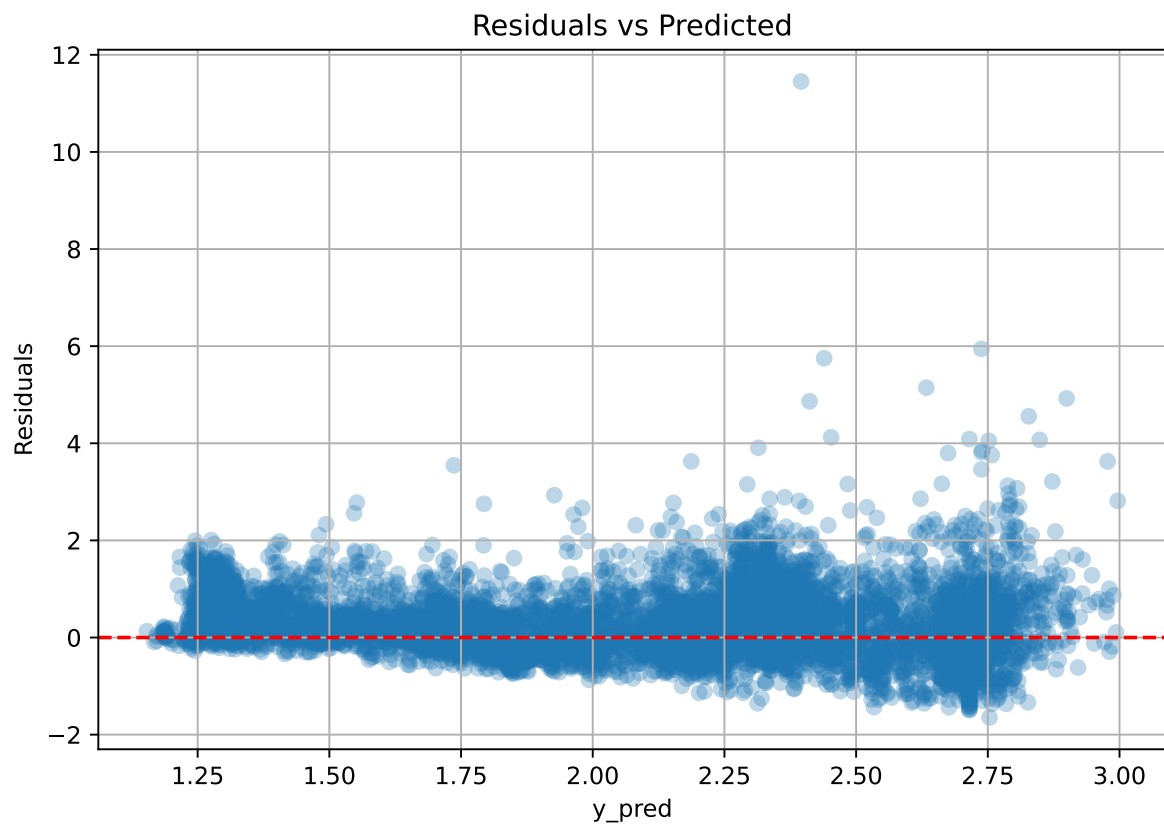
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.015634079450956996, 'reg__num_leaves': 31, 'reg__subsample': 0.5629794149496933, 'reg__colsample_bytree': 0.5883994738140688, 'reg__min_child_samples': 29, 'reg__reg_alpha': 2.379640099550527, 'reg__reg_lambda': 21.327422116899765, 'reg__min_split_gain': 0.02118350108342173, 'reg__path_smooth': 1.5960696842919038, 'reg__min_child_weight': 0.3361789811282654, 'reg__feature_fraction': 0.5357310771581536, 'reg__bagging_fraction': 0.5438943058276906, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 98}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.01836644350925355, 'reg__num_leaves': 29, 'reg__subsample': 0.5016385856045923, 'reg__colsample_bytree': 0.5747699288602104, 'reg__min_child_samples': 35, 'reg__reg_alpha': 1.191177434654606, 'reg__reg_lambda': 22.940255703105173, 'reg__min_split_gain': 0.140461540733559, 'reg__path_smooth': 2.4208056140640277, 'reg__min_child_weight': 0.08394819722261451, 'reg__feature_fraction': 0.566329527971069, 'reg__bagging_fraction': 0.5593114713667351, 'reg__bagging_freq': 1, 'reg__max_bin': 86, 'reg__min_data_per_group': 54}

2.4.18 Scatter



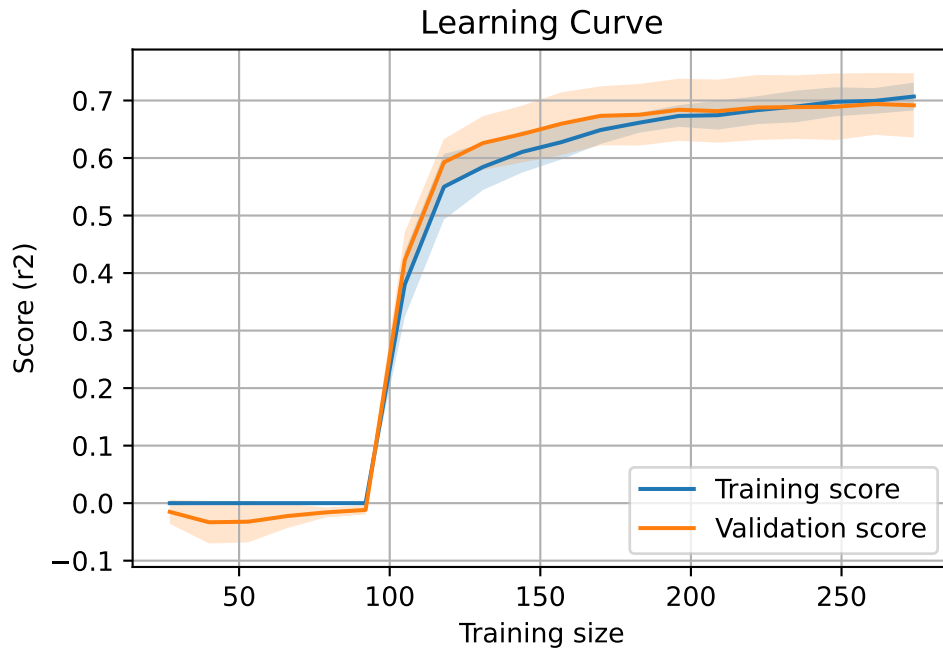
2.4.19 Scatter residuals



2.4.20 Learning curve — Group: 6_3

`Len(group)` : 561, `Len(training)` : 393,

`Len(training_real)` : 276, `Len(test_of_training)` : 117, `Len(test)` : 168

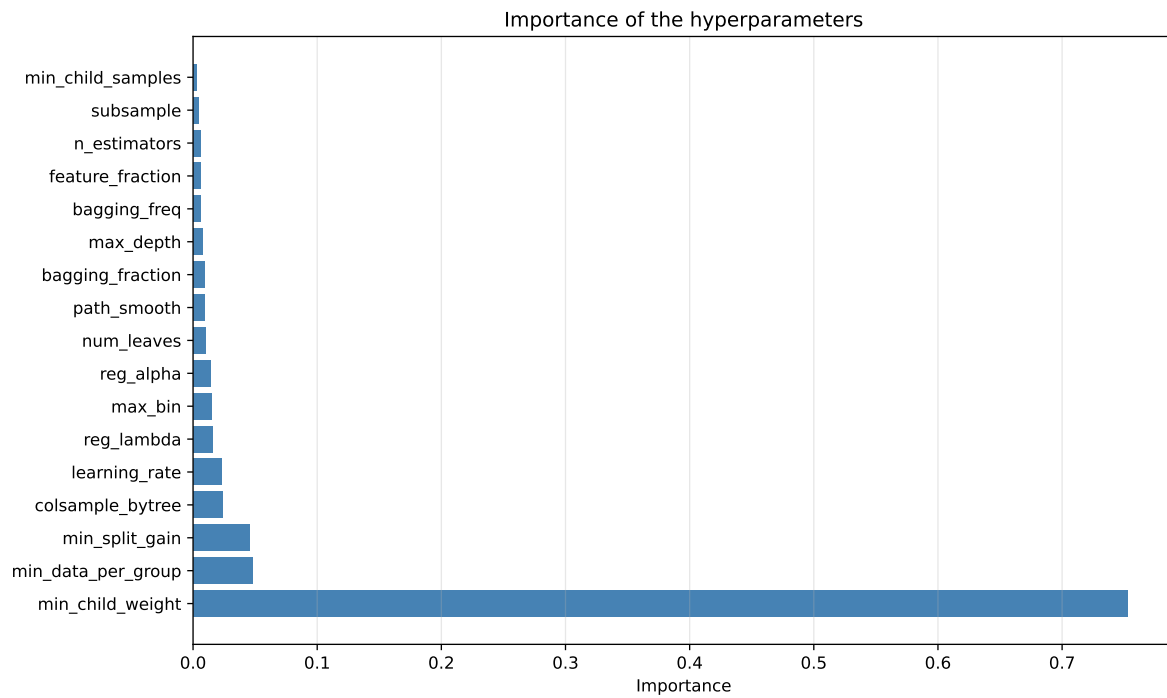
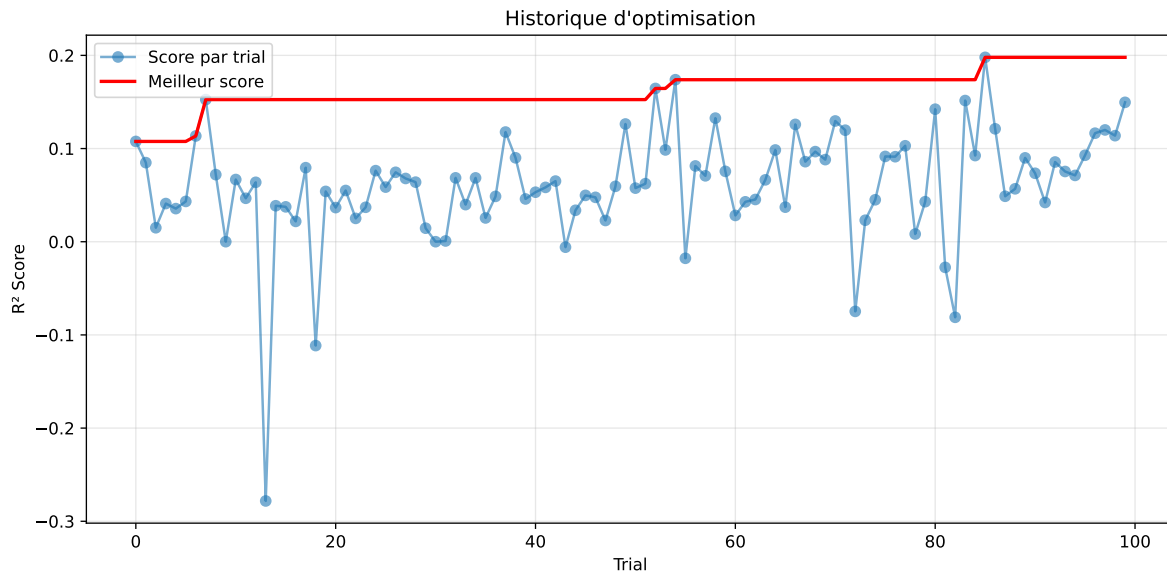


2.4.21 Gridsearch — Group: 6_4

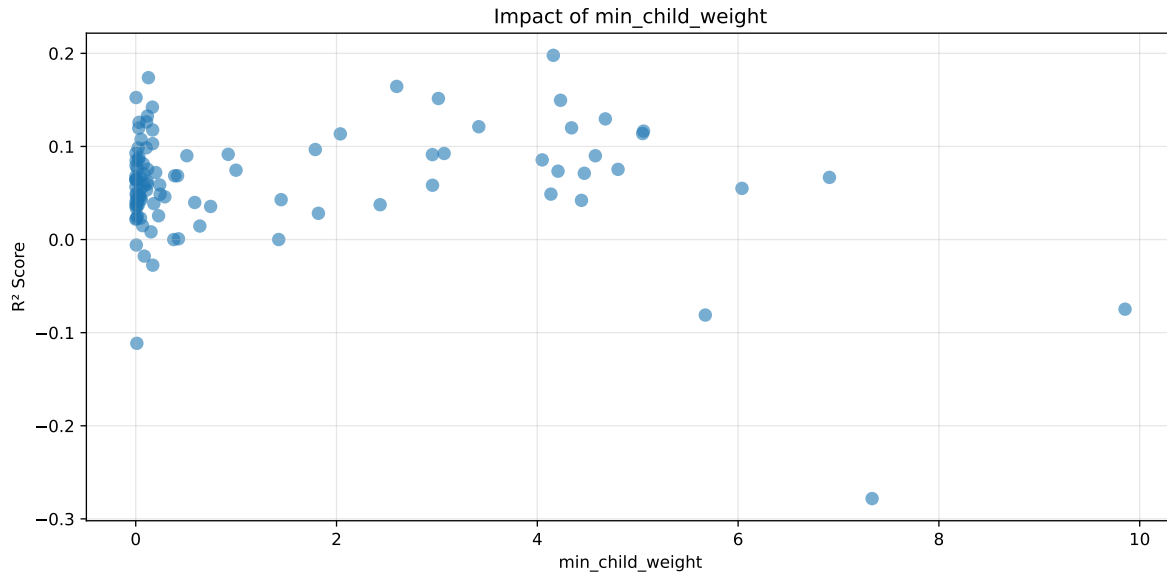
Len(group) : 792, Len(training) : 555, Len(test) : 237

Distribution y_train vs y_test: y_train: mean=2.486, std=0.782, min=1.253, max=7.383
y_test: mean=2.410, std=0.685, min=1.398, max=5.144

===== TOP Importance FEATURES =====
feature importance r_THI 22 r_umidity 15 r_NH3 12 r_temperature 12 r_CO2 5



Paramètres avec >5% d'importance : ['min_child_weight']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3041	0.2864	0.3328	0.0737	0.1979	0.3377	0.0514	1.1794	1.0618
2	0.3857	0.3504	0.4058	0.0369	0.1739	0.5069	0.1565	1.4466	1.1007
3	0.3391	0.31	0.3729	0.0303	0.1645	0.4723	0.1623	1.5235	1.0939
4	0.4033	0.3615	0.437	0.0324	0.1525	0.5273	0.1659	1.4588	1.1156
5	0.3984	0.3573	0.4277	0.0497	0.1515	0.489	0.1317	1.3686	1.1152
6	0.3713	0.3344	0.3935	0.0476	0.1496	0.467	0.1326	1.3965	1.1105
7	0.3879	0.347	0.4227	0.052	0.1422	0.5064	0.1595	1.4596	1.118
8	0.5331	0.4664	0.5424	0.0261	0.1327	0.5859	0.1195	1.2563	1.1431
9	0.2843	0.2585	0.3074	0.0753	0.1296	0.2802	0.0217	1.0841	1.0997
10	0.3914	0.3472	0.4132	0.0373	0.1264	0.5185	0.1713	1.4935	1.1272
---	---	---	---	---	---	---	---	---	---
100	0.6266	0.4758	0.587	0.036	-0.2782	0.622	0.1461	1.3071	1.3169

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.059353496142165126, 'reg__num_leaves': 28, 'reg__subsample': 0.7399071931315705, 'reg__colsample_bytree': 0.598944245760749, 'reg__min_child_samples': 37, 'reg__reg_alpha': 1.440331152786021, 'reg__reg_lambda': 15.118217008353444, 'reg__min_split_gain': 2.504457354521705, 'reg__path_smooth': 3.2060605293136173, 'reg__min_child_weight': 4.159567315624752, 'reg__feature_fraction': 0.7845615398174807, 'reg__bagging_fraction': 0.5922882738192479, 'reg__bagging_freq': 7, 'reg__max_bin': 86, 'reg__min_data_per_group': 105}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.04211598722296186, 'reg__num_leaves': 44, 'reg__subsample': 0.6895600300804395, 'reg__colsample_bytree': 0.6276543506912615, 'reg__min_child_samples': 11, 'reg__reg_alpha': 3.8001409987590624, 'reg__reg_lambda': 43.07307497473142, 'reg__min_split_gain': 0.5292867140874252, 'reg__path_smooth': 2.4040739412660077, 'reg__min_child_weight': 0.12641529739912138, 'reg__feature_fraction': 0.8468788499465207, 'reg__bagging_fraction': 0.5258727190282612, 'reg__bagging_freq': 6, 'reg__max_bin': 113, 'reg__min_data_per_group': 62}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.04037667932332429, 'reg__num_leaves': 44, 'reg__subsample': 0.6934476725997897, 'reg__colsample_bytree': 0.6310193544466794, 'reg__min_child_samples': 39, 'reg__reg_alpha': 1.0744488025499688, 'reg__reg_lambda': 44.09191306634996, 'reg__min_split_gain': 1.3461817157714313, 'reg__path_smooth': 2.420278734378551, 'reg__min_child_weight': 2.5996239385135804, 'reg__feature_fraction': 0.8507646659860566, 'reg__bagging_fraction': 0.5632816910507557, 'reg__bagging_freq': 6, 'reg__max_bin': 79, 'reg__min_data_per_group': 71}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.011602196445674303, 'reg__num_leaves': 47, 'reg__subsample': 0.6678874472647827, 'reg__colsample_bytree': 0.7606822468837124, 'reg__min_child_samples': 37, 'reg__reg_alpha': 1.7262354640018018, 'reg__reg_lambda': 25.510361495707844, 'reg__min_split_gain': 0.06316236723672652, 'reg__path_smooth': 1.5247076668629216, 'reg__min_child_weight': 0.0035265754928676977, 'reg__feature_fraction': 0.892288791532948, 'reg__bagging_fraction': 0.5211641476787839, 'reg__bagging_freq': 3, 'reg__max_bin': 70, 'reg__min_data_per_group': 107}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.05145714974043675, 'reg__num_leaves': 41, 'reg__subsample': 0.7451513518243108, 'reg__colsample_bytree': 0.5981721608445519, 'reg__min_child_samples': 39, 'reg__reg_alpha': 0.5711521222631146, 'reg__reg_lambda': 13.41102736629687, 'reg__min_split_gain': 1.6884646339538276, 'reg__path_smooth': 3.217317032963119, 'reg__min_child_weight': 3.0148935914135735, 'reg__feature_fraction': 0.7775805751413178, 'reg__bagging_fraction': 0.6016393547403072, 'reg__bagging_freq': 7, 'reg__max_bin': 88, 'reg__min_data_per_group': 103}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.06353856460236951, 'reg__num_leaves': 23, 'reg__subsample': 0.7048797547610223, 'reg__colsample_bytree': 0.5886843528384769, 'reg__min_child_samples': 36, 'reg__reg_alpha': 1.43104643449947, 'reg__reg_lambda': 10.369244929970648, 'reg__min_split_gain': 1.762946903941396, 'reg__path_smooth': 3.499458819043419, 'reg__min_child_weight': 4.231441283800734, 'reg__feature_fraction': 0.8089122629351525, 'reg__bagging_fraction': 0.5900081517960826, 'reg__bagging_freq': 4, 'reg__max_bin': 168, 'reg__min_data_per_group': 80}

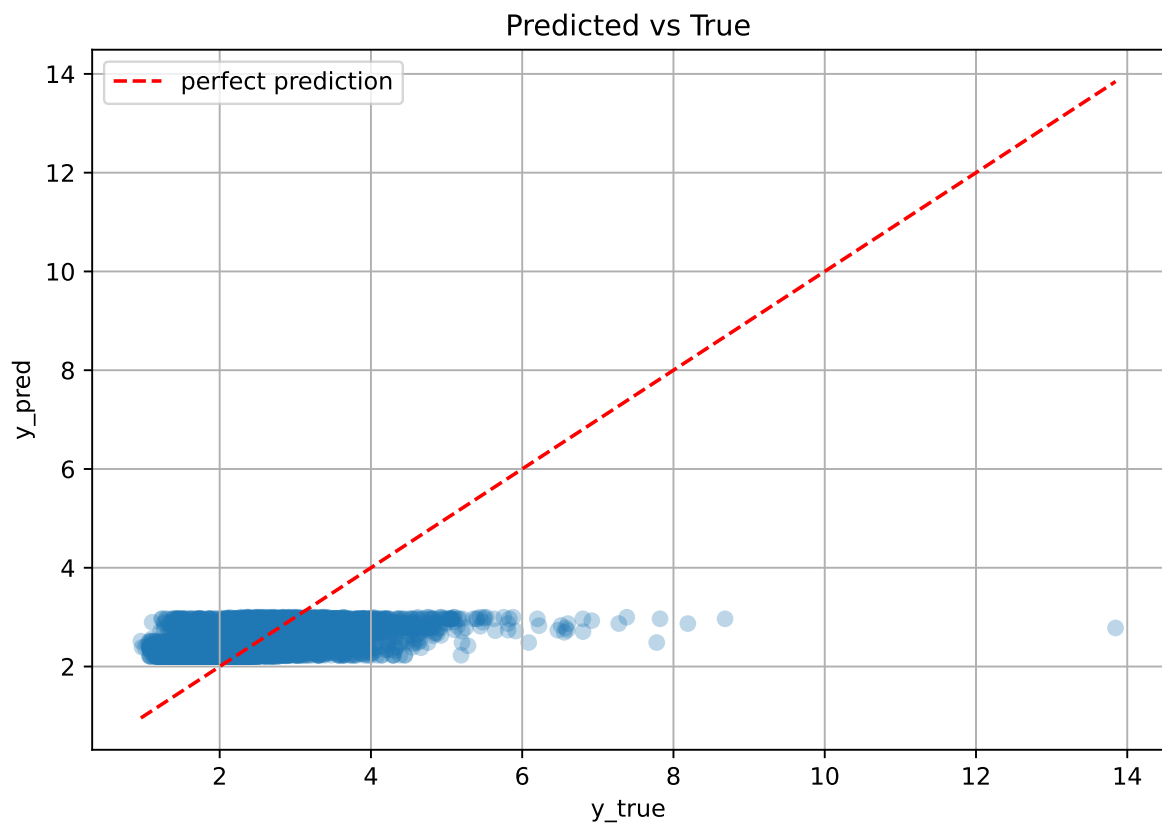
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.051871124886455476, 'reg__num_leaves': 50, 'reg__subsample': 0.5989848915941294, 'reg__colsample_bytree': 0.5957399557149811, 'reg__min_child_samples': 37, 'reg__reg_alpha': 0.5844956979297669, 'reg__reg_lambda': 18.93490686216329, 'reg__min_split_gain': 1.3939393860307323, 'reg__path_smooth': 3.1952497161582314, 'reg__min_child_weight': 0.16592824509749948, 'reg__feature_fraction': 0.8082727511195282, 'reg__bagging_fraction': 0.5575193990484251, 'reg__bagging_freq': 4, 'reg__max_bin': 158, 'reg__min_data_per_group': 102}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.039229853879369905, 'reg__num_leaves': 43, 'reg__subsample': 0.60044389497268, 'reg__colsample_bytree': 0.6347591858556644, 'reg__min_child_samples': 12, 'reg__reg_alpha': 1.0754790800619611, 'reg__reg_lambda': 22.666930415520817, 'reg__min_split_gain': 0.5124200452707244, 'reg__path_smooth': 2.382409937607977, 'reg__min_child_weight': 0.11641777246412685, 'reg__feature_fraction': 0.8534590498846857, 'reg__bagging_fraction': 0.5284906606724068, 'reg__bagging_freq': 7, 'reg__max_bin': 154, 'reg__min_data_per_group': 69}

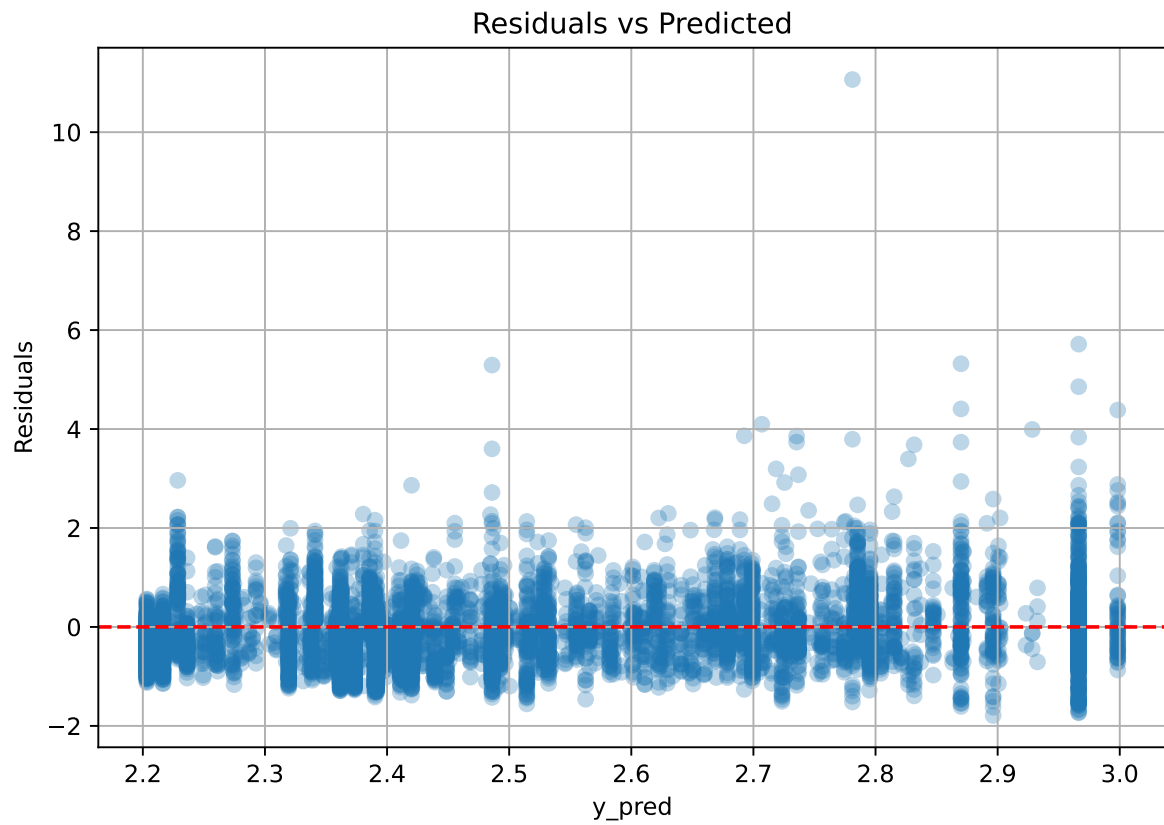
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.05355518768798095, 'reg__num_leaves': 36, 'reg__subsample': 0.7430705748637721, 'reg__colsample_bytree': 0.6869590354657595, 'reg__min_child_samples': 38, 'reg__reg_alpha': 4.82318048904421, 'reg__reg_lambda': 17.309172354491015, 'reg__min_split_gain': 1.4479863398637387, 'reg__path_smooth': 2.7545701550361428, 'reg__min_child_weight': 4.677628536806921, 'reg__feature_fraction': 0.7408774836937814, 'reg__bagging_fraction': 0.5523070597076193, 'reg__bagging_freq': 7, 'reg__max_bin': 67, 'reg__min_data_per_group': 93}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.039697129147568164, 'reg__num_leaves': 43, 'reg__subsample': 0.5367699970319759, 'reg__colsample_bytree': 0.7149950516068657, 'reg__min_child_samples': 18, 'reg__reg_alpha': 1.2752296145654078, 'reg__reg_lambda': 44.82124784203192, 'reg__min_split_gain': 1.1190239657819205, 'reg__path_smooth': 0.8934809259747545, 'reg__min_child_weight': 0.108093245060729, 'reg__feature_fraction': 0.8656397087144626, 'reg__bagging_fraction': 0.5749793791388255, 'reg__bagging_freq': 6, 'reg__max_bin': 216, 'reg__min_data_per_group': 65}

2.4.22 Scatter



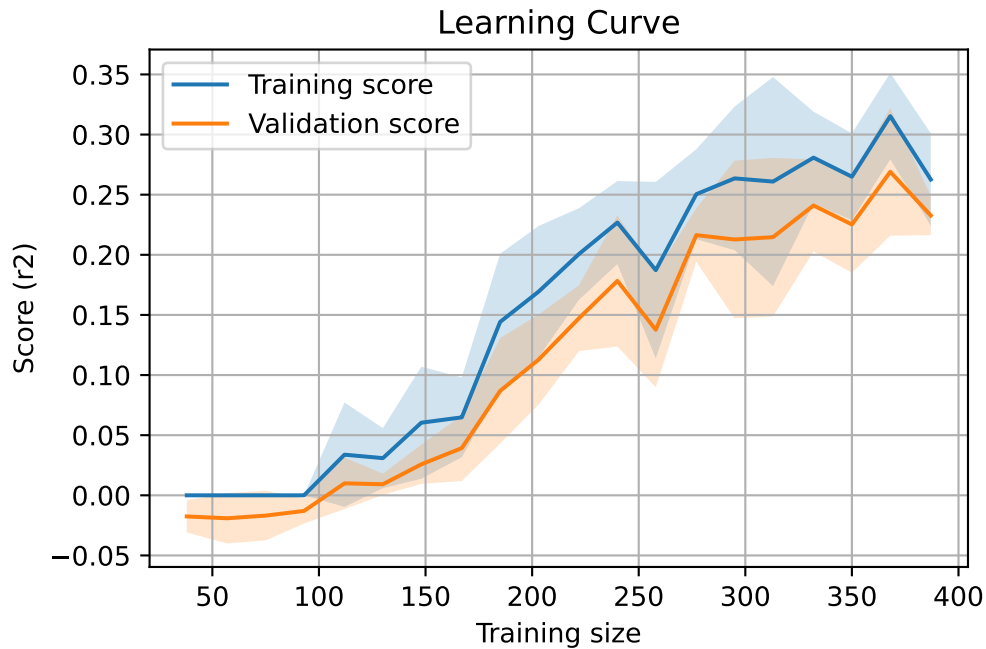
2.4.23 Scatter residuals



2.4.24 Learning curve — Group: 6_4

`Len(group)` : 792, `Len(training)` : 555,

`Len(training_real)` : 389, `Len(test_of_training)` : 166, `Len(test)` : 237



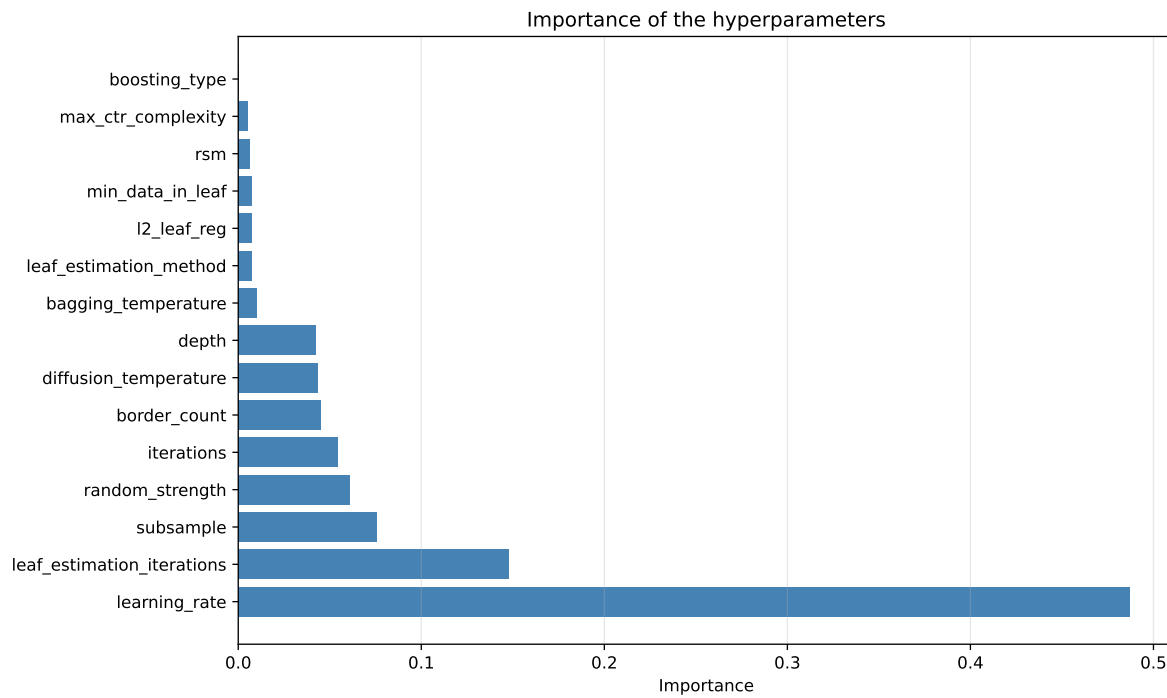
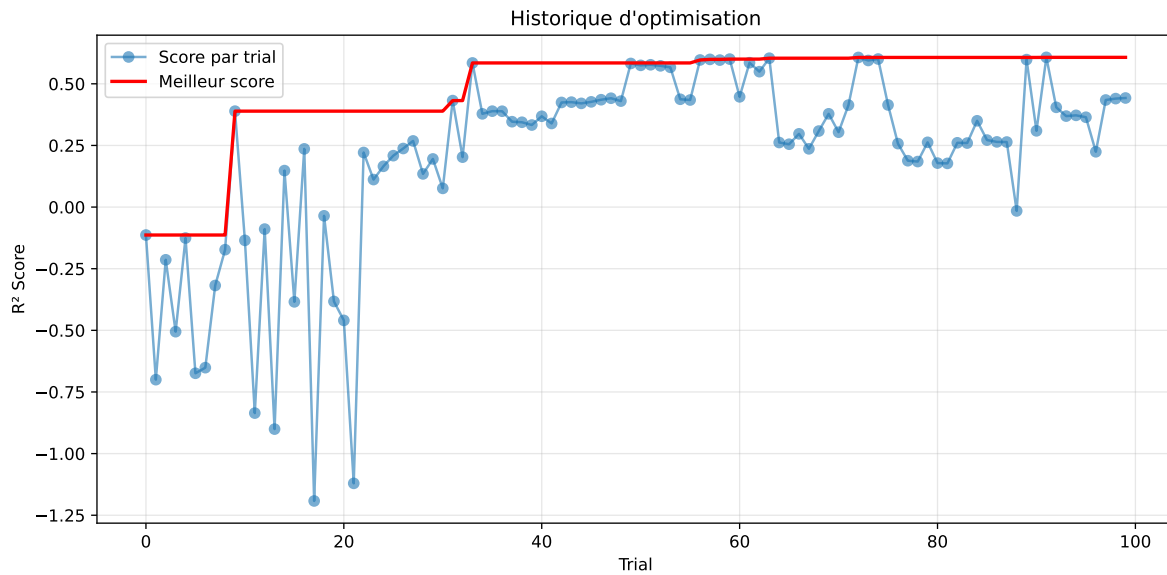
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 1_3

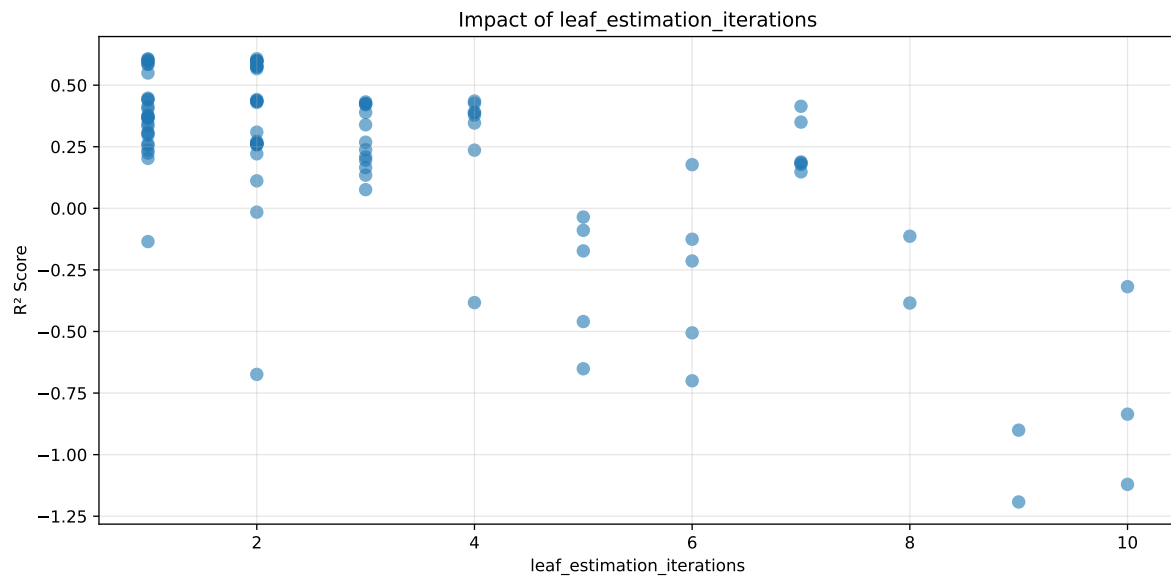
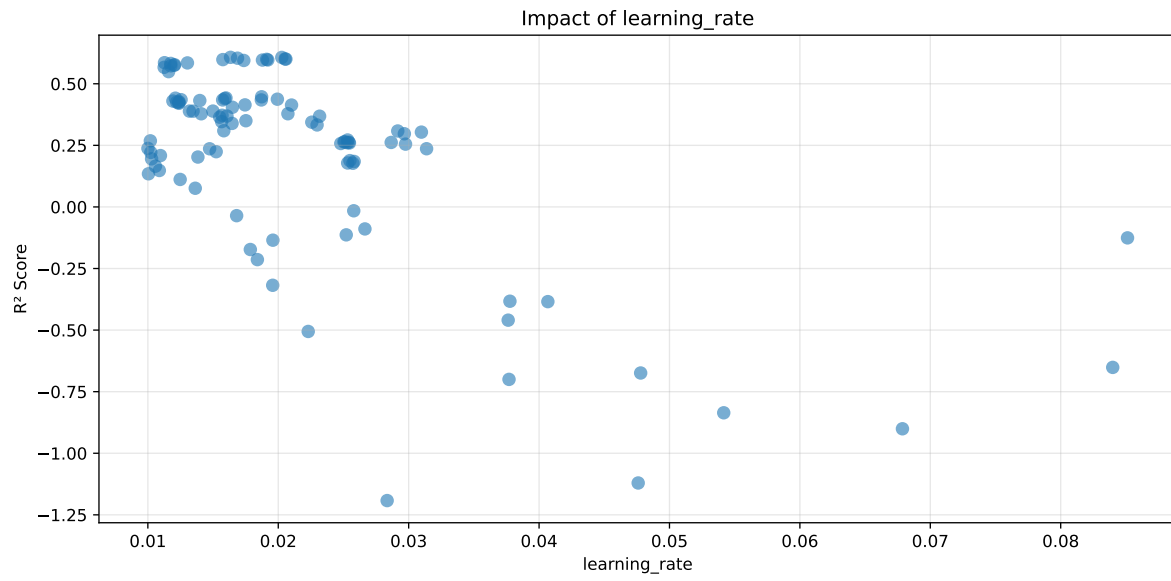
Len(group) : 795, Len(training) : 557, Len(test) : 238

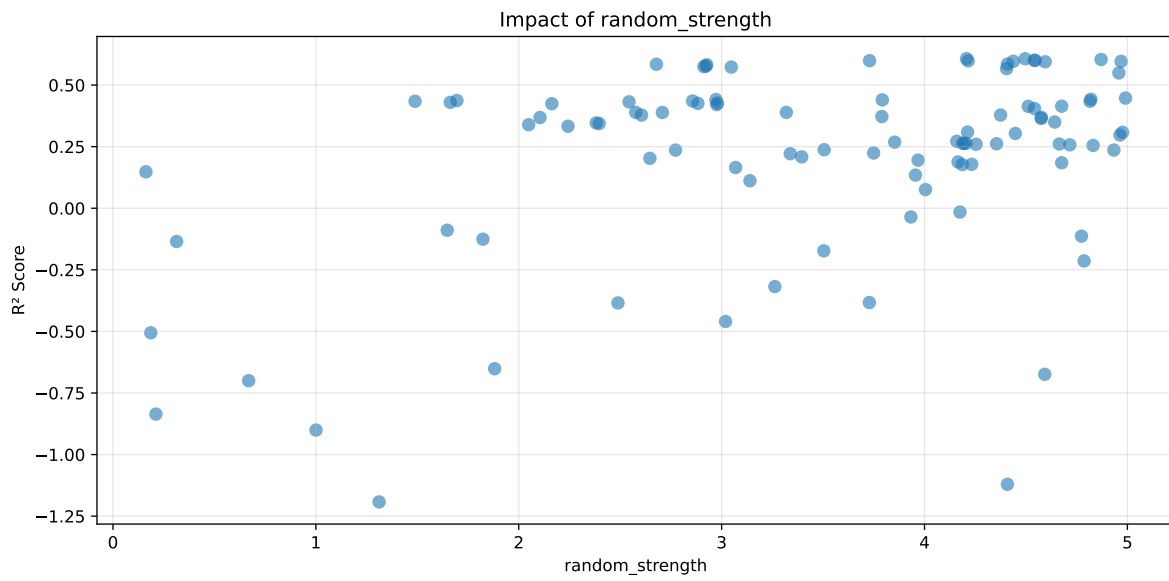
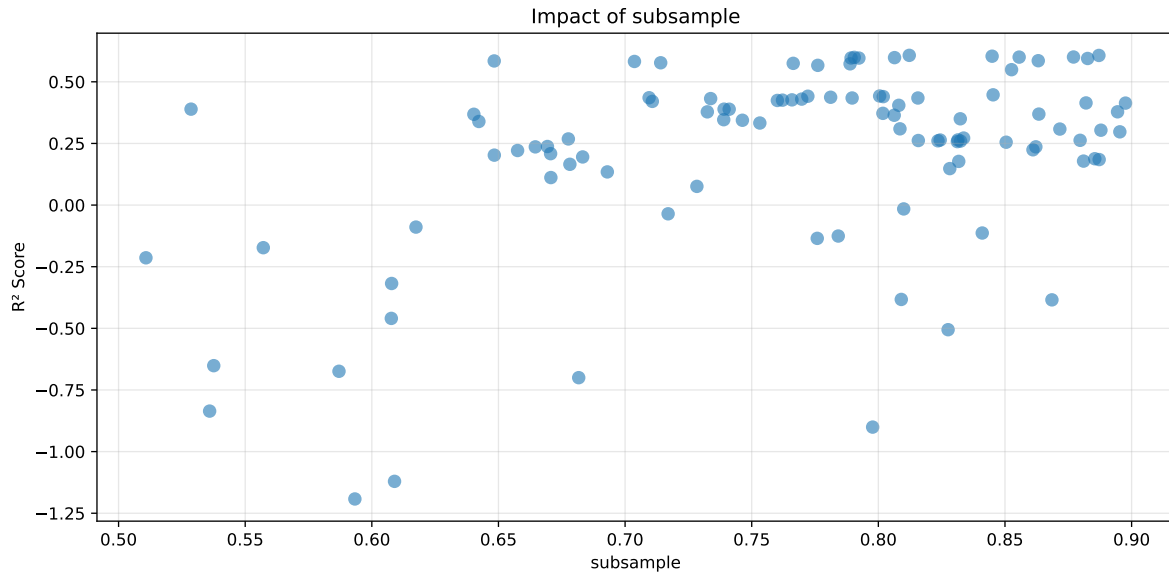
Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986
y_test: mean=2.492, std=0.625, min=1.466, max=5.447

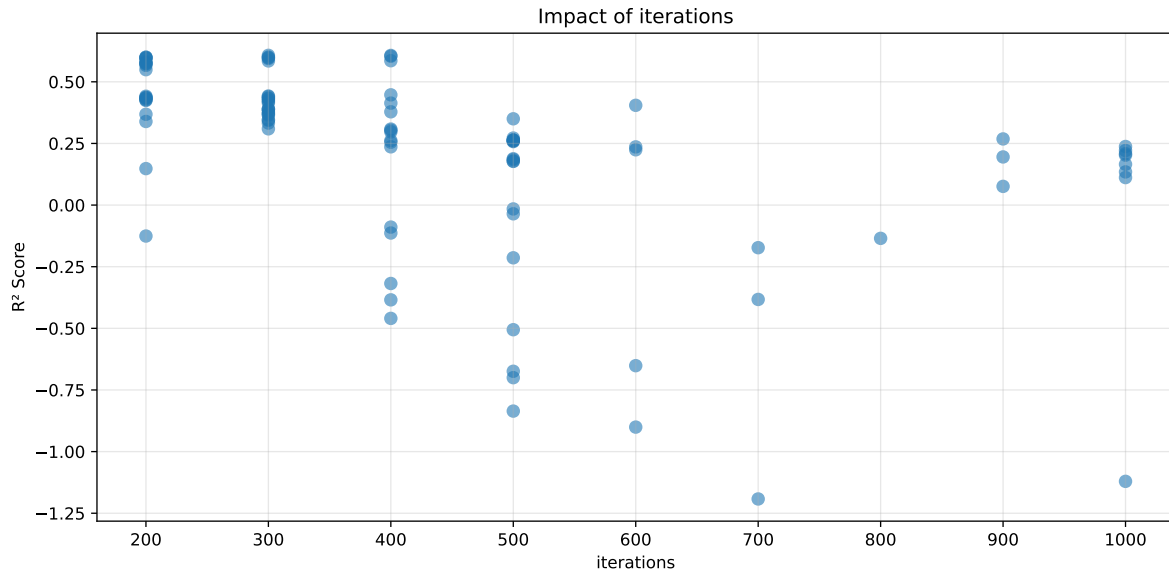
===== TOP Importance FEATURES =====
feature importance r_umidity 33.822746 r_temperature 25.783107 r_THI 23.736613 r_CO2
10.091710 r_NH3 6.565823



Paramètres avec >5% d'importance : ['learning_rate', 'leaf_estimation_iterations', 'subsample', 'random_strength', 'iterations']

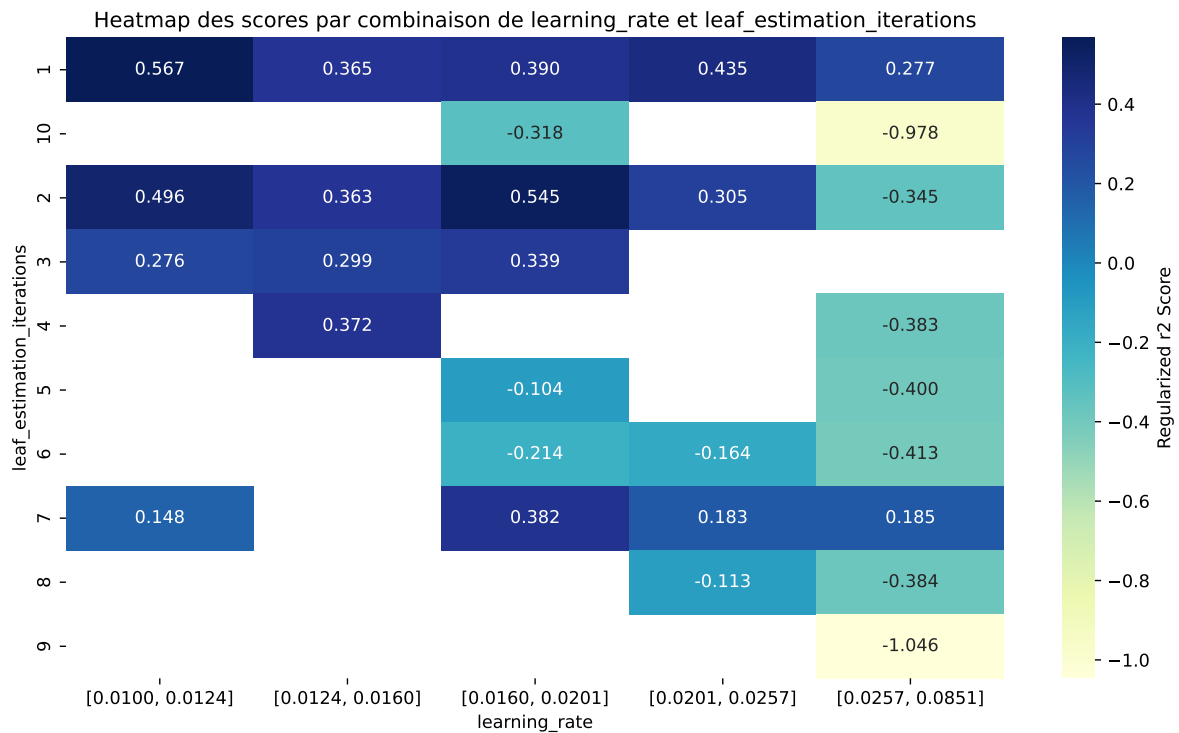






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6358	0.6073	0.5729	0.044	0.6073	0.5863	-0.021	0.9654	1.0468
2	0.637	0.607	0.5746	0.0441	0.607	0.5964	-0.0105	0.9827	1.0494
3	0.6337	0.6039	0.5702	0.0441	0.6039	0.5837	-0.0202	0.9666	1.0494
4	0.6264	0.6006	0.5613	0.0437	0.6006	0.5755	-0.025	0.9583	1.043
5	0.6272	0.6002	0.568	0.0443	0.6002	0.5771	-0.0231	0.9615	1.045
6	0.6246	0.5991	0.5646	0.044	0.5991	0.576	-0.0231	0.9614	1.0425
7	0.6218	0.5981	0.5608	0.0401	0.5981	0.5768	-0.0212	0.9645	1.0396
8	0.6251	0.5967	0.5636	0.044	0.5967	0.5767	-0.02	0.9665	1.0476
9	0.622	0.5963	0.558	0.0412	0.5963	0.5682	-0.028	0.953	1.0432
10	0.6169	0.5946	0.5567	0.0406	0.5946	0.574	-0.0206	0.9653	1.0374
---	---	---	---	---	---	---	---	---	---
100	0.9774	0.6158	0.6674	0.0537	-1.1921	0.6724	0.0566	1.0919	1.5872

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.016340846206202724, 'reg__l2_leaf_reg': 34.46636468957077, 'reg__bagging_temperature': 2.6090109273304356, 'reg__random_strength': 4.207117230134838, 'reg__subsample': 0.812206510120946, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.688900228892764, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7652.223121848539}

Rank 2: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.02027092748127471, 'reg__l2_leaf_reg': 31.15923360070071, 'reg__bagging_temperature': 3.180701841632232, 'reg__random_strength': 4.496023011896468, 'reg__subsample': 0.8870829664278792, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7374945762614505, 'reg__border_count': 96, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8586.407850486963}

Rank 3: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01687576352906594, 'reg__l2_leaf_reg': 30.620212347698256, 'reg__bagging_temperature': 3.006110995119826, 'reg__random_strength': 4.870404614286976, 'reg__subsample': 0.8449198372827527, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7499467115225014, 'reg__border_count': 112, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4818.958261635568}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.020519196425578652, 'reg__l2_leaf_reg': 29.916476398512014, 'reg__bagging_temperature': 3.3088125654095037, 'reg__random_strength': 4.541116956254418, 'reg__subsample': 0.8770574924785259, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':

0.8033508003878119, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9560.509824234348}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.020585381746324044, 'reg__l2_leaf_reg': 19.01887108355322, 'reg__bagging_temperature': 2.6058931329837463, 'reg__random_strength': 4.54538910059301, 'reg__subsample': 0.8555978024555937, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7449726596740843, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8336.79912496856}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.019111339804367796, 'reg__l2_leaf_reg': 19.600354378457595, 'reg__bagging_temperature': 2.6983094863186183, 'reg__random_strength': 3.7291908117854806, 'reg__subsample': 0.7904914370923102, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7445984532864901, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5033.225716431913}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.015751747768765024, 'reg__l2_leaf_reg': 40.75594348307028, 'reg__bagging_temperature': 3.69185724787734, 'reg__random_strength': 4.215188492640772, 'reg__subsample': 0.8063933556615537, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7740266581591385, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7883.8704745858695}

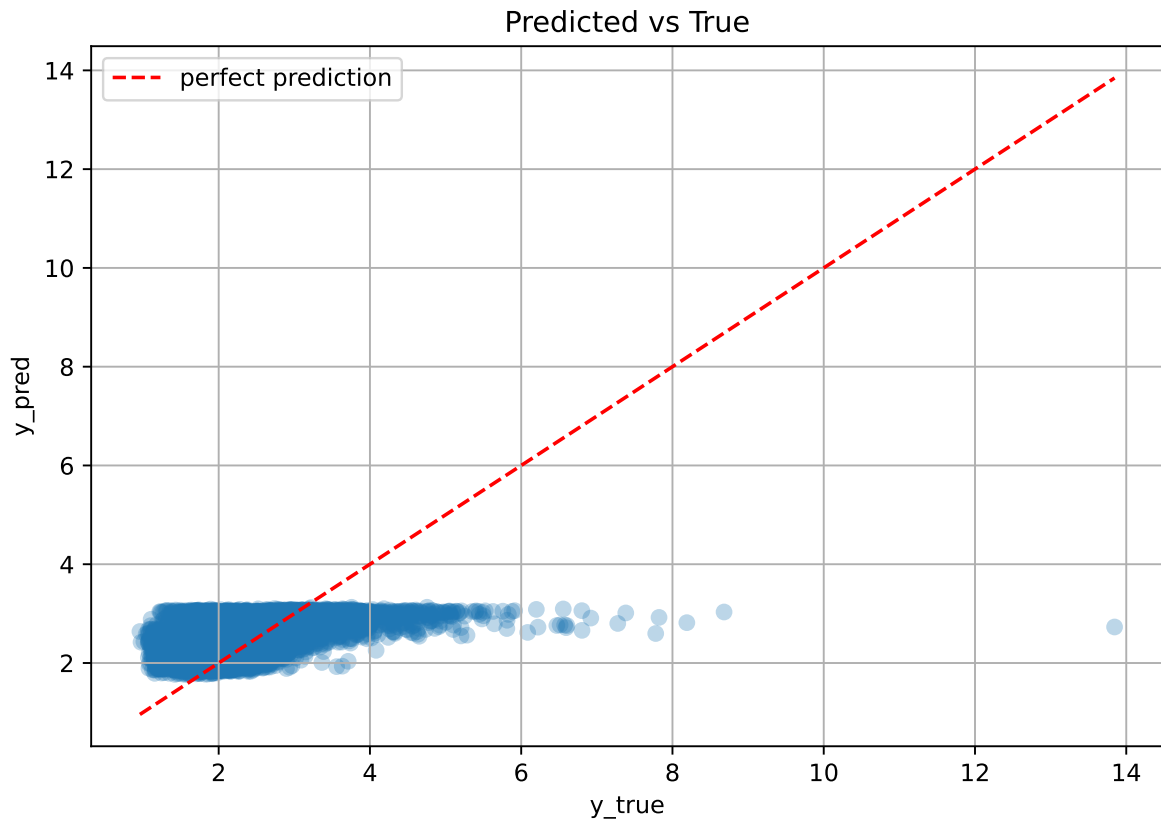
Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.019198292273439152, 'reg__l2_leaf_reg': 19.125427353196223, 'reg__bagging_temperature': 2.549840659313249, 'reg__random_strength': 4.437879928293631, 'reg__subsample': 0.7892286053700585, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6879888226393532, 'reg__border_count': 100, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8263.624433564066}

Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.018788066105301898, 'reg__l2_leaf_reg': 18.240064801980235, 'reg__bagging_temperature': 2.639952762725272, 'reg__random_strength': 4.968938006239508, 'reg__subsample': 0.7923209029582415, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7507044263333306, 'reg__border_count': 104, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8257.069957415279}

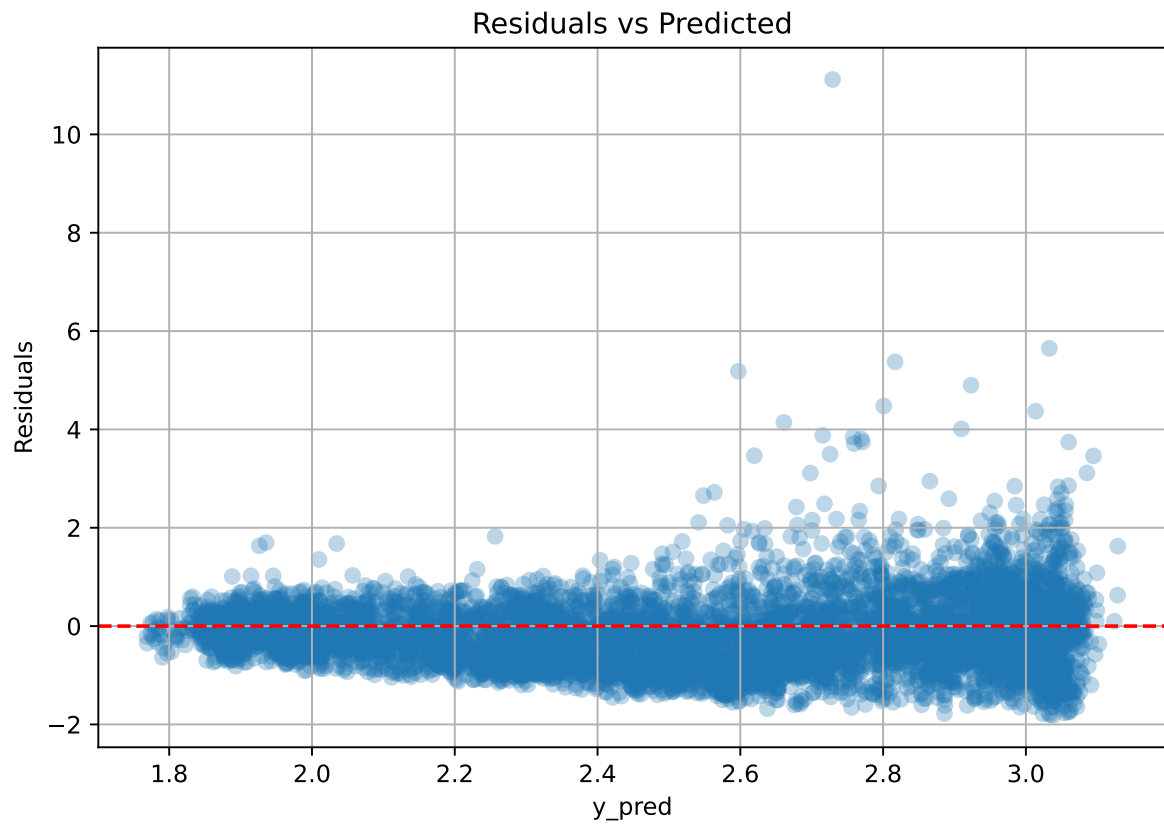
Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.017358925565948882, 'reg__l2_leaf_reg': 30.246149615360498, 'reg__bagging_temperature': 3.1998773514788423,

```
'reg__random_strength': 4.5947805033361, 'reg__subsample': 0.8826421992753037,  
'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':  
0.7336249066903063, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_tempera-  
ture': 9635.240916083643}
```

2.5.2 Scatter



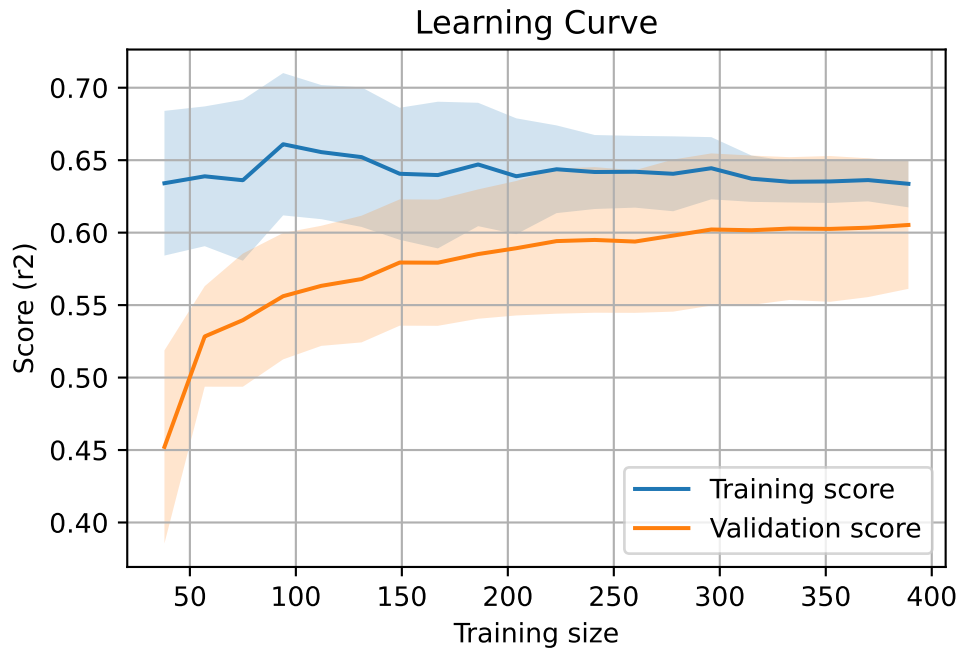
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 1_3

`Len(group)` : 795, `Len(training)` : 557,

`Len(training_real)` : 390, `Len(test_of_training)` : 167, `Len(test)` : 238

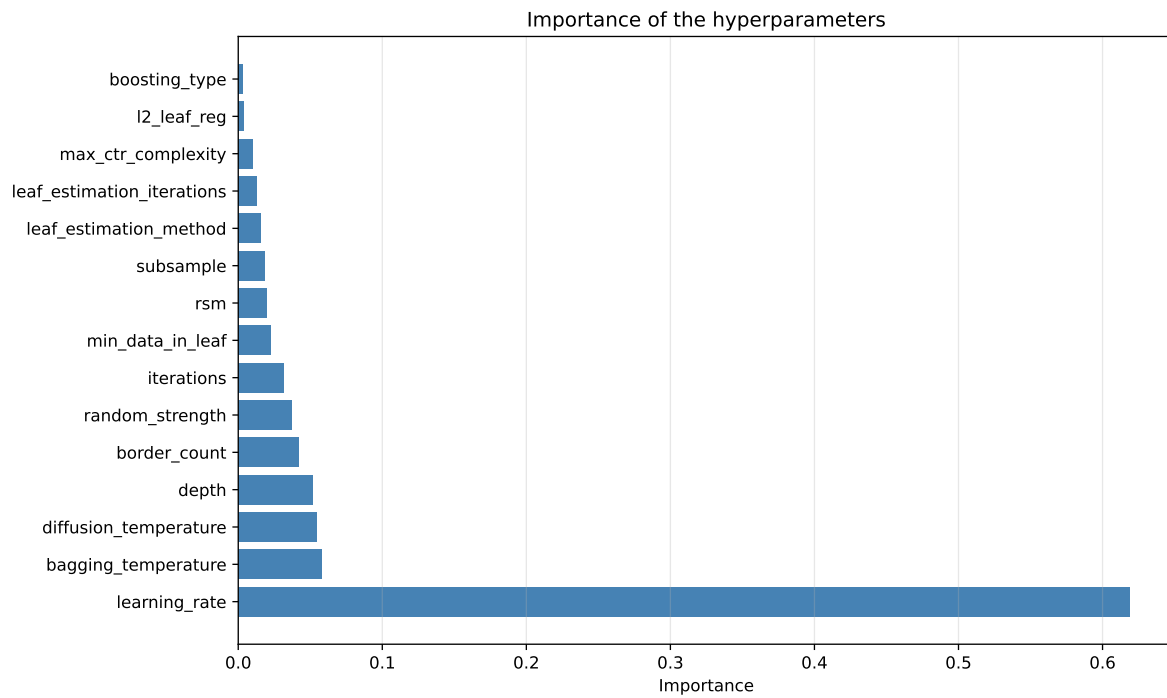
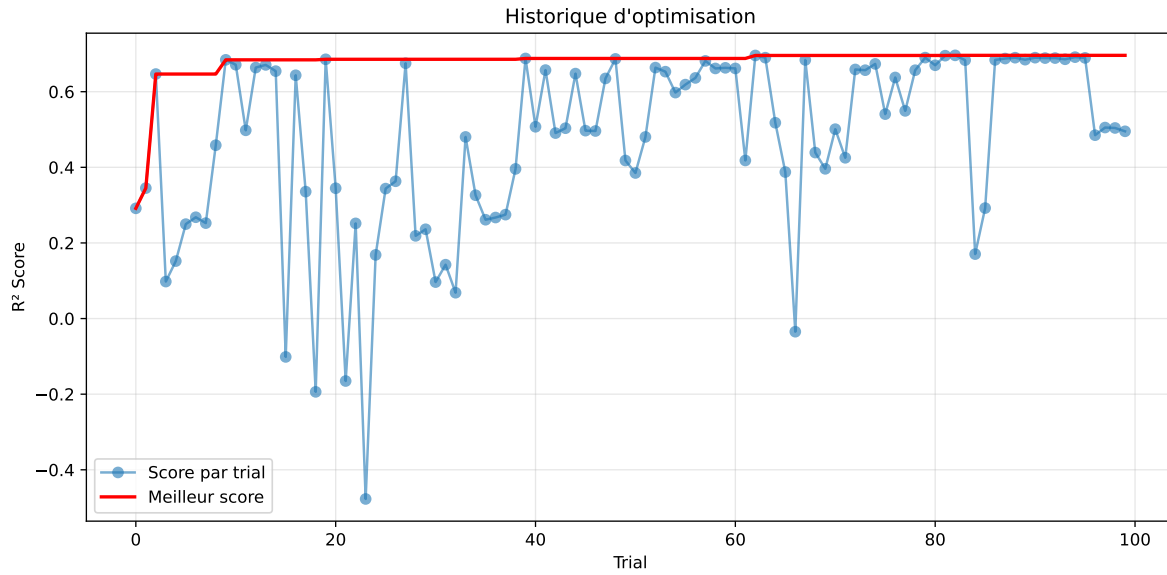


2.5.5 Gridsearch — Group: 2_11

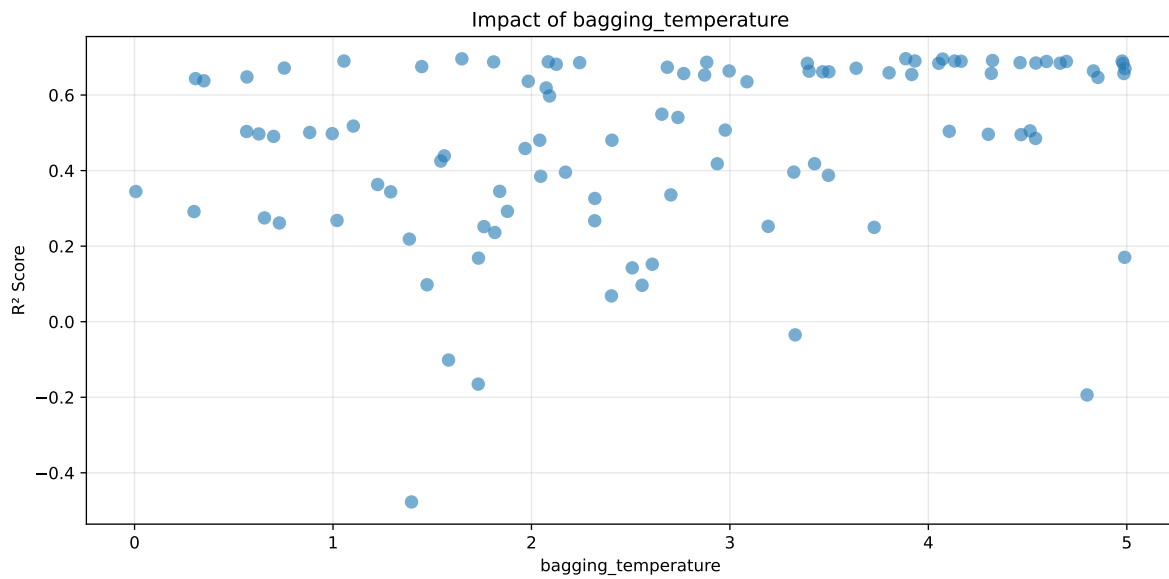
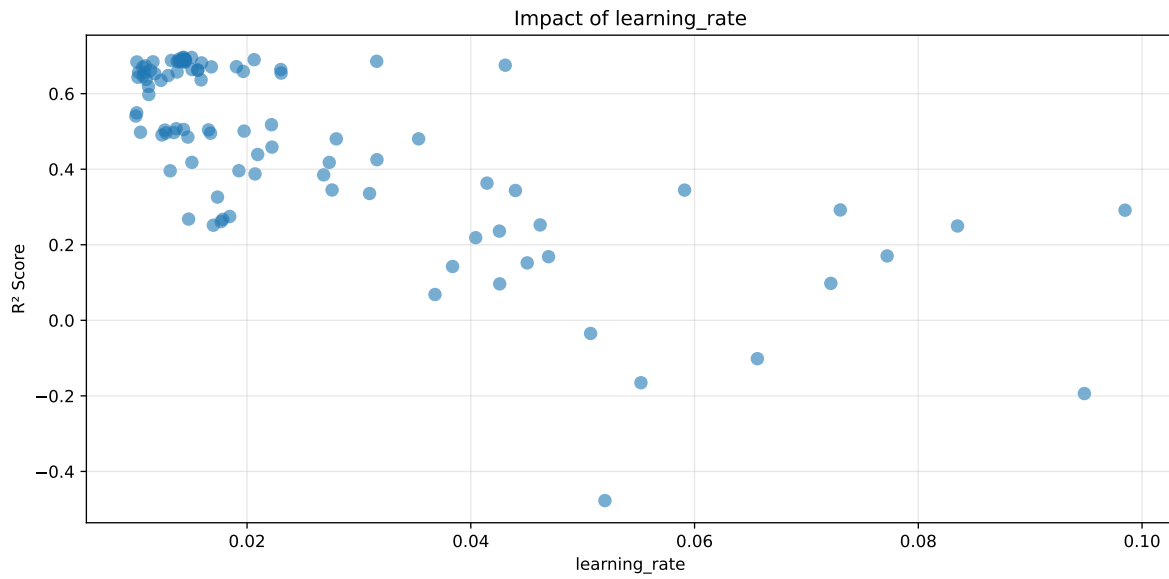
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

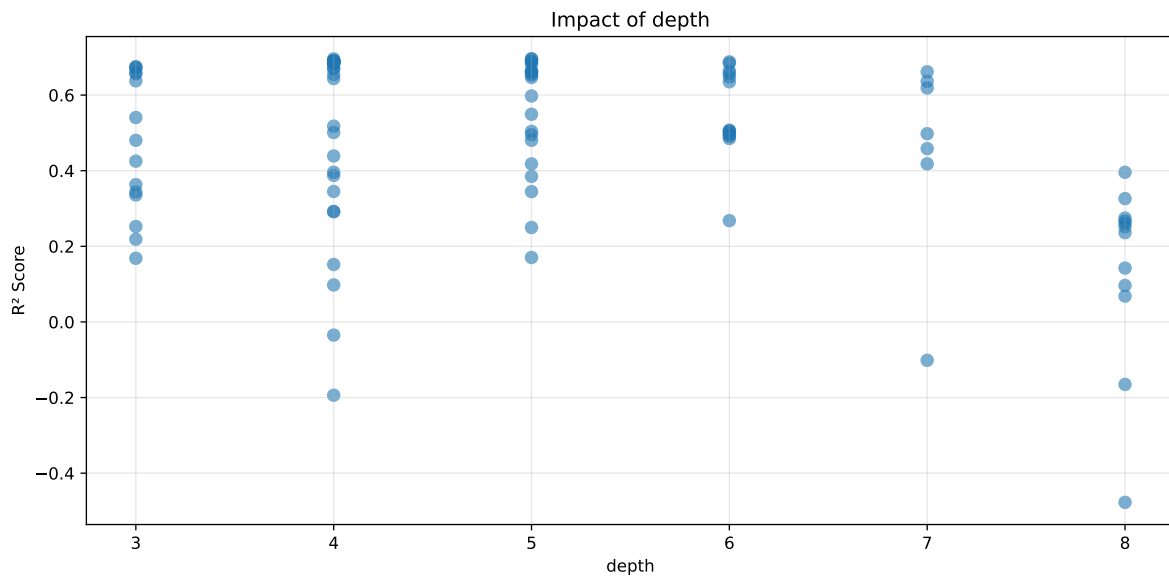
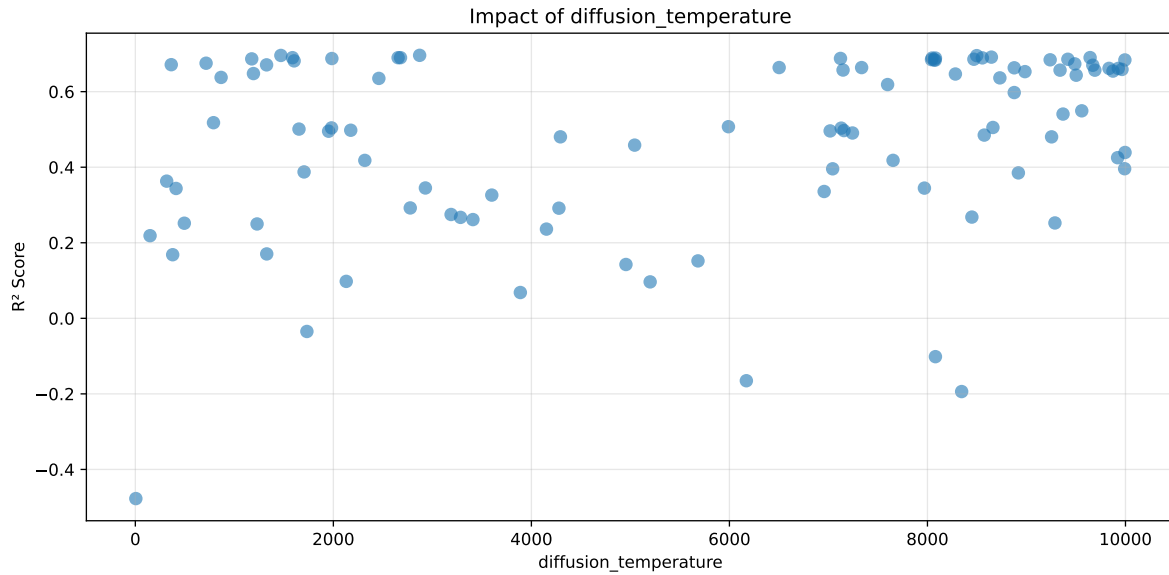
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682
y_test: mean=2.561, std=0.947, min=1.120, max=13.845

===== TOP Importance FEATURES =====
feature importance r_NH3 34.370815 r_umidity 32.900274 r_CO2 14.763376 r_temperature
9.762610 r_THI 8.202926



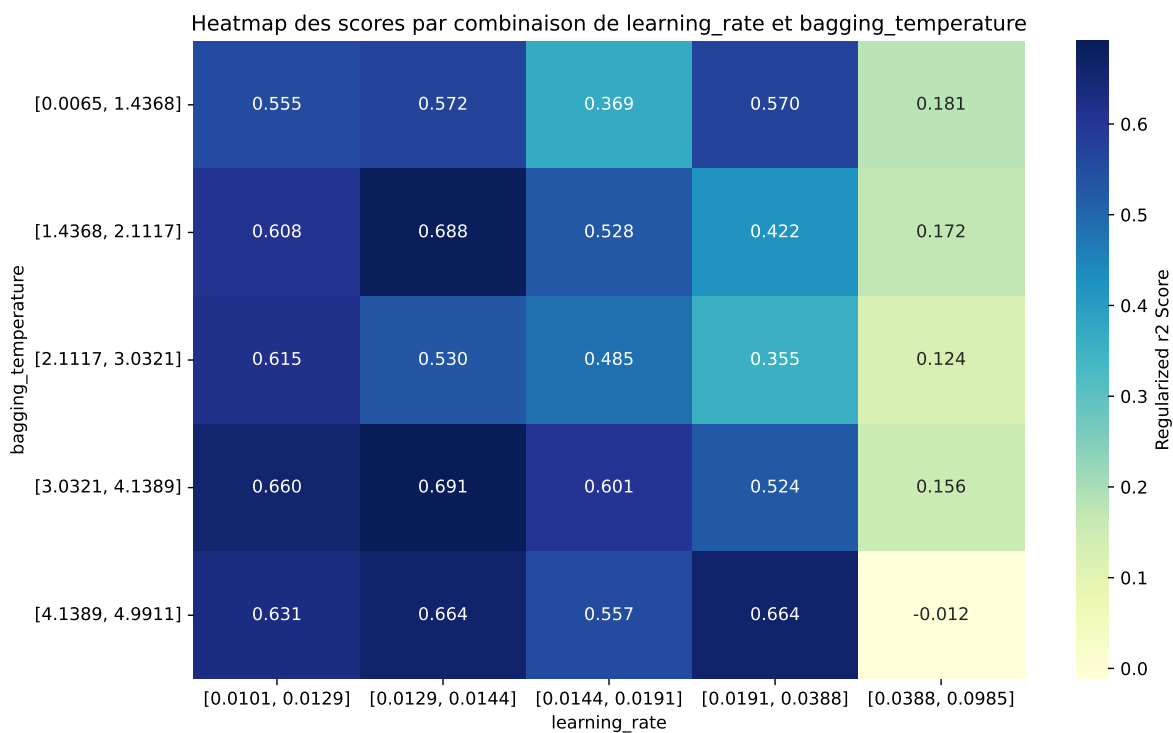
Paramètres avec >5% d'importance : ['learning_rate', 'bagging_temperature', 'diffusion_temperature', 'depth']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.727	0.6961	0.5981	0.016	0.6961	0.607	-0.0891	0.8719	1.0443
2	0.7241	0.6959	0.5974	0.0187	0.6959	0.6047	-0.0912	0.869	1.0405
3	0.7264	0.6953	0.5975	0.0167	0.6953	0.6059	-0.0894	0.8714	1.0447
4	0.7147	0.6914	0.5935	0.0171	0.6914	0.6006	-0.0908	0.8686	1.0338
5	0.7174	0.6902	0.5919	0.017	0.6902	0.6016	-0.0886	0.8716	1.0395
6	0.7123	0.69	0.5935	0.0175	0.69	0.6005	-0.0895	0.8703	1.0323
7	0.7145	0.69	0.5935	0.0183	0.69	0.6004	-0.0896	0.8701	1.0356
8	0.7144	0.6899	0.5957	0.0175	0.6899	0.6008	-0.0891	0.8709	1.0354
9	0.7124	0.6894	0.5939	0.0196	0.6894	0.604	-0.0854	0.8761	1.0334
10	0.7127	0.689	0.5931	0.0197	0.689	0.6039	-0.085	0.8766	1.0345
---	---	---	---	---	---	---	---	---	---
100	0.9526	0.7143	0.5979	0.0182	-0.4771	0.6046	-0.1098	0.8464	1.3336

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.014288405500122992, 'reg__l2_leaf_reg': 42.20600076881129, 'reg__bagging_temperature': 3.886166806614542, 'reg__random_strength': 3.2273552176930185, 'reg__subsample': 0.8342514356665152, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.8705764461063474, 'reg__border_count': 116, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2871.3480338062836}

Rank 2: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.015045290450284568, 'reg__l2_leaf_reg': 46.789822682572336, 'reg__bagging_temperature': 1.6492995899260459, 'reg__random_strength': 3.5964531385276977, 'reg__subsample': 0.8567000105561434, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7246170458722531, 'reg__border_count': 110, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1469.911295675369}

Rank 3: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.014348509180365144, 'reg__l2_leaf_reg': 41.65625068451749, 'reg__bagging_temperature': 4.071999411399027, 'reg__random_strength': 3.6027910048303573, 'reg__subsample': 0.8382490872878823, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.8707607800413667, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8497.755581031604}

Rank 4: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.013937710597395934, 'reg__l2_leaf_reg': 45.44149767668125, 'reg__bagging_temperature': 4.323292806624798, 'reg__random_strength': 2.612354068167349, 'reg__subsample': 0.8469394427487629, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6680966610523049, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8645.65539656192}

Rank 5: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.014452825118478767, 'reg__l2_leaf_reg': 41.64674775275989, 'reg__bagging_temperature': 3.93231590871045, 'reg__random_strength': 3.2287619780338543, 'reg__subsample': 0.8338601502329599, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8006981977808181, 'reg__border_count': 116, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 9642.688885639292}

Rank 6: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.014320281759136149, 'reg__l2_leaf_reg': 45.13589689906273, 'reg__bagging_temperature': 4.132356790151724, 'reg__random_strength': 3.8636825784214146, 'reg__subsample': 0.7806635328815273, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6676988909099456, 'reg__border_count': 123, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2678.0201876439232}

Rank 7: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.020636008210220994, 'reg__l2_leaf_reg': 46.8388112204199, 'reg__bagging_temperature': 1.0552185331632766, 'reg__random_strength': 3.60667522722715, 'reg__subsample': 0.8167167345041789,

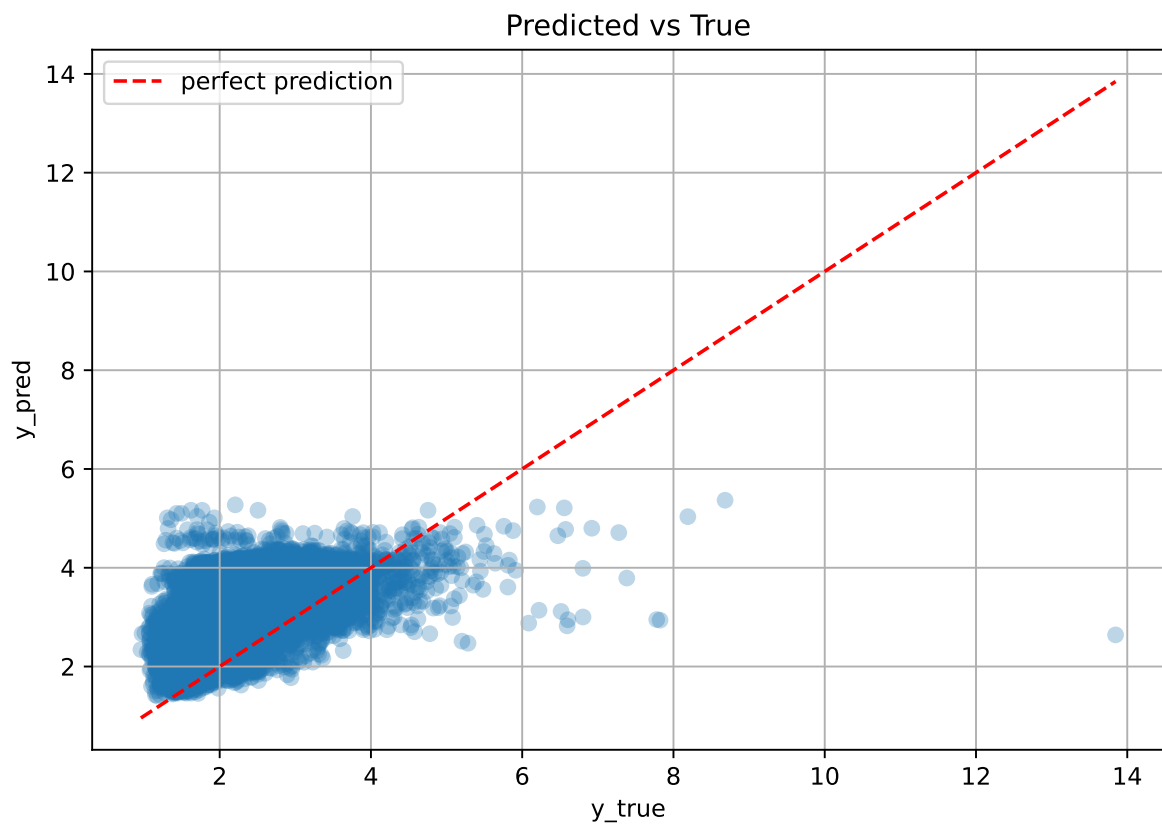
'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6980070840041838, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1586.9136637098072}

Rank 8: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.014498912963966892, 'reg__l2_leaf_reg': 42.17455335066343, 'reg__bagging_temperature': 4.976591704241121, 'reg__random_strength': 3.8306231950538274, 'reg__subsample': 0.8399864534378556, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7965015140463607, 'reg__border_count': 123, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2653.8978785090485}

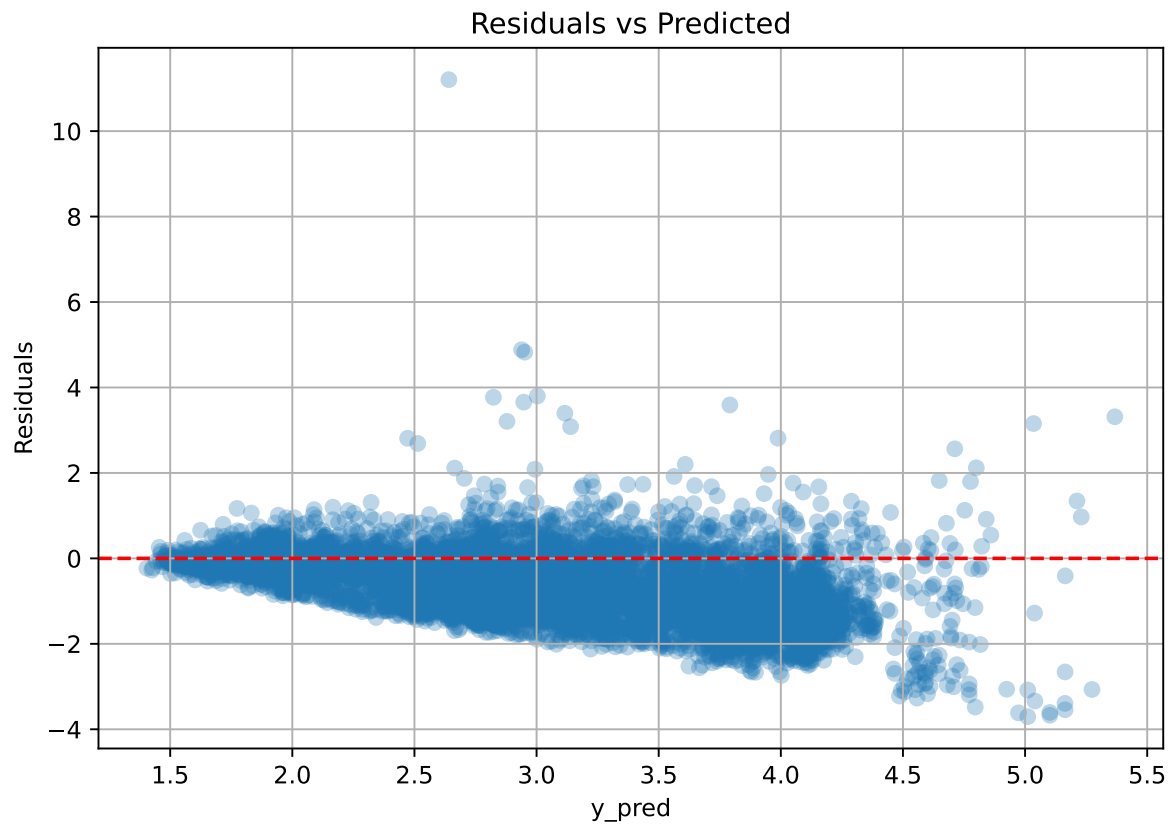
Rank 9: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.014427416212033748, 'reg__l2_leaf_reg': 44.939139980840665, 'reg__bagging_temperature': 4.165085932878247, 'reg__random_strength': 3.8348508239210055, 'reg__subsample': 0.8328080833037338, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8292882633123776, 'reg__border_count': 110, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8554.691953703186}

Rank 10: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.014424945910291974, 'reg__l2_leaf_reg': 44.889928876772665, 'reg__bagging_temperature': 4.695739867475764, 'reg__random_strength': 3.8107930071533183, 'reg__subsample': 0.8320437977797657, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.828138594820312, 'reg__border_count': 110, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8047.092427074998}

2.5.6 Scatter



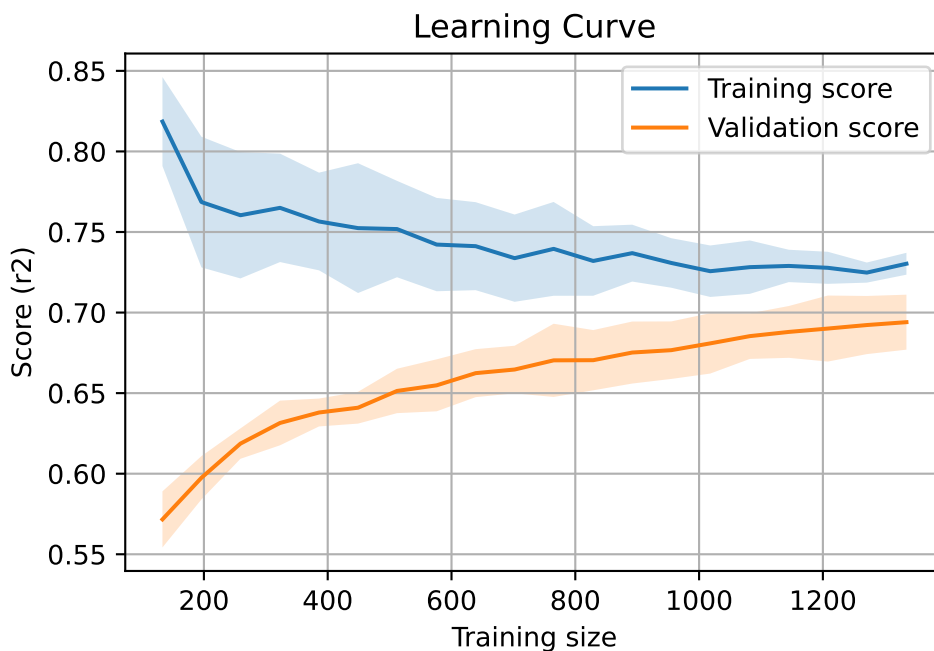
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2_11

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817



2.5.9 Gridsearch — Group: 3_4

Len(group) : 568, Len(training) : 398, Len(test) : 170

Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804
y_test: mean=1.833, std=0.729, min=1.098, max=7.822

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils_tags.py:338: DeprecationWarning:

The following error was raised: 'CatBoostRegressor' object has no attribute '__sklearn_tags__'

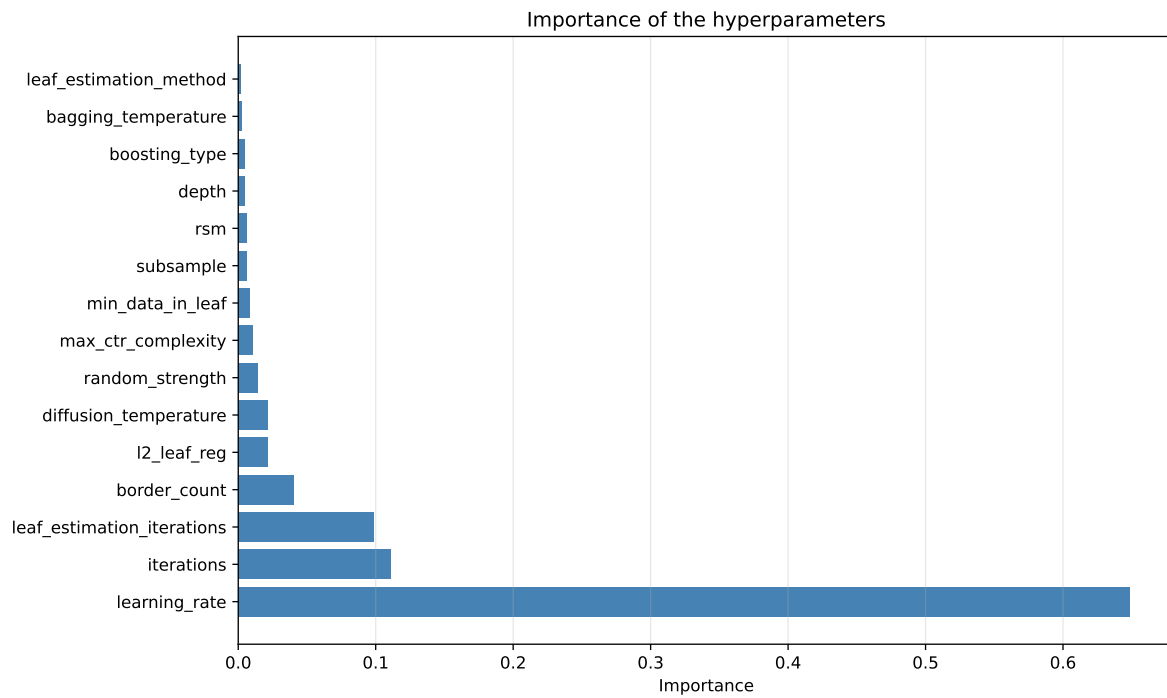
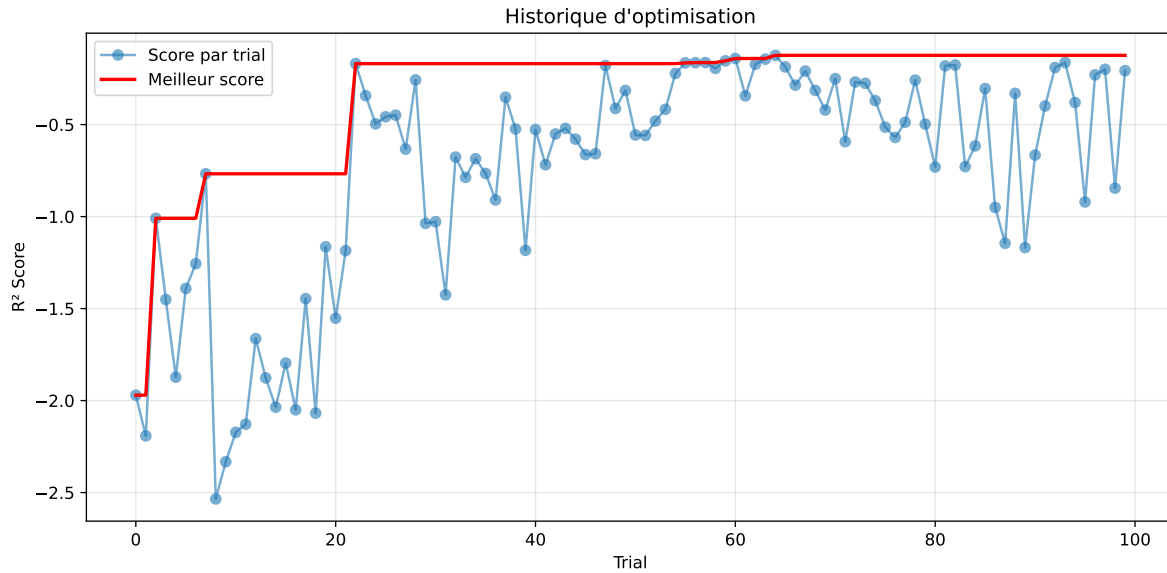
C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils_tags.py:338: DeprecationWarning:

The following error was raised: 'CatBoostRegressor' object has no attribute '__sklearn_tags__'

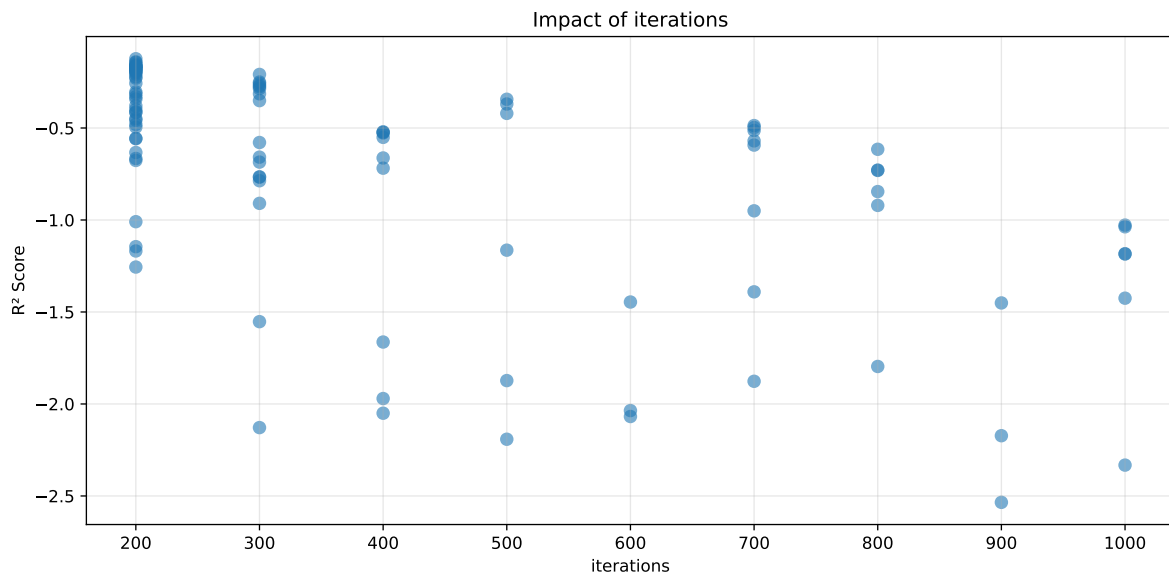
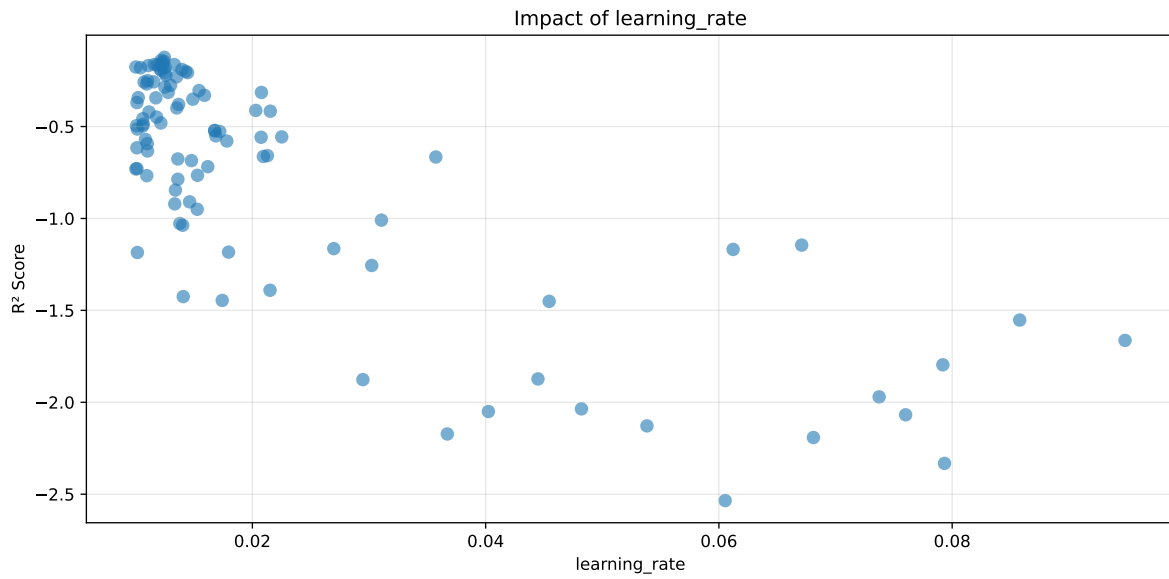
C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils_tags.py:338: DeprecationWarning:

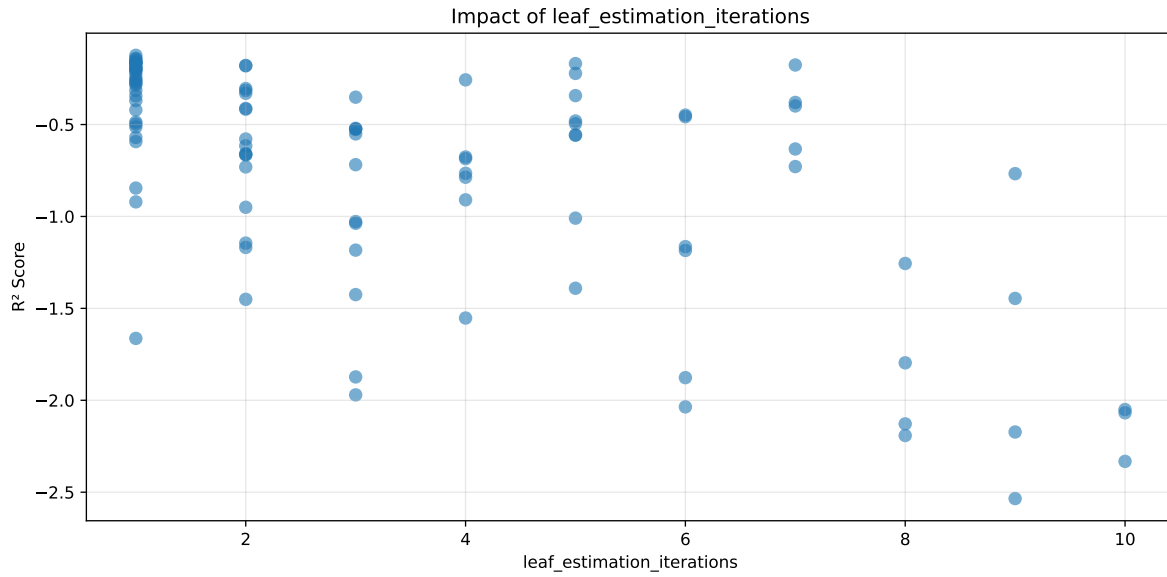
The following error was raised: 'CatBoostRegressor' object has no attribute '__sklearn_tags__'

===== TOP Importance FEATURES =====
feature importance r_umidity 31.223913 r_NH3 21.796857 r_CO2 18.844277 r_THI 14.513937
r_temperature 13.621016



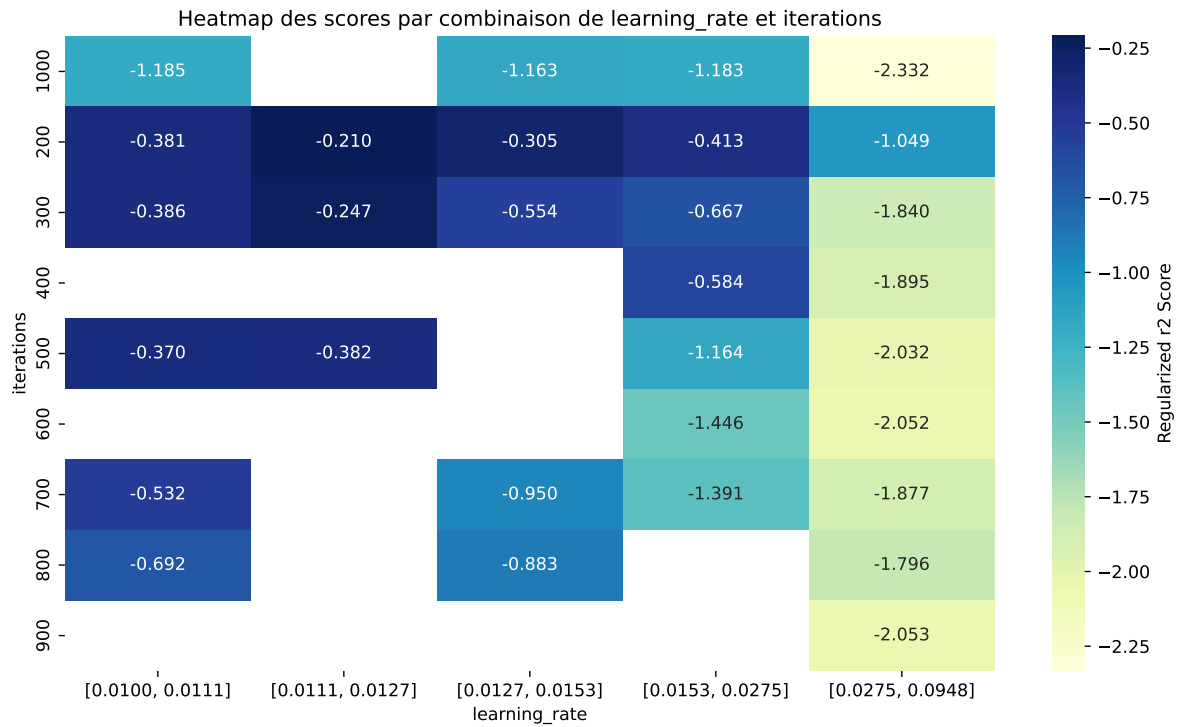
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'leaf_estimation_iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.2924	0.2231	0.1008	0.0325	-0.1235	0.1307	-0.0924	0.5858	1.3108
2	0.336	0.2566	0.1275	0.0312	-0.1405	0.1505	-0.1062	0.5863	1.3095
3	0.3399	0.2592	0.1269	0.0355	-0.1442	0.1519	-0.1072	0.5863	1.3113
4	0.332	0.2512	0.127	0.0317	-0.1533	0.1544	-0.0968	0.6147	1.3221
5	0.3558	0.2694	0.135	0.0339	-0.1626	0.1492	-0.1201	0.554	1.3207
6	0.3599	0.2728	0.1396	0.0335	-0.1628	0.1613	-0.1114	0.5915	1.3193
7	0.3707	0.2817	0.138	0.037	-0.1634	0.1559	-0.1258	0.5533	1.316
8	0.3653	0.277	0.1358	0.036	-0.1644	0.1533	-0.1237	0.5534	1.3187
9	0.3709	0.281	0.1343	0.0432	-0.1685	0.1778	-0.1032	0.6328	1.3199
10	0.3698	0.2795	0.1351	0.0328	-0.1722	0.1527	-0.1268	0.5464	1.3232
---	---	---	---	---	---	---	---	---	---
100	0.928	0.3509	0.1713	0.0813	-2.5347	0.3029	-0.048	0.8632	2.6447

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.012451550603763972, 'reg__l2_leaf_reg': 26.966125479413943, 'reg__bagging_temperature': 3.9092910998052757, 'reg__random_strength': 1.8639915055913192, 'reg__subsample': 0.690662414691521, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7191650679135837, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 30.431681653252966}

Rank 2: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.012171361619924133, 'reg__l2_leaf_reg': 27.574741932627422, 'reg__bagging_temperature': 3.685672376944333, 'reg__random_strength': 2.7699375542885987, 'reg__subsample': 0.6844099420320054, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7613572027354554, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 967.93543168774}

Rank 3: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.012404356309417112, 'reg__l2_leaf_reg': 27.09104276635847, 'reg__bagging_temperature': 3.8052021685355766, 'reg__random_strength': 2.824175035498349, 'reg__subsample': 0.6746915124085484, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7805430389706731, 'reg__border_count': 81, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 726.566686369612}

Rank 4: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.012295847404301228, 'reg__l2_leaf_reg': 26.890977493275905, 'reg__bagging_temperature': 4.01494217969702, 'reg__random_strength': 2.769210267931187, 'reg__subsample': 0.7802681704672395, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':

0.7636794617472523, 'reg__border_count': 94, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 864.9594102875117}

Rank 5: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011591573932046403, 'reg__l2_leaf_reg': 26.940830895925707, 'reg__bagging_temperature': 3.830105499545046, 'reg__random_strength': 1.9565990673530678, 'reg__subsample': 0.8233481588518963, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8975842435628048, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 889.392588850043}

Rank 6: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.013306616671387678, 'reg__l2_leaf_reg': 24.283559897128242, 'reg__bagging_temperature': 4.799691365201878, 'reg__random_strength': 1.7375912162132177, 'reg__subsample': 0.7419762740968235, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7908680577614583, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 105.86588508285153}

Rank 7: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.012182596308408362, 'reg__l2_leaf_reg': 26.249871700895483, 'reg__bagging_temperature': 3.9677847132243174, 'reg__random_strength': 2.0507604389219405, 'reg__subsample': 0.8981871486477465, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.896538428349012, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 953.5273520149833}

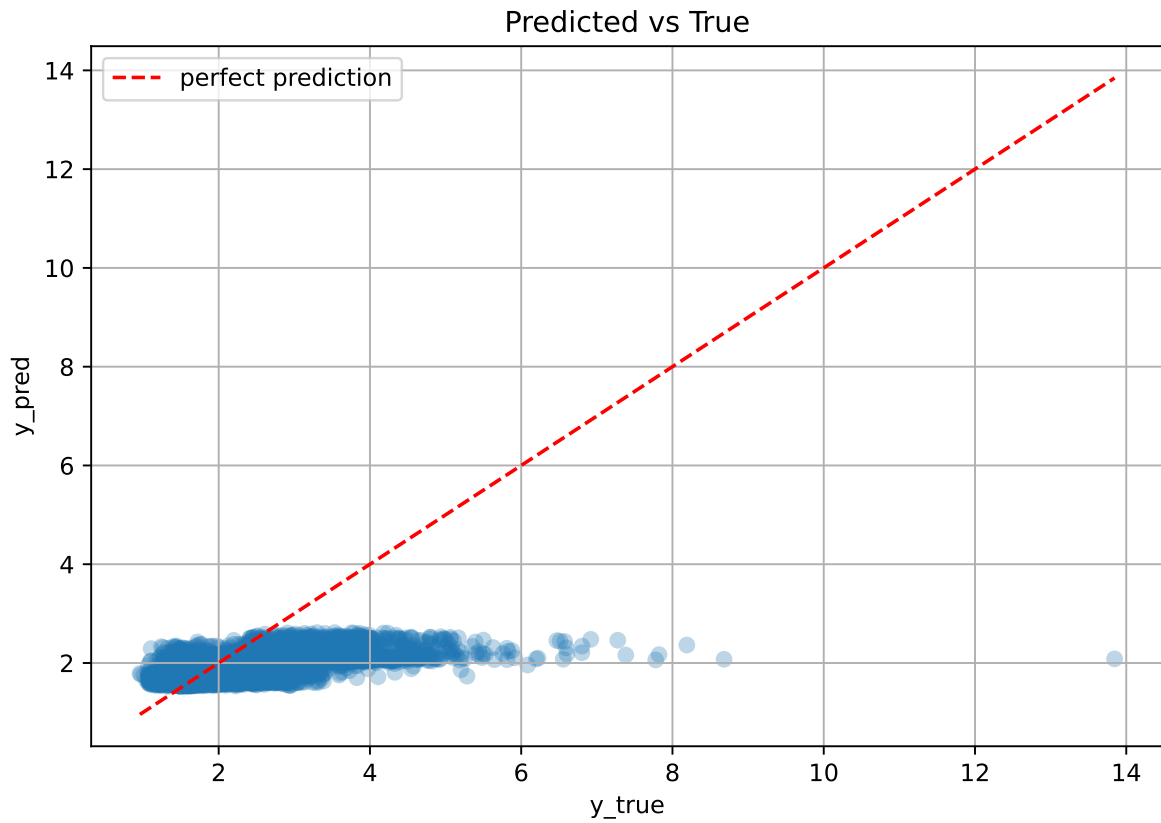
Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011846088757914566, 'reg__l2_leaf_reg': 25.495481575013212, 'reg__bagging_temperature': 3.963340522762298, 'reg__random_strength': 2.0806647496598933, 'reg__subsample': 0.8280444618106181, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8964834693826365, 'reg__border_count': 120, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 883.847266325288}

Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01107314718221797, 'reg__l2_leaf_reg': 36.244060845679655, 'reg__bagging_temperature': 3.1669941860065522, 'reg__random_strength': 2.0516805111565364, 'reg__subsample': 0.7602447101707748, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8995320410431044, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 148.37653160702132}

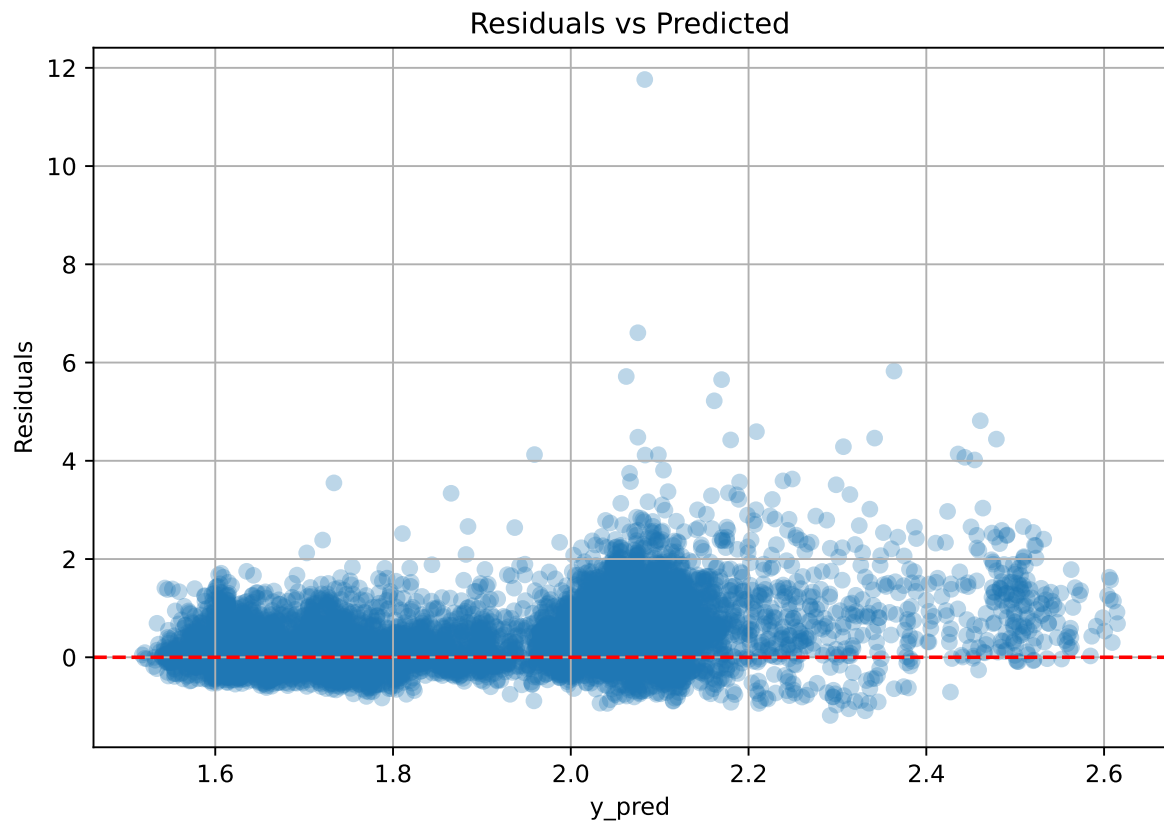
Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01228870467624894, 'reg__l2_leaf_reg': 25.842573394384328, 'reg__bagging_temperature': 4.009895526845408,

```
'reg__random_strength': 1.8929331670504403, 'reg__subsample': 0.708702240951353,  
'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':  
0.7597271712281001, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_tempera-  
ture': 679.9457059937015}
```

2.5.10 Scatter



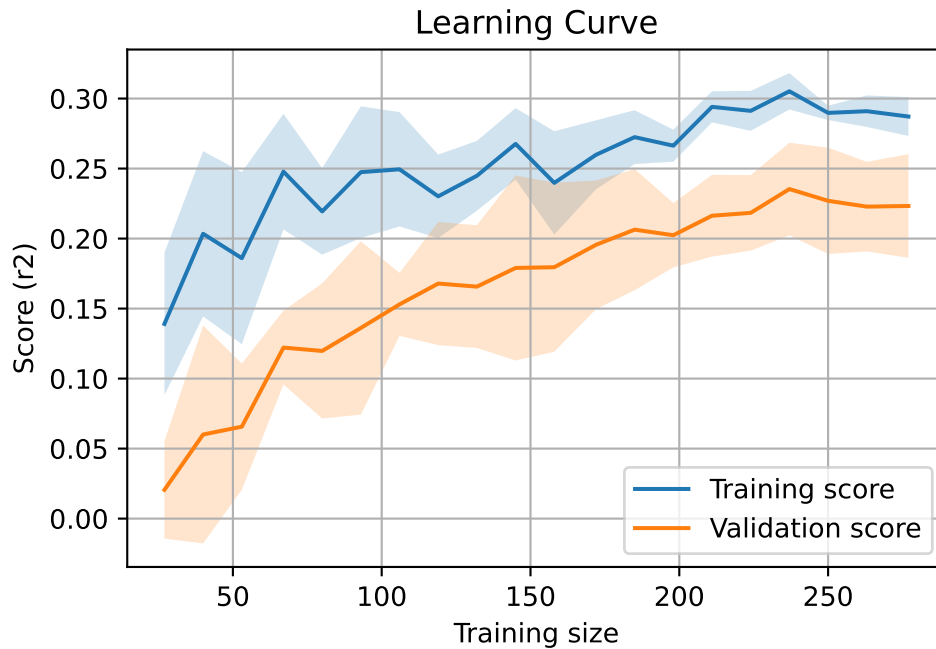
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 3_4

`Len(group)` : 568, `Len(training)` : 398,

`Len(training_real)` : 279, `Len(test_of_training)` : 119, `Len(test)` : 170

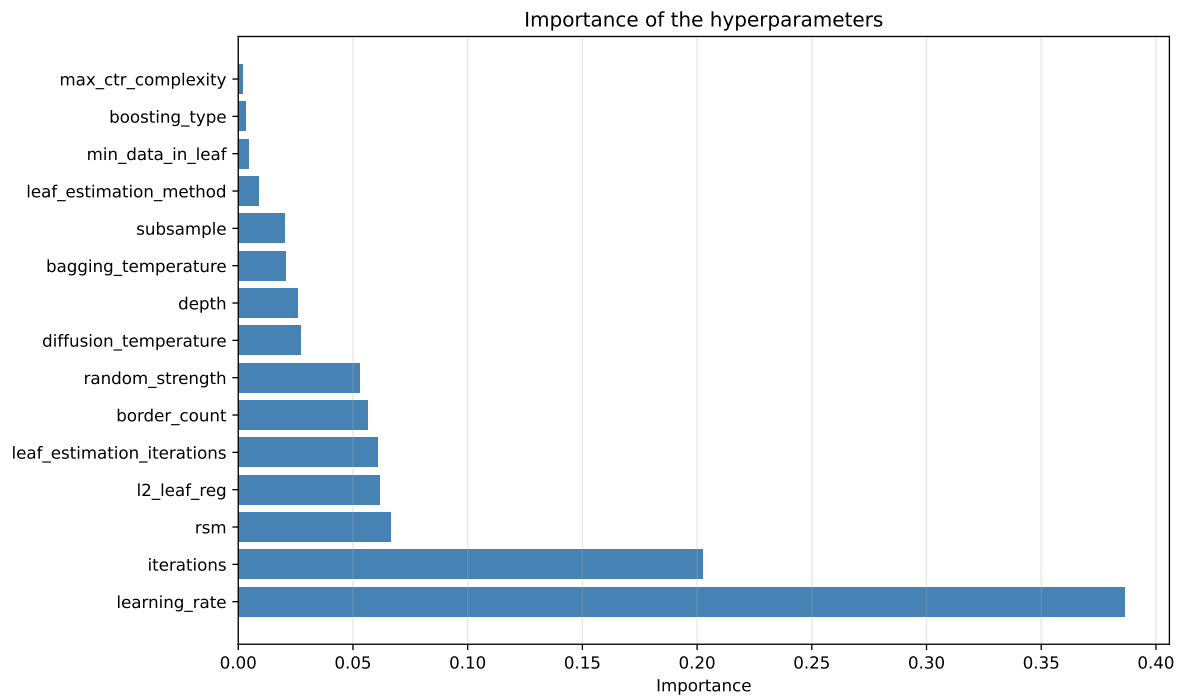
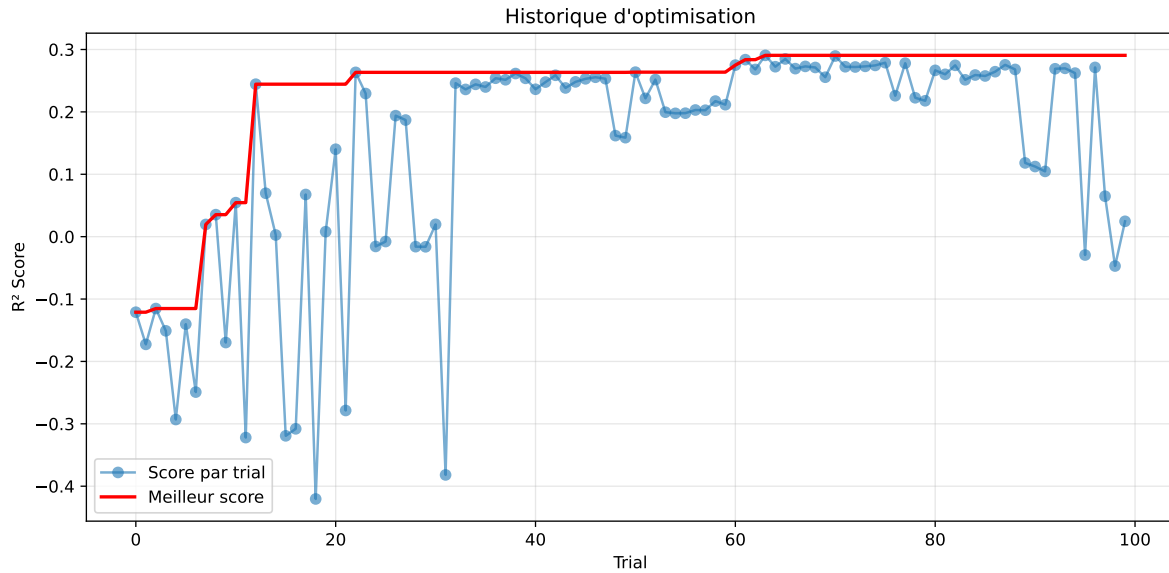


2.5.13 Gridsearch — Group: 4_5

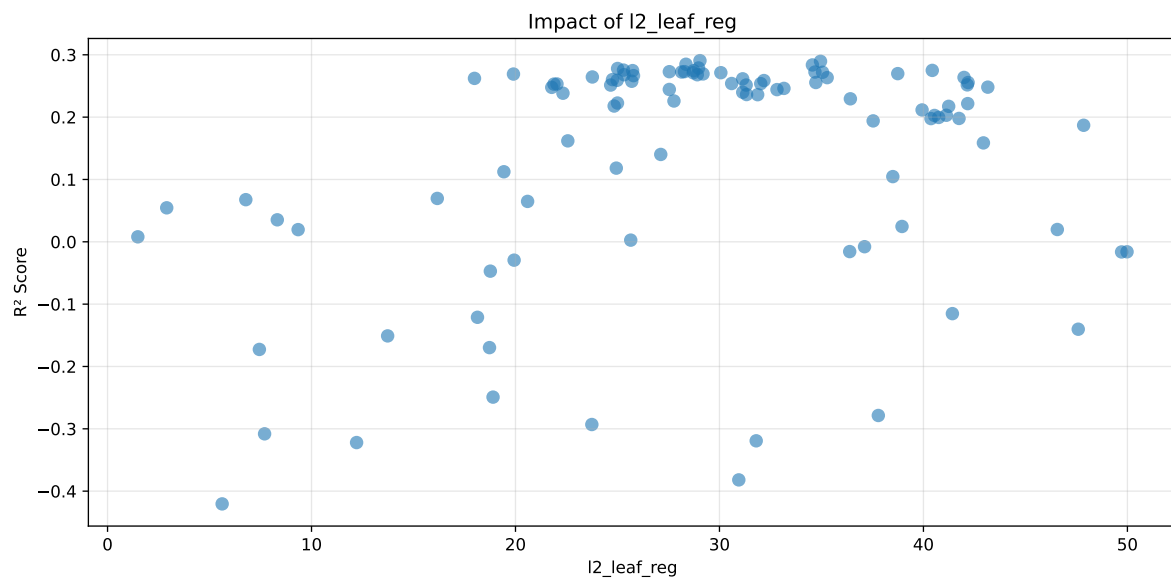
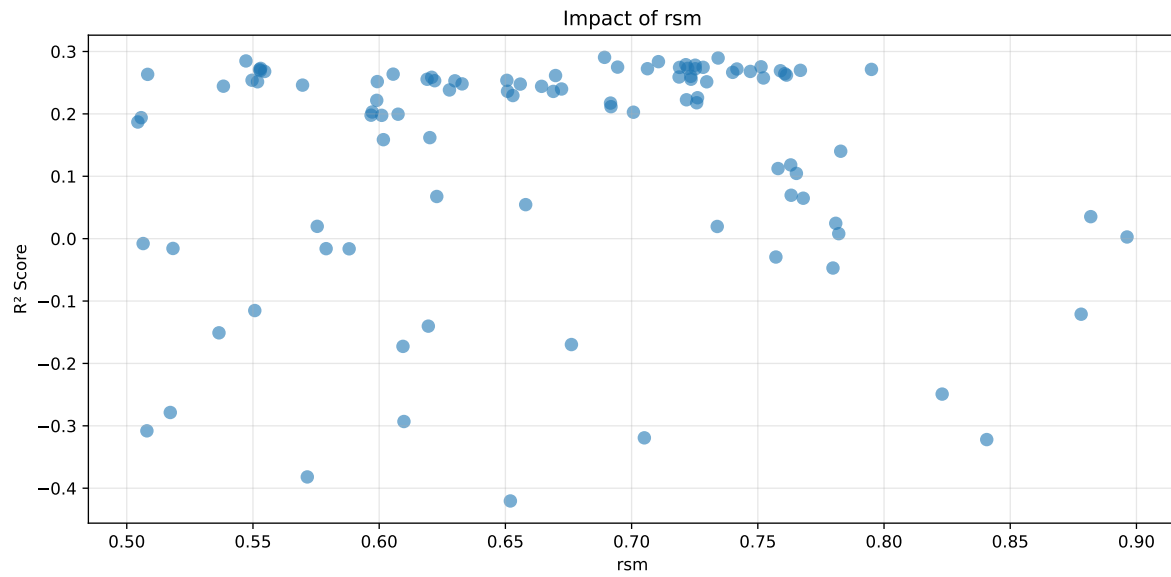
Len(group) : 2726, Len(training) : 1909, Len(test) : 817

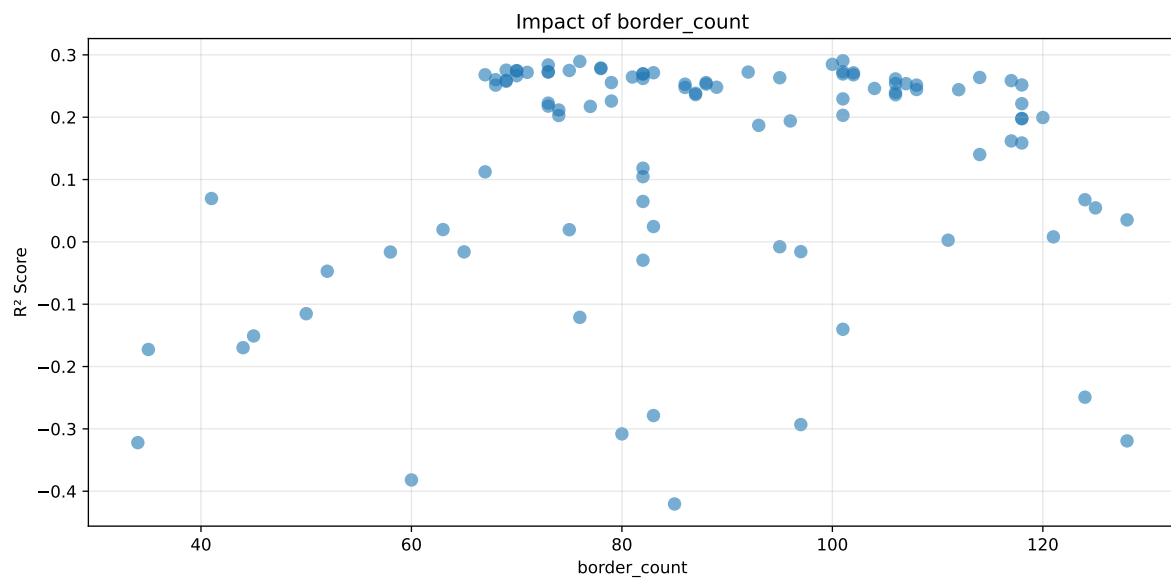
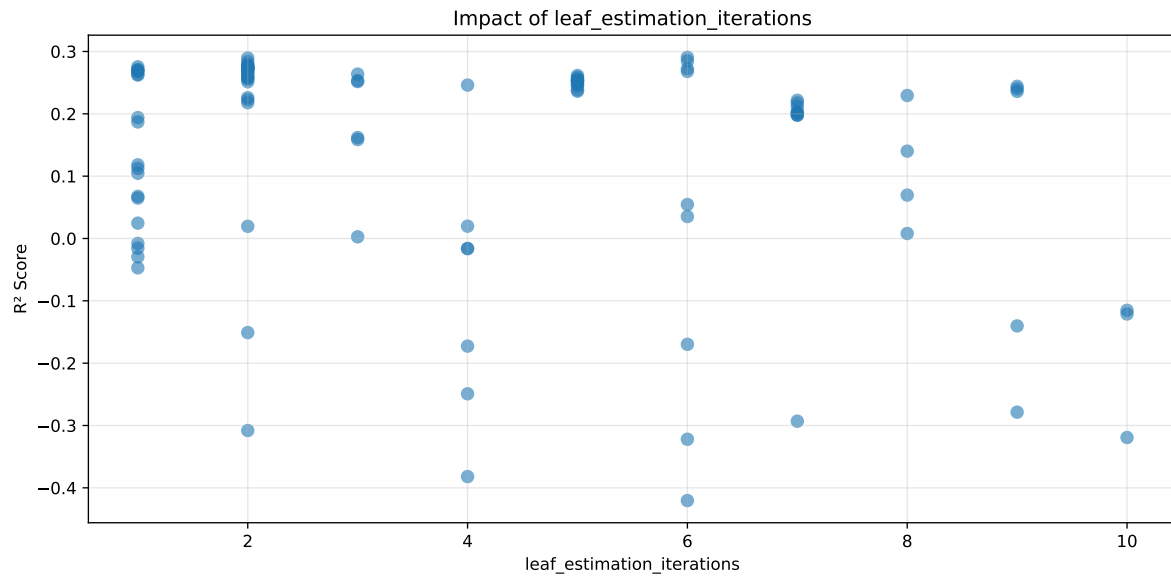
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813
y_test: mean=2.212, std=0.659, min=1.071, max=4.858

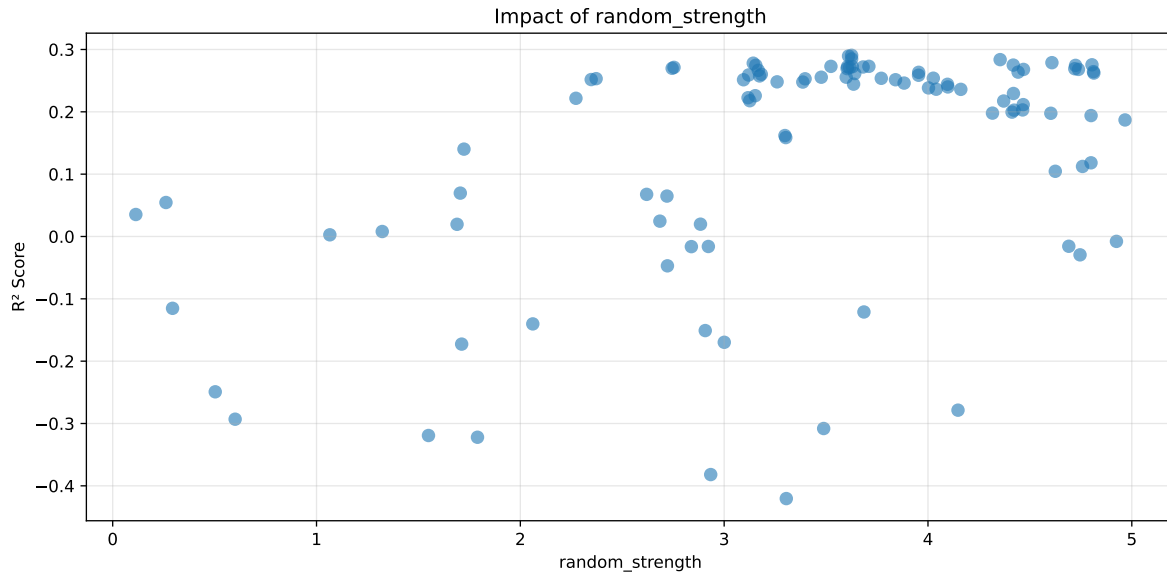
===== TOP Importance FEATURES =====
feature importance r_umidity 62.123047 r_temperature 16.814533 r_THI 8.688861 r_NH3
6.635211 r_CO2 5.738348



Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'rsm', 'l2_leaf_reg', 'leaf_estimation_iterations', 'border_count', 'random_strength']

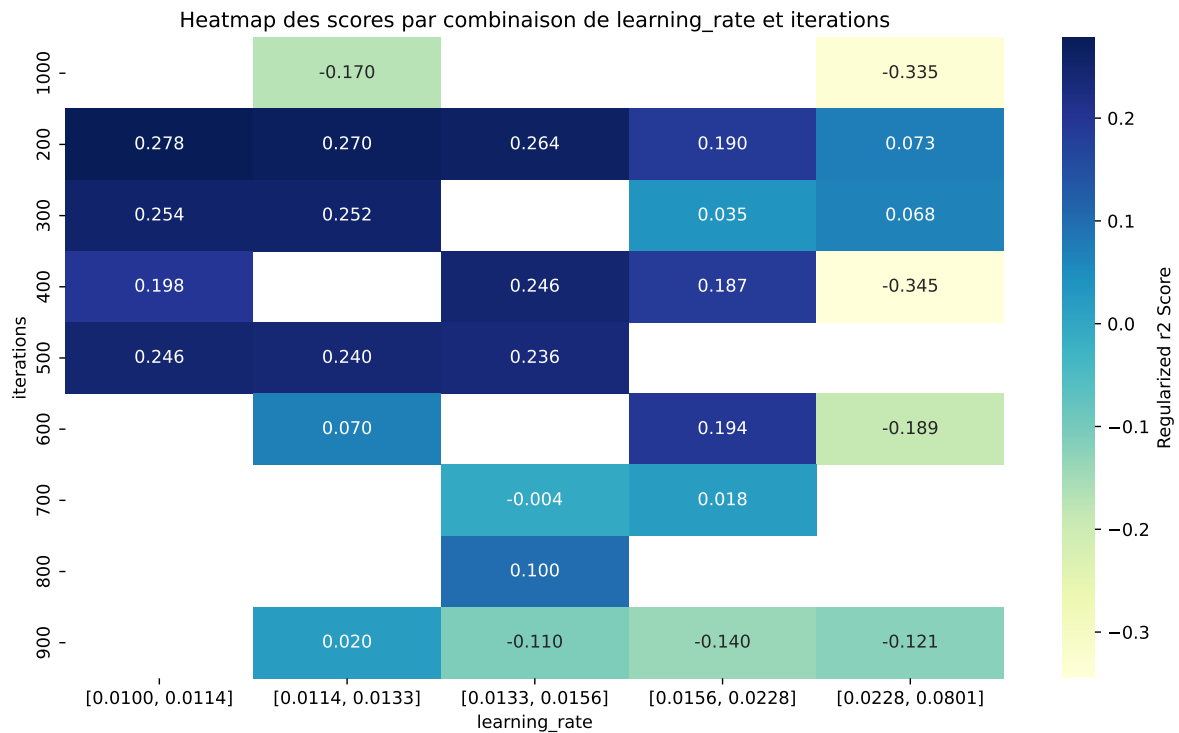






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4289	0.4058	0.4666	0.0198	0.2906	0.4583	0.0524	1.1291	1.0568
2	0.4366	0.4121	0.4678	0.0188	0.2896	0.4718	0.0597	1.1449	1.0595
3	0.4188	0.3964	0.4552	0.0202	0.2849	0.4554	0.0589	1.1486	1.0563
4	0.4324	0.4076	0.4651	0.0186	0.2837	0.4691	0.0614	1.1507	1.0608
5	0.4494	0.421	0.4784	0.0191	0.2789	0.486	0.065	1.1545	1.0675
6	0.4546	0.4252	0.4826	0.02	0.278	0.4858	0.0606	1.1425	1.0692
7	0.4538	0.424	0.482	0.021	0.2755	0.4922	0.0681	1.1607	1.0701
8	0.4681	0.4359	0.4989	0.0198	0.275	0.5037	0.0678	1.1555	1.0738
9	0.4552	0.4251	0.4825	0.0213	0.2746	0.4903	0.0652	1.1535	1.0708
10	0.4502	0.4209	0.4793	0.0189	0.2745	0.4835	0.0626	1.1487	1.0696
---	---	---	---	---	---	---	---	---	---
100	0.7412	0.5476	0.6015	0.026	-0.4205	0.6168	0.0692	1.1265	1.3536

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.011343265761855724, 'reg__l2_leaf_reg': 29.043678516197858, 'reg__bagging_temperature': 3.414184888641062, 'reg__random_strength': 3.6254233976863666, 'reg__subsample': 0.595309681681758, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6892801166423206, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1514.5037882441256}

Rank 2: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01131779821126295, 'reg__l2_leaf_reg': 34.95459325637103, 'reg__bagging_temperature': 2.856024472610275, 'reg__random_strength': 3.6102501645564007, 'reg__subsample': 0.645388131117521, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7342751387139075, 'reg__border_count': 76, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2007.4207255716628}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.011185587391425453, 'reg__l2_leaf_reg': 28.357317353852693, 'reg__bagging_temperature': 2.892484426295976, 'reg__random_strength': 3.6254644505644507, 'reg__subsample': 0.749154951211871, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.547218489243057, 'reg__border_count': 100, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 3281.369278672683}

Rank 4: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.011244926838050768, 'reg__l2_leaf_reg': 34.556485333706654, 'reg__bagging_temperature': 4.859147596155583, 'reg__random_strength': 4.354021225957991, 'reg__subsample': 0.721257900809539, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':

0.7106286960548492, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1890.0309878674034}

Rank 5: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.011379571981026555, 'reg__l2_leaf_reg': 28.975878094776238, 'reg__bagging_temperature': 4.350462772013735, 'reg__random_strength': 4.608101707466867, 'reg__subsample': 0.7462802927612451, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7214882820877745, 'reg__border_count': 78, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1811.5514296157978}

Rank 6: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.011188919908420471, 'reg__l2_leaf_reg': 25.006027031964553, 'reg__bagging_temperature': 4.983701303964616, 'reg__random_strength': 3.142680253951686, 'reg__subsample': 0.6450649620169228, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7251827772129287, 'reg__border_count': 78, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1867.4738819742324}

Rank 7: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.014226376065863475, 'reg__l2_leaf_reg': 25.295648988098907, 'reg__bagging_temperature': 4.220306617801372, 'reg__random_strength': 4.804596769853594, 'reg__subsample': 0.781054679263419, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7512766512440274, 'reg__border_count': 69, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1135.806433353106}

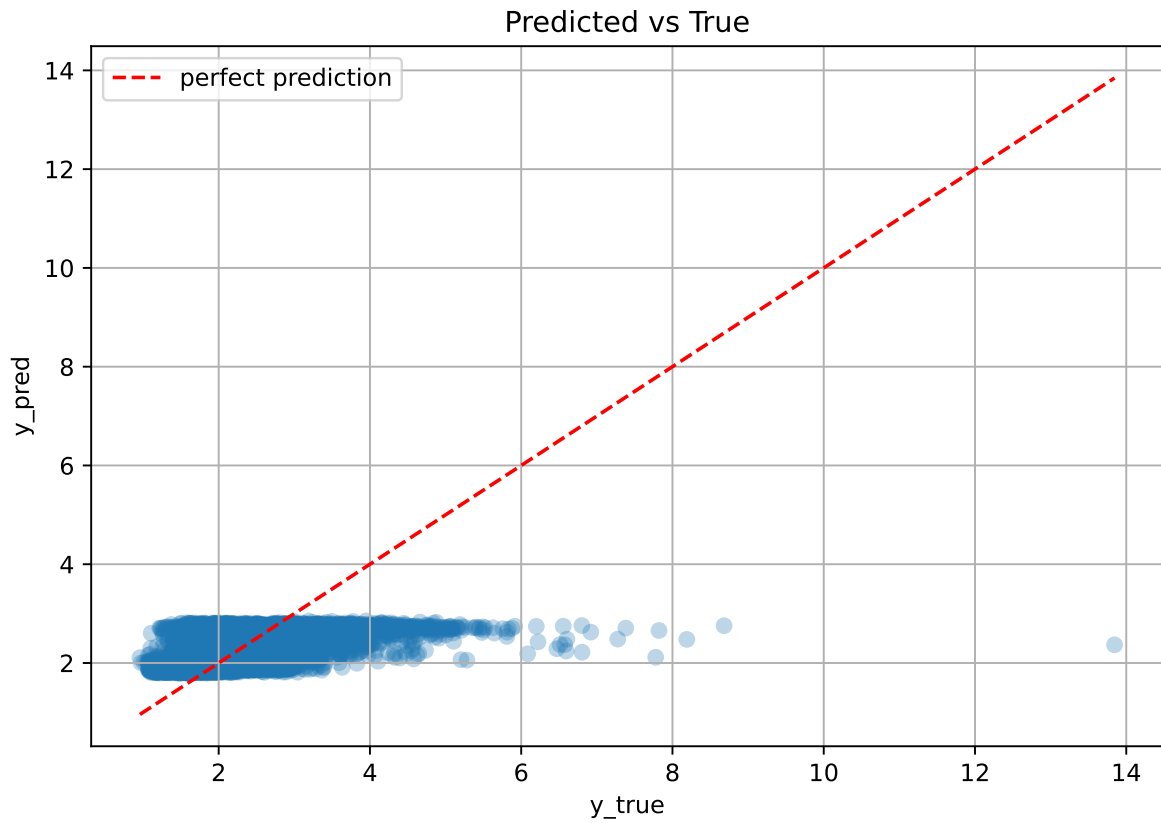
Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.017899859462960025, 'reg__l2_leaf_reg': 40.433464163751005, 'reg__bagging_temperature': 2.8303885613392716, 'reg__random_strength': 4.418423325389085, 'reg__subsample': 0.6513320868880342, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.694451148893424, 'reg__border_count': 75, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1815.2663809550663}

Rank 9: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.011293213700195377, 'reg__l2_leaf_reg': 25.740877944672008, 'reg__bagging_temperature': 4.984943424923714, 'reg__random_strength': 3.1552086459708555, 'reg__subsample': 0.7817699042718909, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7283329711328648, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1089.0700760483119}

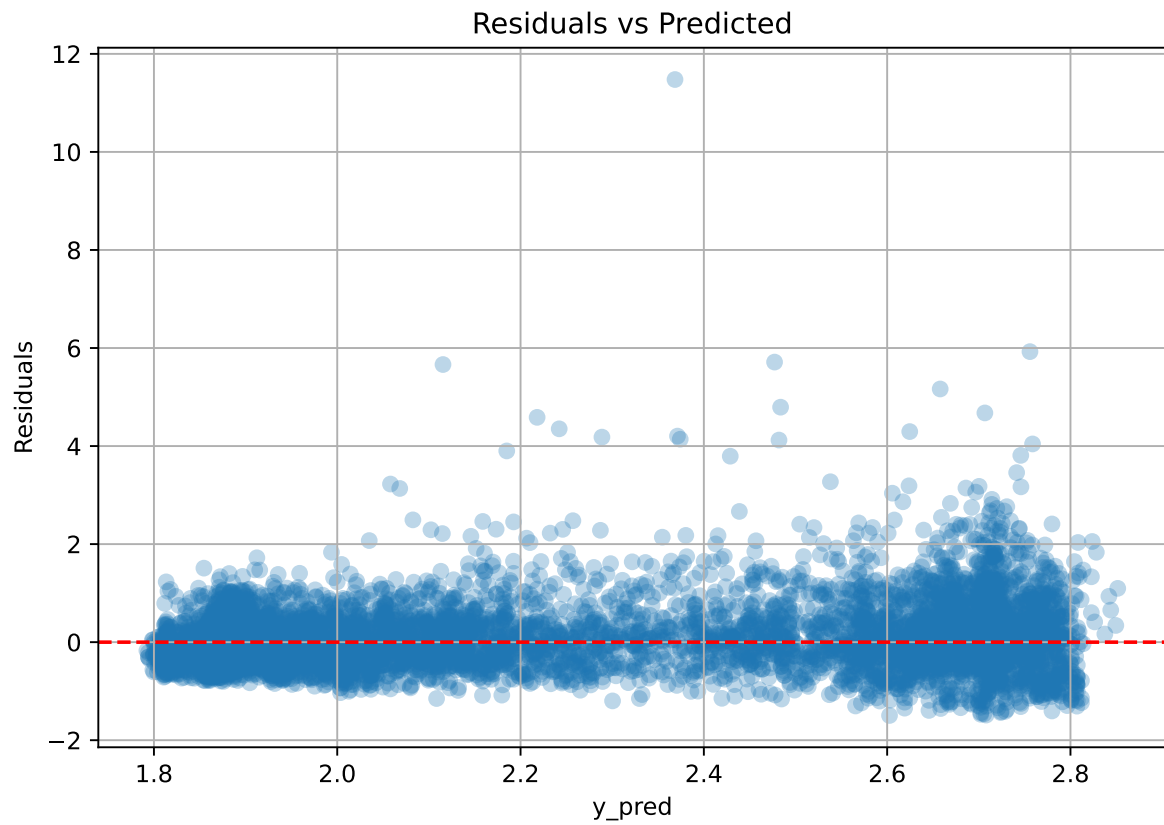
Rank 10: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.011346458014543578, 'reg__l2_leaf_reg': 28.736045570952026, 'reg__bagging_temperature': 4.542943422953848,

'reg__random_strength': 4.723948191408412, 'reg__subsample': 0.7501501914412058,
'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':
0.7189919761516738, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_tempera-
ture': 1848.5404918578834}

2.5.14 Scatter



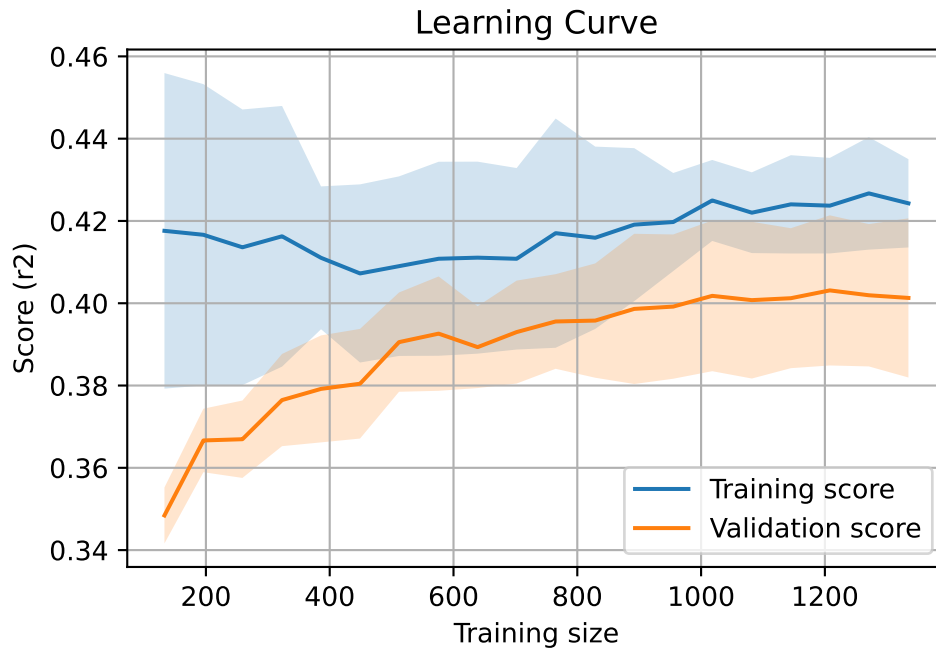
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 4_5

`Len(group)` : 2726, `Len(training)` : 1909,

`Len(training_real)` : 1337, `Len(test_of_training)` : 572, `Len(test)` : 817

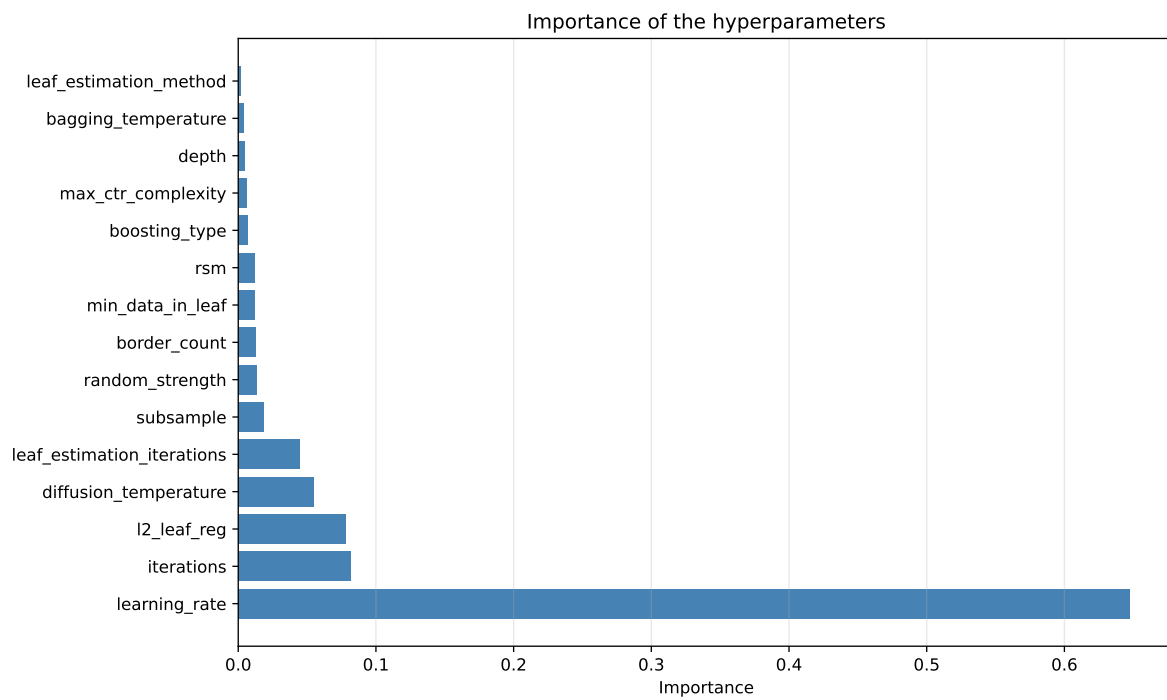
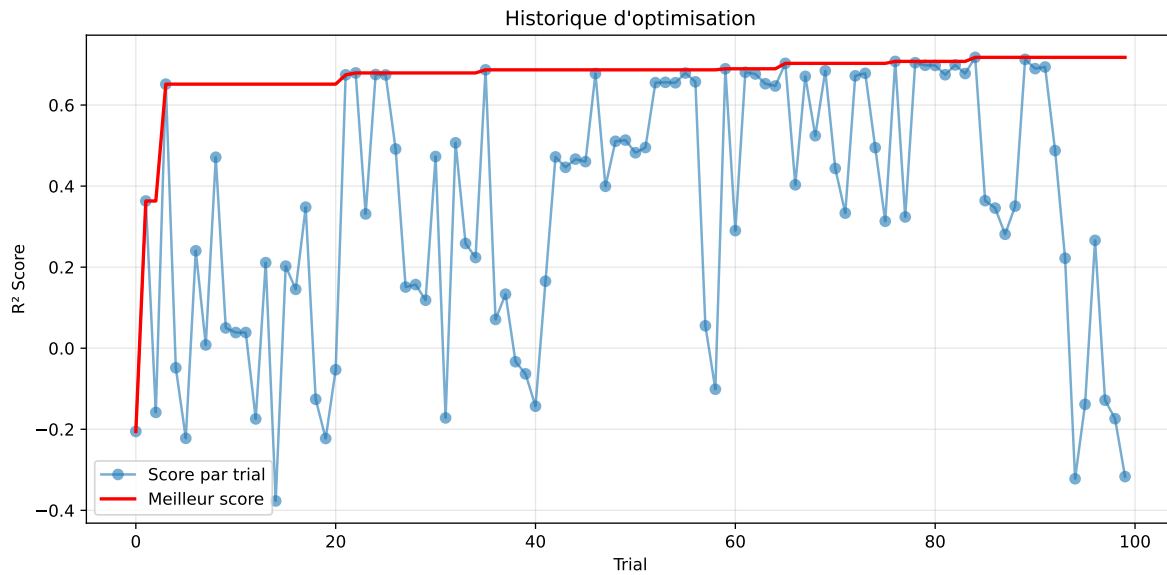


2.5.17 Gridsearch — Group: 6_3

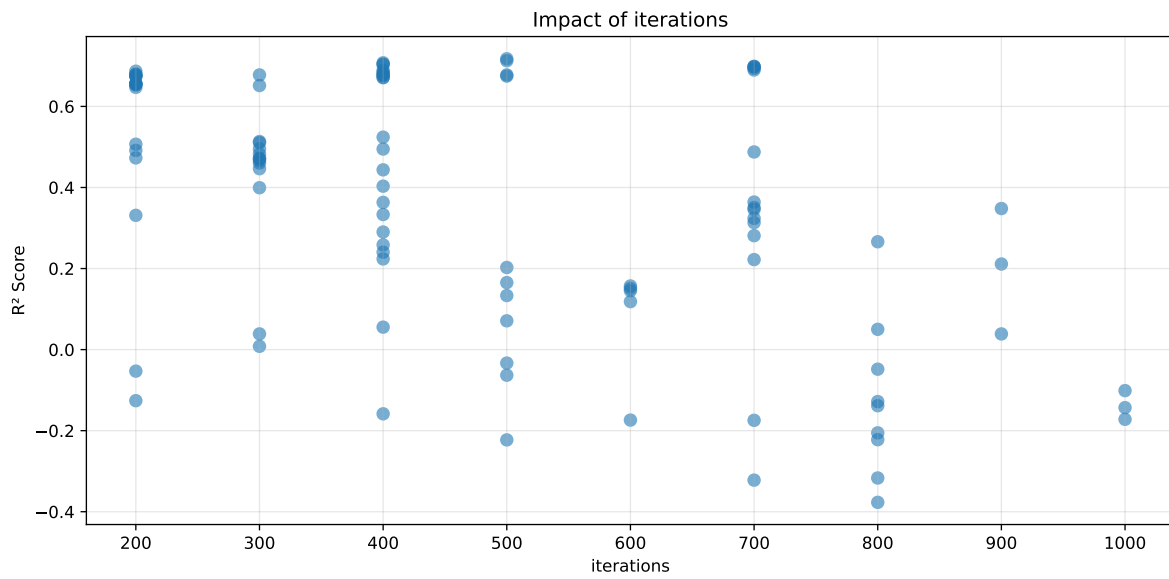
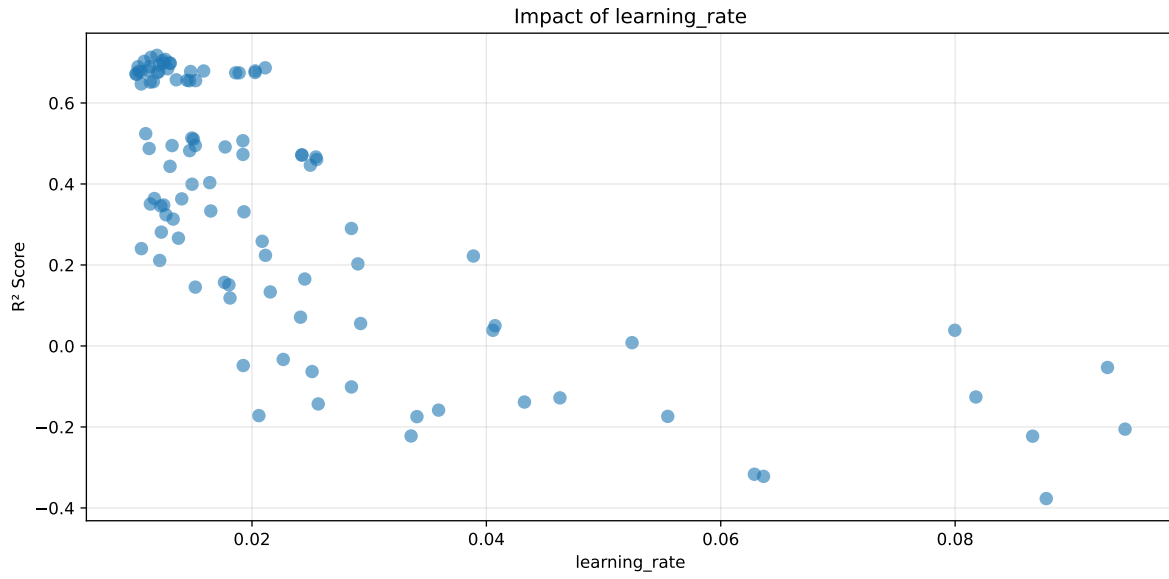
Len(group) : 561, Len(training) : 393, Len(test) : 168

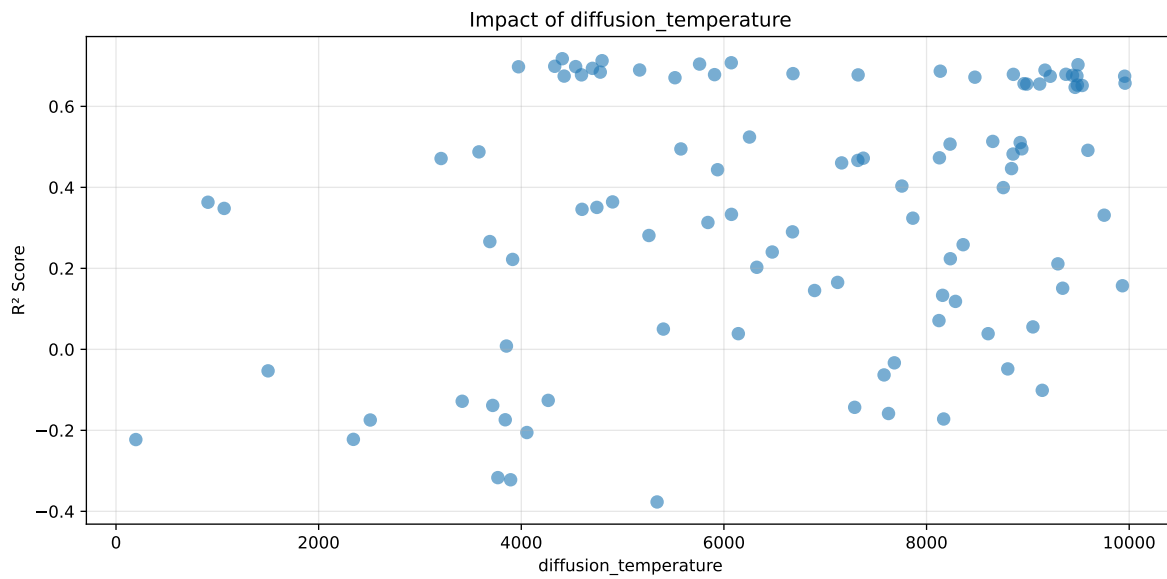
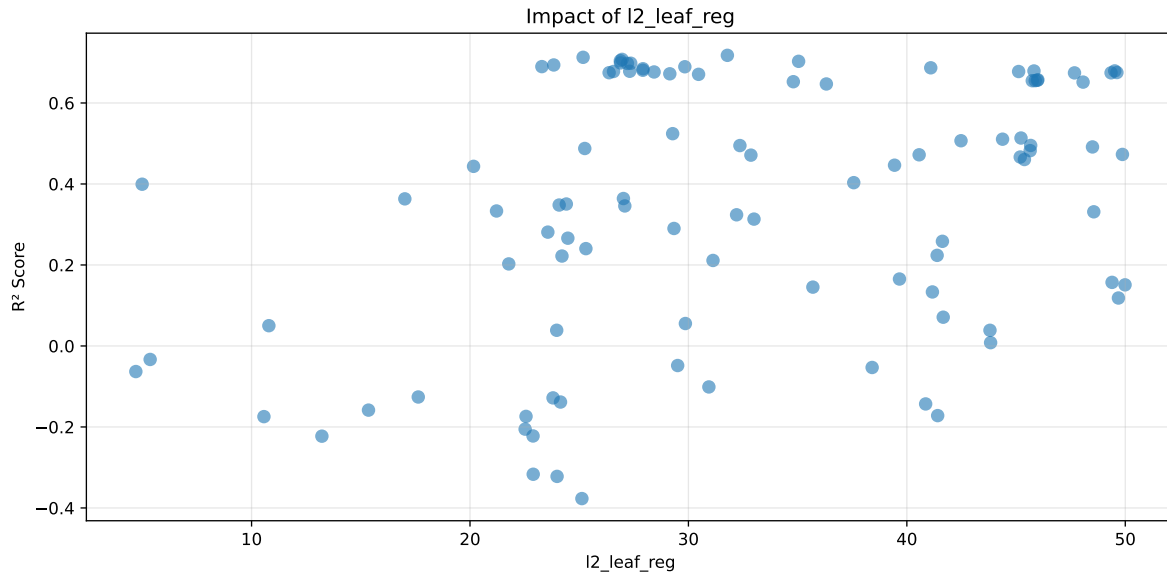
Distribution y_train vs y_test: y_train: mean=1.935, std=0.720, min=1.041, max=6.086
y_test: mean=2.031, std=0.744, min=1.078, max=4.912

===== TOP Importance FEATURES =====
feature importance r_umidity 36.440228 r_NH3 29.834637 r_temperature 14.517596 r_THI
14.058151 r_CO2 5.149388



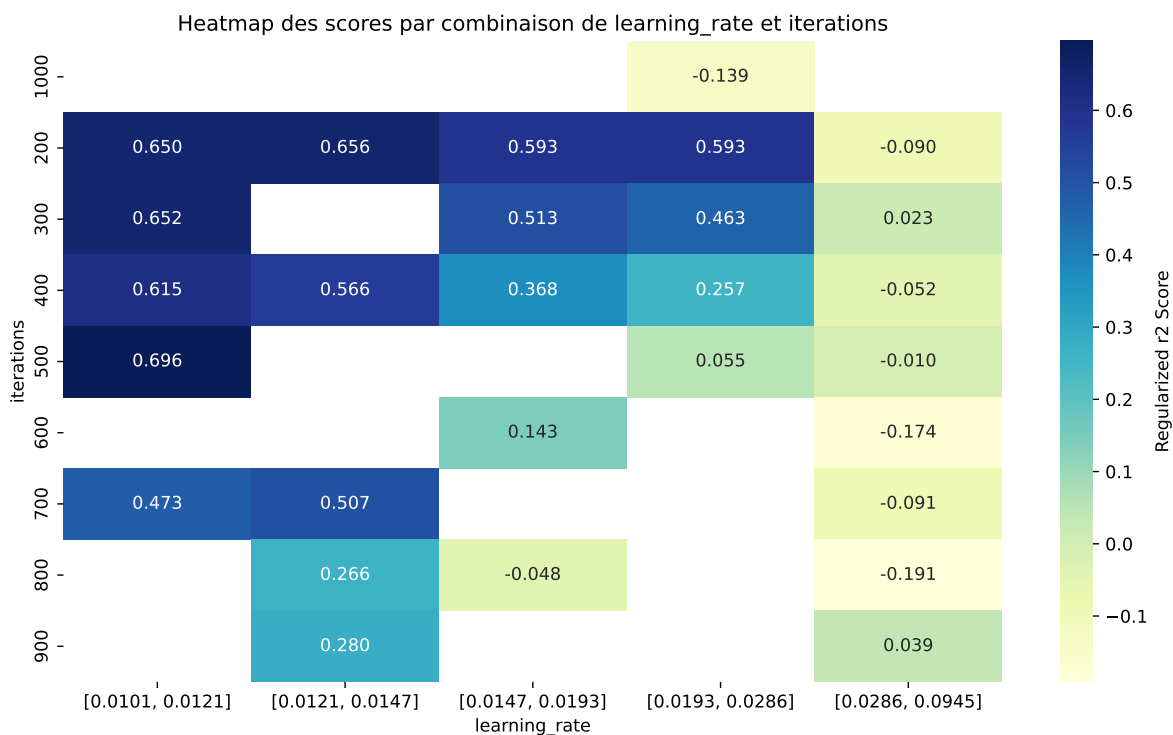
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'l2_leaf_reg', 'diffusion_temperature']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7528	0.7177	0.681	0.0562	0.7177	0.7027	-0.015	0.9792	1.049
2	0.7364	0.7128	0.6635	0.0571	0.7128	0.6843	-0.0285	0.96	1.0332
3	0.7348	0.7077	0.6664	0.063	0.7077	0.6839	-0.0238	0.9664	1.0383
4	0.7288	0.7044	0.6595	0.0584	0.7044	0.6807	-0.0237	0.9663	1.0346
5	0.7363	0.7029	0.6606	0.0604	0.7029	0.6853	-0.0176	0.975	1.0475
6	0.7261	0.699	0.661	0.0616	0.699	0.6857	-0.0132	0.9811	1.0389
7	0.7224	0.6982	0.6624	0.0589	0.6982	0.6924	-0.0058	0.9917	1.0347
8	0.7234	0.6977	0.6634	0.0605	0.6977	0.69	-0.0077	0.9889	1.0368
9	0.7076	0.6939	0.6528	0.0592	0.6939	0.6797	-0.0142	0.9795	1.0198
10	0.6994	0.6898	0.6414	0.0622	0.6898	0.6678	-0.022	0.9681	1.0139
---	---	---	---	---	---	---	---	---	---
100	0.9992	0.7699	0.7907	0.0683	-0.3769	0.8014	0.0315	1.041	1.2979

Hyperparameters:

Rank 1: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.011889981915968508, 'reg__l2_leaf_reg': 31.78366529354839, 'reg__bagging_temperature': 0.603363449514041, 'reg__random_strength': 2.819254981117896, 'reg__subsample': 0.7149226367615429, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5027190592569692, 'reg__border_count': 40, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4404.851421446738}

Rank 2: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011372065646723462, 'reg__l2_leaf_reg': 25.17819600942486, 'reg__bagging_temperature': 0.5745182146348502, 'reg__random_strength': 2.3619609866870865, 'reg__subsample': 0.7133247117323069, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5754286729178292, 'reg__border_count': 49, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4797.387267088872}

Rank 3: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.012592166243700013, 'reg__l2_leaf_reg': 26.96134589062165, 'reg__bagging_temperature': 0.4792209117872202, 'reg__random_strength': 2.9423077708185925, 'reg__subsample': 0.6999181068524056, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5347948654064446, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6072.596664203245}

Rank 4: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.012365542074574746, 'reg__l2_leaf_reg': 26.88404317405875, 'reg__bagging_temperature': 0.4495978206326785, 'reg__random_strength': 2.808326399758934, 'reg__subsample': 0.6696058785236532, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5101301018463628, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5759.680279903354}

Rank 5: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.010802331953394131, 'reg__l2_leaf_reg': 35.04411693607151, 'reg__bagging_temperature': 0.9377500628825819, 'reg__random_strength': 3.247593734987966, 'reg__subsample': 0.6396315725791537, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5741296836000477, 'reg__border_count': 44, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 9494.754314721984}

Rank 6: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.01298587209364199, 'reg__l2_leaf_reg': 26.889651102407445, 'reg__bagging_temperature': 0.37561439581514944, 'reg__random_strength': 2.775287293284992, 'reg__subsample': 0.7177666224672257, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5003693711207583, 'reg__border_count': 52, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4329.975453822939}

Rank 7: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.012438351873969765, 'reg__l2_leaf_reg': 27.35907164093984, 'reg__bagging_temperature': 0.09607402432636003, 'reg__random_strength': 2.8807726349374034, 'reg__subsample': 0.663573083803011,

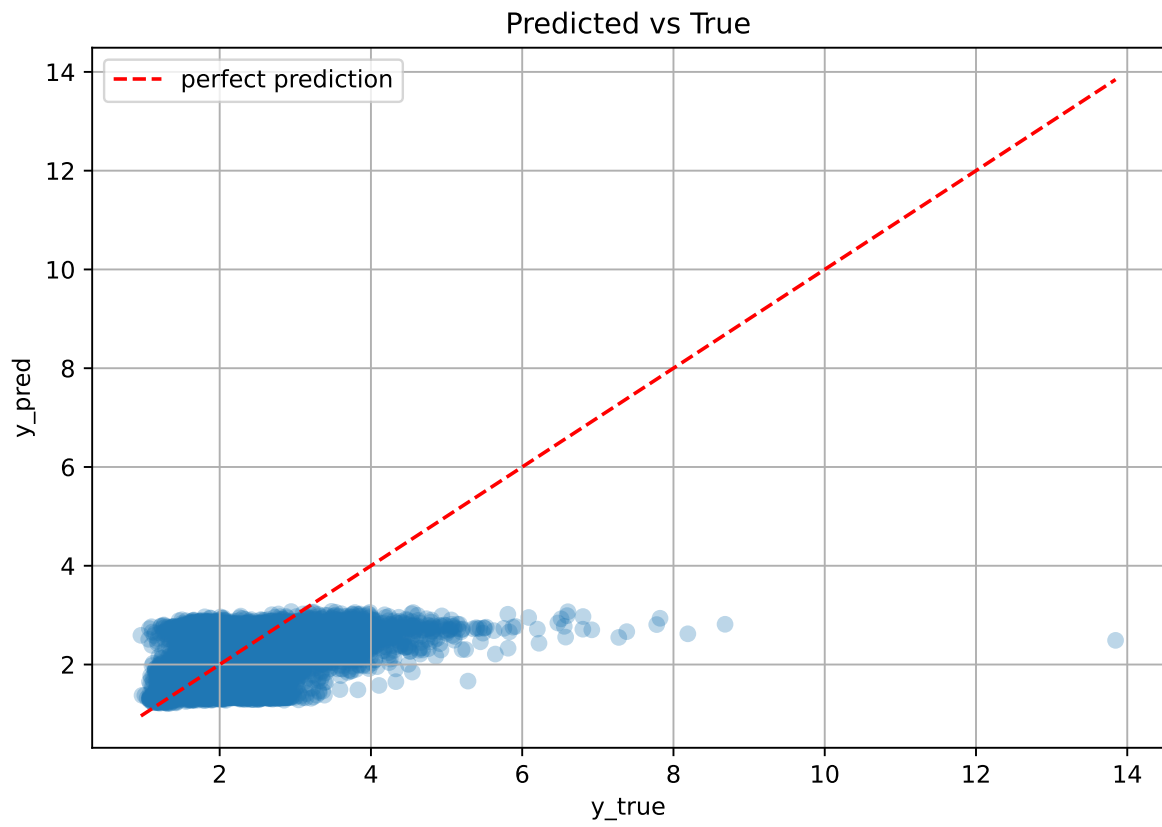
'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5804338657695565, 'reg__border_count': 38, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4536.553422321931}

Rank 8: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.01302387966460026, 'reg__l2_leaf_reg': 27.206905182966565, 'reg__bagging_temperature': 0.4742377541859817, 'reg__random_strength': 2.7872498721855026, 'reg__subsample': 0.6116791568598802, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5068600382079087, 'reg__border_count': 40, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3972.5499124146318}

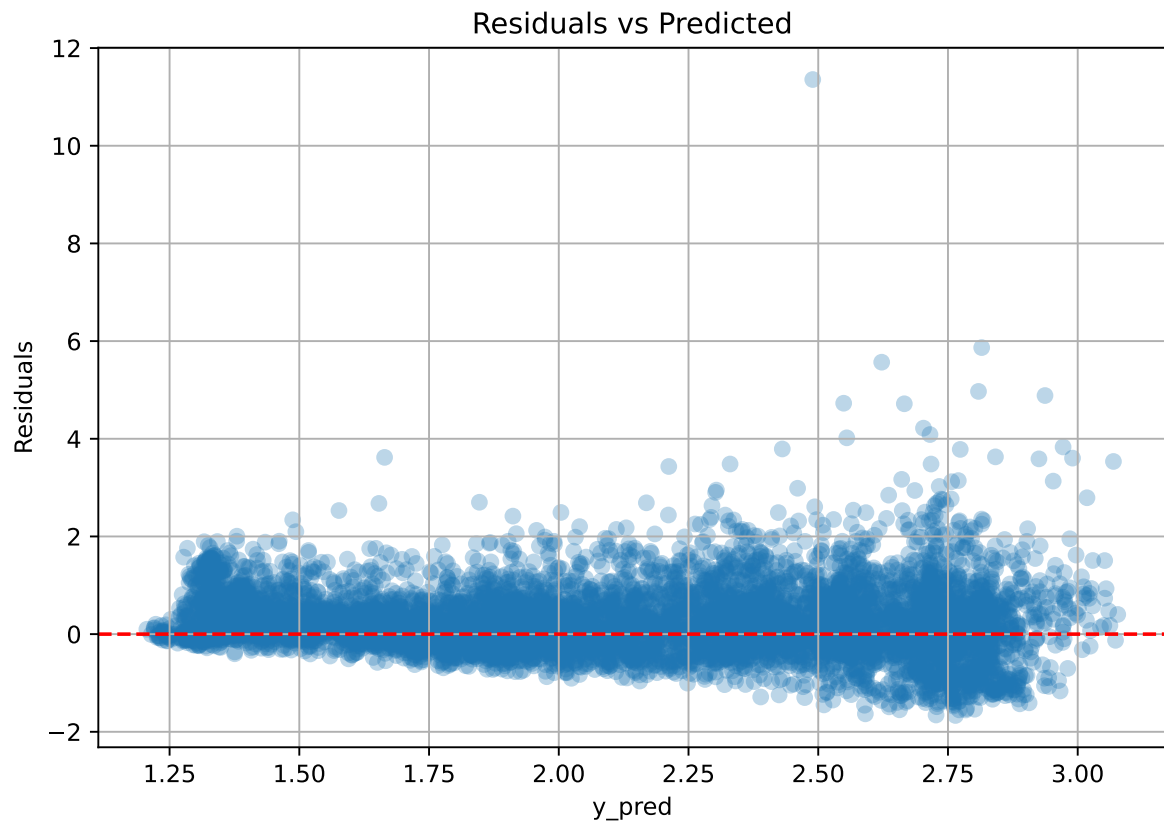
Rank 9: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.012102809751643842, 'reg__l2_leaf_reg': 23.833686719384744, 'reg__bagging_temperature': 0.0018644331638377198, 'reg__random_strength': 2.2871457751288258, 'reg__subsample': 0.7085680915648752, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5239758987779858, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4702.1201071987225}

Rank 10: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.01127184398830959, 'reg__l2_leaf_reg': 23.290607611021088, 'reg__bagging_temperature': 0.233951521148555, 'reg__random_strength': 2.430379993533753, 'reg__subsample': 0.7222899213448323, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5820369746254013, 'reg__border_count': 33, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5168.032538951926}

2.5.18 Scatter



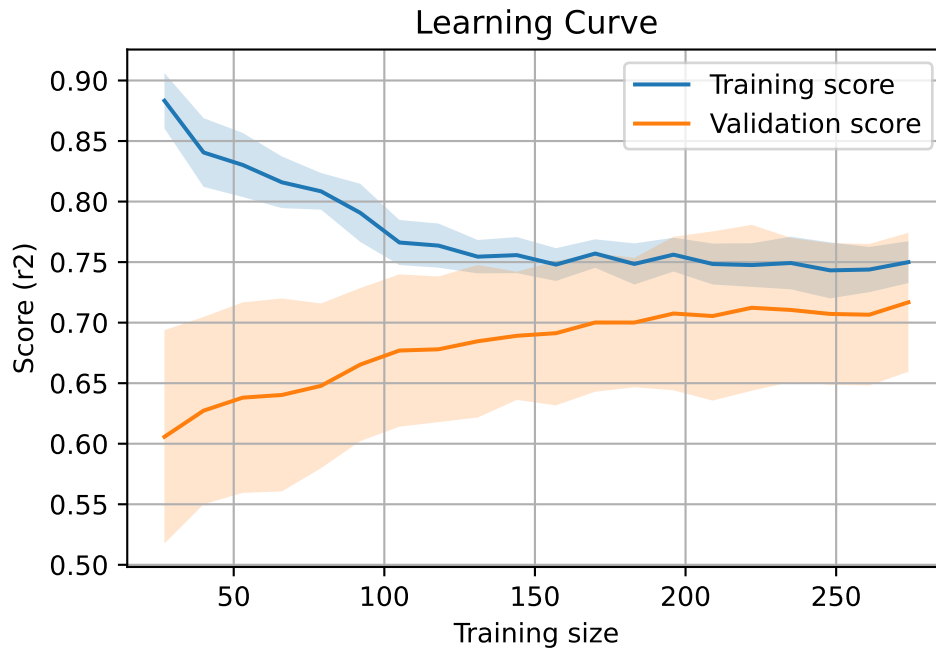
2.5.19 Scatter residuals



2.5.20 Learning curve — Group: 6_3

`Len(group)` : 561, `Len(training)` : 393,

`Len(training_real)` : 276, `Len(test_of_training)` : 117, `Len(test)` : 168

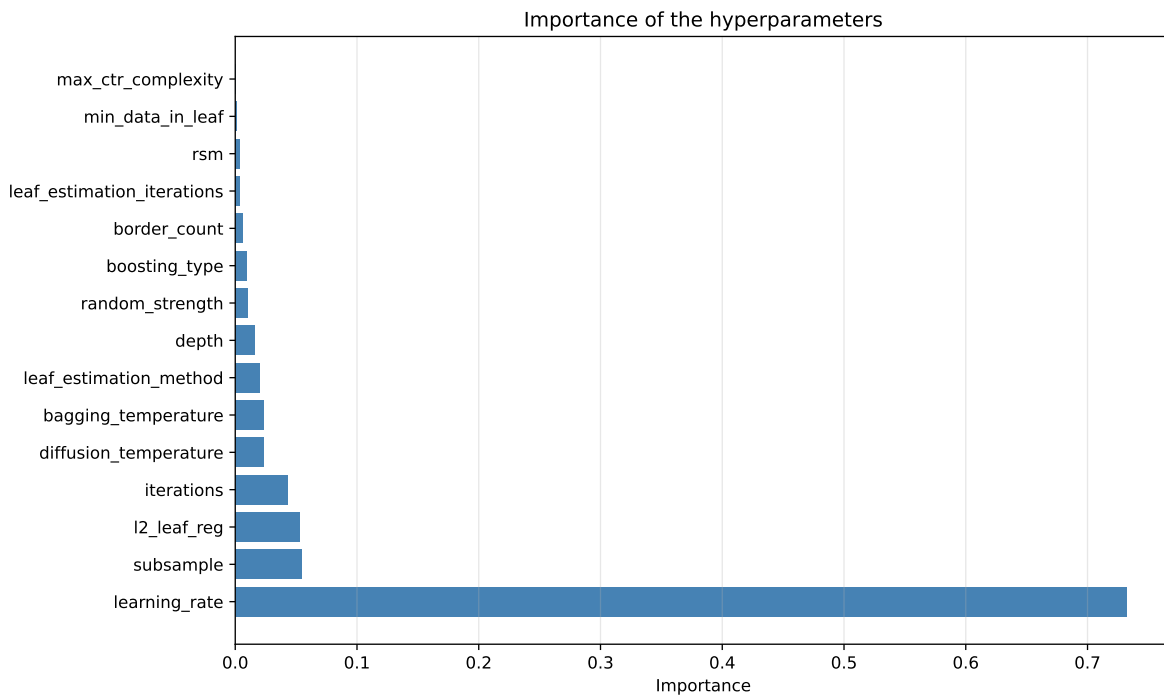
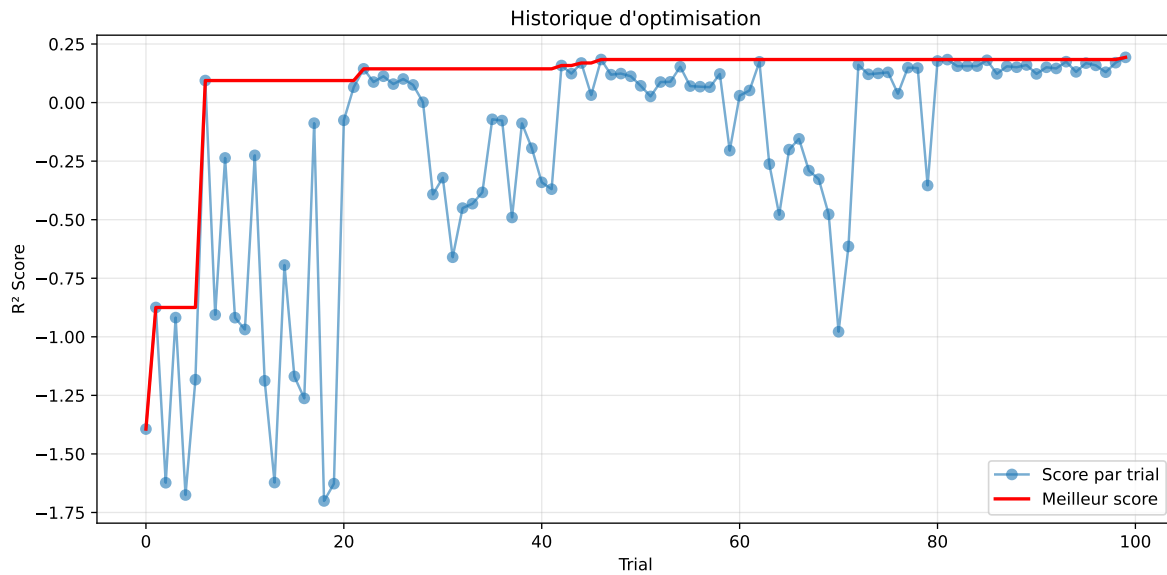


2.5.21 Gridsearch — Group: 6_4

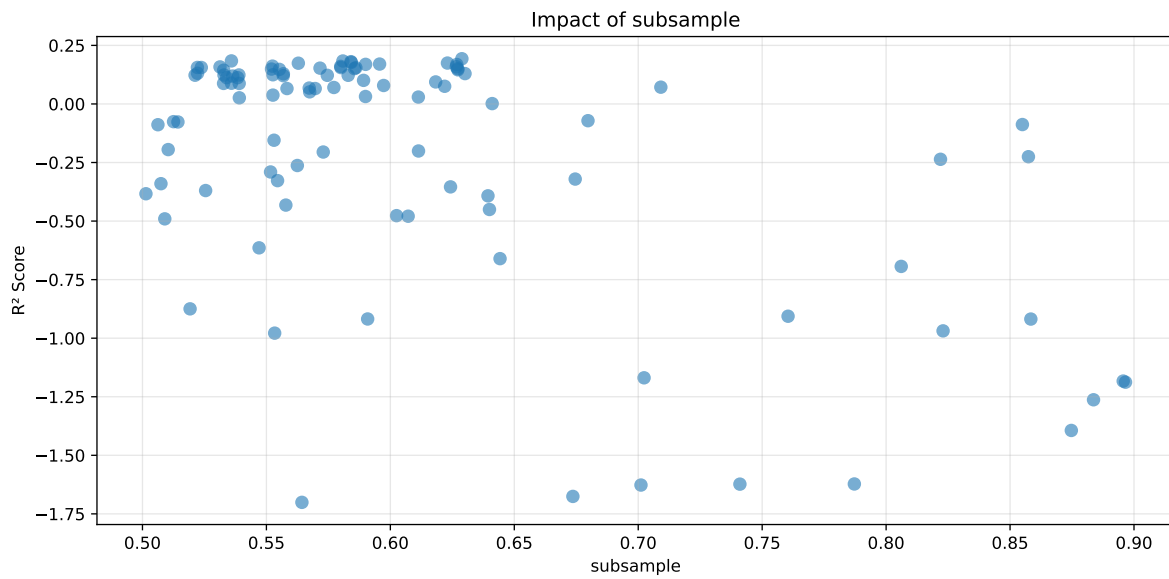
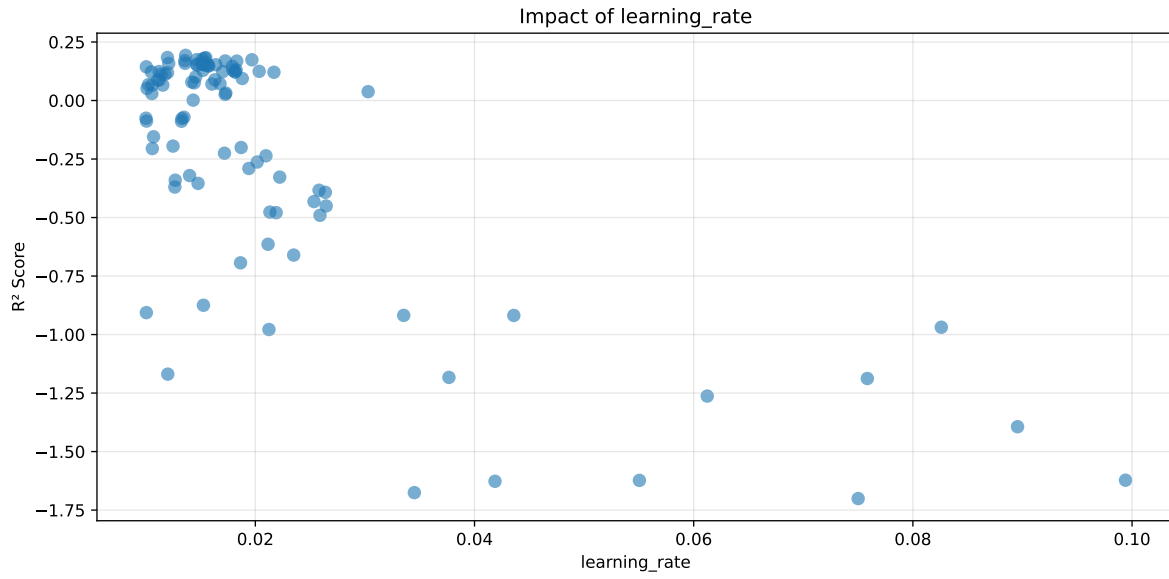
Len(group) : 792, Len(training) : 555, Len(test) : 237

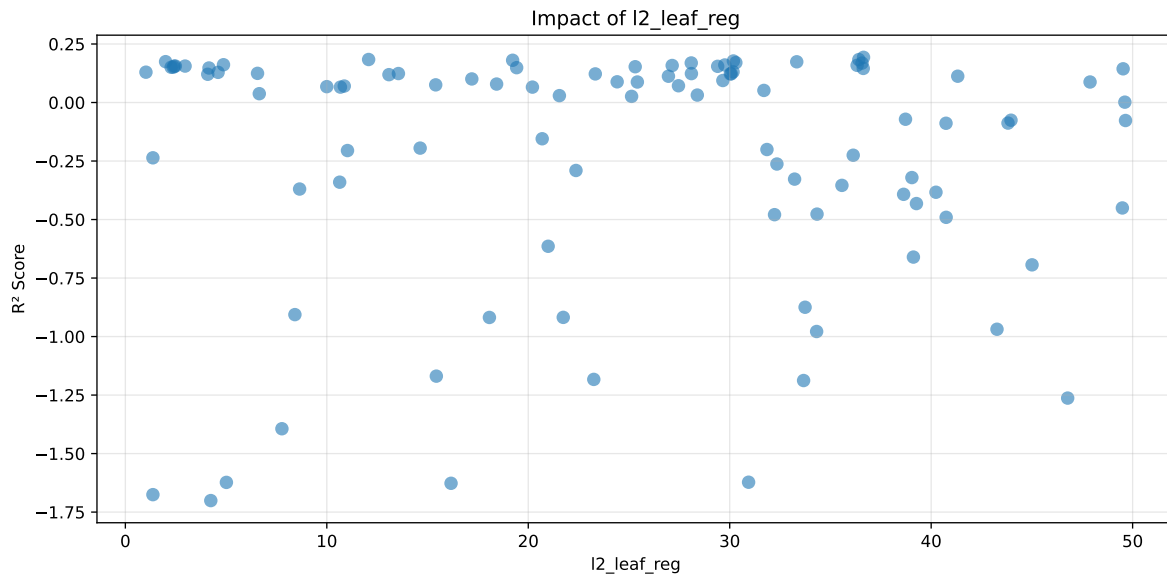
Distribution y_train vs y_test: y_train: mean=2.486, std=0.782, min=1.253, max=7.383
y_test: mean=2.410, std=0.685, min=1.398, max=5.144

===== TOP Importance FEATURES =====
feature importance r_THI 36.658253 r_umidity 25.610962 r_temperature 16.640608 r_NH3
12.569073 r_CO2 8.521105



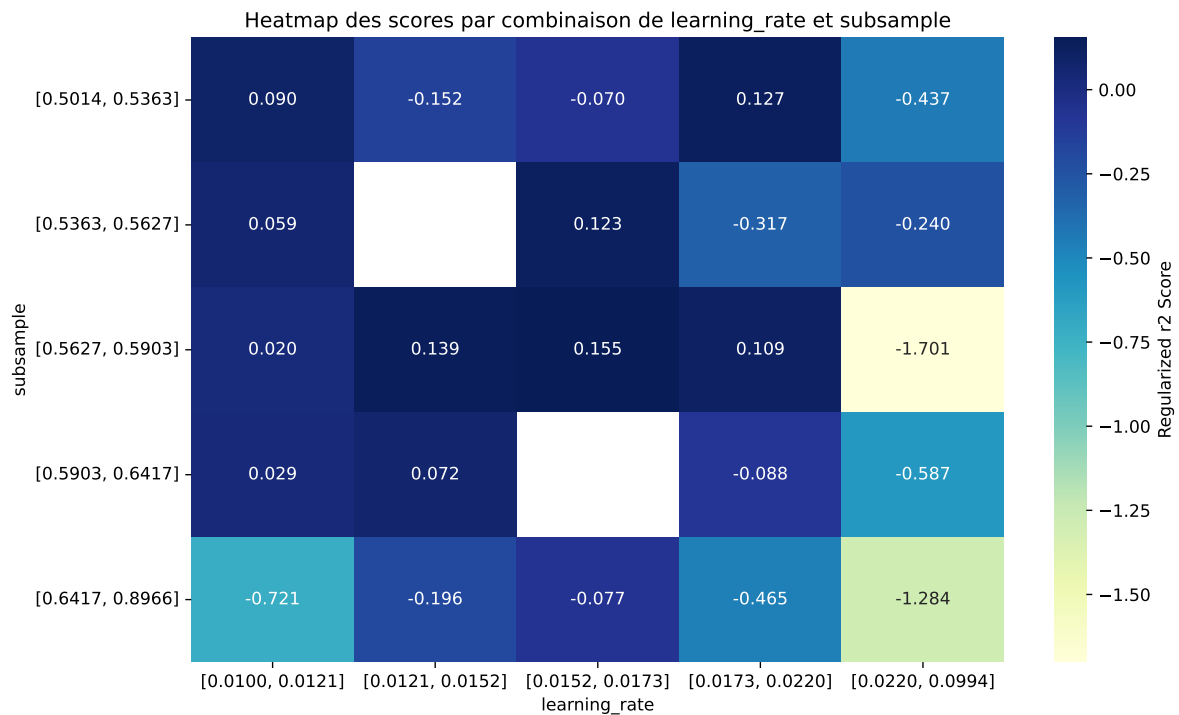
Paramètres avec >5% d'importance : ['learning_rate', 'subsample', 'l2_leaf_reg']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4154	0.3784	0.4308	0.028	0.193	0.4547	0.0764	1.2018	1.098
2	0.4764	0.4276	0.4916	0.0221	0.1838	0.5094	0.0818	1.1914	1.114
3	0.4698	0.422	0.4849	0.0215	0.1834	0.49	0.0679	1.1609	1.1131
4	0.4359	0.3933	0.4417	0.0162	0.1805	0.4695	0.0761	1.1936	1.1082
5	0.4654	0.4174	0.4764	0.0216	0.1774	0.4835	0.0661	1.1585	1.115
6	0.4778	0.4273	0.4916	0.0231	0.1746	0.5111	0.0838	1.1962	1.1183
7	0.523	0.4648	0.5336	0.024	0.174	0.5596	0.0948	1.204	1.1251
8	0.4241	0.3818	0.4344	0.0221	0.1703	0.4608	0.079	1.2069	1.1108
9	0.5338	0.473	0.5415	0.0292	0.169	0.5382	0.0652	1.1378	1.1285
10	0.3964	0.3584	0.408	0.0194	0.1684	0.4522	0.0938	1.2617	1.106
---	---	---	---	---	---	---	---	---	---
100	0.9465	0.5053	0.5974	0.0378	-1.7008	0.6783	0.173	1.3424	1.8732

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.013635272553812949, 'reg__l2_leaf_reg': 36.64234572993806, 'reg__bagging_temperature': 0.6118435926405911, 'reg__random_strength': 3.297333350991634, 'reg__subsample': 0.6288608701759486, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.814890601750946, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6735.93020168503}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011979870369437855, 'reg__l2_leaf_reg': 12.07532023427039, 'reg__bagging_temperature': 1.0420595392970085, 'reg__random_strength': 4.243288564725191, 'reg__subsample': 0.5358914587098058, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7533294800387779, 'reg__border_count': 59, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 2967.625994265676}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.015475182720350937, 'reg__l2_leaf_reg': 36.40928632700703, 'reg__bagging_temperature': 0.583771488220366, 'reg__random_strength': 3.2324015975780576, 'reg__subsample': 0.5808531002275298, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7894277147710959, 'reg__border_count': 93, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 2269.5467796441553}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.015391169686587117, 'reg__l2_leaf_reg': 19.22461171838946, 'reg__bagging_temperature': 0.5653042444412395, 'reg__random_strength': 3.807026079420817, 'reg__subsample': 0.5841069386820785, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':

0.8165635116520547, 'reg__border_count': 50, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 783.5520446554347}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.015182835017643746, 'reg__l2_leaf_reg': 30.182637785022845, 'reg__bagging_temperature': 0.1792136415267379, 'reg__random_strength': 3.863971989024323, 'reg__subsample': 0.5842404118866663, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7885444991105095, 'reg__border_count': 96, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6222.755347713225}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.014654960658835786, 'reg__l2_leaf_reg': 1.996677994404481, 'reg__bagging_temperature': 0.46259495520446486, 'reg__random_strength': 3.3149208122592904, 'reg__subsample': 0.6230397086694539, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8185944043807446, 'reg__border_count': 100, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6634.505166080895}

Rank 7: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01968399648841981, 'reg__l2_leaf_reg': 33.32712646043669, 'reg__bagging_temperature': 1.4003766946666651, 'reg__random_strength': 4.712634866040517, 'reg__subsample': 0.5628825313725708, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8160044033893279, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4899.719216834033}

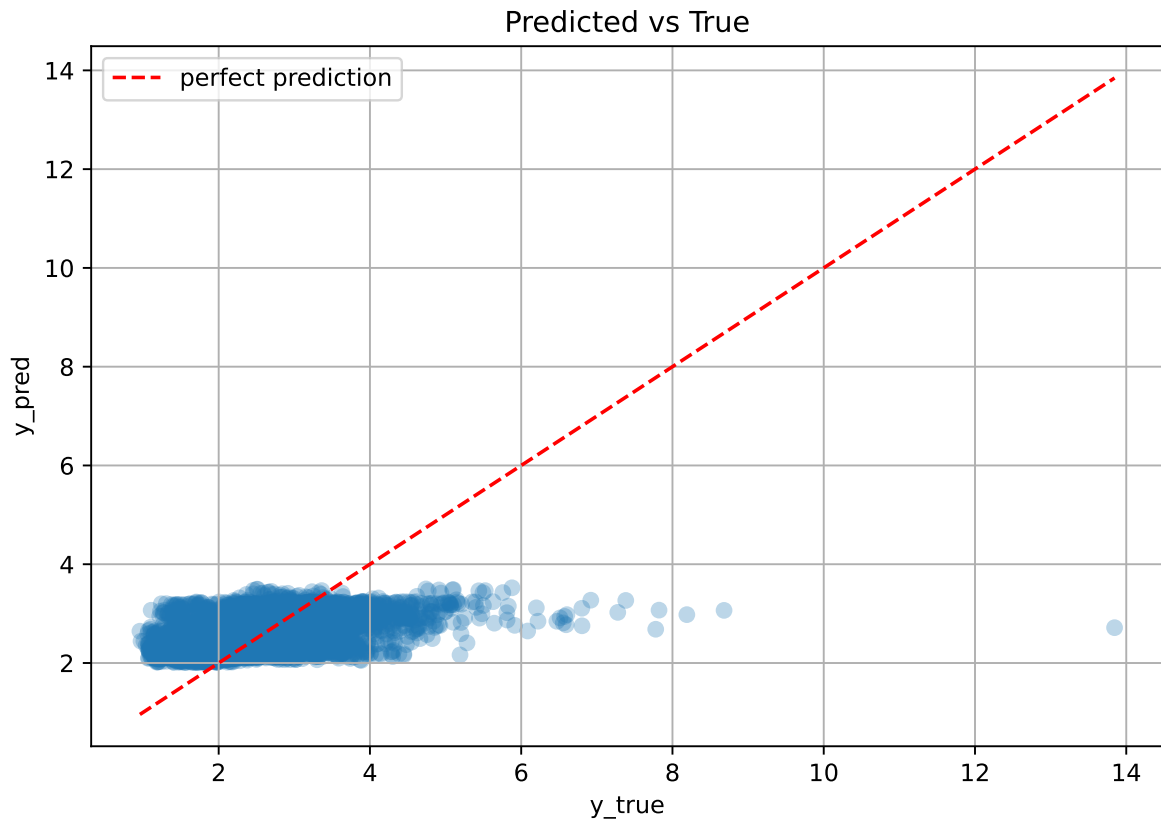
Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.013564016950666913, 'reg__l2_leaf_reg': 30.303878764746713, 'reg__bagging_temperature': 0.5708388375091266, 'reg__random_strength': 3.314289639209913, 'reg__subsample': 0.5956635953679545, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7905986405115214, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 2210.1486628925913}

Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.017249845900458086, 'reg__l2_leaf_reg': 28.090799655018635, 'reg__bagging_temperature': 0.42738354438713455, 'reg__random_strength': 4.22703044649823, 'reg__subsample': 0.5900165967898157, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.828373367313353, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1974.982492505941}

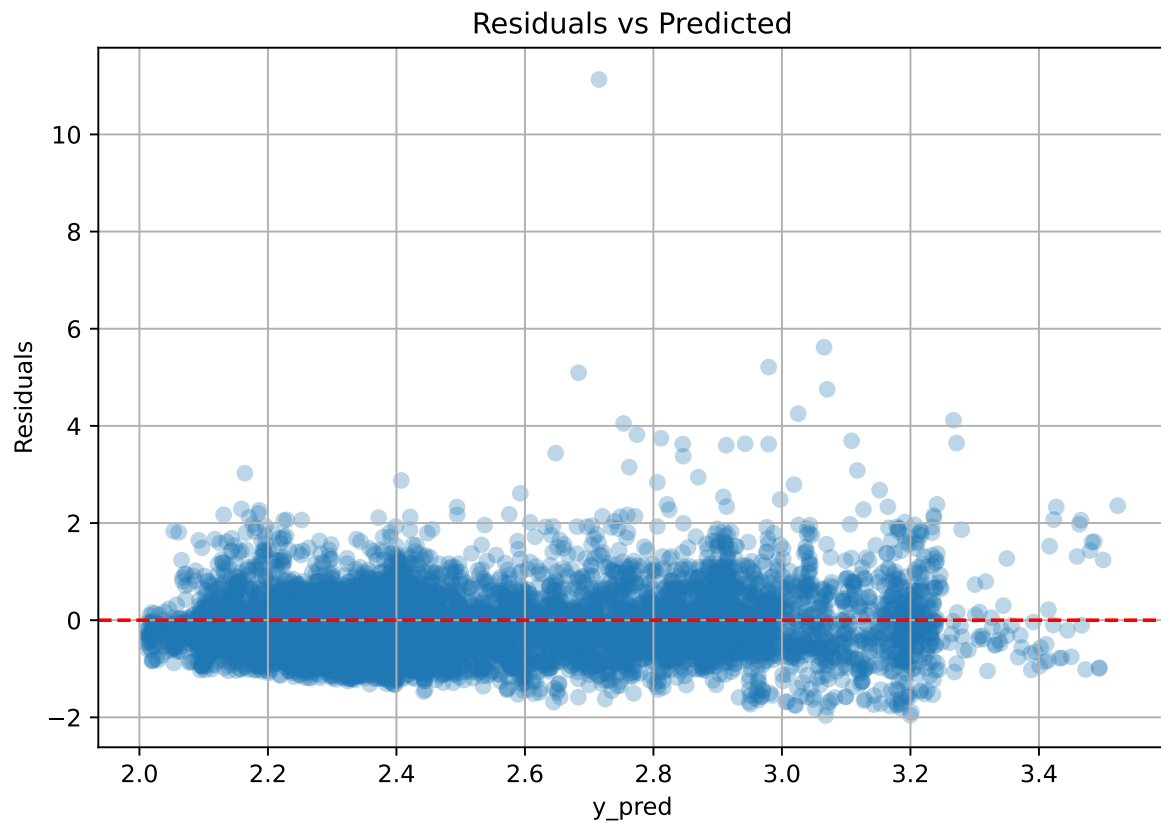
Rank 10: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01830202187022547, 'reg__l2_leaf_reg': 36.56776006612284, 'reg__bagging_temperature': 0.5353253600357182,


```
'reg__random_strength': 3.1608357065309467, 'reg__subsample': 0.626806683204238,  
'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':  
0.7904949408753189, 'reg__border_count': 100, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-  
ture': 2254.095192901545}
```

2.5.22 Scatter



2.5.23 Scatter residuals



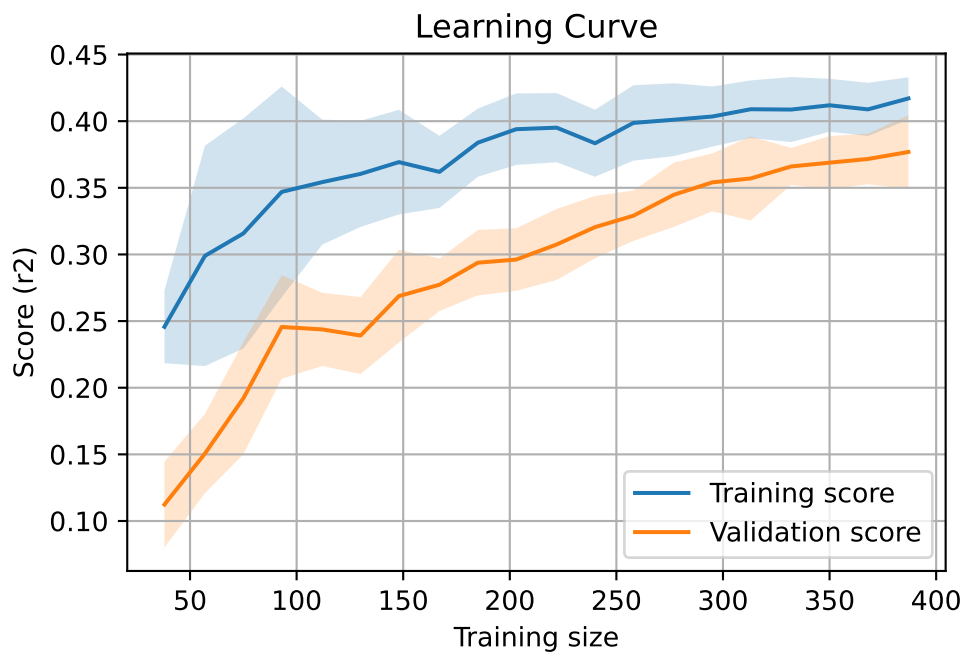
2.5.24 Learning curve — Group: 6_4

`Len(group)` : 792, `Len(training)` : 555,

`Len(training_real)` : 389, `Len(test_of_training)` : 166, `Len(test)` : 237

C:\Users\titou\anaconda3\Lib\site-packages\joblib\externals\loky\process_executor.py:782: Use

A worker stopped while some jobs were given to the executor. This can be caused by a too sho



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	1_3	557	238	795	0.563	0.551	0.044	0.498	0.5	-0.05	0.908	1.022	nan
Linear	2_11	1909	817	2726	0.463	0.463	0.017	0.394	0.394	-0.069	0.852	1.0	nan
Linear	3_4	398	170	568	0.307	0.248	0.036	0.187	0.203	-0.045	0.818	1.241	nan
Linear	4_5	1909	817	2726	0.431	0.404	0.026	0.457	0.457	0.054	1.133	1.069	nan
Linear	6_3	393	168	561	0.625	0.677	0.067	0.616	0.623	-0.054	0.921	0.924	nan
Linear	6_4	555	237	792	0.172	0.181	0.042	0.26	0.264	0.083	1.459	0.952	nan
Linear	MEAN	5721	2447	8168	0.427	0.42	0.039	0.402	0.407	-0.014	1.015	1.035	nan
LGBM	1_3	557	238	795	0.567	0.54	0.044	0.503	0.519	-0.021	0.961	1.05	0.54
LGBM	2_11	1909	817	2726	0.705	0.672	0.015	0.57	0.58	-0.091	0.864	1.05	0.6715
LGBM	3_4	398	170	568	0.121	0.114	0.047	0.04	0.066	-0.048	0.575	1.06	0.0797
LGBM	4_5	1909	817	2726	0.392	0.373	0.022	0.423	0.439	0.066	1.177	1.052	0.2755
LGBM	6_3	393	168	561	0.703	0.689	0.056	0.646	0.702	0.013	1.019	1.02	0.6891
LGBM	6_4	555	237	792	0.304	0.286	0.074	0.333	0.338	0.051	1.179	1.062	0.1979
LGBM	MEAN	5721	2447	8168	0.465	0.446	0.043	0.419	0.441	-0.005	0.963	1.049	0.409
CatBoost	1_3	557	238	795	0.636	0.607	0.044	0.573	0.586	-0.021	0.965	1.047	0.6073
CatBoost	2_11	1909	817	2726	0.727	0.696	0.016	0.598	0.607	-0.089	0.872	1.044	0.6961
CatBoost	3_4	398	170	568	0.292	0.223	0.032	0.101	0.131	-0.092	0.586	1.311	-0.1235
CatBoost	4_5	1909	817	2726	0.429	0.406	0.02	0.467	0.458	0.052	1.129	1.057	0.2906
CatBoost	6_3	393	168	561	0.753	0.718	0.056	0.681	0.703	-0.015	0.979	1.049	0.7177
CatBoost	6_4	555	237	792	0.415	0.378	0.028	0.431	0.455	0.076	1.202	1.098	0.193
CatBoost	MEAN	5721	2447	8168	0.542	0.505	0.033	0.475	0.49	-0.015	0.956	1.101	0.397

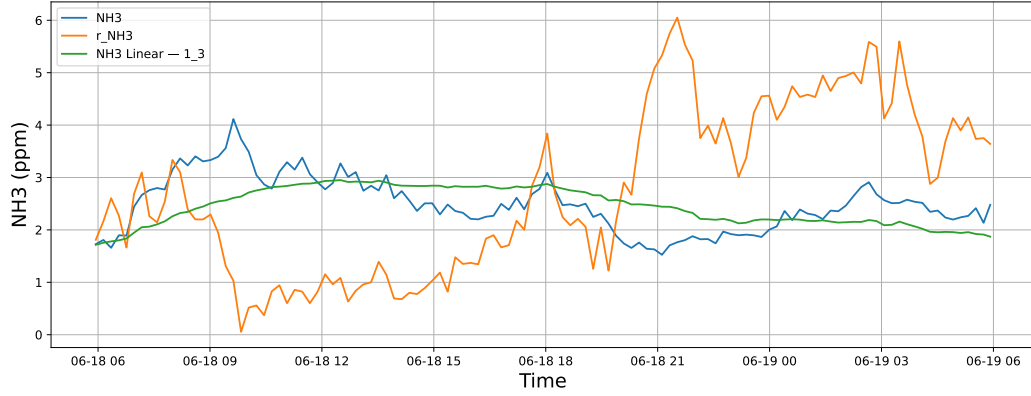
3 Plot

3.1 Standalone plots

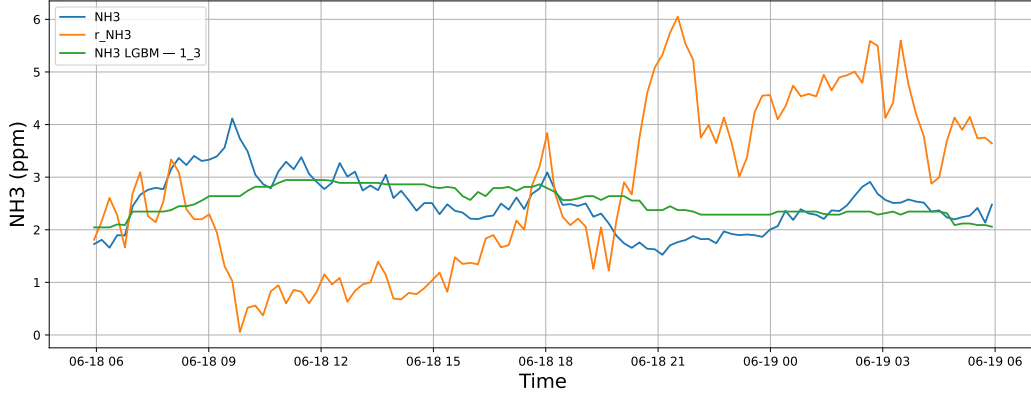
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

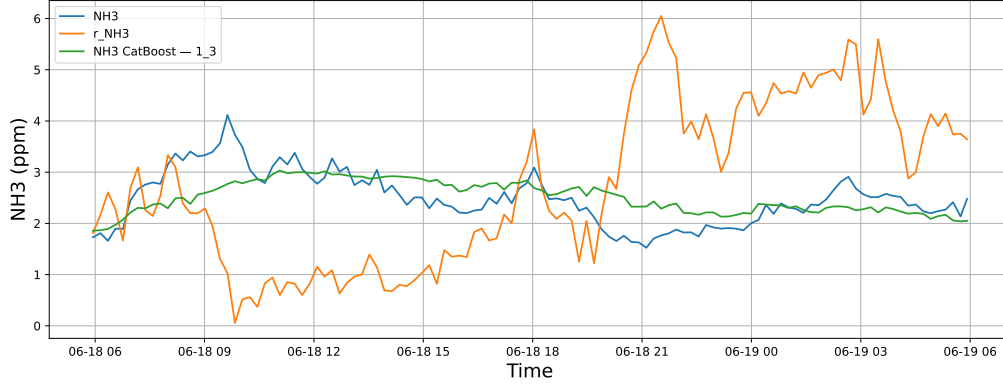
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

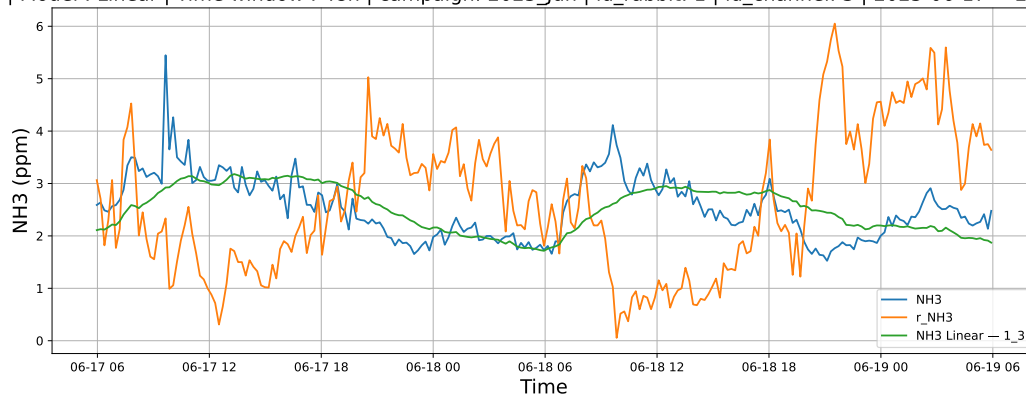


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



3.1.1.2 Time window : 48

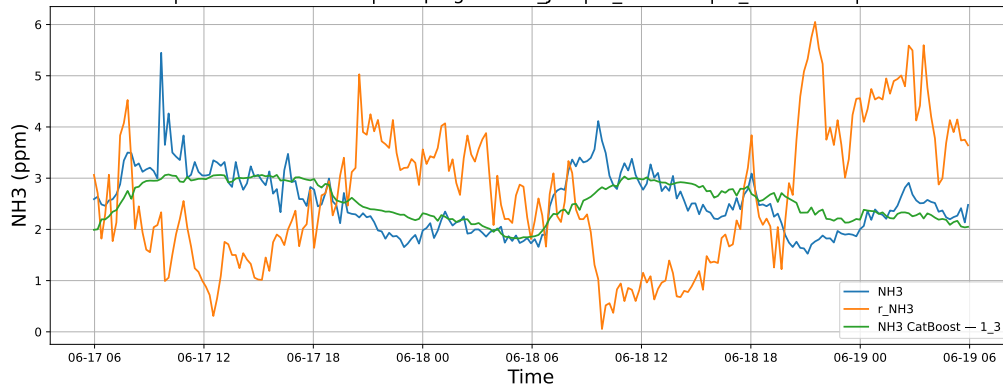
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

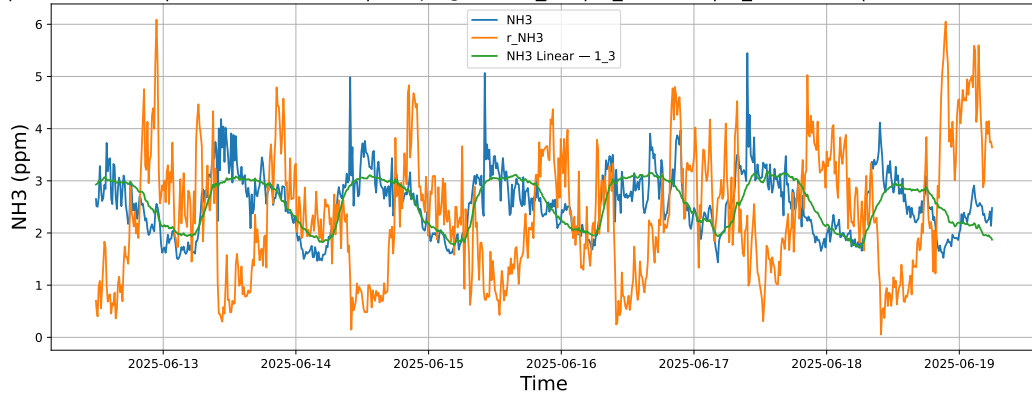


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

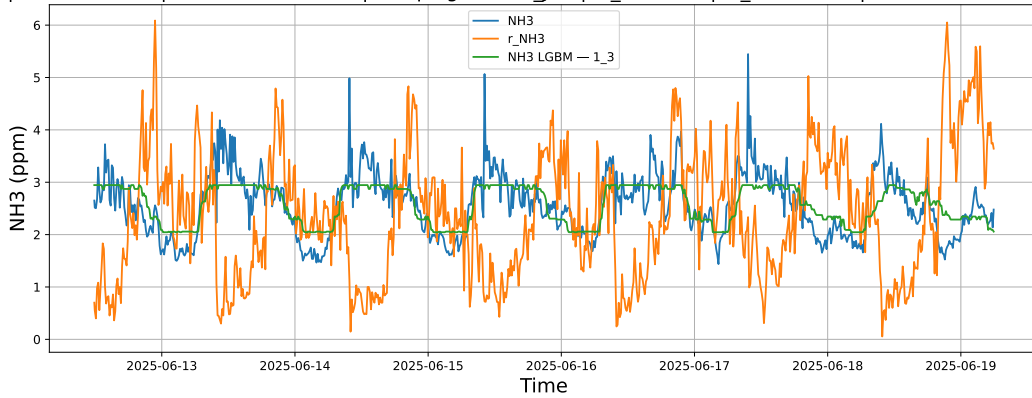


3.1.1.3 Time window : ALL

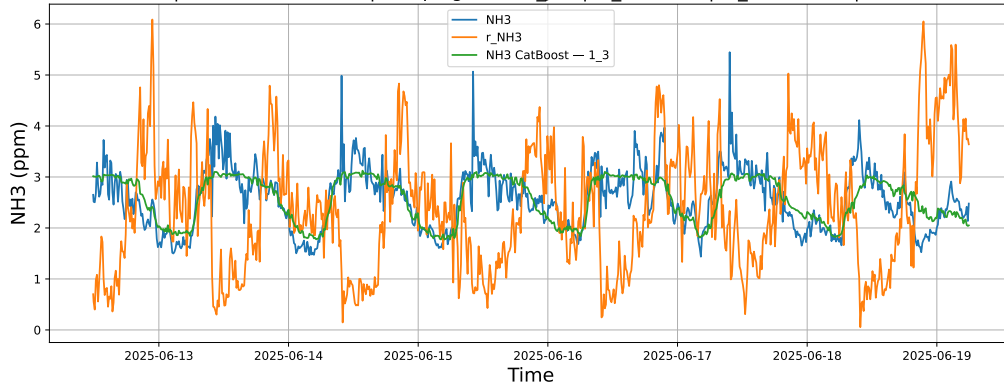
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



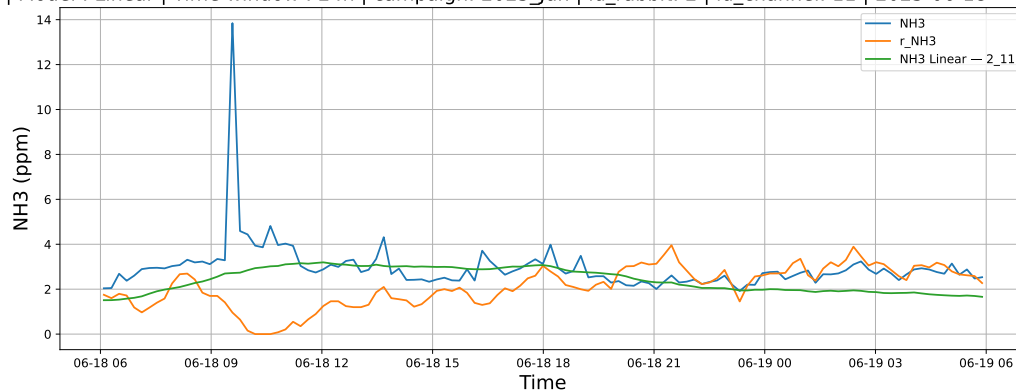
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



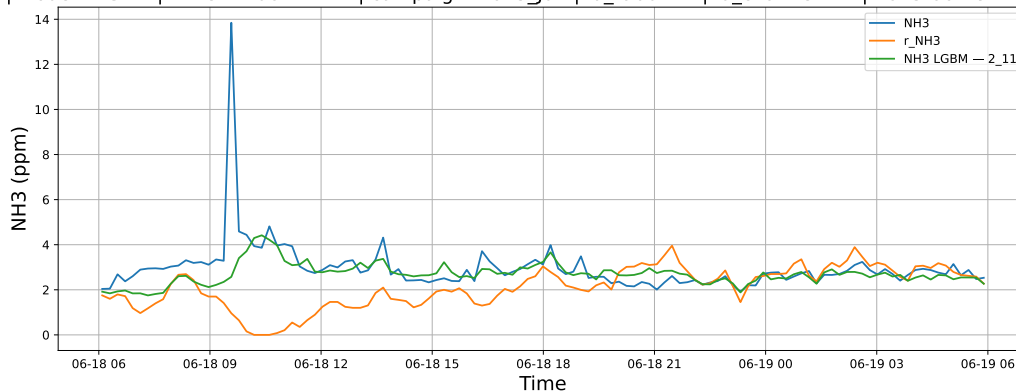
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

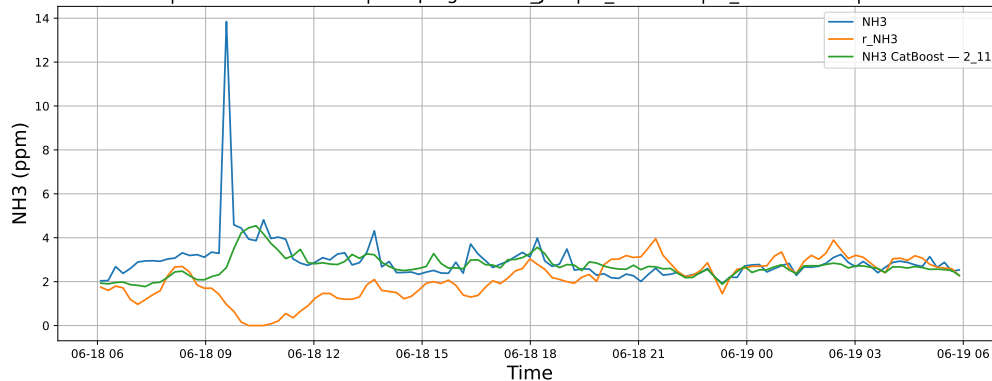
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

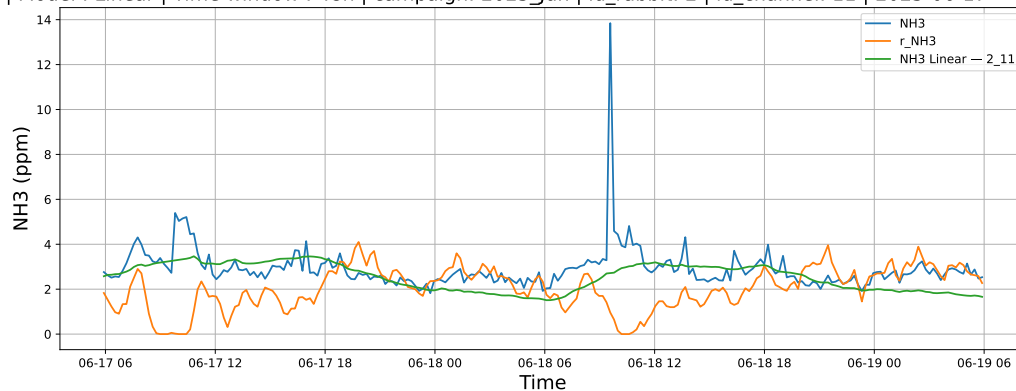


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

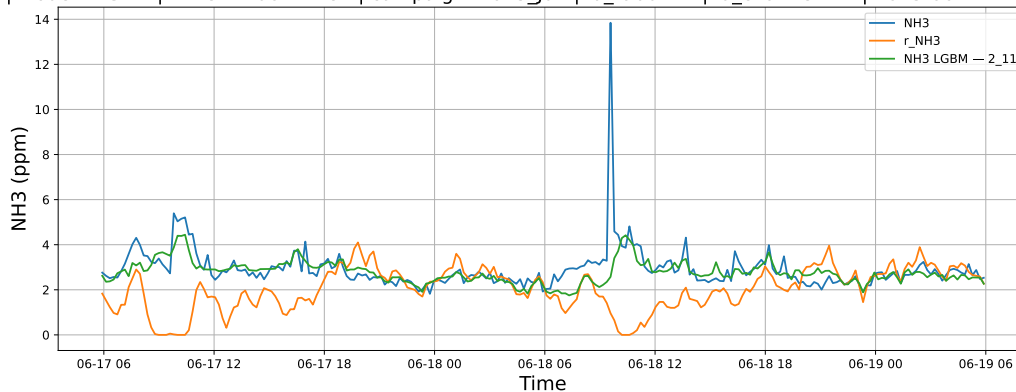


3.1.2.2 Time window : 48

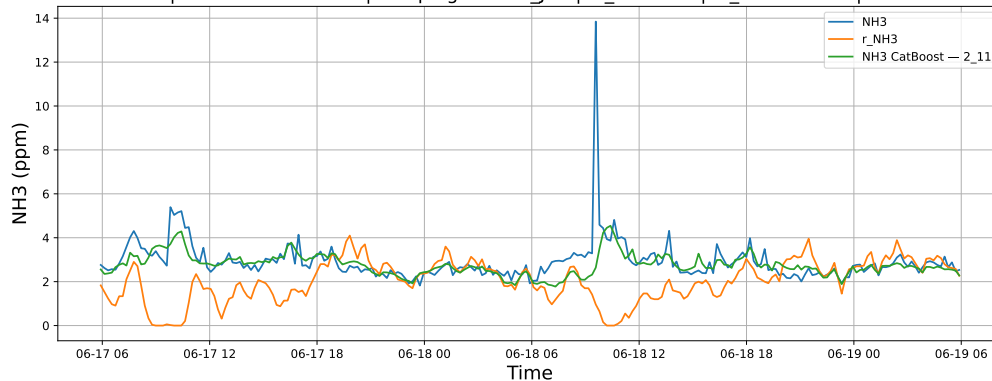
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

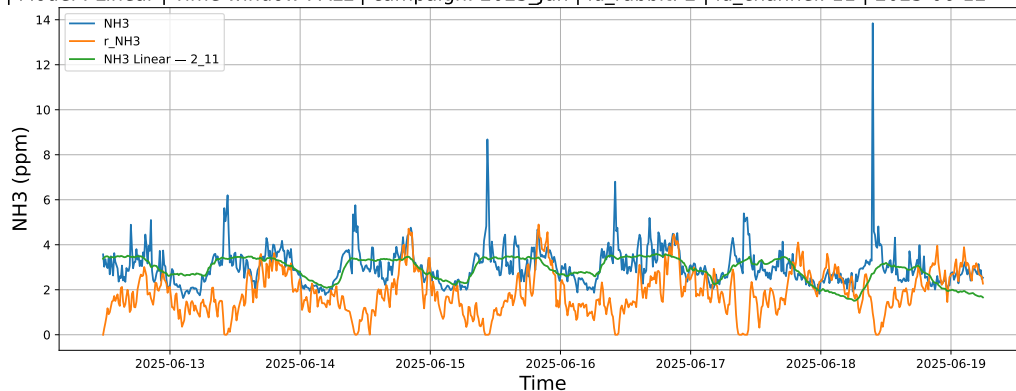


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

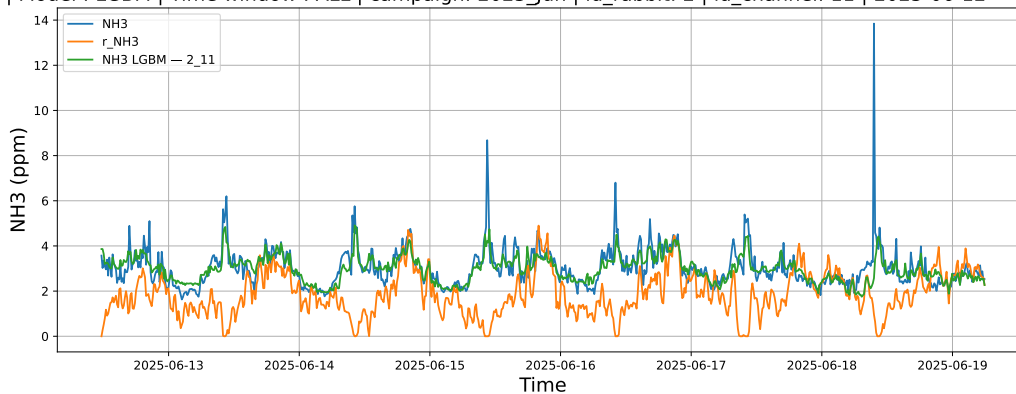


3.1.2.3 Time window : ALL

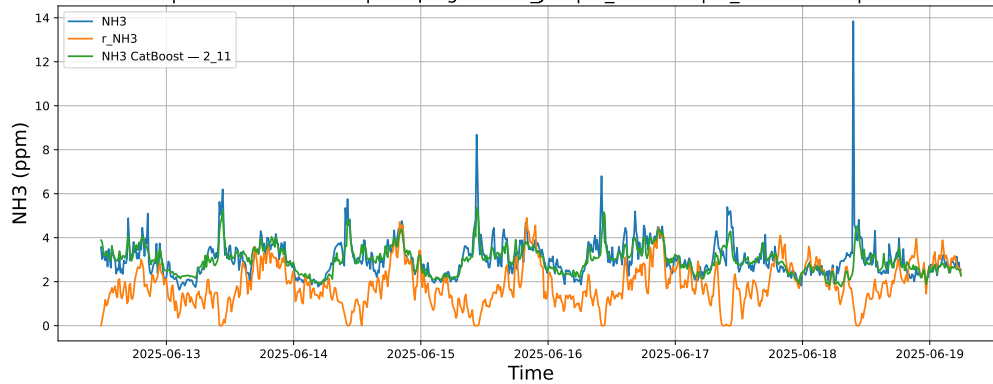
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



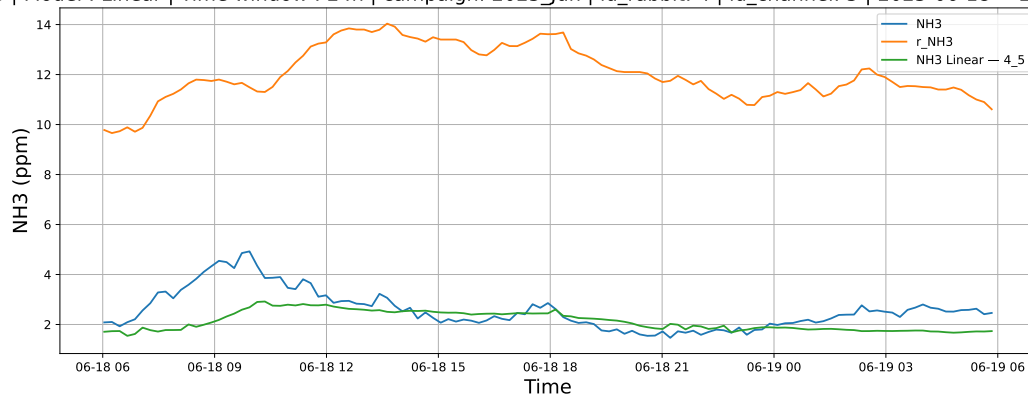
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



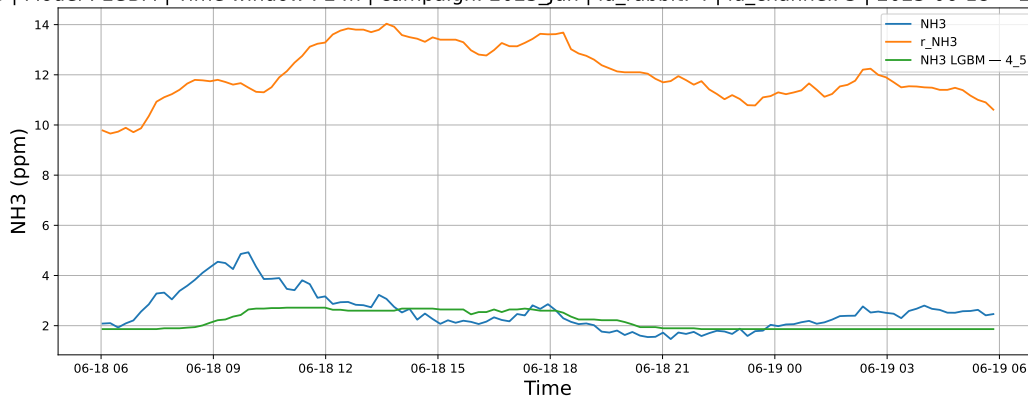
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

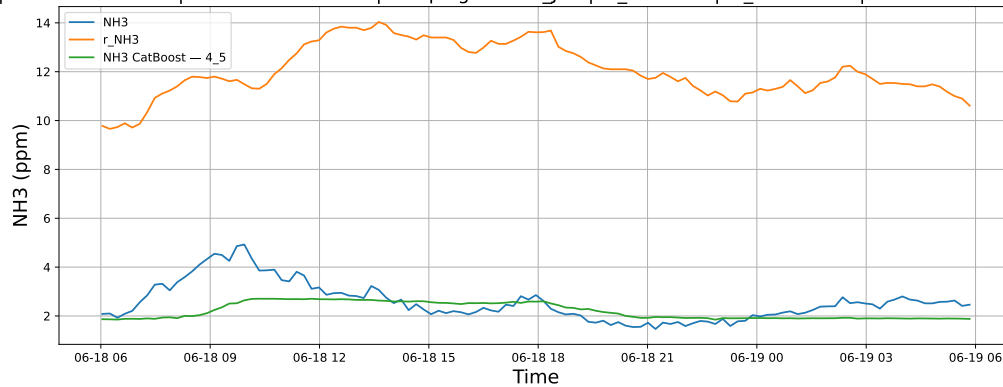
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

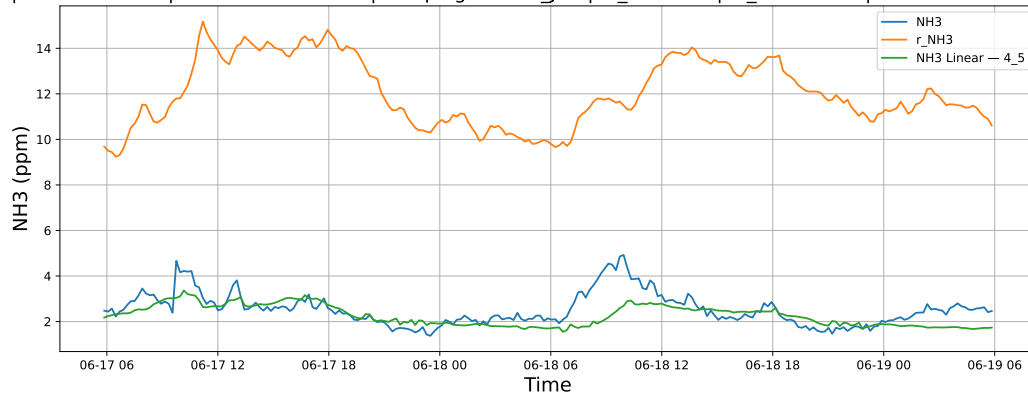


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

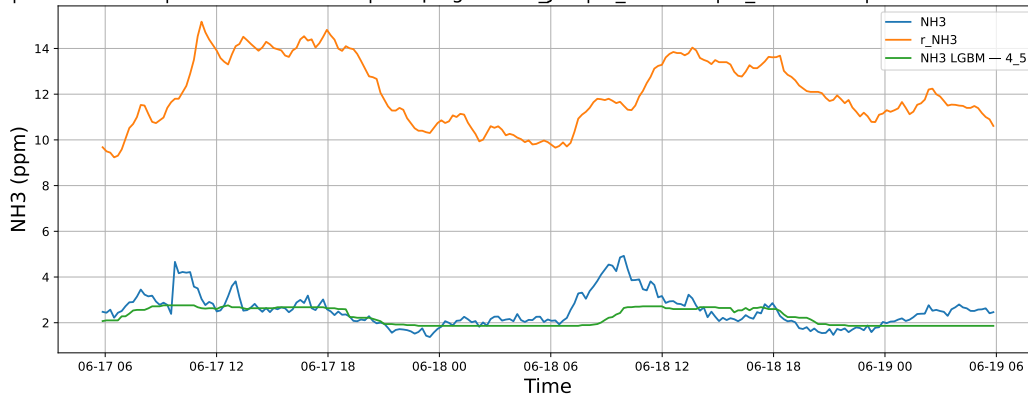


3.1.3.2 Time window : 48

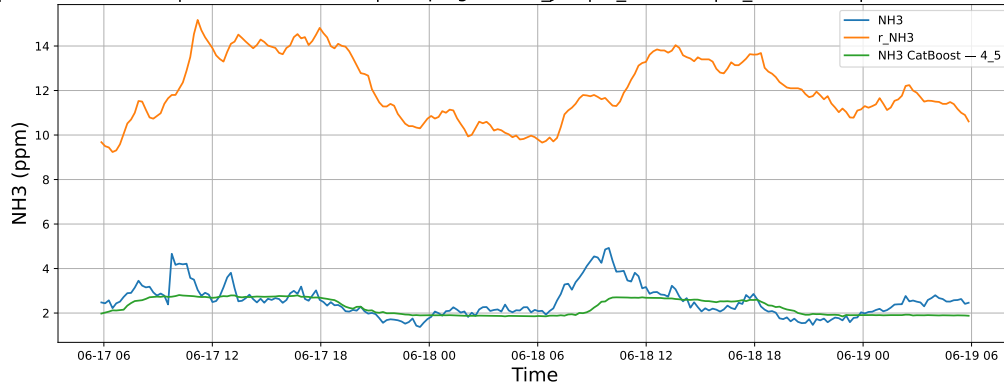
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

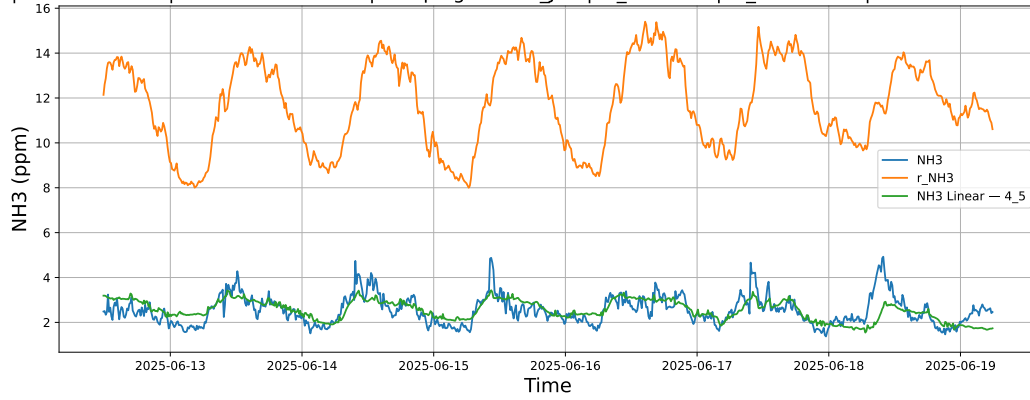


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

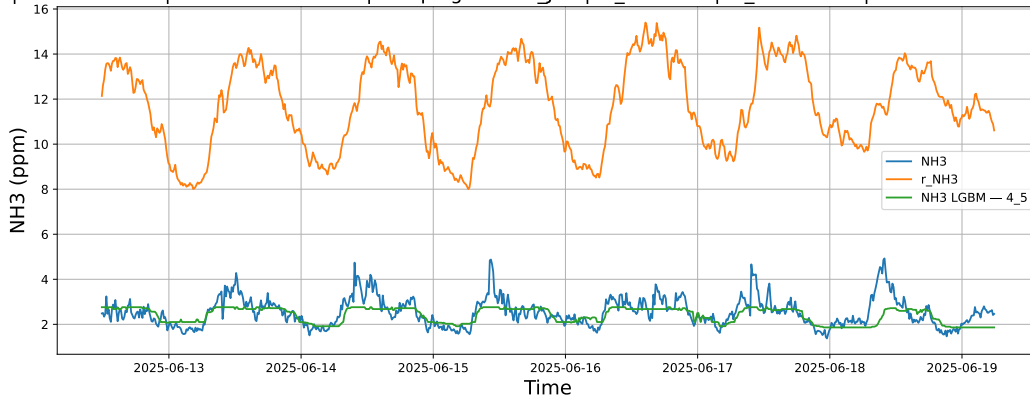


3.1.3.3 Time window : ALL

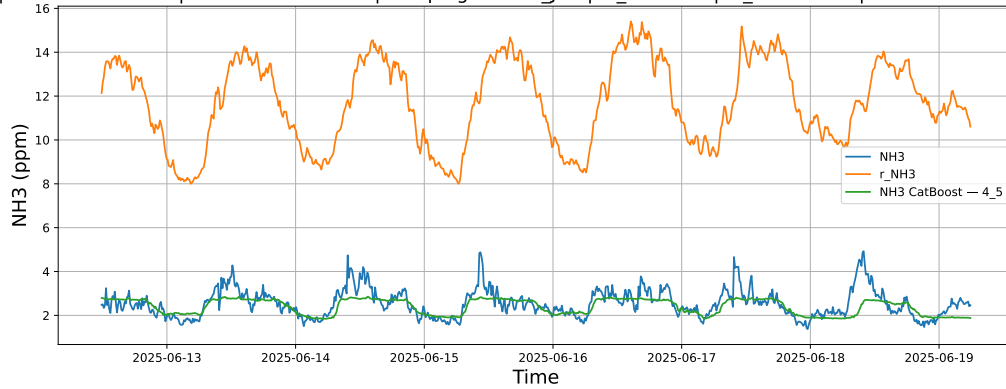
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



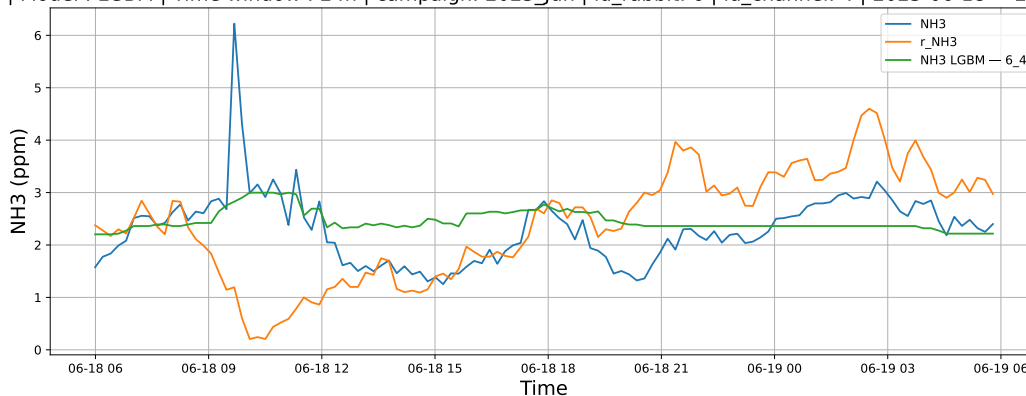
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

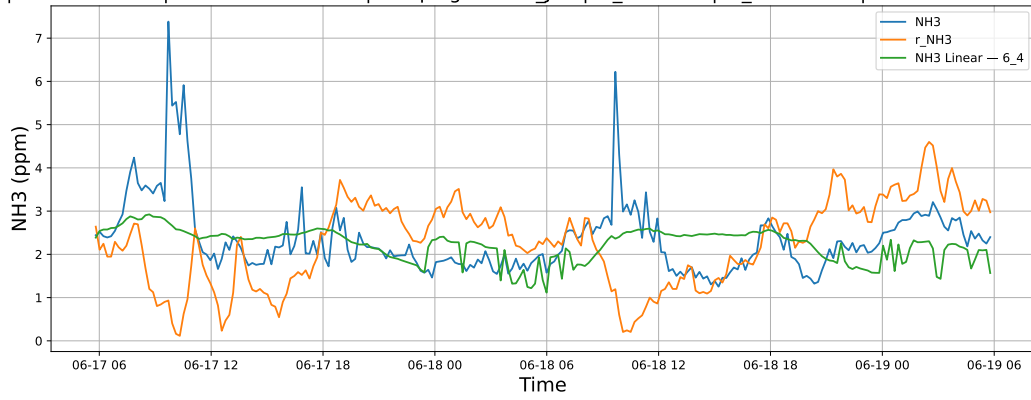


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

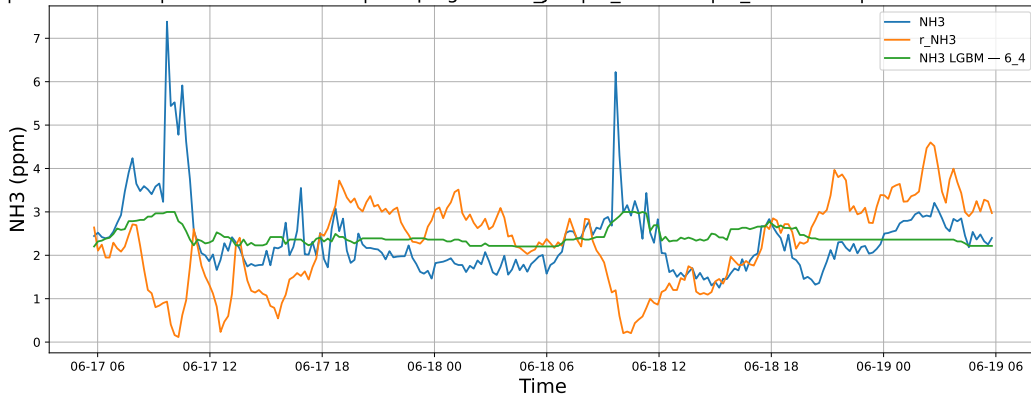


3.1.4.2 Time window : 48

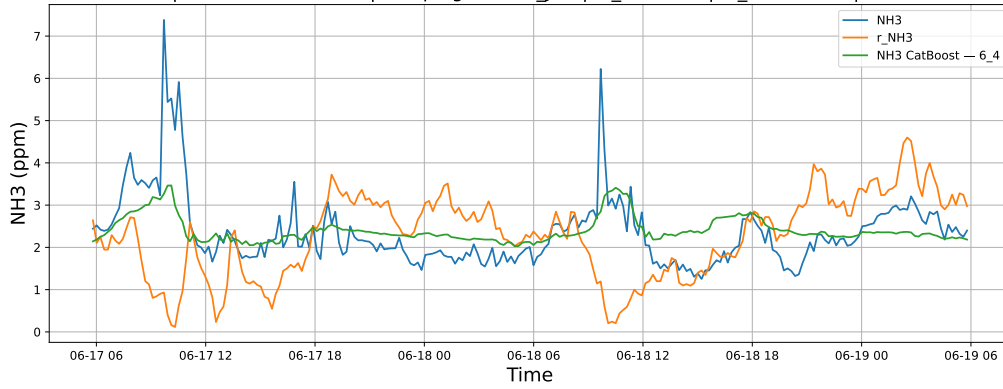
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

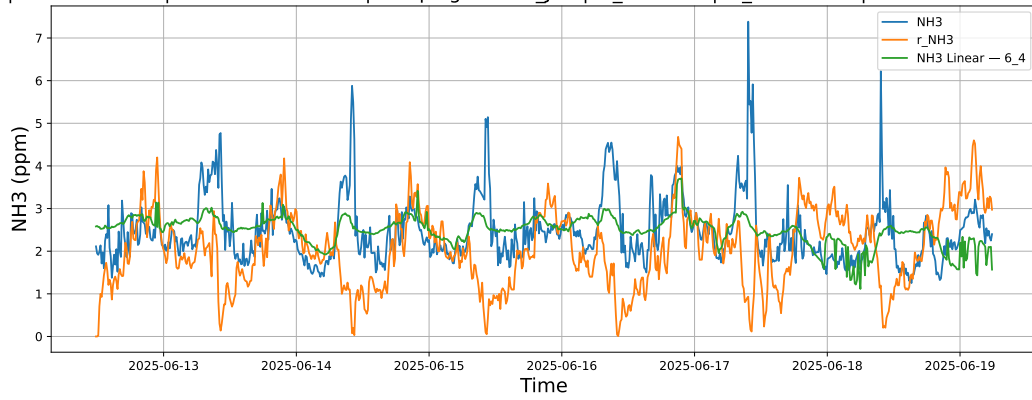


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

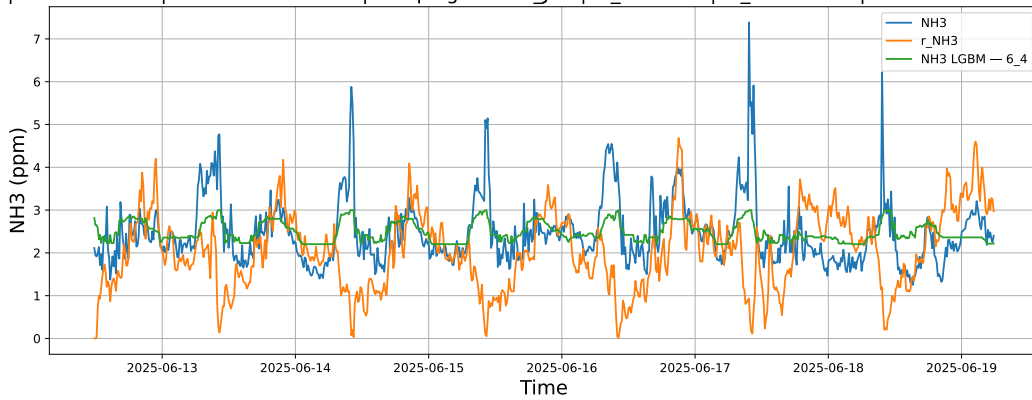


3.1.4.3 Time window : ALL

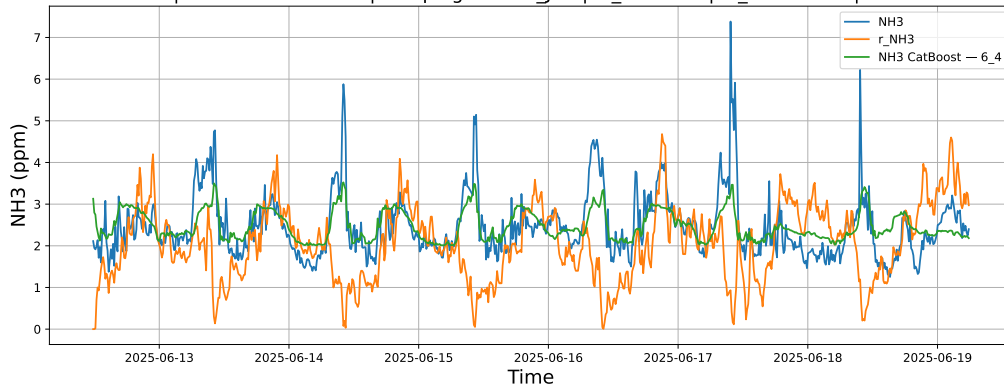
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



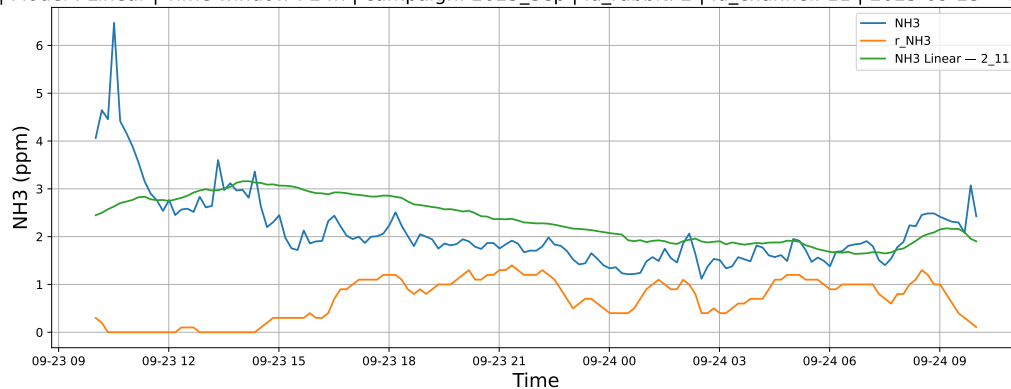
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



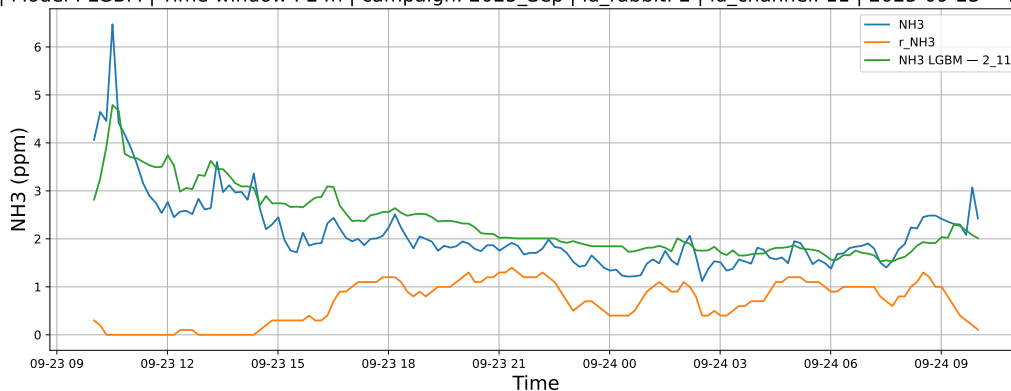
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

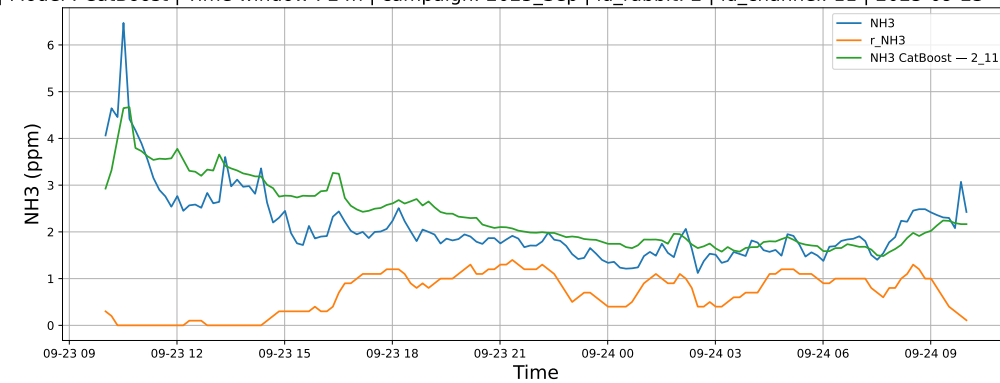
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

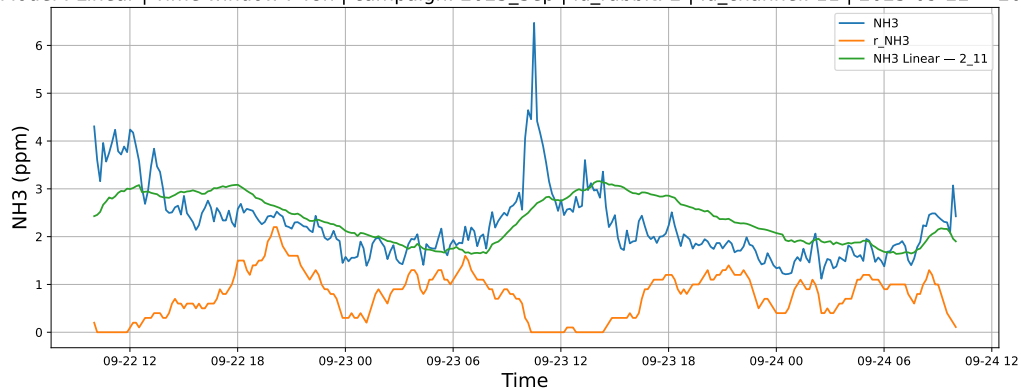


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

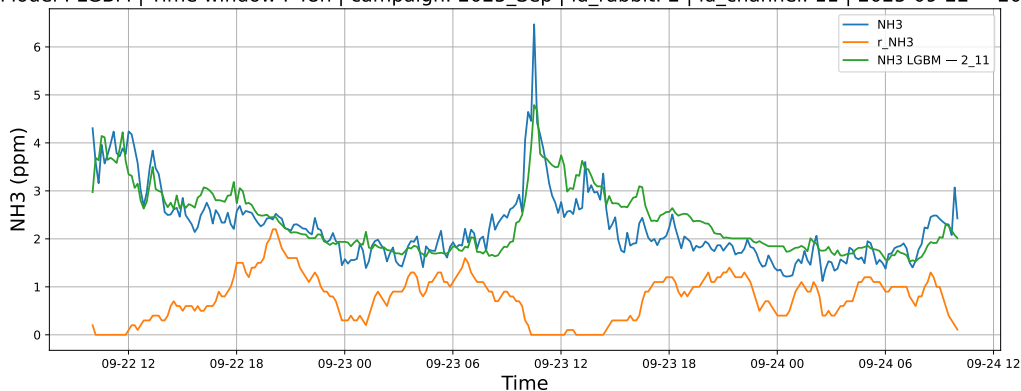


3.1.5.2 Time window : 48

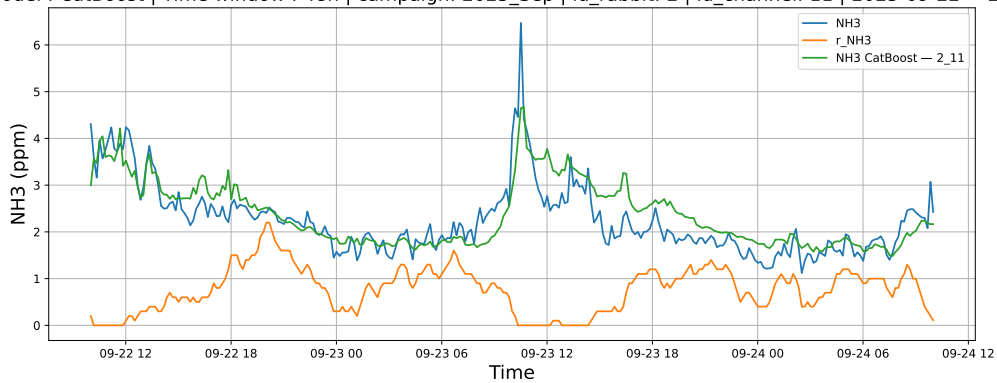
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

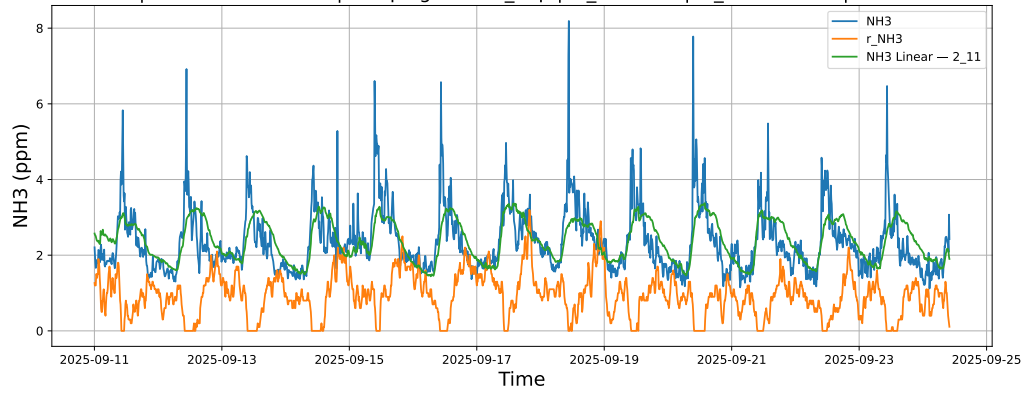


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

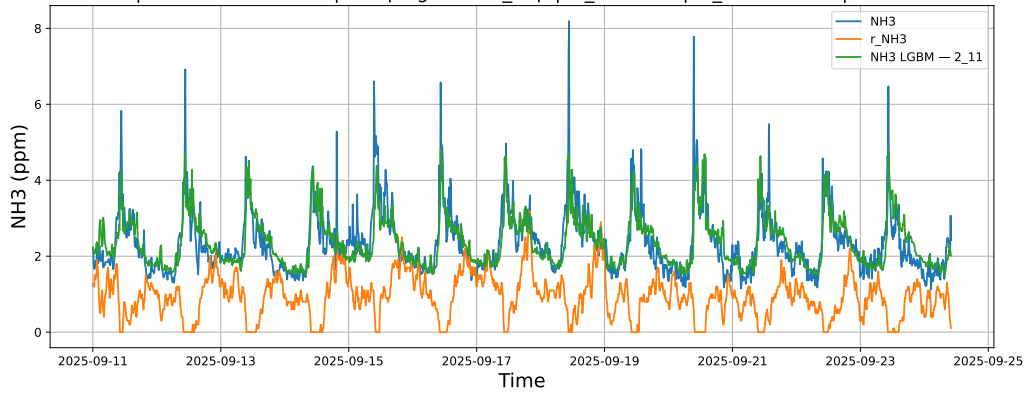


3.1.5.3 Time window : ALL

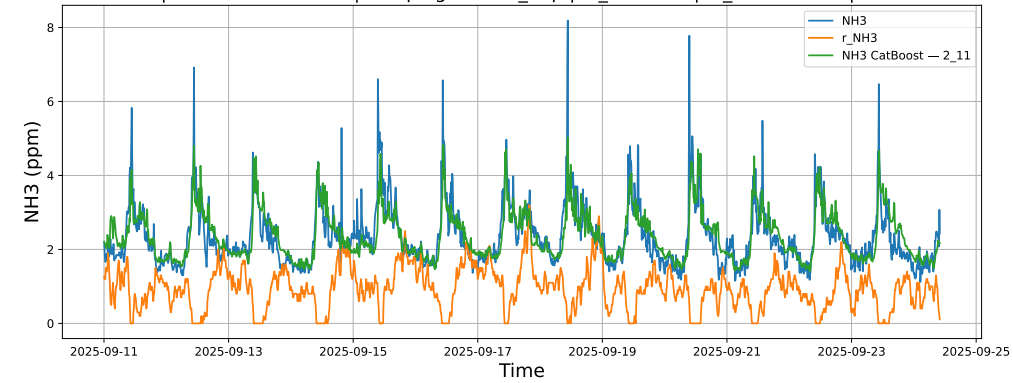
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



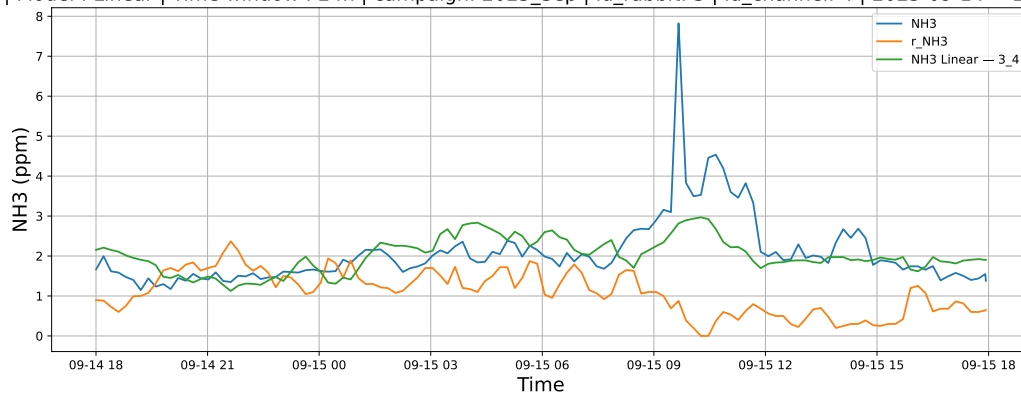
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



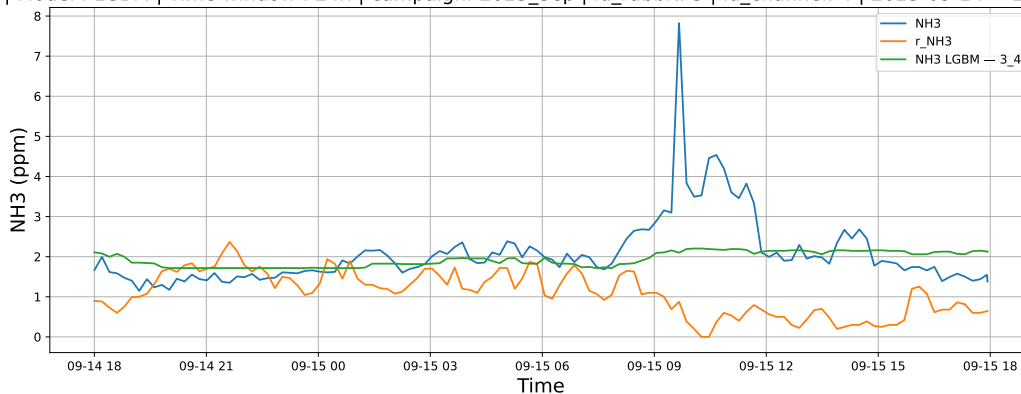
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

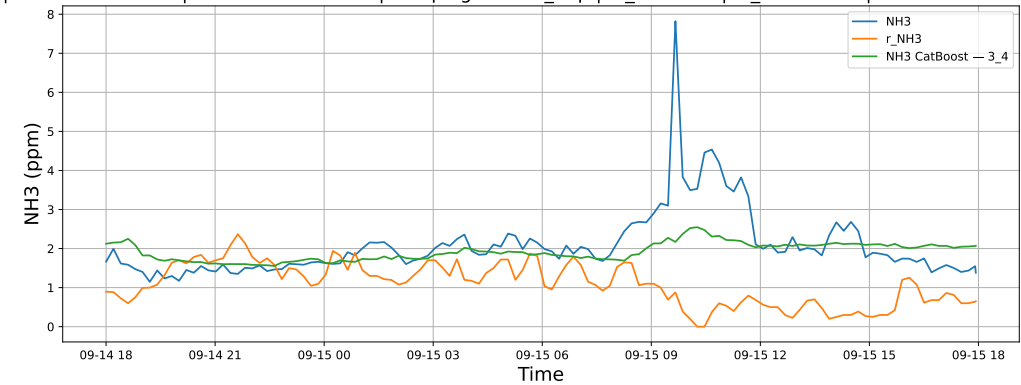
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

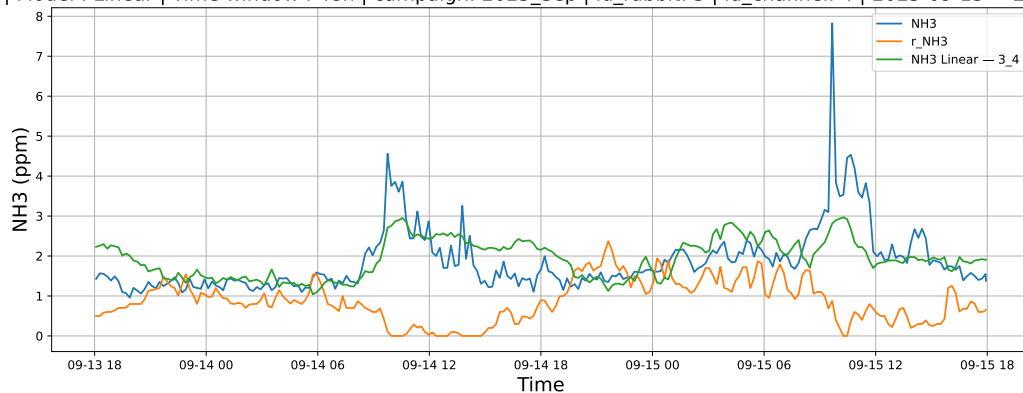


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

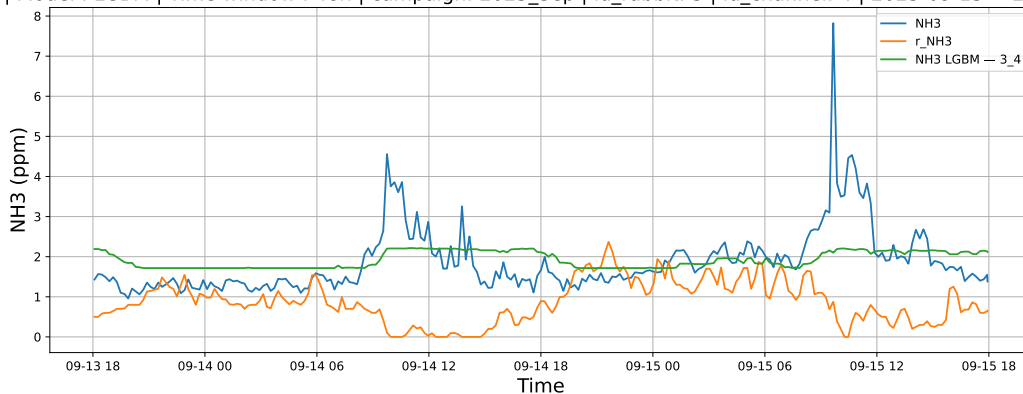


3.1.6.2 Time window : 48

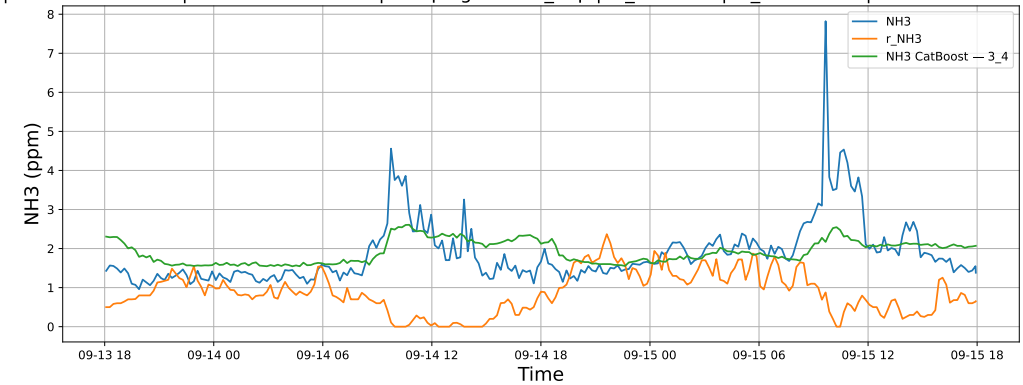
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

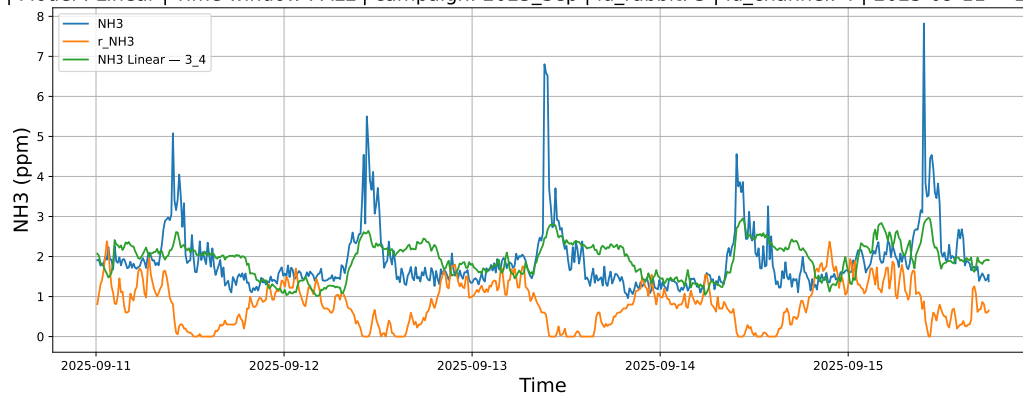


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

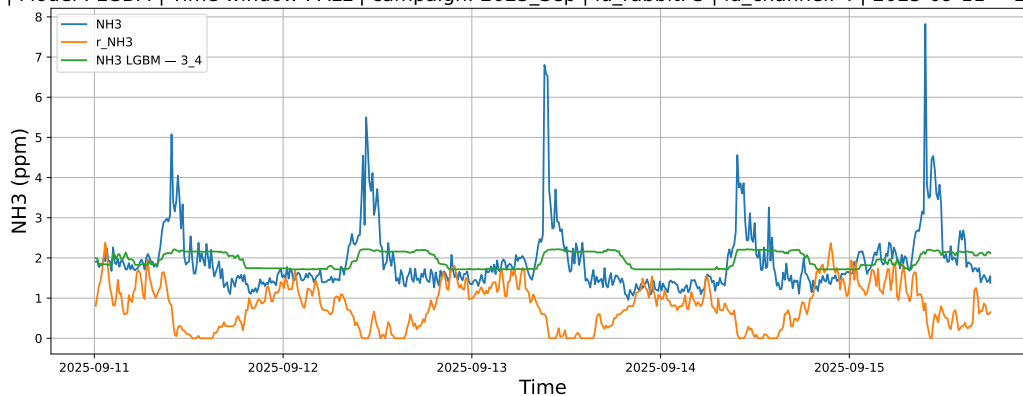


3.1.6.3 Time window : ALL

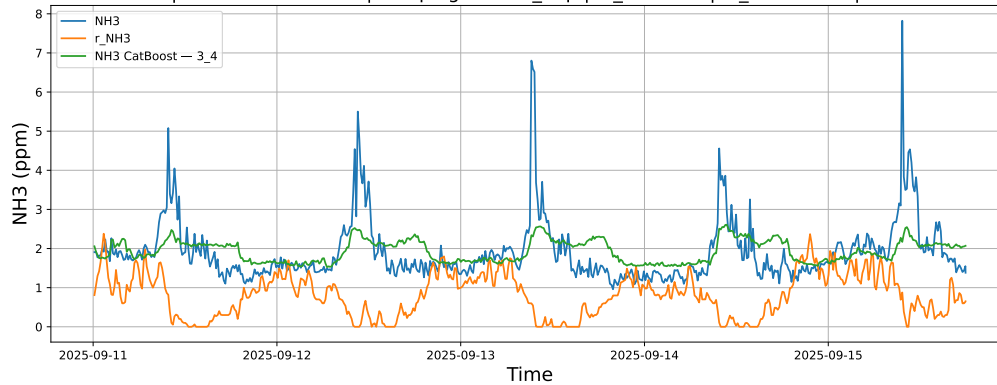
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



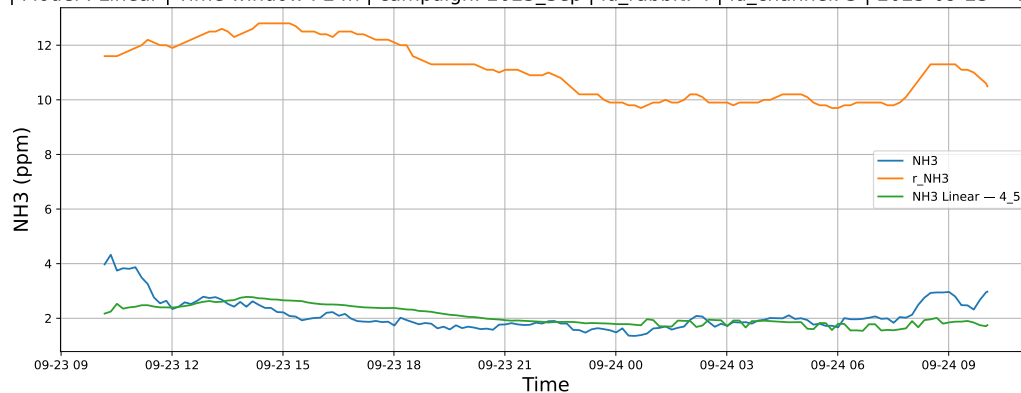
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



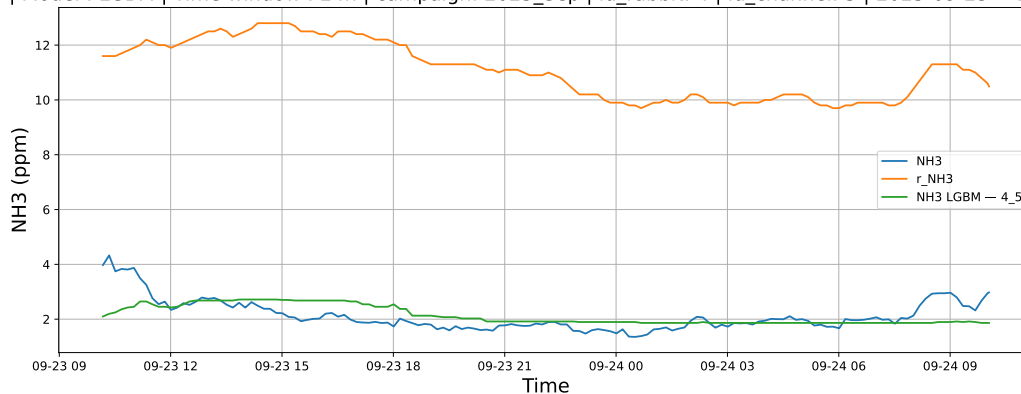
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

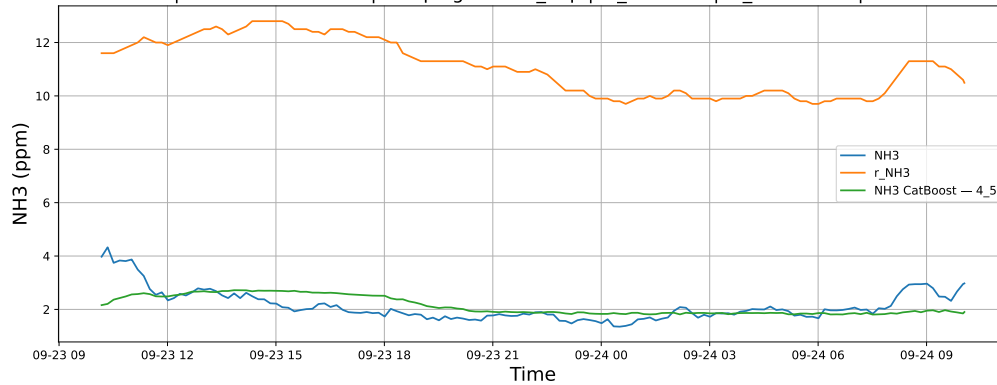
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

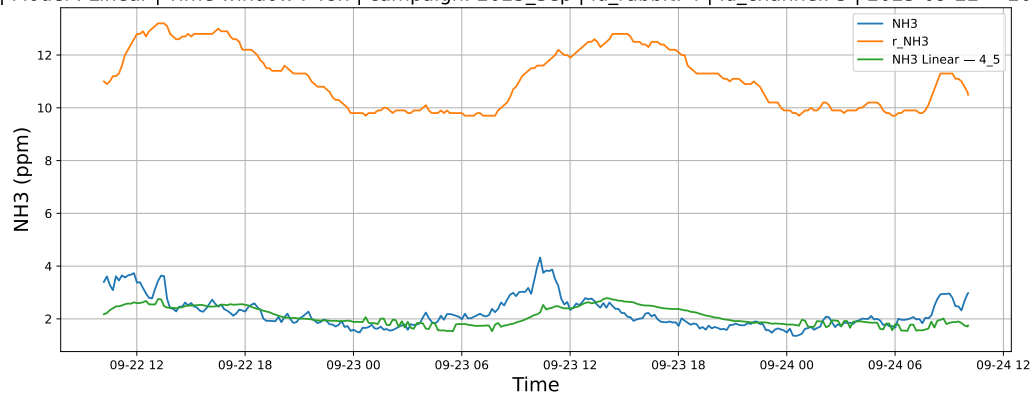


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

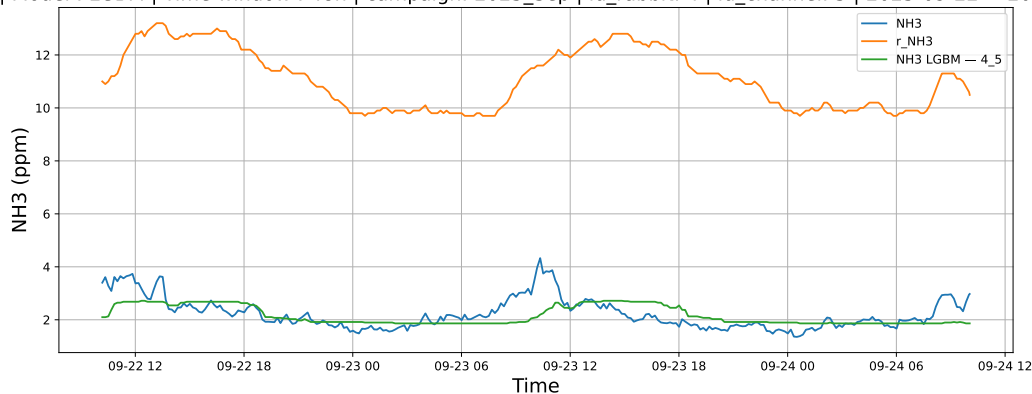


3.1.7.2 Time window : 48

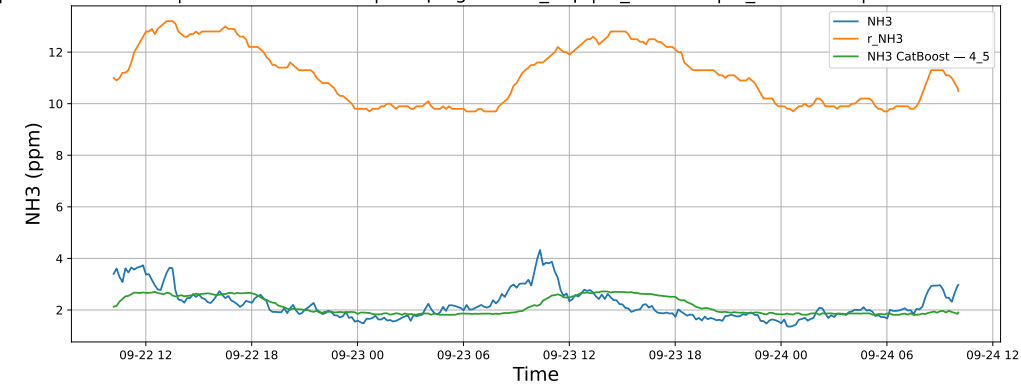
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

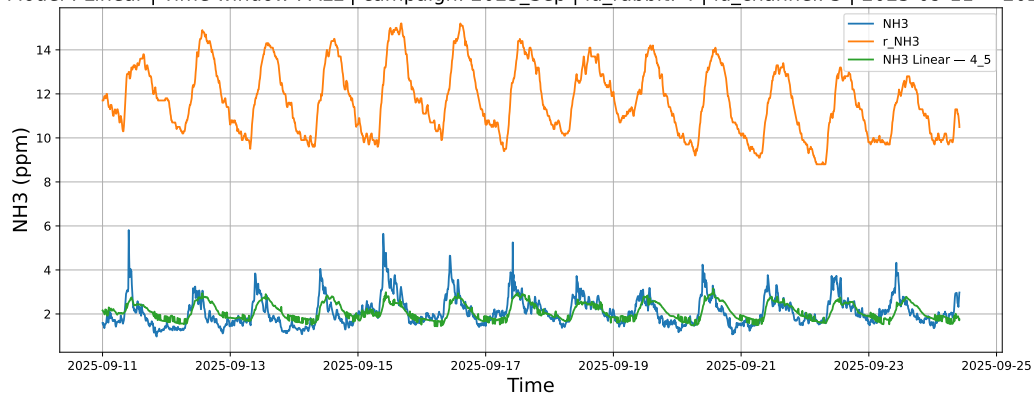


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

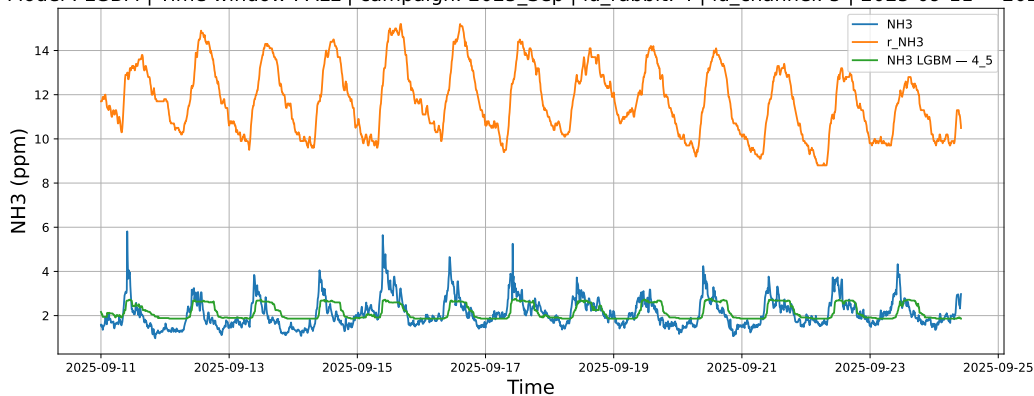


3.1.7.3 Time window : ALL

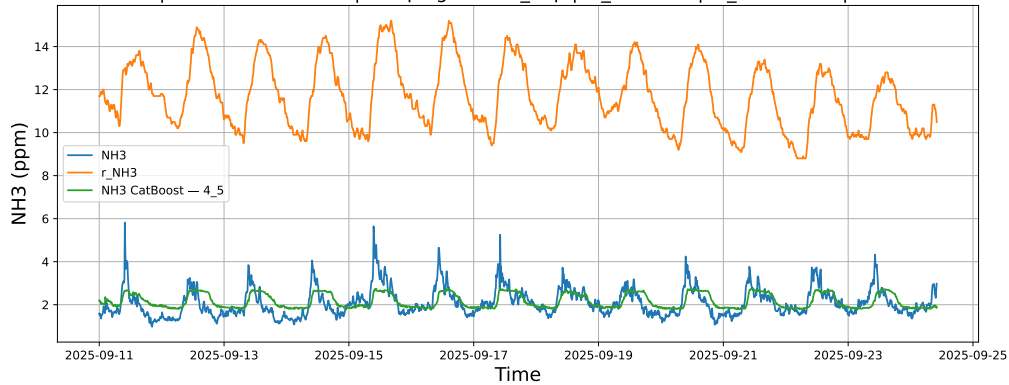
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



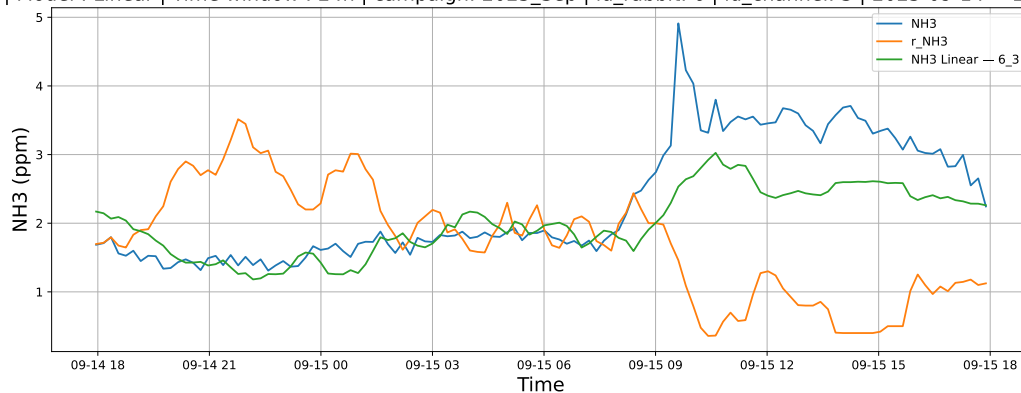
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



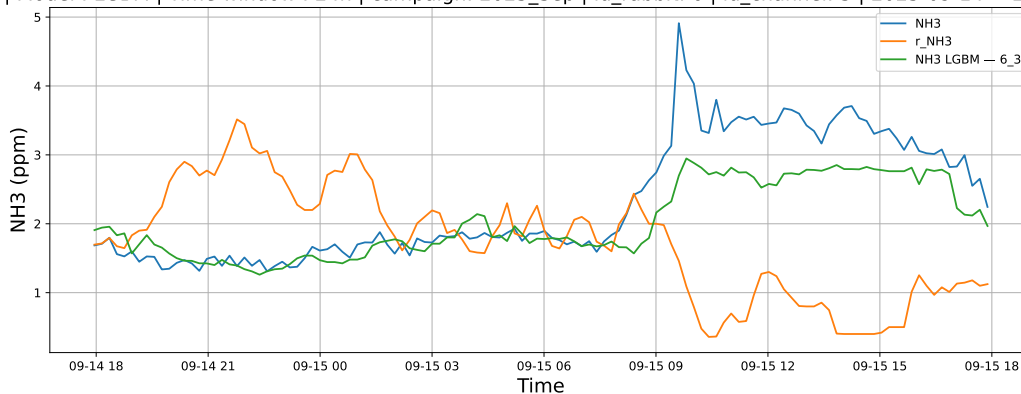
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

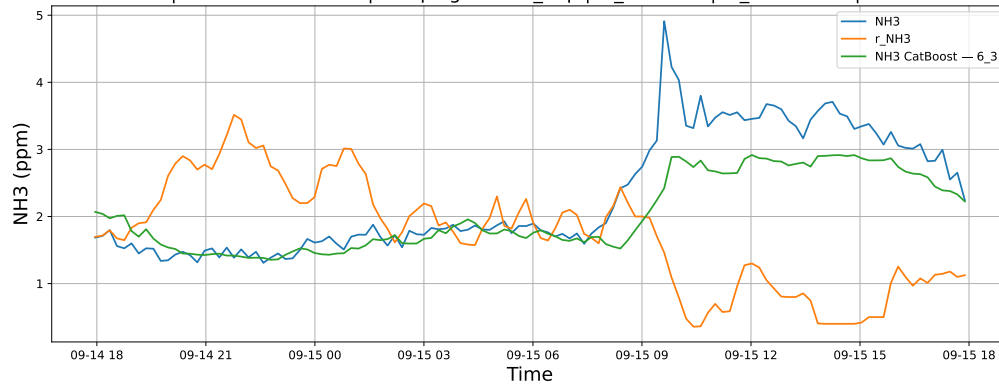
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

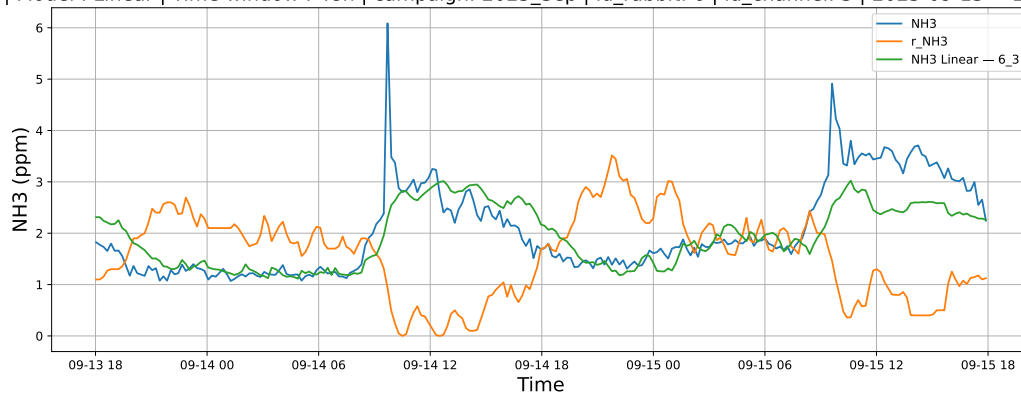


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

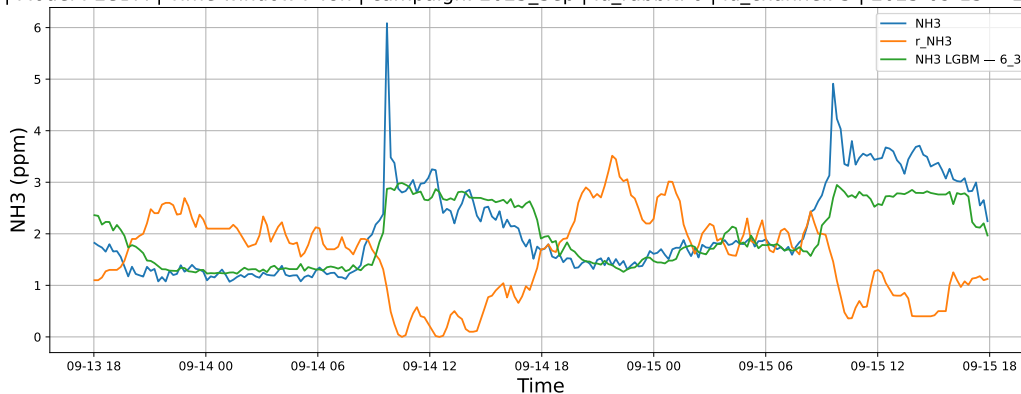


3.1.8.2 Time window : 48

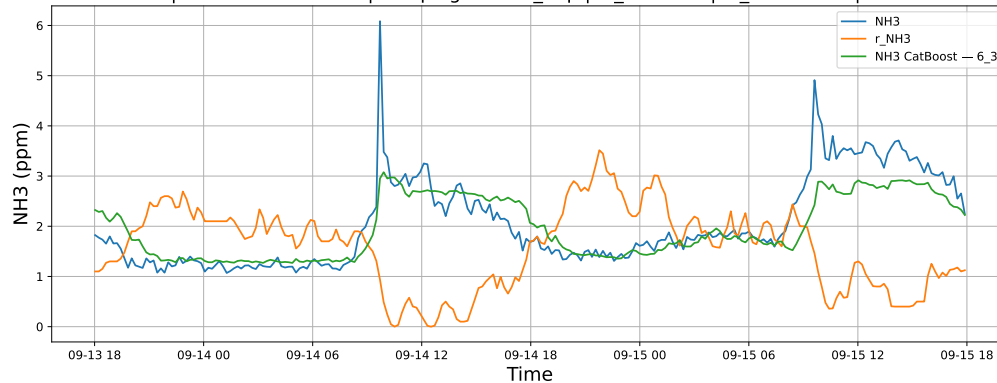
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

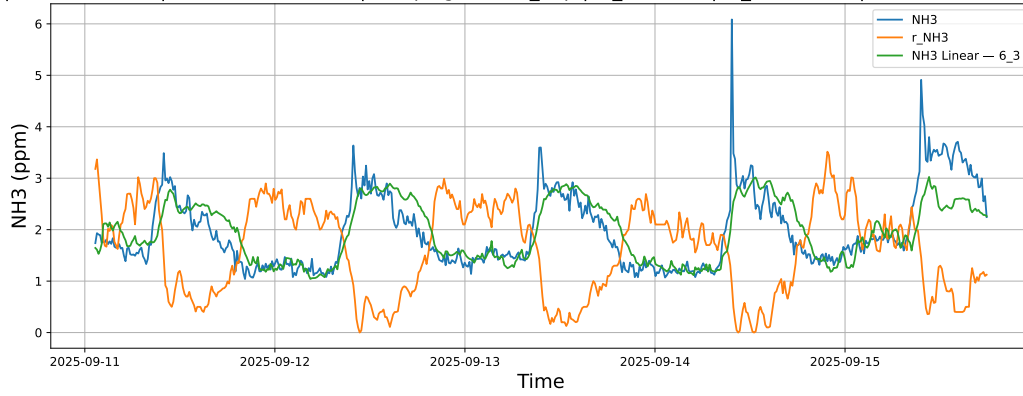


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

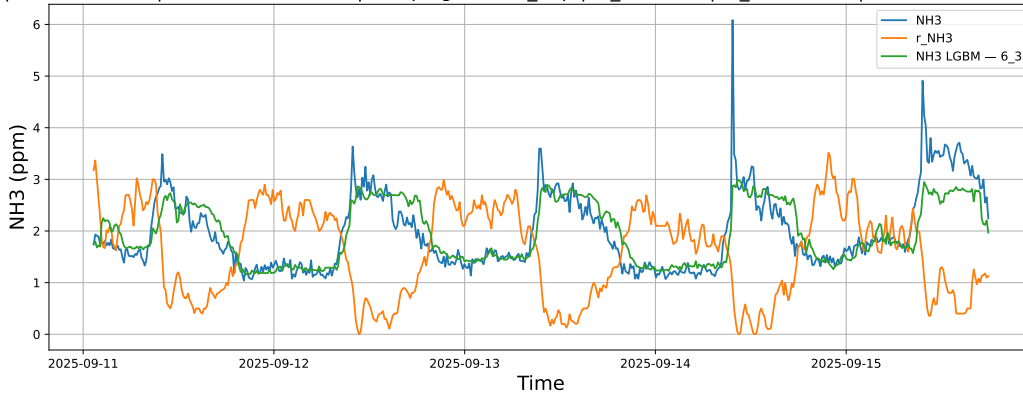


3.1.8.3 Time window : ALL

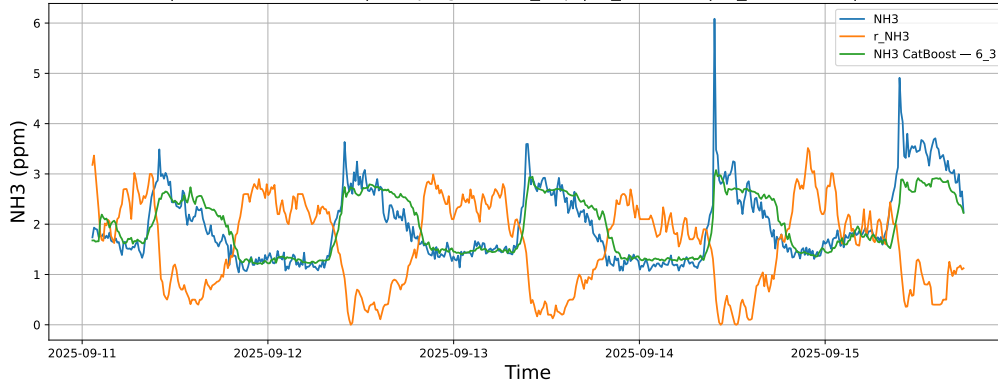
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

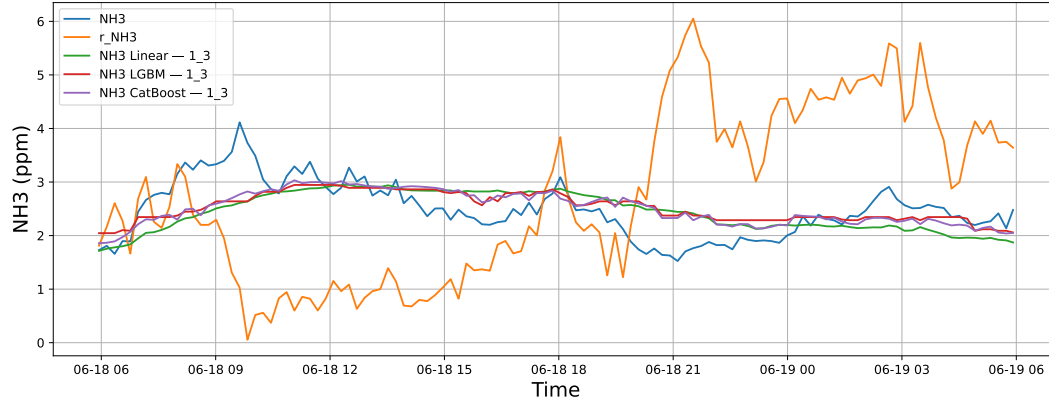


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

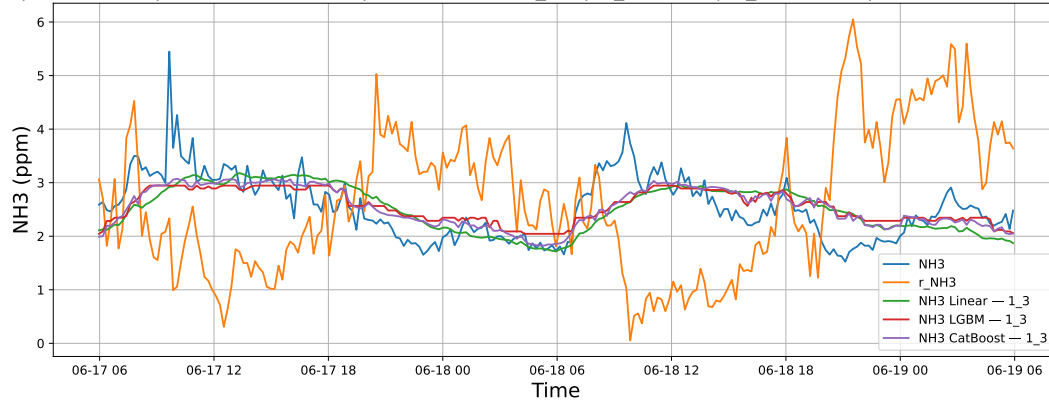
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



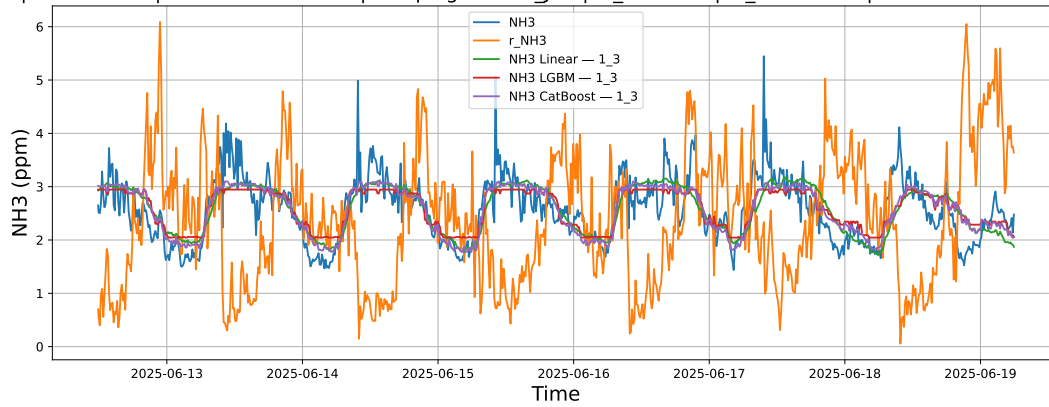
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

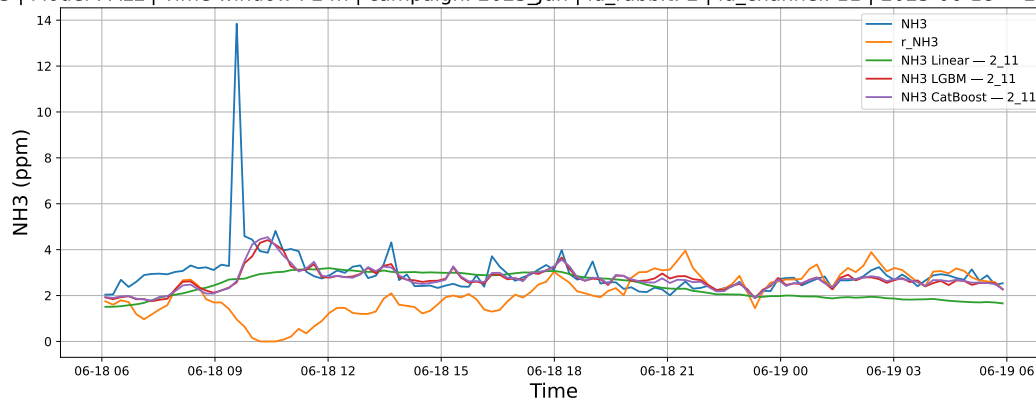
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

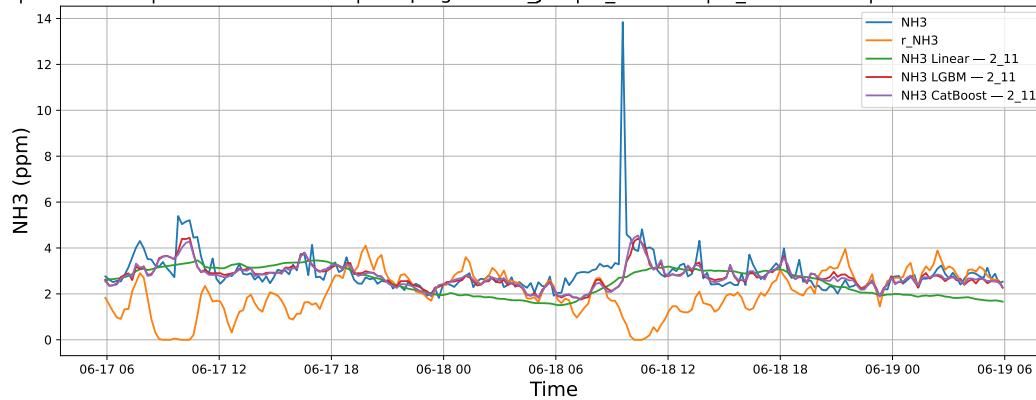
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



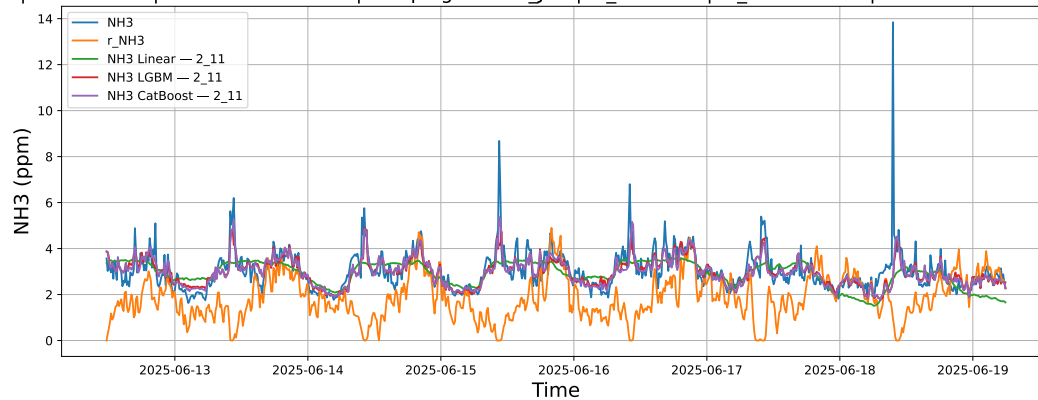
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

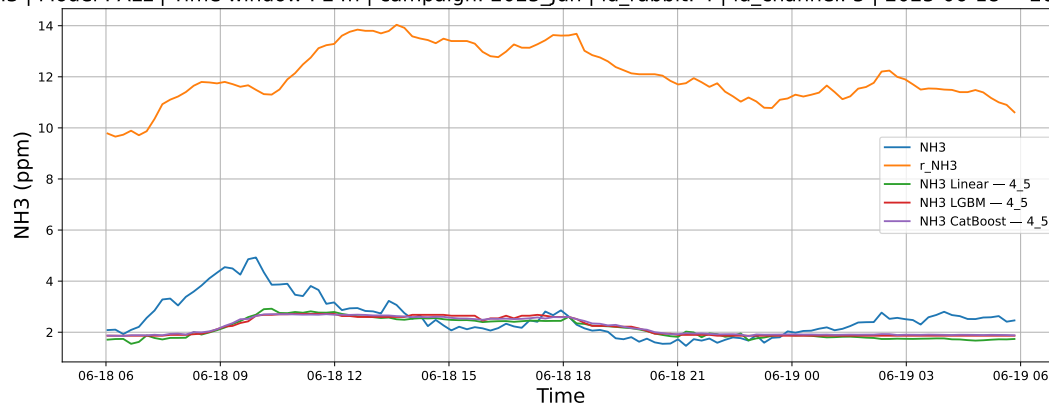
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

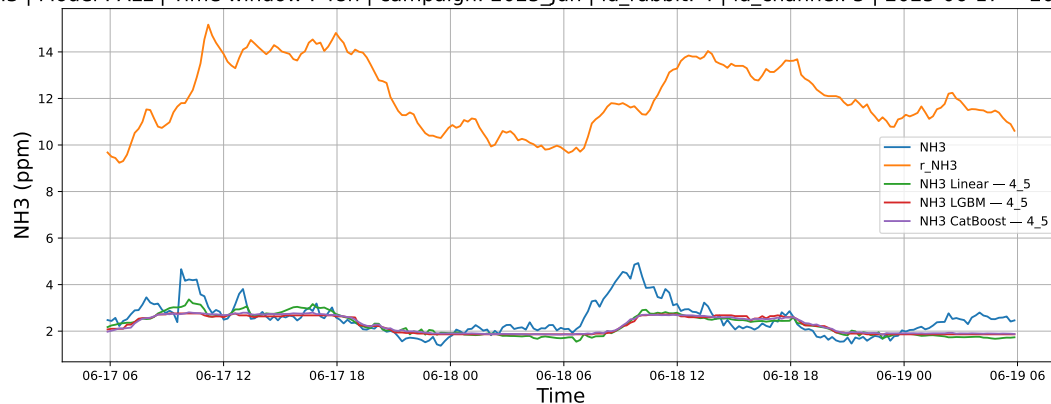
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



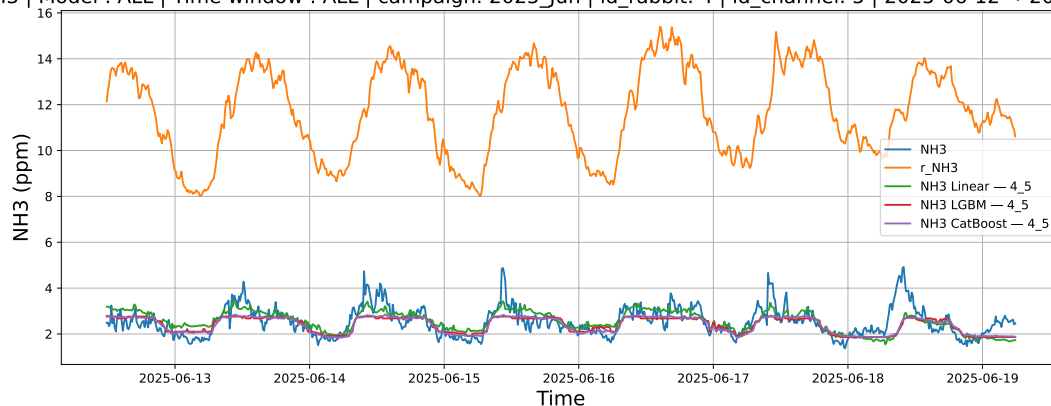
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

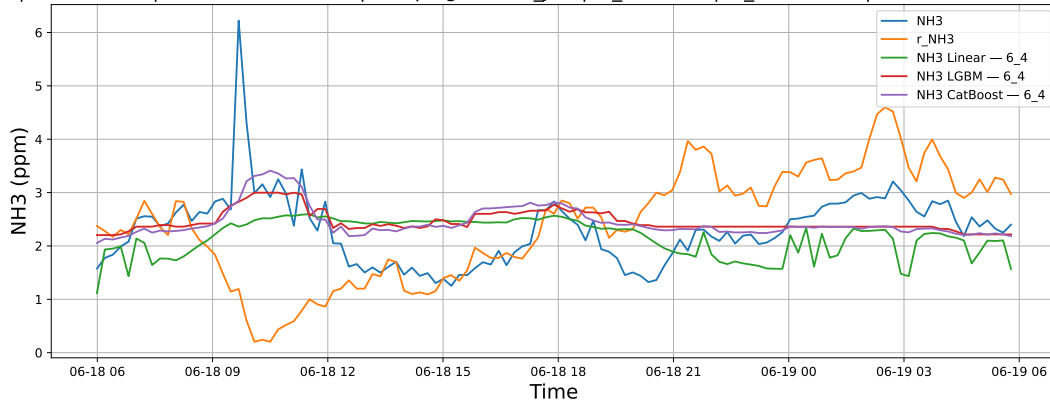
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

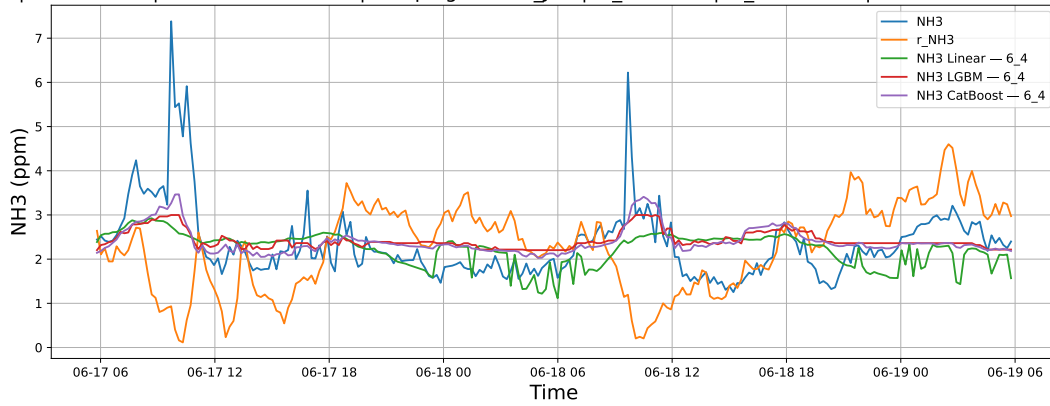
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



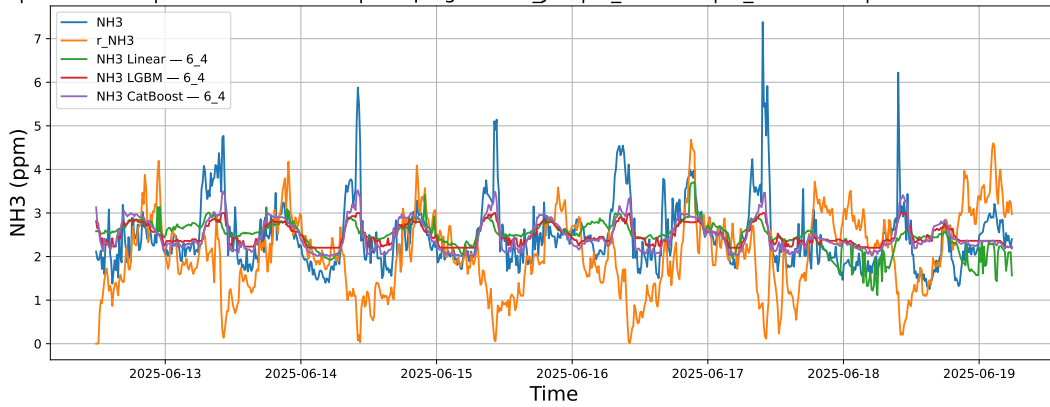
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

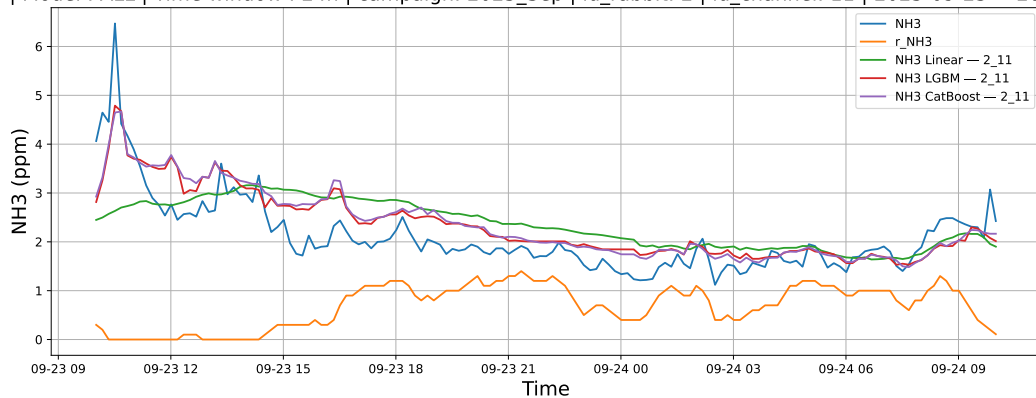
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

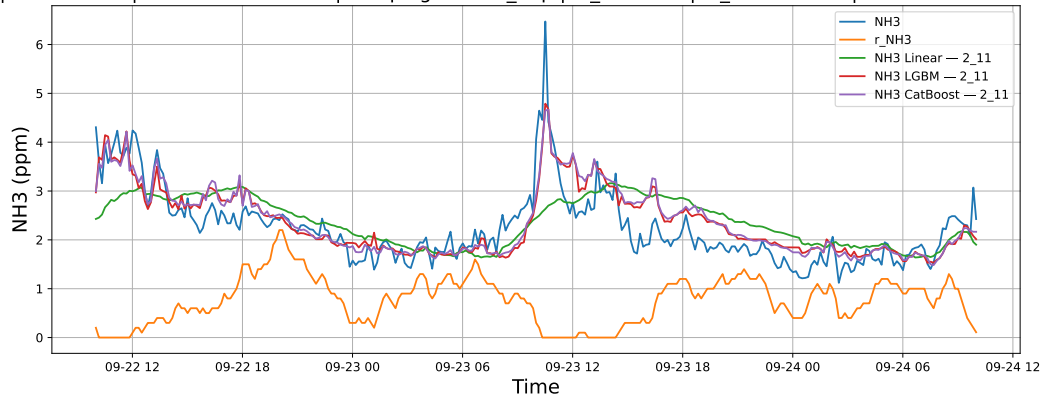
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



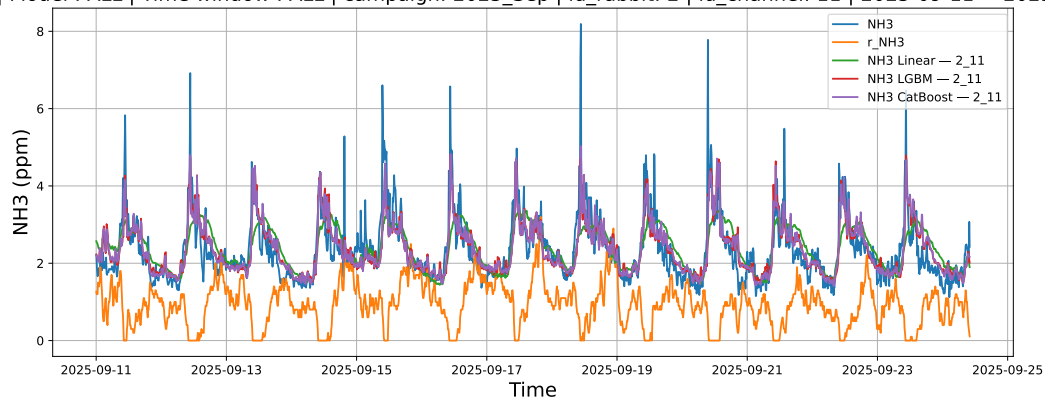
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

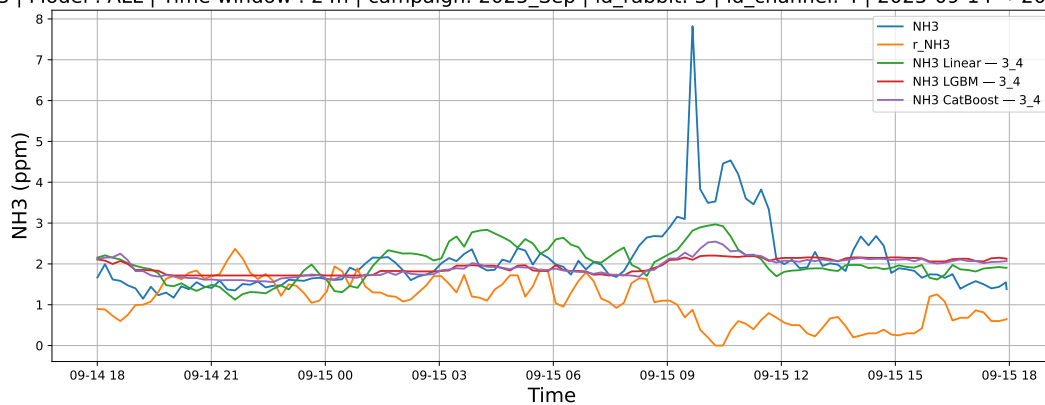
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

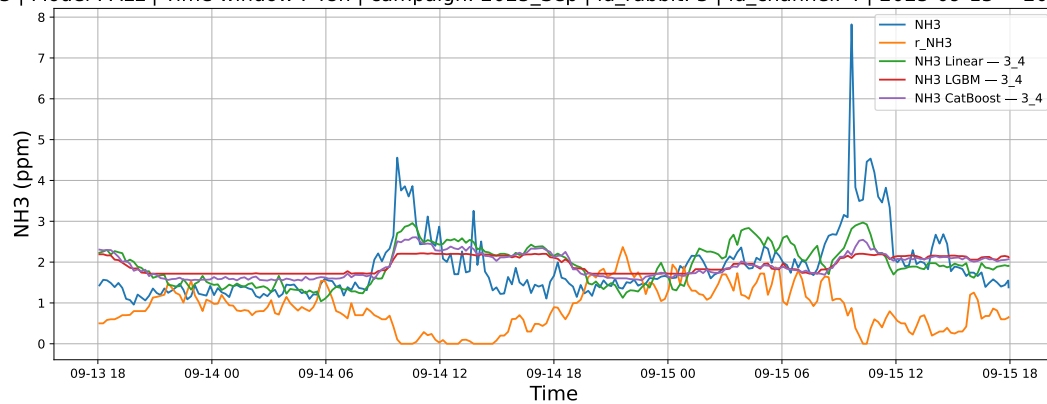
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



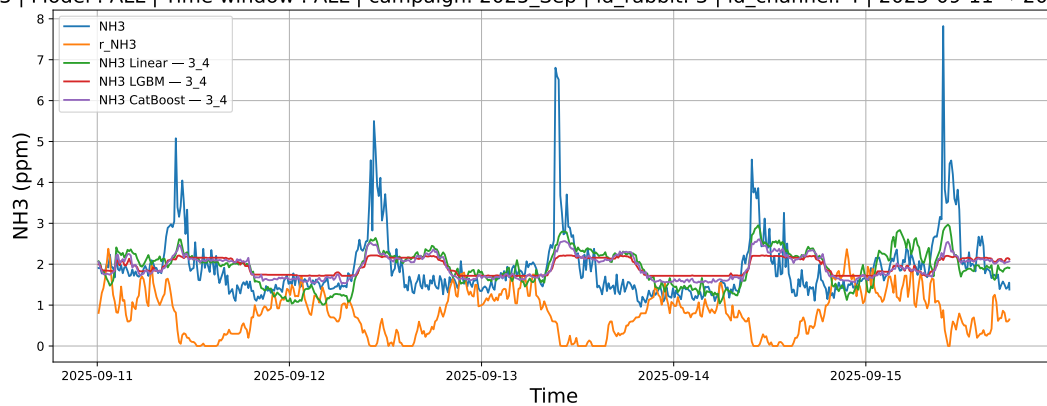
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

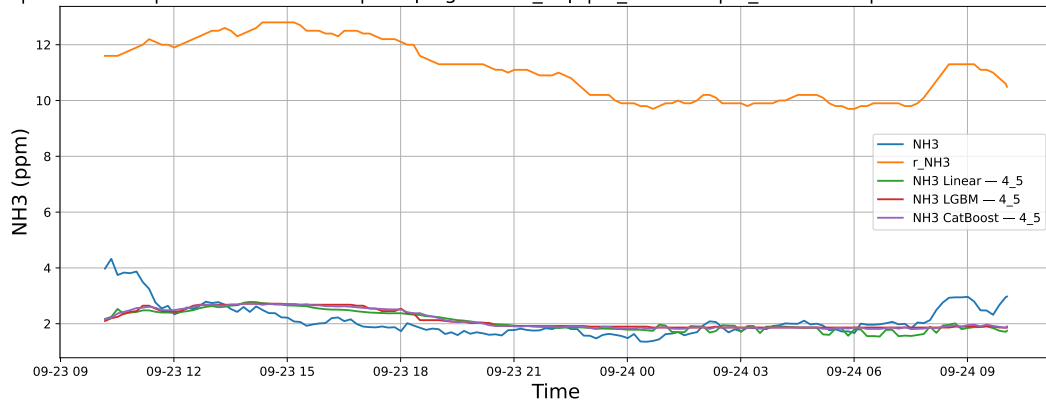
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

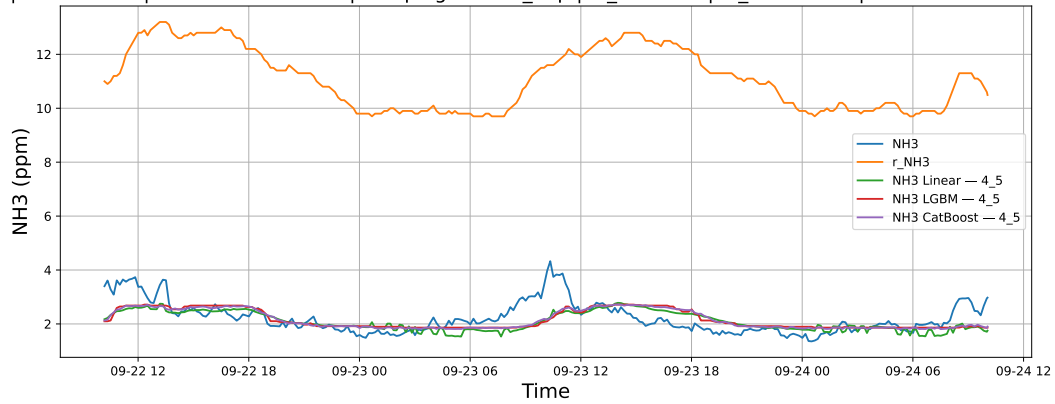
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



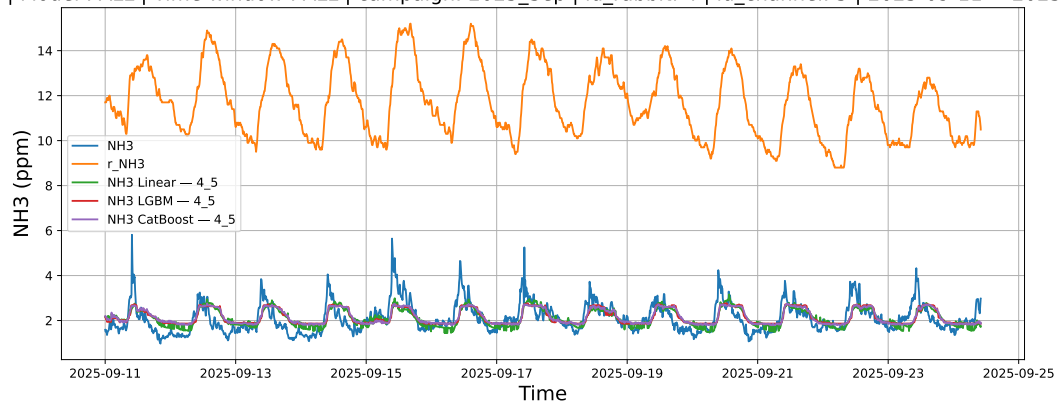
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

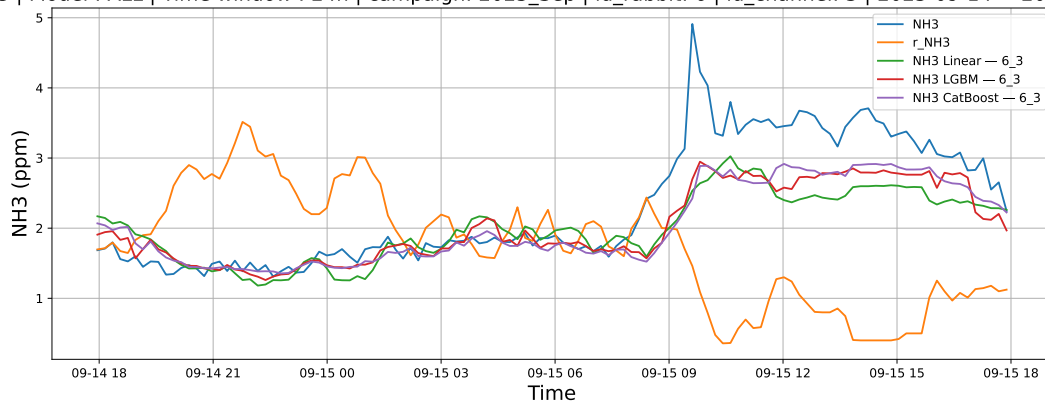
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

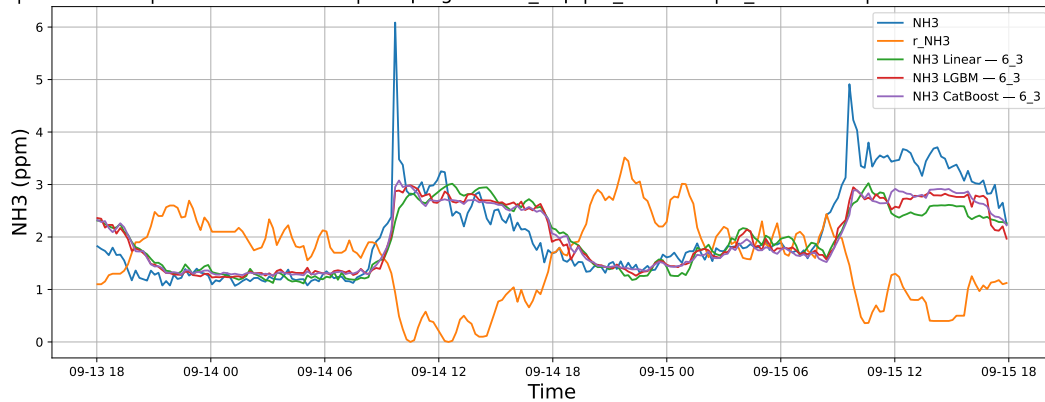
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

